

Tesis Doctoral

Aspects of Lattice Cosmology



VNIVERSITAT
DE VALÈNCIA

preparada por

Nicolás Agenor Loayza Romero

IFIC (Universitat de València – CSIC)

Departamento de Física Teórica

Programa de Doctorado en Física – 3126

València, Julio 2025

Director: Dr. Daniel G. Figueroa

Daniel García Figueroa, Científico Titular del Consejo Superior de Investigaciones Científicas, **certifica**:

Que la presente memoria **Aspects of Lattice Cosmology** ha sido realizada bajo su dirección en el Institut de Física Corpuscular, centro mixto de la Universitat de València y del CSIC, por **Nicolás Agenor Loayza Romero** y constituye su tesis para optar al grado de Doctor en Física.

Y para que así conste, en cumplimiento de la legislación vigente, presenta en el Departamento de Física Teórica de la Universitat de València la referida Tesis Doctoral firmando el presente certificado.

Valencia, a 28 de Julio del 2025,

Daniel G. Figueroa

Agradecimientos

Une oasis d'horreur dans un désert d'ennui!

Charles Baudelaire

Roberto Bolaño iniciaba su gran obra póstuma, *2666*, con la misma cita aquí expuesta. No pienso compararme con él, pero de alguna manera para mí la física es un “oasis de horror” en medio de este “desierto de tedio”. Horror por lo desconocido y arduo de este camino, no sólo por la dureza del mismo, sino porque el destino no está claro. Sin embargo, en el desierto de tedio de nuestra realidad, los momentos de descubrir un concepto, entender un mecanismo u obtener un resultado, son esos pequeños oasis que este camino nos ofrece; oasis como gotas de agua a caminantes sedientos en el abrasivo sol de un verano valenciano.

Quiero empezar agradeciendo a mi supervisor Dani. Gracias por tu confianza desde el inicio, incluso cuando parecía que el COVID quería, a toda costa, que no llegase a Valencia. Gracias por tus enseñanzas, ya sean académicas, pero sobre todo humanas. Gracias por permitirme creer un poco más en mí, aunque aún tengo dudas; y sobre todo, gracias por estar siempre pendiente del aspecto humano en el, a veces, despiadado ambiente académico.

Thanks a lot to all my collaborators, those that have helped me in any way in these last years; and in general those belonging to the *CosmoLattice* community. Thanks to Adrien, Mauro, Catarina, Ander, Joanes (Eskerrik Asko), Jorge, Paco, Ben, Toby, Ken and a lot more people. I am especially grateful to Raphael Flauger for hosting me during my stay at University of California San Diego and being part of the project.

Gracias al IFIC y todos quienes lo conforman; gracias a Nuria, Pilar, Olga y el resto de miembros de SOM y de IFIC. Gracias por esas pequeñas y grandes ayudas que siempre estuvieron dispuestos a brindarnos.

Gracias a mis amigos del IFIC; estudiantes, postdocs y más. Gracias a mis compañeros de SOM e IFIC que pasaron o siguen aquí: gracias a Fernando, Jorge, Drona, Stefan, Carmen, Catarina, Fabian E, Salva, Peera, Pietro, Androniki, Baibhab, Paula, Miryam, Ruth, David B, David R, Melissa, Emanuela, Juliana, Miguel, Ken, Fabian K, Federica, Rasmi, Rebeca, Omar, Gonzalo, Victor, Mariluz y muchísimas personas más que hicieron que la rutina sea menos tediosa. Sobre todo agradezco a aquellos que alguna vez me salvaron la vida, otros que me hicieron parte de la suya y otros que simplemente eran la primera palabra del día que escuchaba; Giacomo, David A y Francesco, muchas gracias por todo.

Gracias a mis amigos, en los cuales Bolívar soñó, JuanSe, Laura y Sharon. Gracias por mostrarme que ser migrante puede ser desgarrador, pero no hay nada que una buena salsa, o un Baile Inolvidable, no pueda arreglar.

Gracias a mis amigos de València, gracias por hacerme sentir parte de la *millor terreta del món* y hacerme conocer la maravillosa diversidad de España; y sobre todo, gracias por ser en ciertos días la vía de escape de esta vida de físico. Gracias a Pablo, Eli, Andrea, Nacho y Beni.

Gracias a mis amigos los físicos de Ecuador, por inspirarme a seguir intentándolo y por hacerme sentir orgulloso de ser parte de una comunidad que la conforman personas como ustedes. Gracias a Joaquín, Camilo, Wilson N, Wilson Y, Raquel, Nico Z, José, Paúl, Anto, Pableinz, Chance, Andy, Pancho, Jenny, Caro, Roberto, Jorge P y todos los que hacen que Ecuador suene ahí donde nadie más ha llegado.

Gracias a mis amigos primordiales, a aquellos con quienes compartí mis primeras ideas de querer ser físico y me apoyaban en las actividades más vergonzosas que se le pueden ocurrir a un adolescente que recientemente había descubierto la física. Gracias por seguir siendo mis amigos después de casi 15 años. Gracias a Diego, Andrés S (Chelo), Andrés F, y gracias a Vanessa y Fabiana por ser parte de este grupo también.

Gracias a mi familia, la que está allá, la que está acá y la que está donde pueda, porque los ecuatorianos somos sobre todo **migrantes**, porque no importa qué tan lejos crezcan las ramas, el árbol siempre sabe dónde está su raíz. Gracias a mis padres, Leonardo y Saida, por hacer de mí la mezcla justa de curiosidad y disciplina para haber logrado esto. Gracias a mis hermanos, a Estefanía por ser literal y figurativamente el espejo en el que me reflejo, mostrándome que se puede; y Agustín por darme esperanzas de que las cosas pueden ser mejores, o al menos ver campeón a Barcelona una vez más. Gracias a mis abuelos, Jorge y Gumercindo, que aunque no estén aquí con nosotros, y a mis abuelas, Rosa y Mélida; espero esto sea motivo de orgullo para todos quienes hacen parte de este linaje piñasiense.

Gracias a mi familia política: Tiyana, Ñusti, Mónica, *Mashi* José y la Abuelita Carmen por también ser parte de este camino. *Gracias... Kanpak ayllupi kimichishkamanta, shinallatak maykunapipash mana rikunayachikpipash, karilla shayarina kashkata rikuchishkamanta tukuy shunkuwan pagui ninimi.*

Finalmente, gracias a Tamia, mi compañera, amiga, confidente y soporte. Gracias por apoyarme y sostenerme. Gracias por estar ahí, y sobre todo gracias por haber tomado la tan difícil decisión de dejar todo atrás para acompañarme en este viaje. Si la física son gotas de rocío que alimentan el oasis de horror, tú eres la lluvia torrencial que mantiene verde mi oasis de paz en este desierto de tedio. *Tamia bonita... Kuyaylla Tamia! Kanpak Kichwa shimipi kay shimikunami ñuka kanmanta karilla kashkata rikuchin. Kanta kuyanimi.*

Perdóname lector
Amistoso lector
Que no me pueda despedir de ti
Con un abrazo fiel:
Me despido de ti
con una triste sonrisa forzada.

Nicanor Parra

Abstract

The early universe is a largely unexplored period in the history of the cosmos. While current observations provide a detailed picture of the late-time universe, many details of the first moments of our universe remain unknown. However, observations from the cosmic microwave background (CMB) supports that the universe began with a phase of accelerated expansion known as inflation. The inflationary paradigm not only solves certain puzzles of the cosmological properties of the universe, but also provides a natural mechanism to source primordial fluctuations that seed the large scale structures in the universe. Eventually, the energy driving inflation must be transferred to relativistic degrees of freedom, initiating the thermal era of the universe. This transition is known as the reheating period, and it may encompass complex dynamics involving non-equilibrium processes and nonlinear interactions. Furthermore, during these first stages of our universe, additional phenomena could take place, like phase transitions and the creation of topological defects. Traditionally, our understanding of the early universe is based on perturbative methods, sometimes relying on strong assumptions like linear dynamics. It is, however, possible that this era was governed by highly nonlinear phenomena such as explosive particle production or field fragmentation, all of which require numerical tools for accurate modeling. These processes are not just dynamically rich, but may also represent efficient sources of gravitational waves (GWs), which are unique messengers of the early universe, carrying information about high-energy physics at scales unreachable by terrestrial experiments. These backgrounds offer a complementary probe of the physics of the early universe at epochs much earlier than CMB decoupling.

In this thesis, we show the necessity to rely on lattice field theory simulations to study the nonlinear dynamics of inflationary and post-inflationary scenarios. We use the *CosmoLattice* code, which allows to evolve interacting scalar and gauge fields in an expanding universe while including nonlinear matter interactions and linear metric perturbations (GWs). The use of lattice simulations is essential to capture the full nonlinear regime of the field dynamics, particularly in scenarios where the backreaction of produced particles on the background becomes significant. Our simulations are performed in 3+1 dimensions and enable the computation of physical observables such

as the power spectrum of gravitational waves, or to follow the energy budget of all fields involved.

We find that the gravitational wave backgrounds (GWB) are sensitive to the details of particle production and can be significantly enhanced in the presence of non-linear phenomena. This sensitivity opens the possibility of using GWB observations to constrain the underlying particle couplings and field content of the early universe. For instance, we explore how different types of couplings could leave distinct imprints in the GWB. Moreover, we demonstrate that efficient energy transfer to daughter fields can occur purely through nonlinear effects, completing the reheating process even in the absence of perturbative decay or direct particle coupling, which is the case of a non-minimal coupling to gravity. We also extend our analysis to axion inflation models with gauge field production, a class of theories that leads to strong excitation of chiral gauge fields due to the interaction of the inflaton with $F\tilde{F}$. In this context, we argue that lattice simulations are not only helpful but necessary to account for the spatial gradients of the inflaton and the local nature of the gauge field instabilities. Our results clarify the impact of these nonlinear effects on the dynamics and discuss the impact it could have on the correct phenomenological consequences of this scenario. Beyond gravitational waves, we discuss that scalar (and in certain scenarios also vector) cosmological perturbations are also generated during the same epochs, which could serve as complementary probes. As an ongoing effort, we present a formalism that aims to follow the production of all cosmological perturbations (scalar, vector, tensors) from the non-linear physics of the early universe.

Our work opens new avenues for probing the physics of the early universe through simulations, enabling precise theoretical predictions in regimes where analytical techniques fail. It provides a robust and flexible computational tool for exploring a wide class of inflationary and post-inflationary scenarios, deepening our understanding of how the universe evolved from a primordial vacuum state to the complex structures we observe today.

Resumen

El universo primitivo es un período en gran medida inexplorado en la historia del cosmos. Aunque las observaciones actuales proporcionan una imagen detallada del universo en épocas tardías, los primeros momentos de nuestro universo siguen siendo desconocidos. Sin embargo, observaciones del fondo cósmico de microondas (CMB por sus siglas en inglés) apoyan la idea que el universo comenzó con una fase de expansión acelerada conocida como inflación. El paradigma inflacionario no solo resuelve ciertos enigmas de las propiedades cosmológicas del universo, sino que también proporciona un mecanismo natural para generar fluctuaciones primordiales que hacen de semillas para la formación de estructuras. Eventualmente, toda la energía que impulsaba la inflación debe transferirse a los grados de libertad relativistas, iniciando la era térmica del universo. Esta transición es conocida como el período de recalentamiento y abarca dinámicas complejas que involucran procesos fuera del equilibrio e interacciones no lineales. Además, durante estas primeras etapas del universo, podrían tener lugar fenómenos adicionales, como transiciones de fase y la creación de defectos topológicos. Tradicionalmente, nuestra comprensión del universo temprano se basa en métodos perturbativos, que a veces dependen de suposiciones fuertes. Es, de hecho, posible que esta era estuviera gobernada por fenómenos altamente no lineales, como la producción explosiva de partículas o la fragmentación de campos, los cuales requieren herramientas no perturbativas para una modelización precisa. Estos procesos no solo son dinámicamente ricos, sino que también constituyen fuentes eficientes de ondas gravitacionales (GWs por sus siglas en inglés), las cuales son mensajeras únicas del universo primitivo, que transportan información sobre física de altas energías a escalas inalcanzables para los experimentos terrestres. Estos fondos constituyen una sonda complementaria de la física del universo primitivo a épocas mucho más tempranas que la recombinación, del cual la imagen más temprana que podemos observar son las anisotropías de temperatura del CMB.

En esta tesis mostramos la necesidad de recurrir a simulaciones de teoría de campos en redes tridimensionales para estudiar la dinámica no lineal de escenarios inflacionarios y post-inflacionarios. Utilizamos el código *CosmoLattice*, que permite evolucionar

campos escalares y de gauge interactuantes en un universo en expansión, incluyendo interacciones no lineales de la materia y perturbaciones métricas lineales (GWs) . El uso de simulaciones en redes tridimensionales es esencial para capturar el régimen completamente no lineal, particularmente en escenarios donde la retroalimentación de las partículas producidas sobre el fondo se vuelve significativa. Nuestras simulaciones se realizan en 3+1 dimensiones y permiten el cálculo de observables físicos como el espectro de ondas gravitacionales, o seguir el presupuesto de energía de todo los campos involucrados.

Encontramos que la señal de los fondos de ondas gravitacionales (GWB por sus siglas en inglés) es sensible a los detalles de la producción de partículas y puede ser significativamente amplificada en presencia de resonancias fuertes. Esta sensibilidad abre la posibilidad de usar observaciones de ondas gravitacionales para restringir los acoplamientos entre partículas y el contenido de campos del universo temprano. Por ejemplo, exploramos cómo distintos tipos de acoplamientos podrían dejar huellas distintivas en el espectro de ondas gravitacionales. Además, demostramos que una transferencia eficiente de energía hacia los campos secundarios puede ocurrir puramente mediante efectos no lineales, completando el proceso de recalentamiento incluso en ausencia de decaimientos perturbativos o acoplamientos directos entre partículas, como en el caso de un acoplamiento no mínimo con la gravedad. También extendemos nuestro análisis a modelos de inflación tipo axión con producción de campos de gauge, una clase de teorías que conduce a una fuerte excitación de campos de gauge quirales debido a la interacción del inflatón con $F\tilde{F}$. En este contexto, argumentamos que las simulaciones en redes tridimensionales no solo son útiles, sino necesarias para describir los gradientes espaciales del inflatón y la naturaleza local de las inestabilidades de los campos de gauge. Nuestros resultados aclaran el impacto de estos efectos no lineales en la dinámica y discuten el impacto que podrían tener sobre las consecuencias fenomenológicas correctas de este escenario. Más allá de las ondas gravitacionales, discutimos que también se generan perturbaciones cosmológicas escalares y vectoriales durante las mismas épocas, las cuales podrían servir como sondas complementarias. Como trabajo futuro, presentamos un formalismo que busca seguir la producción de todas las perturbaciones cosmológicas (escalares, vectoriales, tensoriales) a partir de la física no lineal del universo temprano.

Nuestro trabajo abre nuevas vías para explorar la física del universo primitivo mediante simulaciones, permitiendo predicciones teóricas precisas en regímenes donde fallan las técnicas analíticas. Proporciona una herramienta computacional robusta y flexible para explorar una amplia clase de escenarios inflacionarios y postinflacionarios, profundizando nuestra comprensión de cómo el universo evolucionó desde un estado de vacío primordial hasta las estructuras complejas que observamos hoy en día.

Resum

L'univers primitiu és un període en gran manera inexplorat en la història del cosmos. Encara que les observacions actuals proporcionen una imatge detallada de l'univers en èpoques tardanes, els primers moments del nostre univers continuen sent desconeguts. No obstant això, observacions del fons còsmic de microones (CMB per les seues sigles en anglés) donen suport a la idea que l'univers va començar amb una fase d'expansió accelerada coneguda com a inflació. El paradigma inflacionari no sols resol uns certs enigmes de les propietats cosmològiques de l'univers, sinó que també proporciona un mecanisme natural per a generar fluctuacions primordials que fan de llavors per a la formació d'estructures. Eventualment, tota l'energia que impulsava la inflació ha de transferir-se als graus de llibertat relativistes, iniciant l'era tèrmica de l'univers. Esta transició és coneguda com el període de recalfament i abasta dinàmiques complexes que involucren processos fora de l'equilibri i interaccions no lineals. A més, durant estes primeres etapes de l'univers, podrien tindre lloc fenòmens addicionals, com a transicions de fase i la creació de defectes topològics. Tradicionalment, la nostra comprensió de l'univers d'hora es basa en mètodes perturbatius, que a vegades depenen de suposicions fortes. És, de fet, possible que esta era estiguera governada per fenòmens altament no lineals, com la producció explosiva de partícules o la fragmentació de camps, els quals requereixen ferramentes no perturbatives per a una modelització precisa. Estos processos no sols són dinàmicament rics, sinó que també constitueixen fonts eficients d'ones gravitacionals (GWs per les seues sigles en anglés), les quals són missatgeres úniques de l'univers primitiu, que transporten informació sobre física d'altres energies a escales inassolibles per als experiments terrestres. Estos fons constitueixen una sonda complementària de la física de l'univers primitiu a èpoques molt més primerenques que la recombinació, del qual la imatge més primerenca que podem observar són les anisotropies de temperatura del CMB.

En esta tesi vam mostrar la necessitat de recórrer a simulacions de teoria de camps en xarxes tridimensionals per a estudiar la dinàmica no lineal d'escenaris inflacionaris i post-inflacionaris. Utilitzem el codi *CosmoLattice*, que permet evolucionar camps escalars i de gauge interactuants en un univers en expansió, incloent-hi interaccions

no lineals de la matèria i pertorbacions mètriques lineals (GWs) . L'ús de simulacions en xarxes tridimensionals és essencial per a capturar el règim completament no lineal, particularment en escenaris on la retroalimentació de les partícules produïdes sobre el fons es torna significativa. Les nostres simulacions es realitzen en 3+1 dimensions i permeten el càlcul d'observables físics com l'espectre d'ones gravitacionals, o seguir el pressupost d'energia de tot els camps involucrats.

Troblem que el senyal dels fons d'ones gravitacionals (GWB per les seues sigles en anglés) és sensible als detalls de la producció de partícules i pot ser significativament amplificada en presència de ressonàncies fortes. Esta sensibilitat obri la possibilitat d'usar observacions d'ones gravitacionals per a restringir els acoblaments entre partícules i el contingut de camps de l'univers primerenc. Per exemple, explorem com diferents tipus d'acoblaments podrien deixar petjades distintives en l'espectre d'ones gravitacionals. A més, vam demostrar que una transferència eficient d'energia cap als camps secundaris pot ocórrer purament mitjançant efectes no lineals, completant el procés de recalfament fins i tot en absència de decaïments perturbatius o acoblaments directes entre partícules, com en el cas d'un acoblament no mínim amb la gravetat. També estenem la nostra anàlisi a models d'inflació tipus axió amb producció de camps de gauge, una classe de teories que conduïx a una forta excitació de camps de gauge quirals a causa de la interacció del inflatón amb $F\tilde{F}$. En este context, argumentem que les simulacions en xarxes tridimensionals no sols són útils, sinó necessàries per a descriure els gradients espacials del inflatón i la naturalesa local de les inestabilitats dels camps de gauge. Els nostres resultats aclariexen l'impacte d'estos efectes no lineals en la dinàmica i discutixen l'impacte que podrien tindre sobre les conseqüències fenomenològiques correctes d'este escenari. Més enllà de les ones gravitacionals, discutim que també es generen pertorbacions cosmològiques escalars i vectorials durant les mateixes èpoques, les quals podrien servir com a sondes complementàries. Com a treball futur, presentem un formalisme que busca seguir la producció de totes les pertorbacions cosmològiques (escalars, vectorials, tensorials) a partir de la física no lineal de l'univers primerenc.

El nostre treball obri noves vies per a explorar la física de l'univers primitiu mitjançant simulacions, permetent prediccions teòriques precises en règims on fallen les tècniques analítiques. Proporciona una ferramenta computacional robusta i flexible per a explorar una àmplia classe d'escenaris inflacionaris i postinflacionaris, aprofundint la nostra comprensió de com l'univers va evolucionar des d'un estat de buit primordial fins a les estructures complexes que observem hui dia.

Preface

Our current observations of the universe at cosmological scales suggest that it is isotropic and homogeneous to a large degree. The leading paradigm to explain such properties is inflation, which posits that the universe underwent a phase of exponential expansion approximately 13.8 billion years ago. The earliest image of the universe to which we have direct access through observations is the Cosmic Microwave Background (CMB). Several questions have been raised regarding the properties of the universe at cosmological scales, among the most notable issues are the flatness problem, the cosmological horizon problem, which is related to the observed correlations of perturbations that could not have been in causal contact. Moreover, the inflationary paradigm not only provides a solution to these issues but also provides a natural framework for generating the matter perturbations that eventually led to the formation of large scale structures in the universe. In general, we refer to the events occurring during this era, between inflation and the onset of the thermal era, as the **Early Universe**. Despite its success in explaining several features of the cosmos, inflation remains a hypothetical framework rather than an established fact. The primary challenge lies in its particle physics realization, in other words how to connect inflation with the known physics of the Standard Model. Many models have been proposed, yet current observations are not precise enough to single out any particular one. Furthermore, most realizations of inflation rely on quantum field theory on a curved background.

Additionally, the success of the theoretical framework of Big Bang Nucleosynthesis (BBN) in explaining the observed abundance of light elements in the universe requires a specific setup. In particular, certain proportions on the lightest Standard Model (SM) degrees of freedom must be present to accurately predict these abundances in agreement with observations. At these energy scales all radiation-like active SM degrees of freedom are in thermal equilibrium. This raises the immediate question of how to connect the inflationary phase with such thermal era of the universe. Specifically, if inflation is driven by one or more degrees of freedom during the early universe, how is the energy stored in these fields transferred to the relativistic species driving the thermal era of the universe? This transition phase, known as reheating, remains largely unconstrained,

as it is difficult to probe it directly. As previously mentioned, the earliest image of the universe we have access to is the CMB, which was formed when the free protons and electrons formed hydrogen atoms and photons began to stream freely, an event that occurred approximately 380,000 years after the end of inflation. Early mechanisms proposed for reheating were based on the perturbative decays of the inflaton into another species. However, it was later understood that non-perturbative phenomena could also be relevant to describe the energy transfer, giving rise to a phase known as preheating.

The early universe is not limited to the phases of inflation and reheating. In particular, phase transitions and the creation of topological defects may have marked key moments in the early universe and offer a window into physics beyond the SM. In this context, one might ask: How can we observe these events if the universe was opaque to electromagnetic radiation? A probable answer lies in Gravitational Waves (GWs). The advent of gravitational wave astronomy, marked by the first-ever direct detection of GWs in 2015 by the LIGO/VIRGO collaboration, has opened vast new possibilities for exploring the universe through GWs. It has been shown that certain phenomena potentially occurring in the early universe could generate large amounts of gravitational radiation, which in principle would reach us today as a GW background coming from all directions. Moreover, the recent observations from Pulsar Timing Array (PTA) provides the first evidence on the potential existence of a gravitational wave background, which are placed at low-frequency regime (nHz order). The origin of these signal, whether astrophysical or cosmological, remains to be seen. Future observatories aim to probe many other regions in the GW frequency spectrum. Therefore, it is essential to accurately describe and predict these gravitational wave backgrounds (GWB).

The physics of the early universe can be non-linear, and in such cases analytical approaches often oversimplify the picture in order to make calculations tractable. Although most of the predictions and knowledge we have acquired about the early universe rely on such approaches, the latter fail in many realistic regimes. The main reason is the inability to solve analytically the coupled system of partial differential equations that govern the dynamics. One way to explore these equations is to solve them numerically on 3+1 dimensional lattices. The set of techniques and tools developed to study the early universe, and more generally real-time field dynamics in a cosmological context, is referred to as **Lattice Cosmology**. The aim of this thesis is to present examples where lattice simulations are essential for describing the non-linear dynamics of inflationary and reheating scenarios, and to show how GWs could help us either observe or constrain these phenomena. To this end, the public code called *CosmoLattice* has been used to simulate the dynamics of interacting fields in expanding backgrounds. During the course of my research as a pre-doctoral student, two new modules of the code were released, in which I participated as a co-developer. The latest

version of the code can be found on the [GitHub](#) repository. Additional information about the code structure and user manual can be found in the [CosmoLattice](#) website.

The thesis is structured in two parts: Part **I** constitutes the main portion of the thesis. This part is divided into seven chapters.

In Chapter **1**, a succinct overview of early universe cosmology is presented, with a particular focus on the physics of inflation and the generation of primordial perturbations; reheating, including both its perturbative and non-perturbative treatments; and the production and evolution of GWs in an expanding background.

Chapter **2** describes the techniques of lattice simulations for interacting fields in an expanding background, discretization procedures for the equations of motion and observables are discussed. The chapter also covers the different time integrators, as well as techniques used to follow the evolution of GWs on the lattice.

In Chapter **3**, we present a proof of principle method to perform particle coupling spectroscopy by observing the gravitational wave background of a certain type of early universe scenario.

Chapter **4** investigates the production of gravitational waves during preheating in a model where a scalar field is coupled via a trilinear interaction.

In Chapter **5**, we explore the possibility of reheating the universe through the indirect coupling of a scalar field to gravity.

Chapter **6** presents the results of a study on the full non-linear dynamics of axion inflation.

In chapter **7**, we present a lattice formulation to study the production and evolution of cosmological perturbations sourced by non-linear early universe dynamics.

Finally, in Chapter **8**, we propose possible future directions for each of the research lines explored, aimed at improving our understanding of the early universe.

Part **II** of the thesis contains a summary of the research results written in Spanish.

List of Publications

This doctoral thesis is based on the following works:

- *Spectroscopy of particle couplings with gravitational waves* [1]
D. G. Figueroa, A. Florio, N. Loayza and M. Pieroni
PRD **106** (2022) 6, [[2202.05805](#)].
- *Gravitational wave production from preheating with trilinear interactions* [2]
C. Cosme, D. G. Figueroa and N. Loayza
JCAP **05** (2023) 023, [[2206.14721](#)].
- *Geometric reheating of the Universe* [3]
D. G. Figueroa and N. Loayza
JCAP **03** (2025) 073, [[2406.02689](#)].
- *The non-linear dynamics of axion inflation: a detailed lattice study* [4]
D. G. Figueroa, J. Lizarraga, N. Loayza, A. Urrio and J. Urrestilla
PRD **111** (2025) 6, [[2411.16368](#)].
- *Cosmological perturbations in the lattice*
D. G. Figueroa, R. Flauger and N. Loayza
Work in progress.

Additional articles published

- *Hairy Black-holes in Shift-symmetric Theories* [5]
P. Creminelli, N. Loayza, F. Serra, E. Trincherini and L. Trombetta
JHEP **08** (2020) 045, [[2004.02893](#)].

Notation

The natural unit system $\hbar = c = k_B = 1$ is used throughout the thesis. This implies the dimensional relation $[\text{Energy}] = [\text{Mass}] = [\text{Temperature}] = [\text{Time}^{-1}] = [\text{Length}^{-1}]$. The metric with signature $\text{diag}(-1, +1, +1, +1)$ is used. The reduced Planck mass is $m_p = \sqrt{\hbar c / (8\pi G)} = \sqrt{1 / (8\pi G)} = 2.435 \times 10^{18} \text{ GeV}$. A variable written in bold format, e.g. $\mathbf{x}$, represents a vector in continuous or discrete space. Implicit summation over repeated indices is assumed in continuous space. In the lattice summation is explicitly denoted.

Abbreviations

Abbreviation	Meaning
SM	Standard Model of particle physics
BSM	Beyond the Standard Model of particle physics
DM	Dark matter
CMB	Cosmic microwave background
GWs	Gravitational waves
GWB	Gravitational Wave Background
CP	Charge parity
C	Charge
P	Parity
T	Time
LSS	Large scale structure
vev	Vacuum expectation value
LIGO	Laser Interferometer Gravitational-Wave Observatory
LISA	Laser Interferometer Space Antenna
ET	Einstein Telescope
BBO	Big Bang Observatory
Λ CDM	Standard model of cosmology
FLRW	Friedmann-Lemaître-Robertson-Walker
BBN	Big Bang Nucleosynthesis
DE	Dark energy
MCMC	Markov chain Monte Carlo
NMC	Non-Minimal Coupled
EWPT	Electroweak phase transition
<i>com</i>	Equation(s) of motion
<i>dof</i>	Degree(s) of freedom

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Part I

Thesis

Chapter 1

Early Universe Cosmology

Humankind has always pursued an explanation for the origin of the universe. Theoretical physics has brought us, as of now, the most comprehensive and observationally grounded understanding of the first moments of the universe. The corpus of our knowledge on what is known as cosmology is based on our understanding of the microscopic world of particle physics and our observational knowledge of the celestial motion of bodies, as described by general relativity. Cosmology has entered its precision era and observations are more precise and carried out continuously.

Current observations of the Cosmic Microwave Background (CMB) [6, 7] support the idea that our universe began with a period of exponential expansion, known as *inflation* [8–18]. Once inflation ended, its energy must have been transferred to relativistic degrees of freedom through a process called *Reheating* [19–22]. This period remains largely unconstrained, as it could occur at any energy scale between inflation, with an upper bound from inflation $E_{\text{inf}} \lesssim 10^{16}$ GeV, and a lower bound from Big Bang Nucleosynthesis, $E_{\text{BBN}} \sim 1$ MeV. This leaves room for a large range of interesting phenomenology that cannot be directly probed, at least not by observing photons. Among the many phenomena that might take place during the universe’s first moments, we can broadly classify distinguish:

- Inflation.
- Reheating, particle production.
- Phase transitions.
- Creation of topological defects.

In this thesis we will focus particularly in both inflation and reheating phenomena.

Among the different probes of the universe, the temperature anisotropies of the CMB have provided the greatest insights of the universe. This corresponds to the remnant of primordial photons that begin to free-stream after recombination, occurring roughly 380,000 years after the end of inflation. This means we cannot directly probe the universe between the end of inflation and recombination by relying solely on electromagnetic signatures. Within this largely unconstrained period lies early universe cosmology, a playground for theoretical physicists searching for imprints of physics that challenge our current understanding of the Standard Model (SM) or of cosmology itself. The early universe phenomena share a key characteristic that makes them interesting phenomenologically, they are inherently very energetic and, therefore, expected to produce a significant amount of GWs. The advent of gravitational wave astronomy and the demonstrated ability of observatories to detect GWs from astrophysical sources suggest that the prospect of detecting them from early universe phenomena should be pursued.

1.1 Cosmological framework

In this chapter, we discuss the basics of the physics of Cosmology, which is based on the *Cosmological Principle*, which states that the universe is homogeneous and isotropic at large scales. Let us study the evolution the universe as described by general relativity (GR). Invoking the *Cosmological Principle*, we can write down the metric ($g_{\mu\nu}$) that corresponds to such symmetry principle, whose most general form is [23]

$$ds^2 = g_{\mu\nu}dx^\mu dx^\nu = -dt^2 + a^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2(d\theta^2 + \sin^2\theta d\phi^2) \right], \quad (1.1)$$

where r, θ, ϕ are the polar coordinates, $a(t)$ is the scale factor, t is cosmic time, and k is the spatial curvature. The evolution of the metric is characterized by the scale factor, which quantifies the size of the observable universe, while the curvature k determines its geometry ($k = 0$ flat, $k > 0$ closed universe, $k < 0$ open universe). Observations of the universe at cosmological scales suggest it is flat ($k = 0$), a point that will be clarified in a later section. The coordinates $\mathbf{r} = (x, y, z)$, which defines the variable $r = \sqrt{x^2 + y^2 + z^2}$, are the so-called comoving coordinates, while the combination $r_{\text{phys}} \equiv a(t) r$ gives the physical coordinates. The scale factor a is not physical *per se*, as the metric exhibits rescaling invariance, meaning it remains unchanged under transformations of the kind $a \rightarrow \lambda a$, $k \rightarrow \lambda^2 k$, and $r \rightarrow r/\lambda$ with λ being a constant. A physical quantity derived from the scale factor is the expansion rate of the universe

given by the Hubble parameter

$$H(t) \equiv \frac{1}{a} \frac{da}{dt} . \quad (1.2)$$

To determine the evolution of the scale factor, we must solve Einstein's equations in the presence of matter. Let us first write the of Einstein's equations for a given energy-momentum tensor $T_{\mu\nu}$,

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi GT_{\mu\nu} , \quad (1.3)$$

where $G_{\mu\nu}$ is the Einstein tensor, $R_{\mu\nu}$ is the Ricci tensor, and $R \equiv g_{\mu\nu}R^{\mu\nu}$ is the Ricci scalar. The energy-momentum tensor $T_{\mu\nu}$ is obtained by varying the matter Lagrangian density with respect to the metric: $T_{\mu\nu} = (1/\sqrt{-g})\delta\mathcal{S}_m/\delta g^{\mu\nu}$. Einstein's equations follow from applying the least action principle to the full action, $\mathcal{S} = \frac{1}{16\pi G} \int dx^4 \sqrt{g}(R + \mathcal{L}_m)$, with respect to the background metric. The *Cosmological Principle* applied to the matter sector requires us to write down an energy-momentum tensor with the appropriate symmetries. This corresponds to a perfect fluid [24]

$$T_{\mu\nu} = pg_{\mu\nu} + (p + \rho)u_\mu u_\nu , \quad (1.4)$$

where p is the isotropic pressure, ρ is the energy density, and u^μ is the four-velocity of the fluid as measured by a comoving observer verifying $u^\mu u_\mu = -1$. To better appreciate the form of the energy-momentum tensor, we can write it as:

$$T^\mu{}_\nu = T_{\alpha\nu}g^{\alpha\mu} = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & -p & 0 & 0 \\ 0 & 0 & -p & 0 \\ 0 & 0 & 0 & -p \end{pmatrix} . \quad (1.5)$$

Einstein's equations imply conservation of the energy-momentum tensor, known as the continuity equation: $\nabla_\mu T^\mu{}_\nu = 0$, where ∇_μ denotes the gravitational covariant derivative given by

$$\nabla_\mu A^\nu = \partial_\mu A^\nu + \Gamma_{\mu\alpha}^\nu A^\alpha , \quad (1.6)$$

where Γ are the Christoffel symbols given by

$$\Gamma_{\mu\alpha}^\nu \equiv \frac{1}{2}g^{\nu\lambda}(\partial_\mu g_{\alpha\lambda} + \partial_\alpha g_{\mu\lambda} - \partial_\lambda g_{\mu\alpha}) . \quad (1.7)$$

The conservation equation together with Einstein's equation (1.3) yields

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p), \quad \text{1st Friedmann equation.} \quad (1.8)$$

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho, \quad \text{2nd Friedmann equation.} \quad (1.9)$$

$$\dot{\rho} + 3H(\rho + p) = 0, \quad \text{Continuity equation.} \quad (1.10)$$

These are not three independent equations, one can be derived from the other two. To keep track of energy units, it is useful to define $m_p^2 = (8\pi G)^{-1}$, which will be the convention used throughout the thesis. To solve the Friedmann equations, one must assume a functional form for the equation of state, $p = p(\rho)$. In scenarios where a perfect fluid is a good approximation, this typically follows a barotropic equation of state: $p/\rho = w = \text{const.}$ For different types of fluids, we have:

$$w = \begin{cases} 0 & \text{Radiation } (E \ll m) \\ \frac{1}{3} & \text{Matter } (E \gg m) \\ -1 & \text{Cosmological constant} \\ w(t) & \text{Generic fluid} \end{cases} \quad (1.11)$$

The corresponding solutions for the evolution of the scale factor are:

$$a(t) \propto \begin{cases} t^{1/2} & \text{Radiation} \\ t^{2/3} & \text{Matter} \\ e^{Ht} & \text{Cosmological constant} \\ e^{\int dt H} & \text{Generic fluid} \end{cases} \quad (1.12)$$

and for the energy density:

$$\rho(a) \propto \begin{cases} a^{-4} & \text{Radiation} \\ a^{-3} & \text{Matter} \\ \text{const} & \text{Cosmological constant} \\ e^{-3 \int da(1+w(a))/a} & \text{Generic fluid} \end{cases} \quad (1.13)$$

Currently, our universe is composed of different amounts of each of these fluids, so it is useful to define the fraction of each relative to the critical energy density ρ_c , the energy required for a flat universe: $\rho_c = 3m_p^2 H_0^2$

$$\Omega_i \equiv \frac{\rho_i}{\rho_c} = \frac{\rho_i}{3m_p^2 H^2} \quad (1.14)$$

Another key ingredient in understanding the universe's evolution is during the radiation era is that its energy components are in thermal equilibrium. This allows us to apply thermodynamic and statistical properties to the species that make up the thermal bath. Particle species follow time-independent distributions given by either the Bose-Einstein (for bosons) or Fermi-Dirac (for fermions) statistics:

$$f(p, T) = \frac{1}{e^{E(p)/T} \pm 1} , \quad (1.15)$$

where $+$ corresponds to fermions and $-$ to bosons, and T is the temperature of the bath. From this distribution, we derive the particle number density

$$n(T) = \frac{g}{(2\pi)^2} \int d^3p \, f(p, T) , \quad (1.16)$$

where g is the number of degrees of freedom; and the corresponding energy density and pressure are

$$\rho(T) = \frac{g}{(2\pi)^2} \int d^3p \, f(p, T) E(p) , \quad (1.17)$$

$$p(T) = \frac{g}{(2\pi)^2} \int d^3p \, f(p, T) \frac{p^2}{3E(p)} . \quad (1.18)$$

In the relativistic limit $T \gg m$, we obtain

$$n = \frac{\zeta(3)}{\pi^2} g T^3 \begin{cases} 1 & \text{bosons} \\ \frac{3}{4} & \text{fermions} \end{cases} , \quad (1.19)$$

and

$$\rho = \frac{\pi^2}{30} g T^4 \begin{cases} 1 & \text{bosons} \\ \frac{7}{8} & \text{fermions} \end{cases} , \quad (1.20)$$

With this result, and knowing the temperature of the photon thermal bath in the universe (i.e., the CMB temperature, T_γ), we can compute the energy density of radiation degrees of freedom:

$$\rho_{\text{rad}} = \frac{\pi^2}{30} g_* T_\gamma^4 , \quad \text{with} \quad g_*(T) = \sum_{\text{bosons}} g_i \left(\frac{T_i}{T_\gamma} \right)^4 + \frac{7}{8} \sum_{\text{fermions}} g_i \left(\frac{T_i}{T_\gamma} \right)^4 \quad (1.21)$$

where g_i and T_i are, respectively, the degrees of freedom and temperature of each species. Finally, a key quantity for tracking the thermal evolution of the universe is the entropy density of the thermal bath, derived from the first law of thermodynamics ($T \, dS =$

$dE + p dV$):

$$s \equiv \frac{S}{V} = \frac{\rho + p}{T} \quad (1.22)$$

Using the continuity equation (1.10), we can show that entropy is conserved, which leads to an important consequence

$$g_{s,*} T_*^3 a^3 = \text{const.} , \quad \text{or} \quad T \propto g_{s,*}^{-1/3} a^{-1} . \quad (1.23)$$

We have now set the basis for the study of cosmology, and can move toward possible explanations for the early universe scenarios of interest. In the next section, we will formulate a particle physics model of Inflation, focusing initially on single scalar field inflation. Afterwards, we will review the many possible mechanisms to reheat the universe and the intricate dynamics they may involve.

1.2 Inflation

Current observations of our universe at large scales supports the idea of an isotropic state. the universe appears to be highly isotropic and homogeneous at large scales. Furthermore, the measured temperature anisotropies in the CMB exhibit correlations between fluctuations that should have been causally disconnected at early times. The question of how the universe became so homogeneous is known as the **Horizon problem**. Secondly, the spatial curvature of the universe is observed to be very small which would require a fine-tuning of the initial conditions. This is referred as the **Flatness problem**.

The leading paradigm that solves all these problems is known as inflation. A period of exponential expansion of the universe dynamically sets the initial conditions for homogeneity and isotropy in the primordial plasma [8, 9, 12, 15]. In this section, we will explore the necessary conditions to sustain inflation and satisfactorily address the aforementioned problems.

To set the stage, let us consider the time derivative of the comoving Hubble radius, which measures the size of the causally connected universe:

$$\frac{d}{dt}(aH)^{-1} = \frac{\dot{a}H + a\dot{H}}{(aH)^2} = -\frac{1}{a}(1 - \epsilon) , \quad (1.24)$$

where we have introduced the slow-roll parameter ϵ , defined as

$$\epsilon \equiv -\frac{\dot{H}}{H^2} = -\frac{d \ln H}{dN} , \quad (1.25)$$

with $dN = d \ln a = H dt$, and $N = \ln a$ is the number of e-folds. From this, it follows that inflation which is the exponential growth of the scale factor, or the shrinking of the comoving Hubble radius, occurs for $\epsilon < 1$. It also implies that during inflation, if $\epsilon \ll 1$, $H \approx \text{const.}$

In order to satisfactorily solve the Horizon problem, or equivalently the fact that temperature anisotropies of the CMB are correlated across cosmological distances, it is necessary to have around 50–60 e-folds [23, 24], depending on the specific details of the model. Additionally, to ensure that inflation lasts for a sufficient number of e-folds, we introduce a second parameter, η :

$$\eta \equiv \frac{d \ln \epsilon}{dN} = \frac{\dot{\epsilon}}{\epsilon H} . \quad (1.26)$$

The two necessary conditions to sustain inflation for long enough are $\epsilon < 1$ and $|\eta| < 1$. A simple realization of these conditions is provided by a slowly rolling scalar field ϕ , which from now on will be referred to as the *inflaton*. Let us consider the action of a scalar field in an expanding background described by the FLRW metric:

$$\mathcal{S} = \int d^4x \sqrt{-g} \left(\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi) \right) . \quad (1.27)$$

We can write down its equation of motion (*eom*) in this background

$$\frac{1}{\sqrt{-g}} \partial_\mu (g^{\mu\nu} \partial_\nu \phi) + \frac{dV(\phi)}{d\phi} = 0 , \quad (1.28)$$

which for the FLRW metric $g^{\mu\nu} = \text{diag}(-1, a^2, a^2, a^2)$ and assuming homogeneity reads as

$$\ddot{\phi} + 3H\dot{\phi} + \frac{\partial V(\phi)}{\partial \phi} = 0 . \quad (1.29)$$

We can also write the Friedmann equations for the matter content given by the action in (1.27)

$$H^2 = \frac{1}{3m_p^2} \left(\frac{1}{2} \dot{\phi}^2 + V(\phi) \right) , \quad (1.30)$$

$$\frac{\ddot{a}}{a} = -\frac{1}{3m_p^2} \left(\dot{\phi}^2 - V(\phi) \right) , \quad (1.31)$$

from which we can identify the energy and pressure densities of the scalar field:

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi), \quad (1.32)$$

$$p_\phi = \frac{1}{2}\dot{\phi}^2 - V(\phi). \quad (1.33)$$

To sustain inflation, we require $w = p/\rho \approx -1$, which is achieved when $V(\phi) \gg \dot{\phi}^2$. That is, the scalar field must be potential-dominated and **slowly rolling** for a sufficiently long time. In addition to the slow-roll parameters introduced earlier, whose name now becomes meaningful, we can define the potential slow-roll parameters

$$\epsilon_V \equiv \frac{m_p^2}{2} \left(\frac{V'(\phi)}{V(\phi)} \right)^2, \quad \text{and} \quad \eta_V \equiv m_p^2 \frac{V''(\phi)}{V(\phi)}, \quad (1.34)$$

where $' \equiv \partial/\partial\phi$. During the slow-roll regime, these parameters satisfy [24]

$$\epsilon_V \approx \epsilon \quad \text{and} \quad \eta_V \approx 2\epsilon - \frac{\eta}{2}. \quad (1.35)$$

1.2.1 Quantum origin of inflationary perturbations

One of the key arguments supporting the inflationary paradigm is the fact that it provides a straightforward mechanism to generate the necessary perturbations that lead to structure formation and explain the observed temperature anisotropies in the CMB. In this section, we will outline the production of perturbations during inflation. Let us start by splitting the metric into a background part plus perturbations $g_{\mu\nu} \equiv \bar{g}_{\mu\nu}(t) + \delta g_{\mu\nu}(t, \mathbf{x})$, where $\bar{g}$ is the background FLRW metric sourced by the homogeneous components of the fields and $\delta g_{\mu\nu}$ is the perturbed part of the metric which is sourced by the matter perturbations. We use the Scalar-Vector-Tensor (SVT) decomposition of the metric into its perturbations. A key property of this decomposition is that, at linear order, scalar, vector, and tensor perturbations evolve independently. We can identify 10 degrees of freedom (*dofs*) in this decomposition [23, 24]

$$\begin{cases} 4 \text{ dofs Scalars :} & \Phi, \Psi, B, E, \\ 4 \text{ dofs Transverse Vectors:} & S_i, F_i, \\ 2 \text{ dofs Transverse-Traceless Tensors:} & h_{ij}. \end{cases} \quad (1.36)$$

Under this consideration the metric reads as

$$\begin{aligned} ds^2 &= g_{\mu\nu} dx^\mu dx^\nu \\ &= -(1 + 2\Phi)dt^2 + 2aB_i dx^i dt + a^2[(1 - 2\Psi)\delta_{ij} + E_{ij}]dx^i dx^j \end{aligned} \quad (1.37)$$

where

$$B_i \equiv \partial_i B - S_i, \quad \text{with} \quad \partial^i S_i = 0 \quad (1.38)$$

and

$$E_{ij} \equiv 2\partial_i \partial_j E + (\partial_i F_j + \partial_j F_i) + h_{ij}, \quad \text{with} \quad \partial^i F_i = 0, \quad \partial^i h_{ij} = h_i^i = 0. \quad (1.39)$$

Cosmological perturbations are not necessarily gauge invariant, and therefore cannot always directly represent physical degrees of freedom unless one makes a gauge choice which can be helpful in order to simplify calculations by choosing an appropriate gauge. An important exception is that tensor perturbations in this decomposition are already gauge invariant. These two *dofs* are identified with gravitational waves and correspond to its two polarization modes.

We begin by considering scalar perturbations in Eq. (1.37), choosing the spatially flat gauge, i.e. setting spatial metric perturbations to zero, and moving to conformal time, $d\tau = a^{-1}dt$. We then have

$$ds^2 = a(\tau)^2 [-(1 + 2\Phi)d\tau^2 + 2\partial_i B dx^i d\tau + \delta_{ij} dx^i dx^j]. \quad (1.40)$$

We also consider the inflaton to be inhomogeneous with fluctuations denoted by $\delta\phi$, for which we have a similar decomposition $\phi(\tau, \mathbf{x}) \equiv \bar{\phi}(\tau) + \delta\phi(\tau, \mathbf{x})$. The equation of motion for the inflaton perturbations, in terms of scalar metric perturbations, reads

$$\delta\phi'' + 2\frac{a'}{a}\delta\phi' - \left(\nabla^2 \delta\phi - a^2 \frac{\partial^2 V(\bar{\phi})}{\partial \bar{\phi}^2} \delta\phi \right) = (\Phi' - \nabla^2 B)\bar{\phi}', \quad (1.41)$$

where $' \equiv d/d\tau$. This equation can be written in terms of single variable, for which we use Einstein equations to express metric perturbations (A, B) in terms of the inflaton perturbations $\delta\phi$. We define new variables $v = a\delta\phi$ and $z = a\frac{\dot{\bar{\phi}}}{H}$. Using the Fourier transform of the field v

$$v(\tau, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} v_k e^{i\mathbf{k}\cdot\mathbf{x}}, \quad (1.42)$$

we arrive at the Mukhanov-Sasaki equation

$$v_k'' + \left(k^2 - \frac{z''}{z} \right) v_k = 0, \quad (1.43)$$

which is exact at all scales as a result of 1st order perturbation theory.

We are particularly interested in the modes generated during inflation. So far, we have developed a classical theory of perturbations, but to set the initial conditions, we now

promote v to a quantum field

$$v \rightarrow \hat{v} = \int \frac{d^3\mathbf{k}}{(2\pi)^3} \left[v_k(\tau) \hat{a}_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{x}} + v_k^*(\tau) \hat{a}_{\mathbf{k}}^\dagger e^{-i\mathbf{k}\cdot\mathbf{x}} \right], \quad (1.44)$$

with creation/annihilation operators satisfying the canonical commutation relations

$$[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{k}'}^\dagger] = (2\pi)^3 \delta(\mathbf{k} - \mathbf{k}'), \quad (1.45)$$

imposing the normalization condition $v_k^* v_k' - v_k'^* v_k = -i$. The initial condition is set for modes deep inside the comoving Hubble radius, $k \gg aH$, corresponding to the quantum vacuum in Minkowski space at $\tau \rightarrow -\infty$. The vacuum satisfies $\hat{a}_{\mathbf{k}}|0\rangle = 0$, and the mode function obeys

$$v_k'' + k^2 v_k = 0, \quad (1.46)$$

as $z''/z \sim H^2$. The solution is the *Bunch-Davies* vacuum [23, 24]

$$\lim_{\tau \rightarrow -\infty} v_k = \frac{1}{\sqrt{2k}} e^{-ik\tau}, \quad (1.47)$$

for which we choose the $-ik\tau$ solution because it corresponds to those modes coming from the infinite past ($\tau \rightarrow -\infty$). To follow the evolution of the fluctuations during inflation ($H = \text{const.}$), we use $z''/z \approx 2/\tau^2$. The solution to Eq. (1.43) is

$$v_k = \frac{e^{-ik\tau}}{\sqrt{2k}} \left(1 - \frac{i}{k\tau} \right). \quad (1.48)$$

To relate primordial fluctuations to CMB anisotropies, we define a gauge-invariant quantity, the comoving curvature perturbation

$$\mathcal{R} = \delta\phi \frac{H}{\dot{\phi}}. \quad (1.49)$$

Importantly, outside the horizon ($k \ll aH$), $\dot{\mathcal{R}} = 0$, meaning the modes freeze. When they re-enter the horizon after inflation, they retain the amplitude from horizon exit. Thus, the longest wavelength modes, those entering the Hubble horizon now, correspond to the earliest perturbations generated during inflation.

Let us discuss the statistics of the primordial scalar perturbations. A key observable is the power spectrum

$$\Delta_{\mathcal{R}}^2 \equiv \frac{k^3}{2\pi^2} P_{\mathcal{R}}(k) \quad \text{with} \quad \langle \mathcal{R}_{\mathbf{k}} \mathcal{R}_{\mathbf{k}'}^* \rangle = (2\pi)^3 \delta(\mathbf{k} + \mathbf{k}') P_{\mathcal{R}}(k). \quad (1.50)$$

The scale dependence of the power spectrum is quantified by the *spectral tilt* n_s

$$n_s - 1 \equiv \frac{d \ln \Delta_{\mathcal{R}}^2}{d \ln k} , \quad (1.51)$$

with $n_s = 1$ indicating scale invariance. The spectrum is often parametrized as

$$\Delta_s^2 = A_s(k_*) \left(\frac{k}{k_*} \right)^{n_s-1} \quad (1.52)$$

where k_* is the pivot scale and A_s is the amplitude. CMB observations are well described by this form. Now, the power spectrum of curvature perturbations at horizon crossing is

$$\langle \mathcal{R}_{\mathbf{k}} \mathcal{R}_{\mathbf{k}'}^* \rangle = \left(\frac{H}{\dot{\phi}} \right)^2 \langle \delta\phi_{\mathbf{k}} \delta\phi_{\mathbf{k}'}^* \rangle \quad (1.53)$$

and in the super-horizon limit

$$\langle \delta\phi_{\mathbf{k}} \delta\phi_{\mathbf{k}'}^* \rangle = (2\pi)^3 \delta(\mathbf{k} + \mathbf{k}') \frac{H^2}{2k^3} . \quad (1.54)$$

Evaluating this at horizon crossing $k = a(t_*)H(t_*)$ gives

$$\langle \mathcal{R}_{\mathbf{k}} \mathcal{R}_{\mathbf{k}'}^* \rangle = (2\pi)^3 \delta(\mathbf{k} + \mathbf{k}') \frac{H_*^2}{2k_*^3} \left(\frac{H_*^2}{\dot{\phi}_*^2} \right) , \quad (1.55)$$

and the power spectrum

$$\Delta_{\mathcal{R}}^2 = \frac{H_*^2}{(2\pi)^2} \frac{H_*^2}{\dot{\phi}_*^2} . \quad (1.56)$$

Following a similar approach, we can show that tensor perturbations are produced in a similar manner. Their power spectrum is

$$\Delta_h^2 = \frac{2}{m_p^2} \left(\frac{H_*}{2\pi} \right)^2 , \quad (1.57)$$

and the total tensor power spectrum is $\Delta_t^2 = 2\Delta_h^2$, accounting for both polarizations. The tensor-to-scalar ratio is

$$r \equiv \frac{\Delta_t^2(k)}{\Delta_s^2(k)} . \quad (1.58)$$

In slow-roll inflation, the observables relate to the potential slow-roll parameters as

$$r = 16\epsilon_V^* \quad \text{and} \quad 1 - n_s = 6\epsilon_V^* - 2\eta_V^* , \quad (1.59)$$

where ϵ_V^*, η_V^* are evaluated at horizon crossing. For example, the latest ACT results [7] use $k_* = 0.05 \text{Mpc}^{-1}$ and measure

$$n_s = 0.974 \pm 0.003 \text{ (68\% CL)} , \quad (1.60)$$

while Planck 2018 and combined BICEP2/Keck Array BK15 data [25–27] sets an upper bound

$$r < 0.032 \text{ (95\% CL)} . \quad (1.61)$$

Finally, the amplitude of scalar fluctuations is measured as

$$A_s = 2.099_{-0.292}^{+0.296} \cdot 10^{-9} \text{ (68\% CL)} . \quad (1.62)$$

The measured values of these observables have helped us to constrain possible scenarios and realizations of the inflationary paradigm, but its true model still remains a mystery. Moreover, inflation unfolds many other issues that have to be addressed. In the following section we will discuss briefly about such issues and comment on possible solutions.

1.2.2 Caveats of the inflationary paradigm

Despite the success of inflation to provide a framework to solve some problems about the universe at cosmological scales, and the mechanism to produce the seeds for structure formation, we do not know its micro physical incarnation. Many models of inflation requires a very high energy scale description, much above the energy scales probed at particle colliders, therefore is sensitive to the possible extension of the the particle physics theory at those scales. Given the fact that we do not have a complete theory up to arbitrary energies, the conservative approach to inflation is to considered it as a low energy Effective Field Theory (EFT). We do not know the details of the theory at high scales, but we at least will try to parametrize our lack of knowledge of it. A necessary condition for inflation is to keep the potential flat enough during enough time, meaning $|\eta| \ll 1$. Let us see if the EFT of a slowly rolling scalar field is protected from these higher order operators. The EFT reads as follows

$$\mathcal{L}_{\text{eff}} = \frac{1}{2}(\partial_\mu \phi)^2 - V(\phi) + \sum_n c_n V(\phi) \frac{\phi^{2n}}{\Lambda^{2n}} + \sum_n d_n \frac{(\partial \phi)^{2n}}{\Lambda^{2n}} + \dots , \quad (1.63)$$

with the additional higher energy operators that could in principle impact both the inflationary potential and the kinetic term. It is important to remark that the lowest order term that will affect the η parameter will be of 6th order. Let us consider the

effect of this additional operator modifying the potential

$$\Delta V = c_1 V(\phi) \frac{\phi^2}{\Lambda^2} . \quad (1.64)$$

we suppose that still our physics is below the cutoff scale $\phi \ll \Lambda$, meaning that the correction is small $\Delta V \ll V$. Nevertheless, the flatness of the potential could be severely affected by this small correction. Let us compute the correction to the second potential slow-roll parameter, η_V ,

$$\Delta\eta_V = \frac{m_p^2}{V} (\Delta V)'' \approx 2c_1 \frac{m_p^2}{\Lambda^2} > 1 . \quad (1.65)$$

For constants $c_1 \sim 1$ and $\Lambda \lesssim m_p$. This supposes a problem for inflationary potential at any energy scale. A possible solution to the problem is to invoke additional symmetries that protect the flatness of the potential, for instance a shift symmetry $\phi \rightarrow \phi + c$, with c a constant. The inflaton's shift symmetry is only broken by non-perturbative effects and hence the flatness of the potential can be ensured ‘naturally’, as originally proposed in [28, 29]. Possible interactions of an axion like particle (ALP) inflaton with other fields become then very restricted. The model known as *axion* inflation is very rich phenomenologically and offers very distinctive signatures that could lead to observations in the near future. In particular these happens due to the coupling of the inflaton to a gauge sector through $\phi F \tilde{F}$, which induces an exponential production of gauge quanta. In chapter 6, we will introduce the model of axion inflation and we will realize the necessity of going beyond the standard analytical approach in order to assess correctly the phenomenological consequences of such model and its implications on our ability to observe them in the future.

1.3 Reheating

Inflation is widely regarded as the leading paradigm to explain the cosmological properties of the universe. Nevertheless, the other cornerstone of modern cosmology, the framework of the Hot Big Bang requires a connection from inflation to the onset to the thermal era of the universe. In particular, certain *dofs* of the SM must be in thermal equilibrium at the appropriate energy scales to allow for the successful synthesis of light chemical elements during Big Bang Nucleosynthesis (BBN). However, inflation dilutes any preexisting particle abundances due to its exponential expansion. Furthermore, these *dofs* should be relativistic and dominate the energy budget of the universe. This means that by the end of inflation, all the energy of the universe is stored in the homogeneous energy density that drove inflation itself. Consequently, there must exist

a mechanism to transfer this energy into these species, which subsequently thermalize to form the radiation-dominated plasma necessary for BBN. This transition period is known as *Reheating*, and many scenarios have been proposed to describe it, ranging from perturbative reheating to more intricate mechanisms involving non-perturbative particle production, which lead to a rich phenomenology. In this section, we will introduce some possibilities that exist for achieving reheating, and in particular, we will discuss the idea of *Preheating*, which arises due to non-perturbative and non-linear dynamics.

1.3.1 Perturbative energy transfer

At the end of inflation, the inflaton field decays into single quanta of other particle species, provided there exists a direct coupling between them. For instance, let us consider a set of couplings between the inflaton and some daughter particle species, which could be either a scalar χ or a spinor ψ

$$\mathcal{L} \subset V_{\text{inf}}(\phi) + \frac{1}{2}g^2\phi^2\chi^2 + h\phi\bar{\psi}\psi . \quad (1.66)$$

These couplings g and h cannot be arbitrarily large, as they may spoil the flatness of the inflationary potential via radiative corrections. Let us consider the simplest functional form of the potential, a quadratic monomial

$$V_{\text{inf}} = \frac{1}{2}m_{\text{inf}}^2\phi^2 , \quad (1.67)$$

where m_{inf} is the mass of the inflaton, fixed by matching CMB observations. After the end of inflation, the field begins to oscillate around the minimum of the potential, as the slow-roll conditions break down. The equation of motion for the inflaton in the absence of couplings and gradients of ϕ is

$$\ddot{\phi} + 3H\dot{\phi} = -m_{\text{inf}}^2\phi , \quad (1.68)$$

whose approximate solution is given by $\phi(t) \approx \Phi(t) \cdot \sin(m_{\text{inf}}t)$, with $\Phi(t) \propto (m_{\text{inf}}t)^{-1}$. While the evolution of its energy density is governed by the continuity equation

$$\dot{\rho}_{\phi} + 3H\rho_{\phi} = -3Hp_{\phi} = -\frac{3}{2}H(m_{\text{inf}}^2\phi^2 - \dot{\phi}^2) , \quad (1.69)$$

where, due to the harmonic behavior of the solution, the right-hand side averages to zero over oscillations. This implies that the field behaves as matter ($p_{\phi} = 0$), and its energy density scales as $\rho_{\phi} \propto a^{-3}$. In the presence of couplings, however, there is an

additional decay channel for the inflaton, which modifies the energy density evolution

$$\dot{\rho}_\phi = -(3H + \Gamma_\phi)\rho_\phi, \quad (1.70)$$

where Γ_ϕ is the decay rate of the inflaton into daughter species. These decay rates, derived from the Lagrangian in Eq. (1.66), can be computed at tree level [30]

$$\Gamma_{\phi \rightarrow \chi\chi} = \frac{g^2 \sigma^2}{8\pi m_{\text{inf}}} , \quad \text{and} \quad \Gamma_{\phi \rightarrow \psi\psi} = \frac{h^2 m_{\text{inf}}}{8\pi} , \quad (1.71)$$

where σ is the minimum's position of the inflationary potential. Rewriting Eq. (1.70), we obtain

$$\frac{d}{dt}(a^3 \rho_\phi) = -\Gamma_\phi \cdot (a^3 \rho_\phi) . \quad (1.72)$$

This shows that the comoving energy density $a^3 \rho_\phi$ decays exponentially, with a characteristic lifetime $\tau_\phi \sim \Gamma_\phi^{-1}$. Initially, the decay rate is subdominant compared to the Hubble expansion ($H \gg \Gamma_\phi$), and the amplitude of the inflaton decreases mainly due to the expansion. Eventually, this trend reverses and the decay dominates, transferring energy to the daughter species, which then thermalize via their self-interactions, like $\sim \lambda \chi^4$, or among themselves.

To estimate the reheating temperature T_{RH} , we define as the moment when the decay and Hubble rates become comparable $H \sim \Gamma_\phi/3$. The energy density of the universe at that moment is

$$\rho(t_{\text{RH}}) = 3m_p^2 H^2 \approx \frac{\Gamma_\phi^2 m_p^2}{3} . \quad (1.73)$$

At this point, the universe consists of relativistic degrees of freedom, so we can use Eq. (1.20) to relate energy density to temperature

$$\rho(t_{\text{RH}}) = \frac{\pi^2}{30} g(T_{\text{RH}}) T_{\text{RH}}^4 \quad \implies \quad T_{\text{RH}} \approx \left(\frac{g(T_{\text{RH}})}{10^3} \right)^{-1/4} (\Gamma_\phi m_p)^{1/2} . \quad (1.74)$$

The above results rely on the assumption that energy transfer from the inflaton to the radiation bath occurs perturbatively. However, this framework overlooks a crucial feature the inflaton behaves as a coherent homogeneous condensate which exhibits collective dynamics. In the next section, we will explore the implications of this behavior and how it may drastically alters the physics of reheating, giving rise to new phenomena through non-perturbative and non-linear processes.

1.3.2 Non-perturbative energy transfer

The perturbative approach relies on the assumption that the decay rate, Γ_ϕ , can be obtained using standard perturbative quantum field theory techniques. However, this assumption breaks down in the presence of non-perturbative effects. In this section, we review the basics of non-perturbative particle production induced by the coherent oscillations of the homogeneous inflaton. Let us consider the dynamics of the inflaton field ϕ after inflation ends. The equation of motion of the inflaton is given by Eq. (1.29). We stick to the choice of a potential of the form $V(\phi) = \frac{1}{2}m_{\text{inf}}^2\phi^2$, whose solution is oscillatory and decaying: $\phi(t) \approx (m_{\text{inf}}t)^{-1} \cdot \sin(m_{\text{inf}}t)$. The perturbative approach relies on the assumption that the decay rate, Γ_ϕ , can be obtained using standard perturbative quantum field theory techniques. However, this assumption breaks down in the presence of non-perturbative effects. In this section, we review the basics of non-perturbative particle production induced by the coherent oscillations of the homogeneous inflaton. Let us consider the dynamics of the inflaton field ϕ after inflation ends. The equation of motion of the inflaton is given by Eq. (1.29). We stick to the choice of a potential of the form $V(\phi) = \frac{1}{2}m_{\text{inf}}^2\phi^2$, for which we recall an oscillatory and decaying solution, $\phi(t) \approx (m_{\text{inf}}t)^{-1} \cdot \sin(m_{\text{inf}}t)$.

Now consider the equation of motion for the daughter field χ coupled to the inflaton via a term $\sim \frac{1}{2}g^2\phi^2\chi^2$

$$\ddot{\chi} - \frac{1}{a^2}\nabla^2\chi + 3H\dot{\chi} + g^2\phi^2\chi = 0. \quad (1.75)$$

Note that we retain the gradient term, since the daughter field starts from a vacuum configuration with no homogeneous mode. We now treat the inflaton as a classical background field, while quantizing the daughter field. We expand $\hat{\chi}(t, \mathbf{x})$ in creation and annihilation operators as

$$\hat{\chi}(t, \mathbf{x}) = \frac{1}{(2\pi)^3} \int d^3k \left(\hat{a}_k \chi_k(t) e^{-i\mathbf{k}\cdot\mathbf{x}} + \hat{a}_k^\dagger \chi_k^*(t) e^{i\mathbf{k}\cdot\mathbf{x}} \right), \quad (1.76)$$

where $\chi_k(t)$ are the mode functions. The equation for each mode becomes

$$\ddot{\chi}_k + 3H\dot{\chi}_k + \left(\frac{\mathbf{k}^2}{a^2} + g^2\phi^2 \right) \chi_k = 0. \quad (1.77)$$

Using $\phi(t) = \Phi(t) \sin(m_{\text{inf}}t)$, we introduce the following redefinitions

$$A_k = \frac{4k^2}{m_{\text{inf}}^2}, \quad q = \frac{4g^2\Phi(t)^2}{m_{\text{inf}}^2}, \quad 2z = m_{\text{inf}}t + \frac{\pi}{2}. \quad (1.78)$$

The equation becomes

$$\chi_k'' + (A_k - 2q \cos 2z)\chi_k = 0 , \quad (1.79)$$

which is known as the Mathieu equation [30], whose solutions exhibit regions of exponential instability. These instability bands correspond to resonant particle production, where $\chi_k \sim e^{2\mu_k z}$. This behavior sharply contrasts with perturbative decay. Another way to understand this phenomenon is to consider how the effective mass of χ varies non-adiabatically due to the inflaton oscillations. The mode frequency is

$$\omega_k(t) = \sqrt{(k/a)^2 + g^2 \Phi^2 \sin^2(m_{\text{inf}} t)} , \quad (1.80)$$

non-adiabatic condition is

$$\frac{d\omega_k}{dt} \gtrsim \omega_k^2 , \quad (1.81)$$

and occurs when

$$\frac{k}{a} > \frac{k_*}{a} = \sqrt{g m_{\text{inf}} \Phi} = \sqrt{2} m_{\text{inf}} q^{1/4} , \quad (1.82)$$

where q is the *resonance parameter* that characterizes the strength of parametric resonance. For $q < 1$, the resonance is narrow and typically inefficient due to cosmic expansion. In contrast, broad resonance ($q \gg 1$) leads to copious particle production, even in an expanding universe in a process known as stochastic resonance [30, 31]. The study of the growth of such perturbations and the decay of the inflaton in the context of an expanding background was carried out numerically, confirming some results of the analytical approach and also showing the necessity of these tools in order to address the dynamics of the non-linear regime. [32–34].

One naturally wonders whether the explosive growth of χ can influence the background dynamics. This is a crucial concern, especially since we treat the inflaton as a homogeneous classical field governing the cosmic evolution. We want to know how long it takes to transfer energy to the daughter species and how quickly the universe becomes radiation dominated, i.e. reheats. To address this, we must study the full dynamics of the coupled fields

$$\ddot{\phi} - \frac{1}{a^2} \nabla^2 \phi + 3H\dot{\phi} + (m_{\text{inf}}^2 + g^2 \chi^2)\phi = 0 , \quad (1.83)$$

$$\ddot{\chi} - \frac{1}{a^2} \nabla^2 \chi + 3H\dot{\chi} + g^2 \phi^2 \chi = 0 . \quad (1.84)$$

These are coupled PDEs with nonlinear and inhomogeneous dynamics, for which no analytical solutions exist in general. Hence, we must rely on numerical techniques. In

chapter 2, we introduce and develop the formalism of numerical Lattice Field Theory simulations, an approach we broadly refer as **Lattice Cosmology**.

Furthermore, preheating, being a highly energetic process, may give rise to observable imprints. As previously discussed, electromagnetic signals from this era are not observable, since photons are tightly coupled to the plasma and scatters all the time. We expect another type of observables could help us shed light on the true mechanism governing this process. Various mechanisms for preheating have been explored. For scalar daughter fields, particle production is typically driven by parametric resonance [30, 31, 35–40]. Other mechanisms include particle production through trilinear or higher-order interactions [41–44], geometric preheating via non-minimal couplings to gravity [3, 45–49], and tachyonic preheating for scalar [50–53] and gauge fields [54–64]. Gauge field excitation via shift-symmetric couplings to axion-like inflatons has also been studied [65–68]. Multi-field (p)reheating scenarios have been extensively considered [1, 69–72], including models with non-minimal gravitational couplings [47, 48, 73–90]. For comprehensive overviews, see [19–22, 91].

In this thesis, we focus on three specific scenarios. In chapter 3, we explore preheating via parametric resonance and reconstruct the GW spectrum to infer particle couplings. In chapter 4, we analyze preheating via trilinear interactions and aim to constrain the couplings from the GW signal. Finally, in chapter 5, we study geometric preheating, where a non-minimal coupling to gravity enables the energy transfer to other fields.

1.4 Gravitational wave backgrounds

Early universe scenarios offer a wide range of phenomenology that cannot be constrained by standard observational means in cosmology, like electromagnetic signatures. The advent of gravitational wave astronomy, marked by the first-ever direct detection of a gravitational wave event, in 2015, from a binary black hole merger has sparked a global effort to develop more sensitive and precise gravitational wave detectors. It has shown outstanding results on the relativistic dynamics of compact binaries, and enabling stringent tests of gravity, see e.g. [92–98]. The possibility of detecting a gravitational wave background (GWB) of cosmological origin is both promising and profound. Such a detection would provide a unique observational window into the very early universe and significantly constrain early universe scenarios. Among the phenomena studied we have: Preheating scenarios where a GWB is produced [63, 67, 72, 99–111], and from various post-inflationary phenomena, like kination-domination [112–118], oscillon dynamics [119–123], strong first order phase transitions [124–131, 131–134], or cosmic defects [135–147].

Large varieties of experiments have been designed and actually planned to perform these measurements, which aim to cover a wide range of frequencies. These experiments range from increasingly precise CMB and pulsar timing array observations [148, 149], to present and proposed GW direct detection experiments [150–160] and atom interferometers [161, 162], altogether spanning ~ 20 decades in frequency, from $\sim 10^{-17}$ Hz to $\sim$ kHz. Furthermore, a high-frequency detection effort aiming to cover the window at $\sim$ MHz frequencies and above [163]. In this section, we will introduce the fundamental physics governing the evolution and production of gravitational waves during the early universe. We will also discuss potential detection prospects and the type of information such a background could reveal.

1.4.1 Gravitational waves: creation and characterization

We will review the physics behind the creation and evolution of gravitational waves in an expanding universe. We first review the definition of gravitational waves (GWs) and their energy density power spectrum in a spatially-flat FLRW metric. GWs are identified with perturbations h_{ij} of the metric which are transverse and traceless, i.e.,

$$ds^2 = -dt^2 + a^2(t)(\delta_{ij} + h_{ij})dx^i dx^j, \quad \text{with } \partial_i h_{ij} = 0 \text{ and } h_{ii} = 0, \quad (1.85)$$

with latin indices denoting $i = 1, 2, 3$. In a FLRW background, the dynamics of GWs are described by equations of motion of the form [164]

$$\ddot{h}_{ij} + 3H\dot{h}_{ij} - \frac{\nabla^2}{a^2}h_{ij} = \frac{2}{m_{\text{p}}^2 a^2}\Pi_{ij}^{\text{TT}}, \quad (1.86)$$

where Π_{ij}^{TT} is the transverse-traceless (TT) part of the anisotropic tensor Π_{ij} , which we define below. The conditions $\partial_i \Pi_{ij}^{\text{TT}} = \Pi_{ii}^{\text{TT}} = 0$ hold $\forall \mathbf{x}, t$. Obtaining the TT part of a tensor in coordinate space is a non-local operation. It is convenient to perform it in Fourier space, where a projector filtering out only the TT degrees of freedom of a tensor can be easily build. The GW source can be written as

$$\Pi_{ij}^{\text{TT}}(\mathbf{k}, t) = \Lambda_{ijkl}(\hat{\mathbf{k}})\Pi_{kl}(\mathbf{k}, t), \quad (1.87)$$

where $\Lambda_{ij,kl}$ is a projection operator defined as

$$\Lambda_{ij,lm}(\hat{\mathbf{k}}) \equiv P_{il}(\hat{\mathbf{k}})P_{jm}(\hat{\mathbf{k}}) - \frac{1}{2}P_{ij}(\hat{\mathbf{k}})P_{lm}(\hat{\mathbf{k}}), \quad (1.88)$$

$$P_{ij}(\mathbf{k}) = \delta_{ij} - \hat{k}_i \hat{k}_j, \quad \hat{k}_i = k_i/k, \quad (1.89)$$

where $\mathbf{k} = (k_1, k_2, k_3)$ is the three-momentum and $k = |\mathbf{k}|$. Thanks to the fact that $P_{ij}\hat{k}_j = 0$ and $P_{ij}P_{jm} = P_{im}$, one can easily see that the transverse-traceless conditions in Fourier space, $k_i\Pi_{ij}(\mathbf{k}, t) = \Pi_{ii}(\mathbf{k}, t) = 0$, are satisfied at any time.

Coming back to the anisotropic stress tensor, $\Pi_{\mu\nu}$, it describes the deviation of an energy momentum tensor $T_{\mu\nu}$ with respect to a perfect fluid. The spatial components read

$$\Pi_{ij} \equiv T_{ij} - pg_{ij}, \quad (1.90)$$

with p the homogeneous background pressure and $g_{ij} = a^2(t)(\delta_{ij} + h_{ij})$ the spatial-spatial part of the FLRW perturbed metric.

The energy density of a stochastic GW background (SGWB) is defined as [164]

$$\rho_{\text{GW}}(t) = \frac{m_{\text{p}}^2}{4} \langle \dot{h}_{ij}(\mathbf{x}, t) \dot{h}_{ij}(\mathbf{x}, t) \rangle_V \quad (1.91)$$

$$\approx \frac{m_{\text{p}}^2}{4V} \int_V \frac{d^3\mathbf{k}}{(2\pi)^3} \dot{h}_{ij}(\mathbf{k}, t) \dot{h}_{ij}^*(\mathbf{k}, t) \quad (1.92)$$

$$\equiv \int \frac{d\rho_{\text{GW}}}{d\log k} d\log k, \quad (1.93)$$

where $\langle \dots \rangle_V$ denotes spatial average over a volume V assumed to encompass all relevant wavelengths of the perturbations h_{ij} . We note that the approximate expression in Eq. (1.92) is only valid in the limit $kV^{1/3} \gg 1$, where $\int_V e^{-i\mathbf{x}(\mathbf{k}-\mathbf{k}')} \rightarrow (2\pi)^3 \delta^{(3)}(\mathbf{k}-\mathbf{k}')$. The energy density per logarithmic interval is then defined as

$$\frac{d\rho_{\text{GW}}}{d\log k} = \frac{m_{\text{p}}^2 k^3}{8\pi^2 V} \int \frac{d\Omega_k}{4\pi} \dot{h}_{ij}(\mathbf{k}, t) \dot{h}_{ij}^*(\mathbf{k}, t), \quad (1.94)$$

where $d\Omega_k$ represents a solid angle measure in momentum space. For stochastic sources the volume average can be replaced by an ensemble average $\langle \dots \rangle$ over realizations of the metric perturbations,

$$\begin{aligned} \rho_{\text{GW}}(t) &= \frac{m_{\text{p}}^2}{4} \langle \dot{h}_{ij}(\mathbf{x}, t) \dot{h}_{ij}^*(\mathbf{x}, t) \rangle \\ &= \frac{m_{\text{p}}^2}{4} \int \frac{d^3\mathbf{k}}{(2\pi)^3} \frac{d^3\mathbf{k}'}{(2\pi)^3} e^{-i\mathbf{x}(\mathbf{k}-\mathbf{k}')} \times \langle \dot{h}_{ij}(\mathbf{k}, t) \dot{h}_{ij}(\mathbf{k}', t) \rangle \\ &\equiv \frac{m_{\text{p}}^2}{8\pi^2} \int \frac{dk}{k} k^3 P_h(k, t), \end{aligned} \quad (1.95)$$

where we have defined the power spectrum of the tensor time derivative in the third line, assuming statistical homogeneity and isotropy,

$$\langle \dot{h}_{ij}(\mathbf{k}, t) \dot{h}_{ij}(\mathbf{k}', t) \rangle = (2\pi)^3 P_h(k, t) \delta^{(3)}(\mathbf{k}-\mathbf{k}'). \quad (1.96)$$

Comparing Eq. (1.96) and (1.95) we can obtain the GW power spectrum,

$$\frac{d\rho_{\text{GW}}}{d\log k} = \frac{m_{\text{p}}^2 k^3}{8\pi^2} P_h(k, t) . \quad (1.97)$$

The GW energy density power spectrum can be written as a fraction of the critical energy density of the universe,

$$\Omega_{\text{GW}} = \frac{1}{\rho_c} \frac{d\rho_{\text{GW}}}{d\log k} . \quad (1.98)$$

In order to obtain the proper expression of Π_{ij}^{TT} , let us consider a theory with real singlet fields and complex scalar fields charged under an Abelian U(1) gauge symmetry, and the corresponding gauge boson. The action is

$$\mathcal{S} = - \int dx^4 \sqrt{-g} \mathcal{L} = - \int dx^4 \sqrt{-g} \left\{ \frac{1}{2} \partial_\mu \phi \partial^\mu \phi + (D_\mu \varphi)^* (D^\mu \varphi) + \frac{1}{4} F_{\mu\nu} F^{\mu\nu} + V(\phi, |\varphi|) \right\} , \quad (1.99)$$

where $\mathcal{S}$ is the action, $\mathcal{L}$ is the Lagrangian of the model, $g = \det(g_{\mu\nu})$ is the determinant of the metric and

$$\phi \in \mathbb{R}, \quad \varphi = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2) \in \mathbb{C}, \quad \text{with} \quad \phi_1, \phi_2 \in \mathbb{R} . \quad (1.100)$$

The U(1) gauge field stress tensor and gauge covariant derivative are defined as

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad \text{and} \quad D_\mu \varphi = \partial_\mu \varphi - ig_A Q_A A_\mu \varphi, \quad (1.101)$$

where g_A is the gauge coupling and Q_A is the U(1) charge of φ . From the Lagrangian in Eq. (1.99), we can write down the stress-energy tensor of the scalar and gauge fields [165]

$$\begin{aligned} T_{\mu\nu} &\equiv - \frac{2}{\sqrt{-g}} \frac{\delta(\sqrt{-g} \mathcal{L})}{\delta g^{\mu\nu}} = g_{\mu\nu} \mathcal{L} - 2 \frac{\delta \mathcal{L}}{\delta g^{\mu\nu}} \\ &= -g_{\mu\nu} \left(g^{\alpha\beta} \left[(D_\alpha \varphi_b)^* (D_\beta \varphi_b) + \frac{1}{2} (\partial_\alpha \phi_a) (\partial_\beta \phi_a) \right] + \frac{1}{4} g^{\alpha\delta} g^{\beta\lambda} (F_{\alpha\beta} F_{\delta\lambda}) + V(\phi_a, \varphi_b) \right) \\ &\quad + (\partial_\mu \phi_a) (\partial_\nu \phi_a) + [2\text{Re}\{(D_\mu \varphi_b)^* (D_\nu \varphi_b)\} + g^{\alpha\beta} F_{\mu\alpha} F_{\nu\beta}] . \end{aligned} \quad (1.102)$$

From here, one can determine the source of Eq. (1.86). From what it matters for the GWs source, we only require the effective form of energy-momentum tensor that will survive to the TT projection which we call Π_{ij}^{eff} [108]

$$\Pi_{ij}^{\text{eff}} = \partial_i \phi_a \partial_j \phi_a + 2\text{Re}\{(D_i \varphi_b)^* (D_j \varphi_b)\} - (E_i E_j + a^{-2} B_i B_j) , \quad (1.103)$$

where a and b label all real and complex fields, and E_i and B_i are the Abelian electric and magnetic fields, defined as

$$E_i \equiv F_{0i}, \quad \text{and} \quad B_i \equiv \frac{1}{2} \epsilon_{ijk} F^{jk}. \quad (1.104)$$

1.4.2 Gravitational waves propagation: frequency and amplitude today

The GW background today is obtained by redshifting the amplitude and frequency of the background from the end of production t_f to today. Following the appropriate expansion history of the Universe, we can apply the redshifting from t_f until the present time t_0 . The GWB spectrum actually is located at some sub-horizon scales $k_{\text{GW}} = \beta_{\text{GW}} a_f H_f$, with $a_f = a(t_f)$ and $H_f = H(t_f)$. We expect the GWB spectrum to peak therefore at a frequency f_{GW} today

$$f_{\text{GW}} \equiv \frac{k_{\text{GW}}}{2\pi a_0} = \frac{\beta_{\text{GW}}}{2\pi} \left(\frac{a_f}{a_{\text{RD}}} \right) \left(\frac{a_{\text{RD}}}{a_0} \right) H_f. \quad (1.105)$$

Let us characterize the expansion history between t_f and the onset of radiation domination (RD) at t_{RD} , with an effective equation of state $\bar{w} = p/\rho$

$$\begin{aligned} \log \left(\frac{\rho_{\text{tot}}^{(\text{RD})}}{\rho_{\text{tot}}^{(\text{f})}} \right) &= -3 \int_{a_f}^{a_{\text{RD}}} \frac{da'}{a'} (1 + w(a')), \\ &= -3(1 + \bar{w}) \log \left(\frac{a_{\text{RD}}}{a_f} \right), \end{aligned} \quad (1.106)$$

for later convenience we characterize this epoch in terms of

$$\epsilon_f \equiv \left(\frac{a_f}{a_{\text{RD}}} \right)^{1-3\bar{w}}. \quad (1.107)$$

We can write the ratio of the scale factors as follows

$$\frac{a_f}{a_{\text{RD}}} = \epsilon_f^{1/4} \left(\frac{\rho_{\text{RD}}}{\rho_f} \right)^{1/4}. \quad (1.108)$$

We can relate the scale factor at the onset of Radiation Domination (RD) and today by invoking the conservation of the entropy density of the fluid,

$$g_{*,s} T^3 a^3 = \text{const.}, \quad (1.109)$$

where $g_{*,s}$ is the entropic number of degrees of freedom that behave as radiation. Hence, we have

$$g_{\text{RD},s} T_{\text{RD}}^3 a_{\text{RD}}^3 = g_{0,s} T_0^3 a_0^3, \quad (1.110)$$

for which we can write

$$\frac{a_{\text{RD}}}{a_0} = \left(\frac{g_{0,s}}{g_{\text{RD},s}} \right)^{1/3} \left(\frac{T_0}{T_{\text{RD}}} \right) \quad (1.111)$$

subsequently we can relate the temperature of the radiation fluid with the energy density of radiation

$$\rho_{*,\text{rad}} \propto g_*(T) T^4 \quad \longrightarrow \quad \frac{T_0}{T_{\text{RD}}} = \left(\frac{g_{\text{RD}}}{g_0} \right)^{1/4} \left(\frac{\rho_{0,\text{rad}}}{\rho_{\text{RD},\text{rad}}} \right)^{1/4}, \quad (1.112)$$

combining the two relations we obtain

$$\frac{a_{\text{RD}}}{a_0} = \left(\frac{g_{s,\text{RD}}}{g_{s,0}} \right)^{-1/3} \left(\frac{g_{\text{RD}}}{g_0} \right)^{1/4} \left(\frac{\rho_{0,\text{rad}}}{\rho_{\text{RD},\text{rad}}} \right)^{1/4} \quad (1.113)$$

We can relate the energy density of radiation $\rho_{\text{f,rad}}$, at the reheating time, with the total energy density. Taking into account that the onset of RD is often defined as the moment when radiation makes at least half of the total energy density $\rho_{\text{RD}} = 2\rho_{\text{rad,RD}}$

$$\frac{a_{\text{RD}}}{a_0} = G_{\text{RD}}^{1/4} (2)^{1/4} \frac{H_{\text{f}}}{\rho_{\text{RD}}^{1/4}} \rho_{\text{rad},0}^{1/4}. \quad (1.114)$$

where G_{RD} is defined

$$G_{\text{RD}} \equiv \left(\frac{g_{s,\text{RD}}}{g_{s,0}} \right)^{-4/3} \left(\frac{g_{\text{RD}}}{g_0} \right). \quad (1.115)$$

Finally, we can relate the total energy density at time t_{f} with the Hubble rate via $\rho_{\text{f}} = 3m_p^2 H_{\text{f}}$. Combining everything together we get

$$f_{\text{GW}} = \frac{k_p}{a_{\text{f}} H_{\text{f}}} \epsilon_{\text{f}}^{1/4} G_{\text{RD}}^{1/4} \left(\frac{H_{\text{f}}}{m_p} \right)^{1/2} \left(\frac{2}{3} \right)^{1/4} \frac{\rho_{\text{rad},0}^{1/4}}{2\pi}, \quad (1.116)$$

with $g_{s,t}$ and g_t the entropic and energy density relativistic degrees of freedom. We can characterize the factor $G_{\text{RD}}^{1/4}$ taking into account that the SM degrees of freedom above the electro-weak scale amount to $g_{s,\text{RD}} = g_{\text{RD}} = 106.75$, so we write $G_{\text{RD}}^{1/4} \simeq 0.39 (g_{s,\text{RD}}/100)^{-1/12}$. The redshift factor finally reads, using $\rho_{\text{rad},0} = 3.37 \times 10^{-51} \text{GeV}^4$,

$$f_{\text{GW}} \simeq 4 \times 10^{10} \epsilon_{\text{f}}^{1/4} \left(\frac{g_{s,\text{RD}}}{100} \right)^{-1/12} \frac{k}{a_{\text{f}} H_{\text{f}}} \left(\frac{H_{\text{f}}}{m_p} \right)^{1/2} \text{Hz}. \quad (1.117)$$

The redshifted GW spectrum amplitude is

$$h_0^2 \Omega_{\text{GW}}^{(0)}(f) = \epsilon_f G_{\text{RD}} h_0^2 \Omega_{\text{rad}}^{(0)} \Omega_{\text{GW}}(k), \quad (1.118)$$

and using the value of $h_0^2 \Omega_{\text{rad}}^{(0)} \approx 4 \times 10^{-5}$, the peak amplitude of the spectrum today is

$$h_0^2 \Omega_{\text{GW}}^{(0)} \Big|_{\text{peak}} \simeq 1.6 \times 10^{-5} \epsilon_f \left(\frac{g_{\text{s,RD}}}{100} \right)^{-1/3} \Omega_{\text{GW}}^{(\text{p})}. \quad (1.119)$$

Here we have parametrized our ignorance on the specific details of reheating on the parameter ϵ_f , but in order to assess correctly the observability of a signal a detailed knowledge of these epoch is necessary. Hence, another reason why lattice simulations are needed in this regard is to allow us to properly evaluate the impact of this on the observation prospects of the GWBs.

Chapter 2

Lattice Cosmology

In the previous chapter, we introduce the physics of the early universe and realized how the assumptions made to treat analytically the dynamics failed to grasp the true behavior of the phenomena. Given the non-perturbative and non-linear physics that may have dominated the early universe, we need a framework to follow this dynamics. In this chapter, we set the basis to perform lattice simulations of real time field dynamics. We start from the basic definitions of a lattice and its reciprocal counterpart, discretization of the theory action, equations of motion (*eoms*) and the proper way to define the initial conditions.

2.1 Field theory in an expanding background

In this section, we lay down the formulation of a field theory where we consider bosonic fields and a U(1) abelian gauge sector. Let us consider the most general field theory composed by scalar singlet fields $\{\phi_a\}$, complex scalar fields charged under U(1) gauge symmetry $\{\varphi_b\}$ and a U(1) gauge vector field A_μ . The action read as

$$\mathcal{S} = - \int dx^4 \sqrt{-g} \left\{ \frac{1}{2} \partial_\mu \phi \partial^\mu \phi + (D_\mu \varphi)^* (D^\mu \varphi) + \frac{1}{4} F_{\mu\nu} F^{\mu\nu} + V(\phi, |\varphi|) \right\} , \quad (2.1)$$

where ϕ and φ are singlet and charged fields correspondingly. The potential $V(\phi, |\varphi|)$ describes the interactions of

$$\phi \in \mathbb{R} \quad , \quad \varphi = \frac{1}{\sqrt{2}} (\varphi_1 + i \varphi_2) . \quad (2.2)$$

We introduce the definition of covariant derivatives and the strength tensor of the U(1) gauge vector field

$$D_\mu \equiv \partial_\mu - ig_A Q_A A_\mu , \quad (2.3)$$

$$F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu , \quad (2.4)$$

where g_A is the coupling to the U(1) sector and Q_A is the charge of a φ fields, and $\mathcal{E}_i$ and $\mathcal{B}_i$ are the electric and magnetic fields, defined as

$$\mathcal{E}_i \equiv F_{0i} , \quad \text{and} \quad \mathcal{B}_i \equiv \frac{1}{2} \epsilon_{ijk} F^{jk} . \quad (2.5)$$

The system the following set of gauge transformations such that the action Eq.(2.1) is invariant

$$\varphi(x) \rightarrow e^{-ig_A Q_A \alpha(x)} \varphi(x) , \quad (2.6)$$

$$A_\mu(x) \rightarrow A_\mu(x) - \partial_\mu \alpha(x) , \quad (2.7)$$

which makes $F_{\mu\nu}$ automatically invariant under gauge transformations.

We want to follow the dynamics of the fields in the context of an expanding background. Let us consider a conformally flat FLRW metric

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu = -a(\eta)^2 d\eta^2 + a(\eta)^2 \delta_{ij} dx^i dx^j \quad (2.8)$$

where $a(\eta)$ is the scale factor measure in α -time, which satisfies that for $\alpha = 0$ cosmic/coordinate time (t) is recovered, or $\alpha = 1$ is conformal time ($d\tau \equiv a^{-1} dt$). We make now the convention for this section that $\dot{f} \equiv df/dt$ and $f' \equiv df/d\eta$. We obtain the following system of coupled PDEs for the full theory

$$\phi'' - a^{-2(1-\alpha)} \vec{\nabla}^2 \phi + (3 - \alpha) \frac{a'}{a} \phi' = -a^{2\alpha} \frac{\partial V}{\partial \phi} , \quad (2.9)$$

$$\varphi'' - a^{-2(1-\alpha)} \vec{D}_A^2 \varphi + (3 - \alpha) \frac{a'}{a} \varphi' = -\frac{a^{2\alpha}}{2} \frac{\partial V}{\partial |\varphi|} \frac{\varphi}{|\varphi|} , \quad (2.10)$$

$$F'_{0i} - a^{-2(1-\alpha)} \partial_j F_{ji} + (1 - \alpha) \frac{a'}{a} F_{0i} = a^{2\alpha} J_i^A = 2a^2 g_A Q_A^{(\varphi)} \text{Im}[\varphi^* \vec{D}_A \varphi] , \quad (2.11)$$

$$\partial_i F_{0i} = a^2 J_0^A = 2a^2 g_A Q_A^{(\varphi)} \text{Im}[\varphi^* D_0^A \varphi] , \quad (2.12)$$

where the last equation is not an *eom* but a constraint equation. In particular, this is the Gauss constraint which resembles the Gauss law of electrodynamics. The evolution of the expanding background is given by the Friedmann equations introduced in the previous chapter, here we will write a generalized form sourced by all the fields that make up the

theory. Let us recall the stress energy-tensor of perfect fluid in an expanding background

$$\bar{T}_{\mu\nu} \equiv (\bar{\rho} + \bar{p})u_\mu u_\nu + \bar{p}g_{\mu\nu} , \quad \implies \begin{cases} \bar{\rho} = a^{-2\alpha}\bar{T}_{00} \\ \bar{p} = \frac{1}{3a^2} \sum_j \bar{T}_{jj} \end{cases} , \quad (2.13)$$

where $\bar{p}$ and $\bar{\rho}$ stands for its homogeneous components. The Friedmann equations can be written as

$$\mathcal{H}^2 \equiv \left(\frac{a'}{a}\right)^2 = a^{2\alpha} \frac{\bar{\rho}}{3m_p^2} , \quad \frac{a''}{a} = \frac{a^{2\alpha}}{6m_p^2} [(2\alpha - 1)\bar{\rho} - 3\bar{p}] . \quad (2.14)$$

If the fluid has a barotropic equation of state $w = \bar{p}/\bar{\rho}$, and $w=\text{const.}$ the solution to the Friedmann equations is

$$a(\eta) = a_i \left(1 + \frac{1}{p}\mathcal{H}_i(\eta - \eta_i)\right)^p , \quad \text{with} \quad p = \frac{2}{3(1+w) - 2\alpha} , \quad (2.15)$$

where $a_i \equiv a(\eta_i)$ and $\mathcal{H}_i \equiv \mathcal{H}(\eta_i)$ for some initial time η_i . If the expansion is self consistent, meaning it is sourced by the volume averages of the matter fields, we have to identify the homogeneous parts of the energy and pressure density derived from the total stress energy tensor of the theory, given by [165]

$$\begin{aligned} T_{\mu\nu} &= -\frac{2}{\sqrt{-g}} \frac{\delta(\sqrt{-g}\mathcal{L})}{\delta g^{\mu\nu}} = g_{\mu\nu}\mathcal{L} - 2\frac{\delta\mathcal{L}}{\delta g^{\mu\nu}} \\ &= -g_{\mu\nu} \left(g^{\alpha\beta} \left[\frac{1}{2} \partial_\alpha \phi_a \partial_\beta \phi_a + (D_\alpha^A \varphi_b)^* (D_\beta^A \varphi_b) \right] + \frac{1}{4} g^{\alpha\delta} g^{\beta\lambda} F_{\alpha\beta} F_{\delta\lambda} + V(\phi, \varphi) \right) \\ &\quad + [(\partial_\mu \phi_a)(\partial_\nu \phi_a) + 2\text{Re}\{(D_\mu^A \varphi_b)^* (D_\nu^A \varphi_b)\}] + g^{\alpha\beta} F_{\mu\alpha} F_{\nu\beta} , \end{aligned} \quad (2.16)$$

from where we identify the energy and pressure density

$$\rho = E_{K,\phi} + E_{G,\phi} + E_{K,\varphi} + E_{G,\varphi} + E_{K,U(1)} + E_{G,U(1)} + V , \quad (2.17)$$

$$p = E_{K,\phi} + E_{K,\varphi} + \frac{1}{3}(E_{G,\phi} + E_{G,\varphi}) + \frac{1}{3}(E_{K,U(1)} + E_{G,U(1)}) - V , \quad (2.18)$$

with the following definitions of the different energy components

$$E_{K,\phi} \equiv \frac{1}{2a^{2\alpha}} \phi'^2 , \quad E_{G,\phi} \equiv \frac{1}{2a^{2\alpha}} (\vec{\nabla}\phi)^2 , \quad (2.19)$$

$$E_{K,\varphi} \equiv \frac{1}{2a^{2\alpha}} |D_0^A \phi|^2 , \quad E_{G,\varphi} \equiv \frac{1}{2a^{2\alpha}} |\vec{D}_A \phi|^2 , \quad (2.20)$$

$$E_{K,U(1)} \equiv \frac{1}{2a^{2+2\alpha}} |\vec{\mathcal{E}}|^2 , \quad E_{G,U(1)} \equiv \frac{1}{2a^4} |\vec{\mathcal{B}}|^2 . \quad (2.21)$$

The Friedmann equations read as

$$\left(\frac{a'}{a}\right)^2 = \frac{a^{2\alpha}}{3m_p^2} \langle E_{K,\phi} + E_{G,\phi} + E_{K,\varphi} + E_{G,\varphi} + E_{K,U(1)} + E_{G,U(1)} + V \rangle, \quad (2.22)$$

$$\begin{aligned} \frac{a''}{a} = & \frac{a^{2\alpha}}{3m_p^2} \langle (\alpha - 2)(E_{K,\phi} + E_{K,\varphi}) + \alpha(E_{G,\phi} + E_{G,\varphi}) \\ & + (\alpha - 1)(E_{K,U(1)} + E_{G,U(1)}) + (\alpha + 1)V \rangle, \end{aligned} \quad (2.23)$$

where $\langle \dots \rangle$ stands for the volume average. This means that the evolution of the background is sourced by the homogeneous part of the matter fields making sure that the volume is large enough such it encompass all the relevant wavelengths of the fields in consideration.

2.1.1 Non Canonical kinetic term

The early universe can exhibit many different extensions of the canonical case. A possibility is the realization of non-canonical kinetic terms. Let us consider, the action for scalar fields in which we will allow for an internal metric in field space $\mathcal{G}_{ab}$, which can be a function of both field amplitudes and derivatives $\mathcal{G}_{ab}[\{\phi\},.]$. We can write the action as

$$\mathcal{S}_{\text{NK}} = - \int d^4x \sqrt{-g} \left\{ \frac{1}{2} g^{\mu\nu} \mathcal{G}_{ab} \partial_\mu \phi_a \partial_\nu \phi_b + V(\{\phi\}) \right\}, \quad (2.24)$$

where $\mathcal{G}_{ab}$ is assumed to be symmetric in field space $\{\phi_c\}$. For $\mathcal{G}_{ab} = \delta_{ab}$, we recover the case of canonically normalized scalar fields terms. In the case $\mathcal{G}_{ab} = \beta \delta_{ab}$ with $\beta \neq 1$ and $\beta > 0$ a positive constant, this situation corresponds to canonically normalized scalar fields, as $\partial_\mu \mathcal{G}_{ab} = 0 \ \forall a, b$ and μ , $\mathcal{G}_{ab,c} = 0 \ \forall a, b, c$, and a simple field re-definition $\phi_a \rightarrow \sqrt{\beta} \phi_a$ brings back the kinetic term into its canonical form. Only if $\mathcal{G}_{ab}$ exhibits an explicit dependence on field amplitudes and/or derivatives, the scalar fields are said to be *non-canonically normalized*. In such a case we assume the metric determinant to be non-vanishing $\det(\mathcal{G}_{ab}) \neq 0$, so that there is an inverse field metric $\mathcal{G}_{ac} \mathcal{G}_{cb}^{-1} = \delta_{ab}$.

The *eom* of the system can be obtained from minimizing Eq. (2.24), leading to

$$\frac{1}{\sqrt{-g}} \partial_\mu (\sqrt{-g} g^{\mu\nu} \mathcal{G}_{ab} \partial_\mu \phi_b) - \frac{1}{2} g^{\mu\nu} \mathcal{G}_{bc,a} \partial_\mu \phi_b \partial_\nu \phi_c - \frac{\partial V}{\partial \phi_a} = 0, \quad (2.25)$$

where $\mathcal{G}_{bc,a} \equiv \partial \mathcal{G}_{bc} / \partial \phi_a$. In a FLRW background, the *eom* of non-canonical fields can be conveniently re-written like

$$\mathcal{G}_{ab} \phi_b'' + (3 - \alpha) \frac{a'}{a} \mathcal{G}_{ab} \phi_b' - a^{-2(1-\alpha)} \mathcal{G}_{ab} (\nabla^2 \phi_b) \quad (2.26)$$

$$+ \gamma_{abc} (\phi_b' \phi_c' - a^{-2(1-\alpha)} \partial_i \phi_b \partial_i \phi_c) + a^{2\alpha} \frac{\partial V}{\partial \phi_a} = 0. \quad (2.27)$$

where we have defined

$$\gamma_{abc} = \left(\mathcal{G}_{ab,c} - \frac{1}{2} \mathcal{G}_{bc,a} \right). \quad (2.28)$$

The system of Eq. (2.26) can be re-casted as

$$\begin{cases} \dot{\phi}_a \equiv \mathcal{G}_{ab}^{-1} \pi_b, \\ \dot{\pi}_a + (3 - \alpha) \frac{a'}{a} \pi_a - (\mathcal{F}_a)_{bc} \pi_b \pi_c = a^{-2(1-\alpha)} [\mathcal{G}_{ab} \nabla^2 \phi_b + \gamma_{abc} \partial_i \phi_b \partial_i \phi_c] - a^{2\alpha} \frac{\partial V}{\partial \phi_a}, \end{cases} \quad (2.29)$$

where

$$(\mathcal{F}_a)_{bc} \equiv \mathcal{G}_{ae} \mathcal{G}_{ec,d}^{-1} \mathcal{G}_{db}^{-1} + \left(\mathcal{G}_{ae,d} - \frac{1}{2} \mathcal{G}_{ed,a} \right) \mathcal{G}_{ec}^{-1} \mathcal{G}_{db}^{-1}, \quad (2.30)$$

Some preheating scenarios are driven by the non-canonical kinetic terms of spectator fields during inflation, where those fields experience a parametric resonance effect. These scenarios have been studied both analytically and with the use of lattice simulations [166–173].

2.2 Field theory in the lattice

Let us consider a cubic lattice with $\mathcal{N}$ points/sites per dimension. We are interested in the dynamics in a 3-dimensional space, hence there are a total of $\mathcal{N}^3$ sites which are labeled as

$$\mathbf{n} = (n_1, n_2, n_3) \quad \text{with} \quad n_i = 0, 1, \dots, \mathcal{N} - 1, \quad (2.31)$$

where latin indices stand for spatial coordinates $i = 1, 2, 3$. The lattice has a length L and lattice spacing δx , which are related by $\delta x = L/\mathcal{N}$. A function defined in the lattice satisfies $f(\mathbf{x}) \forall \mathbf{x} = \mathbf{n} \delta x \in \mathbb{R}^3$. In a static background, $\mathbf{x}$ correspond to physical spatial coordinate while in an expanding background correspond to the comoving spatial coordinates. The functions defined in the lattice have periodic boundary conditions satisfying $f(\mathbf{n} + \hat{i} \mathcal{N}) = f(\mathbf{n})$ where we have define the unit vector $\hat{i}$ which are the basis vectors that span the coordinate space. For a lattice in coordinate space there exist a

lattice in the reciprocal space where Fourier modes live, which are labeled as

$$\tilde{\mathbf{n}} = (\tilde{n}_1, \tilde{n}_2, \tilde{n}_3) \quad \text{with} \quad \tilde{n}_i = -\frac{\mathcal{N}}{2} + 1, -\frac{\mathcal{N}}{2}, \dots, -1, 0, 1, \dots, \frac{\mathcal{N}}{2} . \quad (2.32)$$

where the discrete Fourier transform is defined as

$$f(\mathbf{n}) = \frac{1}{\mathcal{N}^3} \sum_{\tilde{\mathbf{n}}} e^{-i\frac{2\pi}{\mathcal{N}}\mathbf{n}\tilde{\mathbf{n}}} f(\tilde{\mathbf{n}}) \quad \Leftrightarrow \quad f(\tilde{\mathbf{n}}) = \sum_{\mathbf{n}} e^{+i\frac{2\pi}{\mathcal{N}}\mathbf{n}\tilde{\mathbf{n}}} f(\mathbf{n}) , \quad (2.33)$$

where we can define the discrete Dirac delta definition from the Fourier transform

$$\sum_{\tilde{\mathbf{n}}} e^{i\frac{2\pi}{\mathcal{N}}\mathbf{n}\tilde{\mathbf{n}}} = \mathcal{N}^3 \delta_{0,\tilde{\mathbf{n}}} \quad (2.34)$$

equivalently to the lattice spacing in coordinate space, in the reciprocal lattice momentum is discretized and there exist a minimal infrared momentum k_{IR} defined as

$$k_{\text{IR}} = \frac{2\pi}{L} = \frac{2\pi}{\mathcal{N}\delta x} , \quad (2.35)$$

where the momentum at each lattice site is defined as $\mathbf{k} = \tilde{\mathbf{n}}k_{\text{IR}}$. There also exists a maximum ultraviolet momentum k_{UV} for each spatial dimension defined as

$$k_{\text{UV}} = \frac{\mathcal{N}}{2} k_{\text{IR}} = \frac{\pi}{\delta x} . \quad (2.36)$$

The actual maximum momentum k_{max} resolved at the reciprocal lattice is the one with magnitude equal to the length of the diagonal segment

$$k_{\text{max}} = \sqrt{k_{1,\text{UV}}^2 + k_{2,\text{UV}}^2 + k_{3,\text{UV}}^2} = \frac{\sqrt{3}}{2} \mathcal{N} k_{\text{IR}} . \quad (2.37)$$

We can define the power spectrum of a field $f(\mathbf{x})$ in the lattice, which characterizes the ensemble average of $\langle f^2 \rangle$ in the continuum

$$\langle f^2 \rangle = \int d \log k \Delta_f(k) , \quad \Delta_f(k) \equiv \frac{k^3}{2\pi^2} \mathcal{P}_f(k) , \quad \langle f_{\mathbf{k}} f_{\mathbf{k}'}^* \rangle = (2\pi)^3 \mathcal{P}_f(k) \delta(\mathbf{k} - \mathbf{k}') . \quad (2.38)$$

In the lattice the ensemble average can be identified, given by Parseval's theorem, with the volume average

$$\langle f^2 \rangle_V = \frac{\delta x^3}{V} \sum_{\mathbf{n}} f^2(\mathbf{n}) = \frac{1}{\mathcal{N}^3} \sum_{\mathbf{n}} f^2(\mathbf{n}) , \quad (2.39)$$

using the discrete Fourier transform

$$\langle f^2 \rangle_V = \frac{\delta x^3}{V} \sum_{\mathbf{n}} f^2(\mathbf{n}) \stackrel{\text{FT}}{=} \frac{1}{\mathcal{N}^6} \sum_{\tilde{\mathbf{n}}} |f(\tilde{\mathbf{n}})|^2 . \quad (2.40)$$

Now we split the sum over all lattice sites $\tilde{\mathbf{n}}$ on the sum over spherical bins that $R(l) = [|\tilde{\mathbf{n}}| - l/2, |\tilde{\mathbf{n}}| + l/2]$ with $l = 1, 2, 3, \dots, [\sqrt{3}\mathcal{N}/2]$, what we call canonical binning. We define the average over a bin in the following manner

$$\langle(\dots)\rangle_{R(l)} = \frac{1}{\#_{R(l)}} \sum_{\tilde{\mathbf{n}} \in R(l)} (\dots) , \quad (2.41)$$

where $\#_{R(l)}$ is the multiplicity of the l th-bin, i.e. number of lattice sites within the bin. We obtain

$$\langle f^2 \rangle = \frac{1}{\mathcal{N}^6} \sum_{|\tilde{\mathbf{n}}|} \#_{R(l)} \langle |f^2| \rangle_{R(l)} , \quad (2.42)$$

where $|\tilde{\mathbf{n}}| = \sqrt{\tilde{n}_1^2 + \tilde{n}_2^2 + \tilde{n}_3^2}$. We define $\Delta \log k(|\tilde{\mathbf{n}}|) \equiv k_{IR}/k(|\tilde{\mathbf{n}}|)$ and $k(|\tilde{\mathbf{n}}|) = k_{IR}|\tilde{\mathbf{n}}|$. We can re-write Eq. (2.42) in the following manner

$$\langle f^2 \rangle = \sum_{|\tilde{\mathbf{n}}|} \Delta \log k(|\tilde{\mathbf{n}}|) \left(\frac{k(|\tilde{\mathbf{n}}|)}{2\pi} \frac{\delta x}{\mathcal{N}^5} \#_{R(l)} \langle |f^2| \rangle_{R(l)} \right) , \quad (2.43)$$

thus we can identify the power spectrum on a lattice as

$$\Delta_f(k(|\tilde{\mathbf{n}}|)) = \frac{k(|\tilde{\mathbf{n}}|)}{2\pi} \frac{\delta x}{\mathcal{N}^5} \#_{R(l)} \langle |f^2| \rangle_{R(l)} . \quad (2.44)$$

We can also approximate the number of modes per bin, i.e. the multiplicity, by the area of the spherical surface of the l -th bin, $\#_{R(l)} \approx 4\pi|\tilde{\mathbf{n}}|^2$, for which we obtain

$$\Delta_f(k(|\tilde{\mathbf{n}}|)) = \frac{k(|\tilde{\mathbf{n}}|)^3}{2\pi^2} \left(\frac{\delta x}{\mathcal{N}} \right)^3 \langle |f^2| \rangle_{R(l)} . \quad (2.45)$$

The formula in Eq. (2.44) is referred as **Type-I** power spectrum while the Eq (2.45) is referred as **Type-II**.

The functions that we have just defined in the lattice will also depend on time, and we will follow their evolution. Therefore we can think of them to depend on an additional variable η , which is defined $\eta = n_0 \delta t$, where $n_0 = 0, 1, 2, 3, \dots$ counts the number of iterations done and δt is the time step. We will use the notation x_n to express that a function/amplitude x is evaluated at the n th-time step.

2.2.1 Lattice derivatives and lattice momenta

In order to write down an action of a field theory in the lattice we have to make sense of derivatives. Therefore, we need a proper definition of derivatives in the lattice for which we use finite differences that are equal to the continuum counterpart up to an error

caused by the discretization, which can be understood in terms of the Taylor expansion of a function around a point. A straightforward definition of a lattice derivative is the neutral lattice derivative given by

$$[\nabla_i^{(0)}]f = \frac{f(n + \hat{i}) - f(n - \hat{i})}{2\delta x} \longrightarrow \partial_i f(x)|_{x=\mathbf{n}\delta x} + \mathcal{O}(\delta x^2), \quad (2.46)$$

the *rhs* shows that the definition of the neutral derivative is equal to the continuum derivative up to an error of order δx^2 . Nevertheless, it fails to resolve for gradients of fields among first neighbors, i.e. it is insensitive to variations at the smallest scale $\sim \delta x$. We can also define the so called forward and backward derivatives

$$[\nabla_i^\pm] = \frac{\pm f(n \pm \hat{i}) \mp f(n)}{\delta x} \rightarrow \partial_i f(x)|_{x \equiv (\mathbf{n} + \hat{i}/2)\delta x} + \mathcal{O}(\delta x^2), \quad (2.47)$$

which is equal to the continuum derivative up to an error of squared lattice spacing only if one identifies the derivative living at mid lattice sites. If one expands the forward/backward derivative at lattices site $\mathbf{n}$, the error is $\mathcal{O}(\delta x)$, while if one does it around $\mathbf{n} + \hat{i}/2$ is $\mathcal{O}(\delta x^2)$. The choice of the derivative operator in the lattice implies the following relation

$$\nabla_i f(\tilde{\mathbf{n}}) = -i k_{L,i}(\tilde{\mathbf{n}}) f(\tilde{\mathbf{n}}). \quad (2.48)$$

where k_L is identified as the lattice momenta. The lattice momenta for the choice of discrete derivatives in Eq. (2.46) and (2.47) are

$$\begin{cases} k_{L,i}^{(0)} = \frac{\sin(2\pi\tilde{n}_i/\mathcal{N})}{\delta x}, \\ k_{L,i}^\pm = 2 \frac{\sin(\pi\tilde{n}_i/\mathcal{N})}{\delta x}, \quad \text{at } \mathbf{n} \pm \frac{\hat{i}}{2}, \\ k_{L,i}^\pm = \frac{\sin(2\pi\tilde{n}_i/\mathcal{N})}{\delta x} \pm \frac{1 - \cos(2\pi\tilde{n}_i/\mathcal{N})}{\delta x}, \quad \text{at } \mathbf{n}, \end{cases} \quad (2.49)$$

which depend on where we consider the derivative of a function to live in the reciprocal space. Given the definition of the forward/backward derivatives a useful operator that we want to define is the Laplacian in spatial coordinates for which we define

$$\nabla^2 f \equiv \sum_{k=1}^3 \nabla_k^+ \nabla_k^- f = \sum_{k=1}^3 \frac{f(\mathbf{n} + \hat{k}) - 2f(\mathbf{n}) + f(\mathbf{n} - \hat{k})}{\delta x^2} \quad \text{at } \mathbf{n}, \quad (2.50)$$

which is equal to the continuum Laplacian up to $\mathcal{O}(\delta x^2)$ error.

2.2.2 Evolution Schemes

In order to solve the field dynamics of the fields in an expanding background one has to write appropriate versions of the *eoms*. Let us denote all the fields involved, independent of how many *dofs* contain each one, as $\{f_i\}$, they will have an associated conjugate momenta $\{\pi_i\}$. In the case of expanding background, the scale factor is the only *dof* sourced by the expectation value of the fields which will be denoted by $a(\eta)$ and its conjugate momentum $\pi_a(\eta) \equiv a'(\eta)$. The system of coupled PDEs can be written as (independent of the possible interactions among *dofs* or the nature of the kinetic term)

$$\pi_a(\eta) = a'(\eta) , \quad (2.51)$$

$$\pi'_a(\eta) = \mathcal{K}_a[a(\eta), E_V(\eta), E_K(\eta), E_G(\eta)] , \quad (2.52)$$

$$\pi_i(\eta) = \mathcal{D}_i[f'_i(\mathbf{x}, \eta), a(\eta); \dots] , \quad (2.53)$$

$$\pi'_i(\mathbf{x}, \eta) = \mathcal{K}_i[f_i(\mathbf{x}, \eta), a(\eta), \pi_a(\eta)] . \quad (2.54)$$

Here, we can identify the functionals $\mathcal{D}$ and $\mathcal{K}$ which are referred as the Drift and the Kernel of evolution respectively. Notice that this scheme is for canonical theories only, as its ‘Drift’ function only depends on derivatives of the field and the scale factor.

There are a plethora of schemes to evolve PDEs, nevertheless we will focus in few of them. Evolvers can be divided in symplectic and non-symplectic. Those that are symplectic have the property that the evolution of the system is restricted to a fixed volume within the phase space, a corollary of these statement is that energy is conserved through the full evolution. A necessary condition for a symplectic evolver is that the Kernels of evolution cannot depend explicitly on the conjugate momenta of the fields. We will see in the following section how we can get rid of the friction term in our *eoms* by properly defining the conjugate momenta. Examples of such evolver are: Leapfrog, and Position and Velocity-Verlet, which we will introduce in the following

Now for simplicity we explain this algorithms in 1+1 dimension and later a generalization of them. Let us consider the equation of motion of a single degree of freedom characterized by the following functional relation

$$\ddot{x}(t) = \mathcal{K}[x(t)] . \quad (2.55)$$

We can write the Taylor expansion of $x(t)$ around $t + \delta t$,

$$x(t + \delta t) = x(t) + \dot{x}(t)\delta t + \frac{1}{2}\mathcal{K}[x(t)]\delta t^2 + \dots , \quad (2.56)$$

we can do the same with the first time derivative and Taylor expand it around $t + \delta t/2$

$$\dot{x}(t + \delta t/2) = \dot{x}(t) + \frac{\delta t}{2} \mathcal{K}[x(t)]\delta t + \dots = \dot{x}(t - \delta t/2) + \mathcal{K}[x(t + \delta t/2)]\delta t + \dots \quad (2.57)$$

By combining these relations we can write an iterative method to obtain the variable x at a time step $t + \delta t$ knowing the value of $x(t)$ and $\dot{x}(t + \delta t/2)$. It looks like

$$\begin{aligned} x(t + \delta t) &= x(t) + \dot{x}(t + \delta t/2)\delta t \\ \dot{x}(t + 3\delta t/2) &= \dot{x}(t + \delta t/2) + \mathcal{K}[x(t + \delta t)]\delta t \\ &\dots \end{aligned} \quad (2.58)$$

We can define the the conjugate momenta of our *dof* as $\pi(t) = \dot{x}(t)$. We realize that the forward time derivative of $x(t)$ can be interpolated (to second order δt^2) to the conjugate momenta living in the intermediate time step

$$\Delta_0^+ x_i \equiv \frac{x(t + \delta t) - x(t)}{\delta t} = \pi_{t+\delta t/2} \cdot \quad (2.59)$$

We can write the scheme as follows

$$\begin{aligned} \text{Given } x(t) = x_n, \dot{x}(t + \delta t/2) = \pi_{n+1/2}, \\ \begin{cases} x_{n+1} = x_n + \pi_{n+1/2} \cdot \delta t \\ \pi_{n+3/2} = \pi_{n+1/2} + \mathcal{K}[x_{n+1}] \cdot \delta t \end{cases} \end{aligned} \quad (2.60)$$

We can notice that after a full time step both the amplitude x and its conjugate momenta are defined at different time steps. This scheme with a “drift-kick” is known as **leapfrog**. The use of this scheme has been vastly used in the context of cosmological evolution and was the original way to evolve such systems [174]. In contrast, if we aim to follow the dynamics of a physical system and measure them at the same time after a full time step the scheme that achieve this is like a ”synchronized leapfrog” which is known as Verlet algorithms. In order to understand, let us consider the same system Eq. (2.55). We will add an extra step to the evolution scheme, such that after a full iteration we end up with both the amplitude and the conjugate momenta evaluated at the same time step. The version “kick-drift-kick” reads as follows

$$\begin{aligned} \text{Given } x(t) = x_n, \dot{x}(t) = \pi_n, \\ \begin{cases} \pi_{n+1/2} = \pi_n + \frac{1}{2} \mathcal{K}[x_n] \cdot \delta t \\ x_{n+1} = x_n + \pi_{n+1/2} \cdot \delta t \\ \pi_{n+1} = \pi_{n+1/2} + \frac{1}{2} \mathcal{K}[x_{n+1}] \cdot \delta t \end{cases} \end{aligned} \quad (2.61)$$

called the Velocity-Verlet evolver. Alternatively, we can follow a different version “drift-kick-drift” which is given by

$$\text{Given} \quad \begin{aligned} x(t) &= x_n, \dot{x}(t) = \pi_n, \\ \begin{cases} x_{n+1/2} = x_n + \frac{1}{2}\pi_n \cdot \delta t \\ \pi_{n+1} = \pi_n + \mathcal{K}[x_{n+1/2}] \cdot \delta t \\ x_{n+1} = x_{n+1/2} + \frac{1}{2}\pi_{n+1} \cdot \delta t \end{cases} \end{aligned} \quad , \quad (2.62)$$

called the Position-Verlet evolver. In case we cannot avoid the evolution kernel to depend on the conjugate momenta, or an explicit time dependent function we can use implicit methods. However, there exist still explicit methods that are able to deal with such problems, these are non-symplectic methods. In order to understand this, let us consider the system

$$\ddot{x}(t) = \mathcal{K}[x(t), \dot{x}(t)] , \quad (2.63)$$

with the initial conditions $x_0 = x(t_0)$ and $\pi_0 = \dot{x}(t_0)$. The updated values are obtained using the following iterative relations

$$x_{n+1} = x_n + \delta t \sum_{i=1}^s c_i \pi^{(i)} , \quad (2.64)$$

with

$$x^{(i)} \equiv x_n + \delta t \sum_{j=1}^s b_{ij} k^{(j)} , \quad k^{(i)} \equiv \mathcal{K}[x^{(i)}, \pi^{(i)}] , \quad (2.65)$$

where the time step δt is divided into smaller steps δt_i that satisfy $\delta t = \sum_{i=1}^s \delta t_i$, with $\delta t_i = c_i \delta t$, $c_i < 1$ and $\sum_{i=1}^s c_i = 1$. The coefficients c_i and b_{ij} are unique and the accuracy of the simulation is given by the number of sub-steps made $\mathcal{O}(\delta t^{2s})$ [175]. The family of such evolvers is known as Runge-Kutta, and depending on the number of smaller time steps taken we can classify them accordingly. The coefficients that make the job are presented by the so called Butcher tableaux

$$\begin{array}{c|cccc} b_{11} & b_{12} & \dots & b_{1s} \\ b_{21} & b_{22} & \dots & b_{2s} \\ \vdots & \vdots & \dots & \vdots \\ b_{s1} & b_{s2} & \dots & b_{ss} \\ \hline c_1 & c_2 & \dots & c_s \end{array} , \quad (2.66)$$

which for the most simple Runge-Kutta evolvers of second and fourth order, denoted RK2 and RK4 respectively, read as

$$\text{RK2 : } \begin{array}{c|cc} & 0 & 0 \\ \hline & 1 & 0 \\ \hline 1/2 & 1/2 & \end{array}, \quad \text{RK4 : } \begin{array}{c|cccc} & 0 & 0 & 0 & 0 \\ \hline & 1/2 & 0 & 0 & 0 \\ & 0 & 1/2 & 0 & 0 \\ & 0 & 0 & 1 & 0 \\ \hline 1/6 & 1/3 & 1/3 & 1/6 & \end{array}. \quad (2.67)$$

These methods can be computationally expensive as they require to store an additional degree of freedom for each field, conjugate momenta and time step subdivision, as you need simultaneously all the $k^{(i)}$ coefficients in order to build the updated value. Let us consider the system $\dot{\pi} = \mathcal{K}[x(t), \dot{x}(t), \ddot{x}(t), \dots]$ and $\dot{x} = \mathcal{D}[x(t), \dot{x}(t), \ddot{x}(t), \dots]$. There exist a family of Runge-Kutta methods that avoid these problem by writing the updated value in the following manner[176]

$$x_{n+1} = y^{(s)}, \quad (2.68)$$

with

$$\begin{aligned} y^{(0)} &= x_n, \\ y^{(i)} &= y^{(i-1)} + B_i \Delta y^{(i)}, \\ \Delta y^{(i)} &= A_i \Delta y^{(i-1)} + \delta t \mathcal{D}[y^{(i-1)}]. \end{aligned} \quad (2.69)$$

This only requires to save one additional degree of freedom for the amplitude and the conjugate momenta respectively, which is $\Delta y^{(i)}$. The coefficients A_i and B_i are also fixed and can be written using the following tableaux

$$\text{RK2 : } \begin{array}{c|cc} & A_i & B_i \\ \hline & 0 & 1/2 \\ & -1/2 & 0 \end{array}, \quad (2.70)$$

It is also possible to write an order $\mathcal{O}(\delta t^3)$ evolver with 4 intermediate steps using the coefficients [177]

$$\text{RK3_4 : } \begin{array}{c|cc} & A_i & B_i \\ \hline & 0 & 0.06688758201974097 \\ & -0.7825460361923583 & 2.876554598956719 \\ & -2.042914325731225 & 0.5534657361343982 \\ & -1.799337253940777 & 0.3912730180961791 \end{array}. \quad (2.71)$$

In particular we will make use of this family of methods for applications where the system is non-symplectic.

2.2.3 Lattice formulation of a Field Theory in an expanding background

In this section, we write down a simple field theory in an expanding background and present its full lattice formulation along with an example of an evolution scheme. Before moving on to the evolution, we must address units and dimensions. Ideally, we want to work with $\mathcal{O}(1)$ numbers in our simulations. Therefore, we express our physical quantities in a dimensionless manner, ensuring they at least start of order one. To this end, we define the following dimensionless variables

$$\tilde{\phi}_a \equiv \frac{\phi_a}{f_*}, \quad d\tilde{\eta} \equiv a^{-\alpha} \omega_* dt, \quad d\tilde{x}_i = \omega_* dx_i, \quad \kappa \equiv \frac{k}{\omega_*}, \quad (2.72)$$

with $' = \partial/\partial\tilde{\eta}$ and $\tilde{\partial}_i = \partial/\partial\tilde{x}_i$. With these definitions, we have, for instance,

$$\tilde{V}(\{\tilde{\phi}_a, \tilde{\varphi}_b\}) \equiv \frac{1}{f_*^2 \omega_*^2} V(\{\phi_a, \varphi_b\}), \quad (2.73)$$

The choice of f_* and ω_* depends on the specific problem. A rule of thumb is to take f_* as the typical field amplitude of the degree of freedom driving the dynamics, while ω_* corresponds to the typical mass scale or inverse time scale of a given phenomena. For instance, in a preheating scenario where the inflaton oscillates around a potential, we choose $f_* = \phi(t_{\text{end}})$, the inflaton amplitude at the end of inflation, and $\omega_*^2 = \partial^2 V / \partial \phi^2|_{\phi=f_*}$, the typical oscillation frequency of the inflaton. The value of α can also be optimized to select appropriate time units for example, setting α such that the oscillation frequency remains constant during expansion, e.g. if the oscillations are driven by a quadratic potential we will choose $\alpha = 0$.

We are now in a position to formulate a lattice version of an interacting scalar field theory. We consider the discretized action, replacing integrals with lattice summations, $\int d\tilde{\eta}(\dots) = \delta\tilde{\eta} \sum_{n_0}(\dots)$ and $\int d^3x(\dots) = \delta\tilde{x}^3 \sum_{\mathbf{n}}(\dots)$, resulting

$$\tilde{S}_L = \delta\tilde{\eta} \delta x^3 \sum_{n_0} \sum_{\mathbf{n}} \left\{ \frac{1}{2} a_{+0/2}^{3-\alpha} \sum_a (\tilde{\Delta}_0^+ \tilde{\phi}_a)^2 - \frac{1}{2} a^{1+\alpha} \sum_{a,k} (\tilde{\nabla}_k^+ \tilde{\phi})^2 - a^{3+\alpha} \tilde{V}(\tilde{\phi}_a) \right\}. \quad (2.74)$$

In the continuum we can define $\tilde{\pi}_i = a^{3-\alpha} \tilde{\phi}'_i$ to eliminate the friction term $(3-\alpha) \frac{a'}{a} \phi'$ from the equations of motion and ensure that the drift depends only on the conjugate

momenta. The system of equations with a self-consistent background reads

$$\tilde{\phi}'_i = a^{-(3-\alpha)} \tilde{\pi}_i, \quad (2.75)$$

$$\tilde{\pi}'_i = a^{1+\alpha} \tilde{\nabla}^2 \tilde{\phi}_i - a^{3+\alpha} \frac{\partial \tilde{V}(\tilde{\phi}_i)}{\partial \tilde{\phi}_i}, \quad (2.76)$$

$$a' = \pi_a, \quad (2.77)$$

$$\pi'_a = \frac{a^{1+2\alpha}}{3} \left(\frac{f_*}{m_p^2} \right) [(\alpha - 2) \tilde{E}_K + \alpha \tilde{E}_G + (\alpha + 1) \tilde{E}_V], \quad (2.78)$$

$$\text{with } \tilde{E}_K \equiv \frac{1}{2a^6} \sum_i \langle (\tilde{\pi}_i)^2 \rangle, \quad \tilde{E}_G \equiv \frac{1}{2a^2} \sum_{i,k} \langle (\tilde{\nabla}_k \tilde{\phi}_i)^2 \rangle, \quad \tilde{E}_V \equiv \langle \tilde{V}(\{\tilde{\phi}_i\}) \rangle. \quad (2.79)$$

An evolution scheme using the Velocity-Verlet algorithm for this system is

$$\begin{aligned} \text{Given } & \{\tilde{\phi}_i, \tilde{\pi}_i, a, \pi_a\} \text{ at } \tilde{\eta}_0, \\ & \left\{ \begin{aligned} \tilde{\pi}_{a,n+1/2} &= \tilde{\pi}_{a,n} + \frac{1}{3} \left(\frac{f_*}{m_p^2} \right) a_n^{1+2\alpha} [(\alpha - 2) \tilde{E}_K + \alpha \tilde{E}_G + (\alpha + 1) \tilde{E}_V]_n \cdot \frac{\delta \tilde{\eta}}{2}, \\ \tilde{\pi}_{n+1/2}^{(i)} &= \tilde{\pi}_n^{(i)} + \left(a_n^{1+\alpha} \tilde{\nabla}^2 \tilde{\phi}_n^i - a_n^{3+\alpha} \frac{\partial \tilde{V}(\tilde{\phi}_n^i)}{\partial \tilde{\phi}_n^i} \right) \cdot \frac{\delta \tilde{\eta}}{2}, \\ a_{n+1} &= a_n + \pi_{a,n+1/2} \cdot \delta \tilde{\eta}, \\ a_{n+1/2} &= \frac{a_n + a_{n+1}}{2}, \\ \tilde{\phi}_{n+1}^{(i)} &= \tilde{\phi}_n^{(i)} + a_{n+1/2}^{-(3-\alpha)} \tilde{\pi}_{n+1/2} \cdot \delta \tilde{\eta}, \\ \tilde{\pi}_{n+1}^{(i)} &= \tilde{\pi}_{n+1/2}^{(i)} + \left(a_{n+1}^{1+\alpha} \tilde{\nabla}^2 \tilde{\phi}_{n+1}^i - a_{n+1}^{3+\alpha} \frac{\partial \tilde{V}(\tilde{\phi}_{n+1}^i)}{\partial \tilde{\phi}_{n+1}^i} \right) \cdot \frac{\delta \tilde{\eta}}{2}, \\ \tilde{\pi}_{a,n+1} &= \tilde{\pi}_{a,n+1/2} + \frac{1}{3} \left(\frac{f_*}{m_p^2} \right) a_{n+1}^{1+2\alpha} [(\alpha - 2) \tilde{E}_K + \alpha \tilde{E}_G + (\alpha + 1) \tilde{E}_V]_{n+1} \cdot \frac{\delta \tilde{\eta}}{2}, \end{aligned} \right. \quad (2.80) \end{aligned}$$

The Hubble constraint reads

$$\tilde{\pi}_a^2 = \frac{1}{3} \left(\frac{f_*}{m_p^2} \right) a^{2(\alpha+1)} [\tilde{E}_K + \tilde{E}_G + \tilde{E}_V]. \quad (2.81)$$

2.2.4 Initial conditions on the lattice

Now let us recall the definition of the expectation value of a scalar field f in terms of its power spectrum given by Eq. (2.38). A scalar field characterized by a potential $V(\phi)$ has a dispersion relation given by

$$\omega_k^2 = (k/a)^2 + m_\phi^2, \quad \text{where } m_\phi^2 \equiv \frac{\partial^2 V(\phi)}{\partial \phi^2}, \quad (2.82)$$

the variance of quantum fluctuations of such a field is given by

$$\mathcal{P}_\phi(k) = \frac{1}{2a^3} \frac{1}{\omega_k} = \frac{1}{2a^3} \frac{1}{\sqrt{(k/a)^2 + m_\phi^2}}. \quad (2.83)$$

Our goal will be to replicate this in the lattice, which will serve as initial conditions for the field and conjugate momenta amplitudes. Let us discuss two aspects which are relevant to properly initialize the power spectrum in the most general manner: the scale factor dependence of the power spectrum and the energy units, which will help us to write them in program units. On one side, we want to know how the power spectrum depends on rescaling of the scale factor. We know that the expectation value $\langle f^2 \rangle = \int d \log k \Delta_f(k)$, c.f. Eq. (2.38), does not depend on the scale factor. As the measure $d \log k = dk/k$ does not depend on the scale factor rescaling, hence the power spectrum Δ_f should be invariant under scale factor rescaling. This forces $\mathcal{P}_f \propto \Delta_f/k^3$ to transform as a^{-3} under such transformations so that Δ_f remains invariant. On the other hand, Let us analyze the units of Eq. (2.38). We have that the energy units¹ of $[\langle f^2 \rangle] = 2$, $[d \log k] = 0$, therefore $[\Delta_f] = 2$. This means that if $[\Delta_f] = 2$ and $[k^3] = 3$, then $[\mathcal{P}_f] = -1$, which is what we get for $\mathcal{P}_\phi$ because the units of $[1/\omega_k] = -1$.

Having all of this in mind, we can write a general form of the power spectrum

$$\mathcal{P}_f(k) = \mathcal{P}_f(k/a; a) \equiv \frac{1}{a^3} \mathcal{F}(k/a) , \quad (2.84)$$

where $\mathcal{F}(k/a)$ is any function of the physical momentum $p = k/a$. Therefore,

$$\langle f^2 \rangle = \int d \log k \frac{(k/a)^3}{2\pi^2} \mathcal{F}(k/a) . \quad (2.85)$$

Now in program units we measure fields in units of f_* ($\tilde{f} = f/f_*$) and time/distance in units of ω_* ($d\tilde{x}_i = \omega_* dx_i$ and $d\eta = \omega_*$), which also allows us to write momentum as $\kappa = k/\omega_*$. Finally, $\mathcal{P}_f(\kappa)$ in program units should be written as $\tilde{\mathcal{P}}_f(\kappa) = E_* \mathcal{P}_f(k)$, where E_* is some scale with $[E_*] = 1$. In program units we have

$$\langle \tilde{f}^2 \rangle = \left(\frac{\omega_*^3}{E_* f_*^2} \right) \int (d \log \kappa) \frac{(\kappa/a)^3}{2\pi^2} \mathcal{F}(\kappa/a) . \quad (2.86)$$

A common choice is $E_* = \omega_*$ such that the factor in front of the expression becomes $(\omega_*/f_*)^2$.

Now let us try the same exercise for the time derivative of the function $\dot{f}$. As in the previous discussion, the power spectrum of the time derivative of a scalar function, $\Delta_{\dot{f}}$, is invariant under transformations of the scale factor. Now, if we allow time to be measured in a more generic choice like conformal time $d\tau = a^{-1}dt$ or α -time $d\eta = a^{-\alpha}dt$, we will have

$$\langle \dot{f}^2 \rangle = \langle f'^2 \rangle a^{-2\alpha} = \int (d \log k) \Delta_{\dot{f}}(k) , \quad (2.87)$$

¹We denote energy units of a variable as $[E] = n$, for instance a mass scale has $[m] = 1$.

where $f' \equiv \partial f / \partial \eta$. The *lhs* of the equation is still scale factor invariant, so the *rhs* should be as well. If we isolate $\langle f'^2 \rangle$, we get

$$\langle f'^2 \rangle = \int (d \log k) \Delta_{f'}(k) , \quad (2.88)$$

with $\Delta_{f'} \equiv a^{2\alpha} \Delta_{\dot{f}}$, which tell us that $\Delta_{f'}$ transforms as $a^{2\alpha}$ to keep $\Delta_{\dot{f}}$ invariant. Finally, Let us analyze the units of Eq. (2.38) again. We have $[\langle f'^2 \rangle] = 4$, $[d \log k] = 0$, therefore $[\Delta_{f'}] = 4$. This means that if $[\Delta_{f'}] = 4$ and $[k^3] = 3$, then $[\mathcal{P}_{f'}] = 1$. Now we write

$$\mathcal{P}_{f'}(k) = \frac{a^{2\alpha}}{a^3} \mathcal{G}(k/a) , \quad (2.89)$$

where $\mathcal{G}(k/a)$ is any function of the physical momentum k/a , therefore

$$\langle f'^2 \rangle = \int d \log k a^{2\alpha} \frac{(k/a)^3}{2\pi^2} \mathcal{G}(k/a) . \quad (2.90)$$

Finally, $\mathcal{P}_{f'}(\kappa)$ in program units should be written as $\tilde{\mathcal{P}}_{f'}(\kappa) = \mathcal{P}_{f'}(k)/E_*$, where E_* is some quantity with $[E_*] = 1$, which we could choose to be $E_* = \omega_*$. In program units we have

$$\langle \tilde{f}'^2 \rangle = a^{2\alpha} \left(\frac{\omega_* E_*}{f_*^2} \right) \int (d \log \kappa) \frac{(\kappa/a)^3}{2\pi^2} \mathcal{G}(\kappa/a) . \quad (2.91)$$

Now, we want to write the expectation value of the square of field fluctuations for a scalar field $\langle \phi^2 \rangle$ in the lattice, from which we recall

$$\langle \phi^2 \rangle_V = \frac{\delta x^3}{V} \sum_{\mathbf{n}} \phi^2(\mathbf{n}) \stackrel{\text{FT}}{=} \frac{1}{\mathcal{N}^6} \sum_{\tilde{\mathbf{n}}} |\phi(\tilde{\mathbf{n}})|^2 , \quad (2.92)$$

from which we obtain Eq. (2.43), recalled here

$$\Delta_\phi(k(|\tilde{\mathbf{n}}|)) = \frac{k(|\tilde{\mathbf{n}}|)}{2\pi} \frac{\delta x}{\mathcal{N}^5} \#_{R(l)} \langle |\phi^2| \rangle_{R(l)} . \quad (2.93)$$

Finally, we want to compare it with $\Delta_\phi(k) = k^3/(2\pi^2)\mathcal{P}_\phi(k)$ resulting in

$$\mathcal{P}_\phi = \frac{\delta x \pi}{k(|\tilde{\mathbf{n}}|)^2 \mathcal{N}^5} \#_{R(l)} \langle |\phi^2| \rangle_{R(l)} . \quad (2.94)$$

In order to initialize the field or conjugate momenta amplitudes, we will set on the lattice samples of a gaussian distribution with zero mean and variance given by

$$\langle \phi^2 \rangle_{R(l)} = \frac{k(|\tilde{\mathbf{n}}|)^2 \mathcal{N}^5}{\delta x \pi \#_{R(l)}} \mathcal{P}_\phi(k) , \quad (2.95)$$

which in program units reads as

$$\langle \tilde{\phi}^2 \rangle = \left(\frac{\omega_*^3}{f_*^2 E_*} \right) \frac{\kappa(|\tilde{\mathbf{n}}|)^2 \mathcal{N}^5}{\delta \tilde{x} \pi \#_{R(l)}} \tilde{\mathcal{P}}_{\tilde{\phi}}(\kappa) , \quad (2.96)$$

and after some rearranging

$$\langle \tilde{\phi}^2 \rangle = \left(\frac{\omega_*^3}{f_*^2 E_*} \right) \frac{4\pi |\tilde{\mathbf{n}}|^2}{\#_{R(l)}} \left(\frac{\mathcal{N}}{\delta \tilde{x}} \right)^3 \tilde{\mathcal{P}}_{\tilde{\phi}}(\kappa) . \quad (2.97)$$

We can input eq. (2.84)

$$\langle \tilde{\phi}^2 \rangle = \left(\frac{\omega_*^3}{f_*^2 E_*} \right) \frac{4\pi |\tilde{\mathbf{n}}|^2}{\#_{R(l)}} \left(\frac{\mathcal{N}}{\delta \tilde{x}} \right)^3 \frac{1}{a^3} \mathcal{F}(\kappa/a) , \quad (2.98)$$

and for the amplitude of $\langle \phi'^2 \rangle$ we get

$$\langle \tilde{\phi}'^2 \rangle = a^{-2\alpha} \left(\frac{\omega_* E_*}{f_*^2} \right) \frac{4\pi |\tilde{\mathbf{n}}|^2}{\#_{R(l)}} \left(\frac{\mathcal{N}}{\delta \tilde{x}} \right)^3 \frac{1}{a^3} \mathcal{G}(\kappa/a) . \quad (2.99)$$

For **Type-II**, the multiplicity of the bins is approximated by $\#_{R(l)} \sim 4\pi |\tilde{\mathbf{n}}|^2$, and then we get

$$\langle \tilde{\phi}^2 \rangle = \left(\frac{\omega_*^3}{f_*^2 E_*} \right) \left(\frac{\mathcal{N}}{\delta \tilde{x}} \right)^3 \frac{1}{a^3} \mathcal{F}(\kappa/a) , \quad (2.100)$$

and

$$\langle \tilde{\phi}'^2 \rangle = a^{-2\alpha} \left(\frac{\omega_* E_*}{f_*^2} \right) \left(\frac{\mathcal{N}}{\delta \tilde{x}} \right)^3 \frac{1}{a^3} \mathcal{G}(\kappa/a) . \quad (2.101)$$

2.3 Gravitational waves in the lattice

2.3.1 Evolving gravitational waves in the lattice

Studying the dynamics of GWs is a numerically expensive task, given that the TT projection is a non-local operation in position space. In Ref. [178], a workaround was proposed to overcome this problem is noting that in Fourier space $\Pi_{ij}^{\text{TT}}(\mathbf{k}, t)$ is just a linear combination of the components of the full tensor $\Pi_{ij}(\mathbf{k}, t)$, and that the Eq. (1.86) is linear in Π_{ij} , one can write the TT tensor perturbations (i.e., GWs) as

$$h_{ij}(\mathbf{k}, t) = \Lambda_{ij,kl}(\hat{\mathbf{k}}) u_{kl}(\mathbf{k}, t) , \quad (2.102)$$

where $u_{ij}(\mathbf{k}, t)$ is the Fourier transform of the solution to the following equation

$$\ddot{u}_{ij} + 3H\dot{u}_{ij} - \frac{\nabla^2}{a^2}u_{ij} = \frac{2}{m_p^2 a^2}\Pi_{ij}^{\text{eff}}, \quad (2.103)$$

where Π_{ij}^{eff} is an effective anisotropic tensor that contains all terms of Π_{ij} with non-vanishing TT projection. In order to evolve numerically the *eom* of the u fields, we re-normalize the field amplitude as

$$\tilde{u}_{ij} = \left(\frac{m_p}{f_*}\right)^2 u_{ij}. \quad (2.104)$$

Numerically, $\tilde{u}$ -fields are evolved by defining a conjugate momenta, $(\pi_{\tilde{u}})_{ij} = a^{3-\alpha}\tilde{u}'_{ij}$, which allows rewriting Eq. (2.103) as

$$\begin{cases} \tilde{u}'_{ij} &= a^{\alpha-3}(\pi_{\tilde{u}})_{ij}, \\ (\pi_{\tilde{u}})'_{ij} &= a^{1+\alpha}\tilde{\nabla}^2\tilde{u}_{ij} + 2a^{1+\alpha}\tilde{\Pi}_{ij}^{\text{eff}}. \end{cases} \quad (2.105)$$

The energy density power spectrum is computed by building the bilinear $h_{ij}h_{ij}^*$, and relating the physical time derivative of the h -fields to the program conjugate momentum

$$\dot{h}_{ij} = \frac{\omega_*}{a^3}\Lambda_{ij,kl}^L \pi_{u,kl} = \frac{\omega_*}{a^3}\left(\frac{f_*}{m_p}\right)^2 \Lambda_{ij,kl}^L \pi_{\tilde{u},kl}. \quad (2.106)$$

Eq. (2.103) can be evolved in configuration space for as long as desired, and only when we wish to obtain the physical degrees of freedom (*dof*) h_{ij} , we Fourier transform its solution, $u_{ij}(\mathbf{x}, t) \rightarrow u_{ij}(\mathbf{k}, t)$, and apply the projector from Eq. (1.88) as in Eq. (2.102). The viability of the method relies on the following observation to compute the GWs, we could first project the TT part of the source Π_{ij} and then solve Eq. (1.86) directly for the physical tensor fields h_{ij} . This would require doing the operation at every time step, making the procedure numerically expensive, as obtaining Π_{ij}^{TT} in real space is a non-local operation. Instead, we can achieve the same result if we commute the operations, we first solve Eq. (2.103) for the un-physical fields u_{ij} , and then apply the TT projector only when we wish to obtain the physical *dof* h_{ij} , as in Eq. (2.102). We can do this because the TT projection and the operator driving the evolution of the h modes are linear transformations of h , therefore these operations commute and can be applied in any order.

As can be seen, the lattice momenta can be either real or complex, depending on the choice of lattice derivative. This extends to the TT projector. In the neutral case, we

define a real one,

$$P_{ij}^0 = \delta_{ij} - \frac{k_{L,i}^0 k_{L,j}^0}{|\mathbf{k}_L^0|^2}, \quad (2.107)$$

$$\Lambda_{ij,kl}^0 = P_{ik}^0 P_{jl}^0 - \frac{1}{2} P_{ij}^0 P_{kl}^0. \quad (2.108)$$

while it is complex for $\mathbf{k}_L^\pm$,

$$P_{ij}^\pm = \delta_{ij} - \frac{(k_{L,i}^\pm)^* k_{L,j}^\pm}{|\mathbf{k}_L^\pm|^2}, \quad (2.109)$$

$$\Lambda_{ij,kl}^\pm = P_{ik}^\pm P_{jl}^{\pm,*} - \frac{1}{2} P_{ij}^\pm P_{kl}^{\pm,*}. \quad (2.110)$$

The complex projectors obey the following properties

$$\begin{aligned} 1) \sum_i k_{L,i}^\pm P_{ij}^\pm &= 0, & 2) \sum_i (k_{L,i}^\pm)^* P_{ij}^\pm &\neq 0, \\ 3) \sum_j k_{L,j}^\pm P_{ij}^\pm &\neq 0, & 4) \sum_j (k_{L,j}^\pm)^* P_{ij}^\pm &= 0, \\ 5) P_{ij}^{\pm,*} &= P_{ji}^\pm, & 6) P_{ij}^\pm(-\tilde{\mathbf{n}}) &= P_{ji}^\pm(\tilde{\mathbf{n}}), \\ 7) P_{ij}^\pm P_{jk}^\pm &= P_{ik}^\pm, & 8) P_{ij}^\pm P_{ki}^\pm &\neq P_{ik}^\pm, \end{aligned} \quad (2.111)$$

The most relevant of which are the idempotence of the projector (property 7) and its hermiticity (property 5). A real projector like P_{ij}^0 obeys a similar set of properties, except that it is symmetric instead of hermitian. A proof of these properties can be found in Ref. [179].

We are interested in the bilinear product $\dot{h}_{ij}(\tilde{\mathbf{n}}) \dot{h}_{ij}^*(\tilde{\mathbf{n}})$. In terms of the u -fields, it can be written as a linear combination of two traces

$$\dot{h}_{ij} \dot{h}_{ij}^* = \text{Tr}(\mathbf{P} \dot{\mathbf{u}} \mathbf{P} \dot{\mathbf{u}}^*) - \frac{1}{2} \text{Tr}(\mathbf{P} \dot{\mathbf{u}}) \text{Tr}(\mathbf{P} \dot{\mathbf{u}}^*). \quad (2.112)$$

Here, $\dot{\mathbf{u}}$ and $\mathbf{P}$ are matrices with elements $(\dot{\mathbf{u}})_{ij} = \dot{u}_{ij}$ and $(\mathbf{P})_{ij} = P_{ij}$. Eq. (2.112) is valid for both real and complex-valued projectors. We define the matrix products $v_{ij} \equiv P_{ik} \dot{u}_{kj}$ and $\tilde{v}_{ij} \equiv P_{ik} \dot{u}_{kj}^*$, and then determine the trace values from (note $\tilde{v} \neq v^*$)

$$\begin{aligned} \text{Tr}(\mathbf{P} \dot{\mathbf{u}} \mathbf{P} \dot{\mathbf{u}}^*) &= v_{11} \tilde{v}_{11} + v_{22} \tilde{v}_{22} + v_{33} \tilde{v}_{33} + v_{12} \tilde{v}_{21} + v_{21} \tilde{v}_{12} + v_{13} \tilde{v}_{31} \\ &\quad + v_{31} \tilde{v}_{13} + v_{23} \tilde{v}_{32} + v_{32} \tilde{v}_{23}, \end{aligned} \quad (2.113)$$

$$\text{Tr}(\mathbf{P} \dot{\mathbf{u}}) = v_{11} + v_{22} + v_{33}, \quad (2.114)$$

$$\text{Tr}(\mathbf{P} \dot{\mathbf{u}}^*) = \tilde{v}_{11} + \tilde{v}_{22} + \tilde{v}_{33}, \quad (2.115)$$

while in general $\tilde{v} \neq v^*$, in the real case $\tilde{v} = v^*$.

2.3.2 Gravitational wave power spectrum in the lattice

The observable we want to recover from simulations is the power spectrum of GWs. Let us recall Eq. (1.91), for which we obtain, in the limit $V \rightarrow \infty$, the spectrum per logarithmic interval in Fourier space

$$\frac{d\rho_{\text{GW}}}{d\log k} = \frac{k^3}{(4\pi)^3 GV} \int \frac{d\Omega_k}{4\pi} \dot{h}_{ij}(\mathbf{k}, t) \dot{h}_{ij}^*(\mathbf{k}, t), \quad (2.116)$$

where $d\Omega_k$ represents a solid angle element in $\mathbf{k}$ -space. The integral average over the spherical shell of radius $|\mathbf{k}|$ mimics in the lattice the stochastic average of Eq. (1.91), as long as GWs are produced by a source with random initial fluctuations. Following Ref. [179], we obtain a discretized version of the aforementioned equation

$$\left(\frac{d\rho_{\text{GW}}}{d\log k} \right) (\tilde{\mathbf{n}}) = \frac{k^3(\tilde{\mathbf{n}}) dx^3}{(4\pi)^3 G \mathcal{N}^3} \left\langle \dot{h}_{ij}(|\tilde{\mathbf{n}}|, t) \dot{h}_{ij}^*(|\tilde{\mathbf{n}}|, t) \right\rangle_{R(\tilde{\mathbf{n}})}, \quad (2.117)$$

$$(2.118)$$

We introduce different versions of the GW energy density power spectrum normalized by the critical energy density. The GW energy density power spectrum normalized by the critical energy density for **Type-I** is

$$\Omega_{\text{GW}}(\tilde{\mathbf{n}}, t) = \frac{1}{\rho_c} \frac{m_p^2 k(l)}{8\pi} \frac{\delta x}{\mathcal{N}^5} \#_l \left\langle \left[\dot{h}_{ij}(\tilde{\mathbf{n}}, t) \right] \left[\dot{h}_{ij}^*(\tilde{\mathbf{n}}, t) \right] \right\rangle_{R(l)}, \quad (2.119)$$

where $k(l) = k_{\text{IR}} l$. In program variables this is expressed as

$$\begin{aligned} \Omega_{\text{GW}}(\tilde{\mathbf{n}}, t) &= \frac{1}{\tilde{\rho}_c} \frac{\kappa(l)}{(8\pi a^{2\alpha})} \left(\frac{\delta \tilde{x}}{\mathcal{N}^5} \right) \left(\frac{f_*}{m_p} \right)^2 \#_l a^{-2(3-\alpha)} \\ &\quad \left\langle [\Lambda_{ij,kl}(\tilde{\mathbf{n}})(\pi_{\tilde{u}})_{kl}(\tilde{\mathbf{n}}, t)] [\Lambda_{ij,mn}(\tilde{\mathbf{n}})(\pi_{\tilde{u}})_{mn}(\tilde{\mathbf{n}}, t)]^* \right\rangle_{R(l)}, \end{aligned} \quad (2.120)$$

while for **Type-II**, the GW energy density power spectrum normalized by the critical energy density is

$$\Omega_{\text{GW}}(\tilde{\mathbf{n}}, t) = \frac{1}{\rho_c} \frac{m_p^2 k^3(l)}{8\pi^2} \frac{\delta x^3}{\mathcal{N}^3} \left\langle \left[\dot{h}_{ij}(\tilde{\mathbf{n}}, t) \right] \left[\dot{h}_{ij}^*(\tilde{\mathbf{n}}, t) \right] \right\rangle_{R(l)}, \quad (2.121)$$

and is expressed in program variables as

$$\begin{aligned} \Omega_{\text{GW}}(\tilde{\mathbf{n}}, t) &= \frac{1}{\tilde{\rho}_c} \frac{\kappa^3(l)}{(8\pi^2 a^{2\alpha})} \left(\frac{\delta \tilde{x}}{\mathcal{N}} \right)^3 \left(\frac{f_*}{m_p} \right)^2 a^{-2(3-\alpha)} \\ &\quad \left\langle [\Lambda_{ij,kl}(\tilde{\mathbf{n}})(\pi_{\tilde{u}})_{kl}(\tilde{\mathbf{n}}, t)] [\Lambda_{ij,mn}(\tilde{\mathbf{n}})(\pi_{\tilde{u}})_{mn}(\tilde{\mathbf{n}}, t)]^* \right\rangle_{R(l)}. \end{aligned} \quad (2.122)$$

Chapter 3

Spectroscopy of particle couplings with gravitational waves

In this chapter, we discuss the possibility to determine particle couplings with GWBs. It is expected that under certain circumstances a series of peaks of different amplitude and frequency, like a stairway, emerge in a GWB spectrum, with each peak corresponding to a different coupling. The detection of such signature opens the possibility to reconstruct couplings (spectroscopy) of particle species involved in high energy phenomena generating GWBs. These stairwaylike signatures may arise in causally produced backgrounds in the early Universe, e.g., from preheating or first order phase transitions. As a proof of principle we study a preheating scenario with an inflaton ϕ coupled to multiple daughter fields $\{\chi_j\}$ with different coupling strengths g_j . As a clear stairway signature is imprinted in the GWB spectrum, we reconstruct the relevant couplings with various detectors. This chapter is based on article [1].

3.1 A proof of principle

In high energy physics, we mostly rely on particle colliders looking for features in statistical observables. In our proposal we rather look for features in GWs spectra. Considering early universe scenarios sourcing GWs, we address the question of how the presence of multiple fields, all contributing to GW production, affect the Gravitational Wave Background (GWB) spectrum. While the source of GWs is a linear superposition of individual contributions, the GWs spectrum is quadratic in the source. Furthermore, the dynamics of the different fields in the multi-field system may differ from their single-field dynamics. Under certain circumstances, namely large separation of scales of the daughter-inflaton couplings or negligible interactions among daughter species, a

sequence of peaks of different amplitude and frequency, a *stairway*, may emerge in the GW spectrum, probing the leading interaction of each individual species. Detecting such signature would allow to reconstruct the relevant couplings (spectroscopy) of the field species involved in high energy phenomena generating GWs.

As a proof of principle, we consider a preheating scenario where an inflaton ϕ oscillates around the minimum of its potential and multiple daughter fields $\{\chi_j\}$ are coupled to the inflaton, each with different coupling strength. When the couplings are large enough, the daughter species are excited in broad parametric resonance, developing spectra that are peaked at scales determined by their coupling strength. By studying the GWB sourced by various daughter fields, we show that, in fact, a stairway of peaks of different amplitude and frequency, emerges in the GWB spectrum. As each peak probes a different coupling, the detection of such signature allows for a reconstruction of the coupling constants. Our working scenario represents a compelling case of the ability of GWBs to probe particle couplings. We consider a preheating scenario where an inflaton ϕ oscillates around the minimum of its potential, while multiple *daughter* fields $\{\chi_j\}$ are coupled to the inflaton, each with different coupling strength. We consider an observationally viable inflationary model inspired by α -*attractors* [180]

$$V(\phi, \{\chi_j\}) = \frac{1}{2}\Lambda^4 \tanh^2\left(\frac{\phi}{M}\right) + \frac{1}{2}g_j^2\chi_j^2\phi^2. \quad (3.1)$$

We fix $M = 5m_p$ and $\Lambda = 0.00564m_p$ for compatibility with CMB constraints, and keep free the coupling constants g_j^2 . The inflaton potential flattens out for $|\phi| \gg M$, and takes the form $V \simeq \frac{1}{2}\omega_*^2\phi^2$ for $|\phi| \ll M$, with $\omega_* = \Lambda^2/M$. We choose quadratic interactions as these are scale-free and serve as a proxy for the leading term in scalar-gauge interactions [181]. Preheating takes place via *broad parametric resonance*, as the oscillations of the homogeneous inflaton induce an exponential growth of the preheat mode functions $\chi_j(k, t)$. The resonance of each species is controlled by their resonance parameter $q_j = g_j^2\phi_*^2/\omega_*^2 \gg 1$, with $\phi_* \simeq 0.95m_p$ the inflaton amplitude at the end of inflation. Modes of each preheat species are exponentially excited during the initial linear regime, up to a co-moving scale $k \lesssim k_l^{(j)} \sim \omega_*q_j^{1/4}$ [30]. Eventually, the preheat fields back-react into the inflaton, and the dynamics become non-linear. As a result, mode-to-mode interactions populate higher momenta beyond the linear threshold, typically up to a peak scale $k_p^{(j)} \sim \omega_*q_j^p$, $p \gtrsim 0.5$ [107]. Modes with $k > k_p^{(j)}$ are exponentially suppressed. The system reaches eventually a stationary state characterized by no significant transfer of energy among species [182–184]. When the couplings are large enough, the daughter species are excited in broad parametric resonance, developing spectra that are peaked at scales determined by their coupling strength. By studying the GW sourced by various daughter fields, we show that, in fact, a stairway of peaks

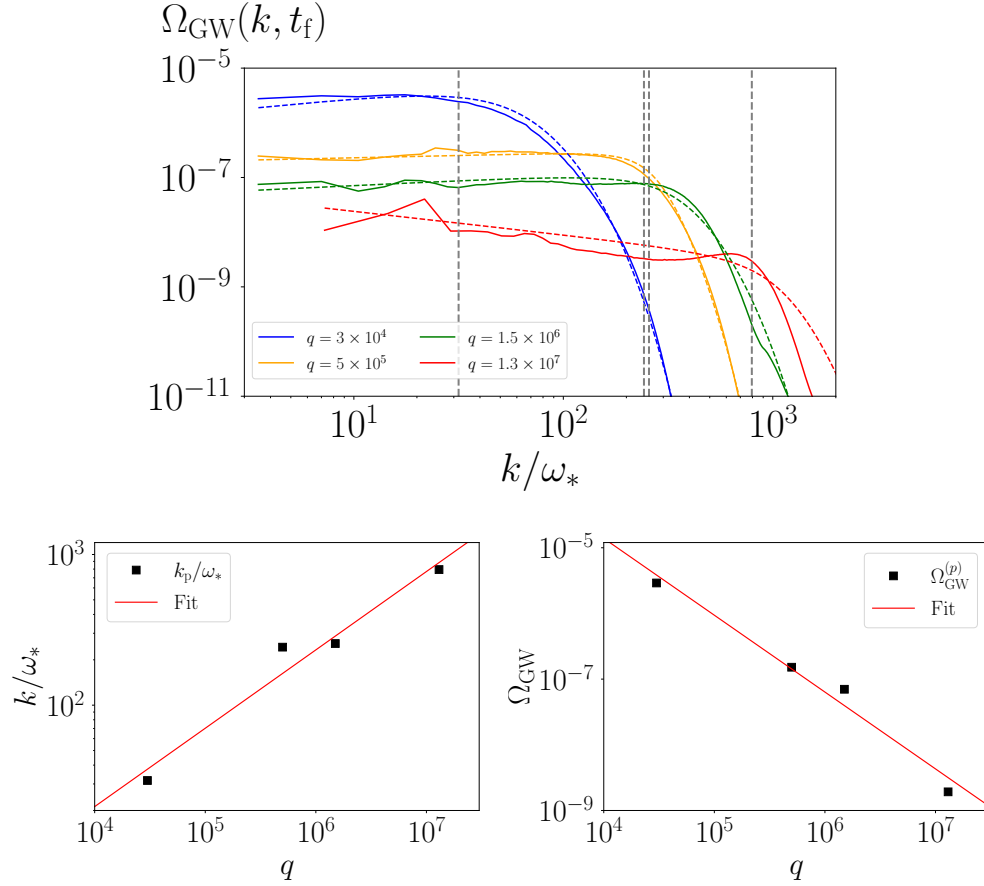


FIGURE 3.1: *Top*: Gravitational wave spectrum evaluated at the end of production (t_f) for single preheat fields with resonance parameters q . Solid lines are the simulations results, and dashed lines are the best fit function from template in Eq. (3.2). *Bottom-Left*: Power-law fit of measured values of the peak momenta of the GWB as a function of the resonance parameter q . *Bottom-Right*: Power-law fit of measured values of the peak amplitude of the GWs as a function of the resonance parameter q .

of different amplitude and frequency, emerges in the GW spectrum [72]. As each peak probes a different coupling, the detection of such signature allows for a reconstruction of the coupling constants. Our working scenario represents a compelling case of the ability of GWs to probe particle couplings.

3.2 Stairway in the gravitational wave spectrum

The dynamics of the fields lead to an anisotropic stress with non-vanishing transverse-traceless (TT) part $\Pi_{ij}^{\text{TT}} = \{\partial_i \phi \partial_j \phi + \partial_i \chi_a \partial_j \chi_a\}^{\text{TT}}$. GWs are governed by Eq. (1.86) and its characterized by its power spectrum given by Eq. (1.98), see Sec. 1.4, which are sourced by all field species as given by Π_{ij}^{TT} . The non-linear dynamics of the system can only be studied with lattice simulations. We run simulations with a single daughter

field for different coupling constants g_J^2 , we have parametrized the peak's position (k_p) and amplitude ($\Omega_{\text{GW}}^{(p)} \equiv \Omega_{\text{GW}}(k_p)$) of single-daughter GW spectra as a function of $q = g^2\phi_*^2/\omega_*^2$. By measuring spectra at a time $\omega_*t_f = 600$, we guarantee (for all couplings used) that the system is in the stationary regime when GW production ceases. In Fig. 3.1, we show the GW spectrum resulted from the parametric resonance of a single rehear field coupled to the inflaton with a characteristic resonance parameter q . We fit each spectrum with the following template function

$$\Omega_{\text{GW}}(k, t_f) = 10^{\log_{10}\Omega_{\text{GW}}^{(p)}} \frac{2 \cdot \left(\frac{k}{k_p}\right)^n}{1 + \exp\left\{\frac{\delta \cdot (k - k_p)}{k_p}\right\}}, \quad (3.2)$$

in order to identify both the position of peak of the spectrum k_p/ω_* and the peak amplitude $\Omega_{\text{GW}}^{(p)}$. We obtain (for $q \gtrsim 10^4$) the following fit as a function of the resonance parameter

$$k_p(q)/\omega_* = (21.22 \pm 7.64) \left(\frac{q}{10^4}\right)^{0.52 \pm 0.08}, \quad (3.3)$$

$$\Omega_{\text{GW}}^{(p)}(q) = (1.2 \pm 0.7) \times 10^{-5} \left(\frac{q}{10^4}\right)^{-1.1 \pm 0.12}, \quad (3.4)$$

It is clear from Eqs. (3.3), (3.4) that sufficiently separated values of q lead to single-daughter spectra with quite different amplitudes and peak frequencies. In realistic scenarios we expect several daughter fields to be present. If these are coupled to the inflaton with different coupling strengths, relevant questions can then be raised: will the final GW spectrum be the superposition of each individual daughter's spectrum? do interference effects affect the final spectral shape? can we expect a series of peaks to emerge at the positions indicated by single daughter field spectra?

In Fig. 3.2 we plot GW spectra from single daughter simulations (left, central and right panels), and superimpose the spectrum from simulations with two (central panel) or three (right panel) daughter fields, with the same couplings as the single daughter cases. The plots are self-explanatory: a *stairway* signature consisting of a series of peaks, as many as the daughter fields, emerges clearly in the multi-daughter spectra. As the energy must be split between the different species, the stairway peak amplitudes are smaller than in the individual daughter field spectra. The position of the stairway peaks, however, is still well predicted by the single-daughter spectra prediction Eq. (3.3). Redshifting the background, the peak frequencies of a signal today is approximated by result in Sec. 1.4.2. Here we have assumed the time to reheating is negligible, i.e. $\epsilon_{\text{RD}}^{1/4} \approx 1$,

$$f_p \simeq 4 \cdot 10^{10} \times \left(\frac{H_f}{m_p}\right)^{1/2} \times \frac{k_p(q)}{a_f H_f} \text{ Hz}. \quad (3.5)$$

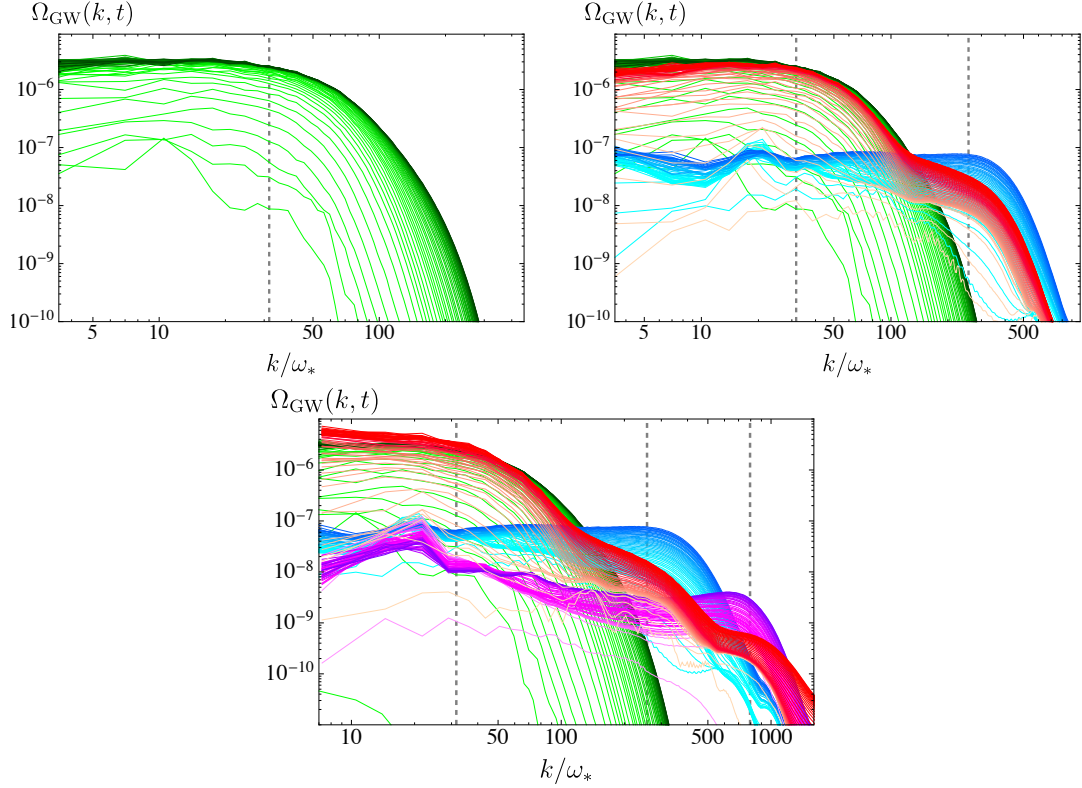


FIGURE 3.2: *Top-left*: GW spectrum from a single daughter field with $q = 3 \times 10^4$. *Top-right*: GW spectra from single daughter field with $q_1 = 3 \times 10^4$ (green) and $q_2 = 1.5 \times 10^6$ (blue). A two-peak spectrum (red) emerges when the two daughter fields are simultaneously coupled to the inflaton with the same couplings as in the single daughter cases. *Bottom*: GW spectra from three single daughter fields for $q_1 = 3 \times 10^4$ (green), $q_2 = 1.5 \times 10^6$ (blue) and $q_3 = 1.3 \times 10^7$ (purple). A three-peak spectrum appears when the three daughter fields are simultaneously coupled to the inflaton with the same couplings as the single daughter cases. In all panels $\Omega_{\text{GW}}(k, t)$ stands for the GW spectrum at the time of production (measured every $\Delta(\omega_* t) = 5$ units up to $\omega_* t_f = 600$), and vertical dashed lines indicate the peak position as predicted by Eq. (3.3) for single daughter simulations.

where $H_f = H(t_f)$ and $a_f = a(t_f)$. To separate clearly the peaks we need $q_j/q_i \gtrsim \text{few} \times 10$. It is also important that interactions among daughter species are negligible during GW production (in our calculations we simply neglected daughter-to-daughter interactions). If daughter-to-daughter interactions were active during GW production, this would affect their dynamics, for example re-distributing power differently among them. The presence of such interactions may therefore affect significantly the peak structure of the resulting stairway in the GW spectrum.

3.3 Particle coupling spectroscopic reconstruction

In order to perform the GW spectrum reconstruction, we use data analysis techniques [185–187]. We reconstruct the multi-peak GW spectra generated with

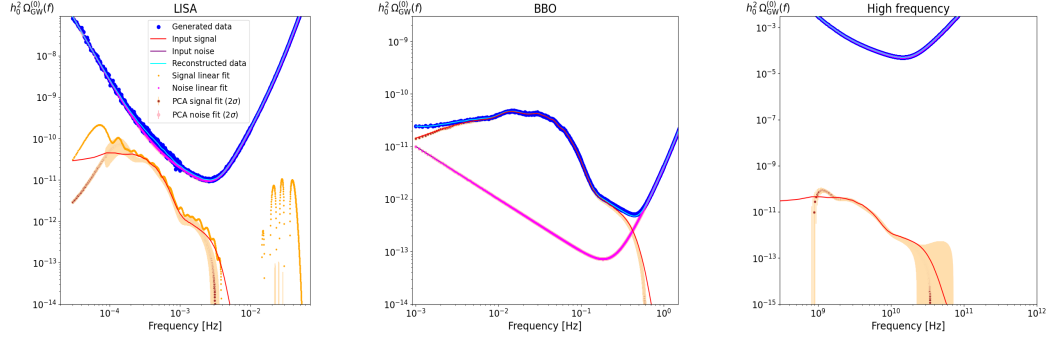


FIGURE 3.3: Signal and noise reconstructions using Principal Component Analysis (PCA) over a two-peak GW signal ($\Omega_{\text{GW}}^{(0)}$ stands for the GW spectrum today) with $q_1 = 3 \cdot 10^4$ and $q_2 = 1.5 \cdot 10^6$, for LISA (left panel), BBO (central panel) and a speculative High Frequency Experiment (HFE, right panel). The dots with error bars represent the PCA fit to the signal and the noise.

the simulations, and turn the reconstruction into constraints on the particle couplings thanks to Eqs. (3.3), (3.5). We generate data sets for the expected GW signals using the results of the simulations as templates to mimic an observation. We first apply the Principal Component Analysis (PCA) method of [186]. This is a model independent procedure for reconstructing an unknown signal by first expanding it onto a general basis of functions, and then dropping the low information components of the Fisher Information Matrix (FIM), $F_{\alpha\beta} \equiv \langle -\frac{\partial^2 \ln \mathcal{L}}{\partial \theta^\alpha \partial \theta^\beta} \rangle$.

Inspecting the fitting formula Eq. (3.3) with the redshift one Eq. (3.5), shows us that the GW peak frequencies around ~ 100 MHz for $q_j \sim 10^4 - 10^6$. These frequencies are way above the range accessible to present GW direct detection experiments like LIGO [188], Virgo [189] and KAGRA [154], planned detectors like LISA [151], ET [159], or even futuristic projects like BBO [190]. While other multi-field scenarios might produce GWBs with stairway signatures at observable frequencies, like hybrid preheating scenarios [63, 64, 68] or first order phase transitions [124–131, 131–134] with gauge fields, any characterization of their potential spectroscopic signatures requires dedicated studies currently unavailable. In order to proceed with a coupling reconstruction, we consider instead, for simplicity, that our multi-peak GWB spectra serve as a template for lower-frequency signal scenarios. We thus shift by hand today’s GW spectra by a *fudge* factor $\alpha_{\text{exp}} \equiv f_{\text{exp}}/f_p$, while leaving their amplitudes intact. The shifted backgrounds peak therefore at some typical frequency f_{exp} of a given experiment. By choosing $\alpha_{\text{LISA}} = 1.11 \cdot 10^{-13}$ and $\alpha_{\text{BBO}} = 1.55 \cdot 10^{-11}$, we analyze the capability of LISA and BBO to reconstruct GWB spectra with stairway structures given by our simulation-generated templates. Furthermore, we also introduce a speculative futuristic high-frequency experiment (HFE) capable of detecting our GWs at their natural high-frequencies. Examples of a 2-peak signal reconstruction are shown in Fig 3.3, performed respectively with: *i*) LISA, using the noise model from [187, 191], *ii*) BBO, using the

noise model from [190, 192], and *iii*) a futuristic high frequency experiment (HFE), the sensitivity of which we construct *ad hoc* in such a way that the signal has roughly the same signal-to-noise ratio as in LISA when shifting the LISA noise curve, see Fig 3.3. The two peaks in the signal are clearly identified in the three cases. By using our procedure, together with alternative methods [185, 187], it is possible to reconstruct the GWB frequency shape in an agnostic way. Once an appropriate shape has been identified, a direct search can be actually used in order to directly constrain the model parameters. For this purpose we use the same likelihood as in Eq.(3.12) of [187], and model a general 2-peaked signal as a sum of power law signals with exponential cut-offs, which is just the extension of the template used in Eq. (3.2):

$$h_0^2 \Omega_{\text{GW}}^{(0)} = \sum_{i=1}^2 10^{\log(h_0^2 \Omega_i^*)} \frac{2 \left(\frac{f}{f_i^*} \right)^{n_i}}{1 + \exp \left\{ \frac{\delta_i (f - f_i^*)}{f_i^*} \right\}}, \quad (3.6)$$

where $h_0 \equiv H_0[\text{Mpc/Km/s}]/100$ characterizes today's Hubble rate, $h_0^2 \Omega_i^*$ denotes the amplitude of the n -th signal in the peak frequency f_i^* , n_i its power-law tilt, and $\delta_i > 0$ controls the strength of the exponential suppression. In Fig. 3.4 we show the results obtained by injecting the same signal used in the left panel of Fig 3.3, and simultaneously fitting for signal and noise with LISA. The parameter space is sampled using **Polychord** [193, 194] via its interface with **Cobaya** [195]. In the left panel we show the reconstruction of the signal and of the noise as compared to the injections, and in the right panel we show the contour plots for the amplitudes and pivot frequencies of the two-peak spectra (after marginalizing over all other parameters). For reference we compare with results obtained with a Fisher matrix approach which only includes the signal parameters. As seen in the figure, the deviation between the two results is minimal, which implies that the inclusion of noise parameters does not significantly affect the reconstruction of the signal. In the three cases we reconstruct the coupling constants as indicated in Table 3.1, for a GW signal obtained from a two-daughter field simulation with $q_1 = 3 \cdot 10^4$ and $q_2 = 1.5 \cdot 10^6$. The determination of the peak frequencies and their errors for each experiment (for BBO and HFE we use a FIM approach), leads to a reconstruction of the resonance parameters inverting Eqs. (3.3) & (3.5), from where we obtain the coupling constants. While the numerical reconstruction of q_i 's and g_i 's is only precise to within the order of magnitude, it is remarkable that all reconstructed values are compatible at 1σ with the injected values. This clearly indicates the great potential for particle coupling spectroscopy that GWB detection may offer in the future.

Realistic early universe scenarios are naturally expected to contain multiple fields, as there is no symmetry or fundamental principle that implies otherwise. The Standard Model (SM), between matter and gauge fields, contains dozen of fields. Beyond the

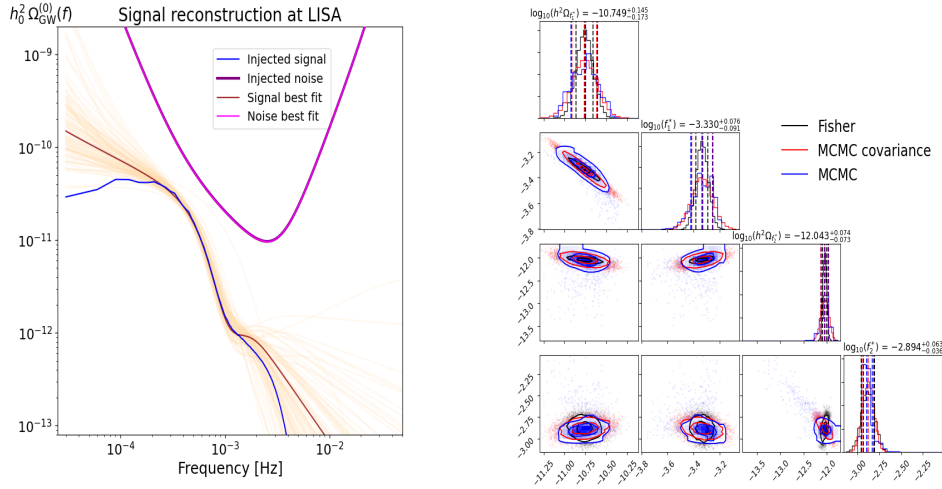


FIGURE 3.4: *Left*: Signal and noise reconstruction of a double peaked signal at LISA (see main text for details). The 2σ signal and noise error bands are shown in orange and pink respectively. *Right*: Corner plot for some of the signal parameters showing the 1 and 2σ regions. The blue curves are the MCMC chains results, the red curves correspond to the covariance matrix obtained from the MCMC chains and in black we show the constraints obtained with a Fisher matrix approach.

	LISA	BBO	HFE
$\log_{10}(f_1^* [\text{Hz}])$	$-3.33^{+0.08}_{-0.09}$	$-1.539^{+0.002}_{-0.002}$	$9.45^{+0.15}_{-0.15}$
$q_1 = 3 \cdot 10^4$	$6.14^{+29.86}_{-4.10} \cdot 10^4$	$1.28^{+2.45}_{-0.62} \cdot 10^4$	$2.83^{+17.88}_{-2.06} \cdot 10^4$
$g_1 = 1.16 \cdot 10^{-3}$	$1.66^{+4.01}_{-0.55} \cdot 10^{-3}$	$0.76^{+0.73}_{-0.18} \cdot 10^{-3}$	$1.13^{+3.56}_{-0.41} \cdot 10^{-3}$
$\log_{10}(f_2^* [\text{Hz}])$	$-2.89^{+0.06}_{-0.04}$	$-0.496^{+0.001}_{-0.001}$	$10.4^{+0.4}_{-0.4}$
$q_2 = 1.5 \cdot 10^6$	$0.43^{+2.8}_{-0.3} \cdot 10^6$	$1.3^{+7.42}_{-0.89} \cdot 10^6$	$1.9^{+108.65}_{-1.78} \cdot 10^6$
$g_2 = 8.2 \cdot 10^{-3}$	$4.39^{+14.3}_{-1.53} \cdot 10^{-3}$	$7.64^{+21.8}_{-2.61} \cdot 10^{-3}$	$9.23^{+263.9}_{-4.32} \cdot 10^{-3}$

TABLE 3.1: Numerical reconstruction of frequency peaks, resonance parameters and coupling constants.

SM (BSM) scenarios only add up to this argument. In this chapter, we discuss the possibility to measure particle couplings through GWs. We focus on the emergence of a sequence of peaks of different amplitude and frequency, a *stairway*, in the GW spectrum produced in scenarios with multiple particle species involved in the GW generation. As a proof of principle we have studied a preheating scenario with an oscillating inflaton ϕ , coupled simultaneously to multiple *daughter* fields $\{\varphi_j\}$, interacting through $g_j^2 \varphi_j^2 \phi^2$ with different coupling strengths. When the daughter species are excited, we show that a stairway signature with as many peaks as fields $\{\varphi_j\}$, is imprinted in the GWB spectrum, with each peak probing the interaction strength of each individual species.

Our working scenario represents a compelling case of the ability of GWs to probe particle couplings. We have assumed a quadratic potential $V \propto \phi^2$ during the inflation

oscillations, so that only daughter field peaks are imprinted. If a different shape was considered, say e.g. $V \propto \phi^p$ with $p \neq 2$ or a linear combination of monomials, inflaton self-resonance will possibly create new peaks in the GW spectrum. For simplicity we have assumed the same type of quadratic interactions $\propto \varphi_j^2 \phi^2$ for all daughter species.

Chapter 4

Gravitational wave production from preheating with trilinear interactions

In this chapter, we investigate the production of gravitational waves (GWs) during preheating with monomial and polynomial inflationary potentials, considering a trilinear coupling $\phi\chi^2$ between a inflaton ϕ and a daughter scalar field χ . We have found that trilinear interactions generate exponential production of χ particles and large amounts of gravitational wave radiation, in form gravitational wave backgrounds (GWB), is generated throughout the process. We study the linear and non-linear dynamics of preheating with lattice simulations, following the production of GWs. These GWBs are peaked at high frequencies which makes them undetectable by current or planned GW observatories. The amount of GWs produced is in any case remarkable, we discuss the prospects for probing the GWB indirectly using constraints on the effective number of relativistic species in the universe N_{eff} . This chapter is based on article [2].

4.1 Introduction

The usual preheating picture is based in quadratic interactions $\phi^2\chi^2$ of the inflaton ϕ with another scalar field χ , the inflaton can never decay completely in this manner [30, 182–184, 196]. In order to complete the inflaton decay, interactions of the form $\phi\chi^n$ should be accounted for. Trilinear interactions with $n = 2$ are the most natural choice, as interactions of this type are always present in particle physics. An example are Yukawa interactions, which represent a coupling between a scalar and two fermions. Three-leg decay via fermions was in fact the first perturbative decay considered of the

inflaton to achieve reheating [197, 198], it was realized later that the inflaton also decays via non-perturbative parametric excitations in that case [199–202]. Three-leg decay via interactions with bosons leads also to non-perturbative parametric excitations [41], and this option is actually expected to take over Yukawa interactions, simply due to Pauli blocking of the Fermions. Trilinear interaction $\phi\chi^2$ between an inflaton ϕ and a scalar daughter field χ represents a natural coupling between an inflationary sector and a matter sector.

In this chapter, we consider in particular the interaction term $\frac{1}{2}\sigma\phi\chi^2$, where σ is a coupling of mass dimensions. We can understand the excitation mechanism in the following manner, if towards the end of inflation the inflaton oscillates coherently around the origin of its potential, the effective oscillation frequency of a χ -field mode function χ_k is given by $\omega_k^2(t) = \frac{k^2}{a^2} + m_\chi^2 + \sigma\phi(t)$, with k the modulus of the comoving momentum, and $a(t)$ the scale factor. If the mass of χ satisfies $m_\chi^2 \ll \sigma|\phi|$, the trilinear coupling will induce a tachyonic mass for χ whenever $\phi < 0$, which happens during half of the time of each inflaton oscillation. In this manner, modes within the infrared (IR) window $k^2 < \sigma|\phi|a^2$ will suffer an exponential amplification every half-period of the inflaton oscillations. This resonant mechanism, dubbed *tachyonic resonance* in Ref. [41], is a very efficient manner of χ particles production, causing preheating to be completed within few oscillations of ϕ .

It could be the case that the field χ is the Standard Model (SM) Higgs. The question of the stability of its vacuum becomes of utmost relevance. It is known that the presence of a trilinear coupling does not affect the stability of the Higgs vacuum during inflation, but it can be relevant during preheating [203]. If the trilinear coupling σ is large enough, the exponentially large occupation number of the Higgs created by tachyonic resonance can destabilize the vacuum. Lattice simulations have been used in order to set upper bounds on the possible value of the trilinear coupling σ , and on the dimensionless coupling of a scale-free interaction $\frac{1}{2}\lambda_{h\phi}\phi^2h^2$, based on demanding a stable evolution of the Higgs during preheating. According to Ref. [44], Higgs vacuum stability is assured whenever the couplings respect the bounds $10^{-10} < \lambda_{h\phi} < 10^{-8}$ and $\sigma < 2 \times 10^8$ GeV.

In this chapter, we consider a field χ with its self-interaction coupling strictly positive¹. The excitation of the field modes χ_k within the IR tachyonic window provides a time-dependent spatial configuration for $\chi(\mathbf{x}, t)$, which in turn is a source of gravitational radiation. We consider different inflaton potentials $V_{\text{inf}}(\phi)$ with minimum at the origin, and study the GW production during preheating in each case, as a function of σ . The periodic tachyonic excitation of the IR modes χ_k resembles to some extent the excitation

¹In other words, χ could be identified with the SM Higgs only if the self-interaction coupling λ of the Higgs remains positive at large scales, something that depends (in the absence of beyond the SM physics) very sensitively on the choice of α_s and the top Yukawa coupling y_t [204, 205].

of gauge fields during preheating after axion-inflation [65, 66], or the case of axion inflation which will be discussed in chapter 6. These scenarios are known to be efficient sources of GWs [67, 68, 111]. We also investigate, if these GWBs could be constrained by upper bounds on the number of relativistic species in the universe [206–210], such that we can place constraints on the coupling σ . Moreover, we speculate the possibility, that due to the nature of the trilinear interaction, the frequencies of the expected GWB could be shifted towards smaller (more observable) frequencies than the GWBs produced in standard $\phi^2\chi^2$ preheating [107].

4.2 Monomial potential

We consider a quadratic potential for the inflaton ϕ around the origin. Monomial potentials $V_{\text{inf}}(\phi) \propto \phi^2$ for inflation are strongly disfavored² by latest cosmological data [26], we can consider consistent inflaton potentials that flatten out towards large field values by developing a plateau, and then adopt a monomial shape around the origin only after inflation ends. As we have seen already, potentials that satisfy this conditions are α -attractors [180] and can be written as $V_{\text{inf}}(\phi) = \frac{1}{2}\Lambda^4 \tanh^2(\phi/M)$, where M and Λ constants have dimensions of mass. In the limit $|\phi| \ll M$, the potential becomes $V_{\text{inf}}(\phi) \simeq \frac{1}{2}m_\phi^2\phi^2$, with $m_\phi^2 \equiv \Lambda^2/M$. Compatibility with CMB observations allow us to choose $M = 5m_p$, which is equivalent to $\Lambda = 0.00564m_p$, and from there to an inflaton mass $m_\phi \simeq 1.6 \cdot 10^{13}$ GeV. Finding m_ϕ from fitting CMB anisotropy observables with an inflationary quadratic potential $V_{\text{inf}}(\phi) = \frac{1}{2}m_\phi^2\phi^2$, leads to a very similar scale for the inflaton mass as in the α -attractor case. So, in practice, we then simply consider a quadratic potential for the inflaton ϕ , and supplement it with a trilinear interaction $\phi\chi^2$ with a daughter scalar field χ , for which we allow self-interactions,

$$V(\phi, \chi) = \frac{1}{2}m_\phi^2\phi^2 + \frac{\sigma}{2}\phi\chi^2 + \frac{\lambda}{4}\chi^4, \quad (4.1)$$

where σ is a coupling with dimensions of mass, and λ is the self-interaction coupling of the daughter field. We neglect a term $\frac{1}{2}m_\chi^2\chi^2$ in Eq. (4.1) as we anticipate that the typical momenta excited $k/a \lesssim \sigma|\phi|$ will be very large, and hence we simply assume that $m_\chi \ll \sigma|\phi|$. It is however necessary to include the χ^4 self-interaction to ensure that the potential is bounded from below and, consequently, only stable dynamics are developed. We then identify a *critical* coupling

$$\lambda_c(\sigma, m_\phi) \equiv \frac{\sigma^2}{2m_\phi^2}, \quad (4.2)$$

²Monomial potentials during inflation are not ruled out if an appropriate non-minimal coupling between the inflaton and the Ricci scalar is also present [211].

so that values $\lambda < \lambda_c$ are not allowed because the potential in Eq. (4.1) would be unbounded from below. The critical value λ_c defines in fact a flat direction $\phi = -\frac{\sigma\chi^2}{2m_\phi^2}$, where $V(\phi, \chi) = 0$. For $\lambda > \lambda_c$ the flat direction is lifted and the potential has a minimum at $\phi = \chi = 0$ where $V(\phi, \chi) = 0$. The requirement that the potential has to be bounded from below imposes a relation between the two couplings, σ and λ . In our analysis, we adopt a value $\lambda = 2\lambda_c$ so that our numerical dynamics are guaranteed to avoid running-away solutions.

4.2.1 Preheating dynamics and Lattice simulations

The dynamics of preheating is given by the shape of the potential around the minimum, our results hold for a whole family of inflationary models with similar effective potential around the origin, as long as there is a leading quadratic term. The monomial inflaton potential that we considered in Eq. (4.1), $V_{\text{inf}}(\phi) = \frac{1}{2}m_\phi^2\phi^2$, is a good approximation to α -attractor potentials $V_{\text{inf}}(\phi) = \frac{1}{2}\Lambda^4 \tanh^2(\phi/M)$ at field amplitudes $|\phi| \ll M$. We perform the analysis by fixing the initial amplitude Φ_i of the inflaton at the beginning of preheating, or the onset of inflaton oscillations, by determining the end of inflation in a quadratic potential $V_{\text{inf}}(\phi) \propto \phi^2$. The inflaton behaves as a homogeneous condensate, which then oscillates around the origin. During this regime, the inflaton excites particles of the χ field transferring exponentially energy into the new modes of the field. In this model, preheating occurs via a trilinear interaction which provides an effective time-dependent mass term for χ with oscillatory sign in time. For each moment $\phi < 0$, which happens during half the time of each oscillation, the trilinear coupling supports a negative squared mass for χ . Hence, the corresponding χ_k modes will be exponentially enhanced every time that the inflaton is negative. This mechanism, known as tachyonic resonance, leads to a very efficient production of χ particles after few oscillations of ϕ [41, 44]. The system is fully characterized by the following set of equations,

$$\ddot{\phi} + 3H\dot{\phi} + m_\phi^2\phi + \frac{\sigma}{2}\chi^2 - \frac{\nabla^2\phi}{a^2} = 0, \quad (4.3)$$

$$\ddot{\chi} + 3H\dot{\chi} + \lambda\chi^3 + \sigma\chi\phi - \frac{\nabla^2\chi}{a^2} = 0, \quad (4.4)$$

where $a(t)$ is the scale factor and $H \equiv \dot{a}/a$ the Hubble rate. We will first consider the linear regime, where the inflaton ϕ is treated as a homogeneous background field. In order to do this, we start by a change of variables

$$d\tilde{x}^\mu = m_\phi dx^\mu, \quad \tilde{\phi} = \frac{\phi}{\Phi_i}, \quad \tilde{\chi} = \frac{\chi}{\Phi_i}, \quad \tilde{H} = \frac{H}{m_\phi}, \quad (4.5)$$

and then move perform a Fourier transformation, where we can quantize the field χ

$$\tilde{\chi}(\mathbf{x}, t) = \int \frac{d^3\mathbf{k}}{(2\pi)^3} [\hat{a}_k \tilde{\chi}_k(t) e^{i\mathbf{k}\cdot\mathbf{x}} + \hat{a}_k^\dagger \tilde{\chi}_k^*(t) e^{-i\mathbf{k}\cdot\mathbf{x}}], \quad (4.6)$$

with $\hat{a}_k^\dagger$, $\hat{a}_k$ standard creation/annihilation operators that satisfy $[\hat{a}_k, \hat{a}_{k'}^\dagger] = (2\pi)^3 \delta(\mathbf{k} - \mathbf{k}')$. The evolution equation of the mode functions $\tilde{\chi}_k$, is written as a linear equation using the Hartree approximation $\tilde{\chi}^3 \rightarrow 3\tilde{\chi} \langle \tilde{\chi}^2 \rangle$ in Eq. (4.4)

$$\tilde{\chi}_k'' + 3\tilde{H}\tilde{\chi}_k' + \left(3q_\chi \langle \tilde{\chi}^2 \rangle + q_3 \tilde{\phi}(t) + \frac{\kappa^2}{a^2}\right) \tilde{\chi}_k = 0, \quad \text{with } \kappa = \frac{k}{m_\phi}, \quad (4.7)$$

where prime ' stands for derivatives with respect to the new time variable, and the variance $\langle \tilde{\chi}^2 \rangle$ can be related to the *power spectrum* of χ as follows

$$\langle \tilde{\chi}^2 \rangle = \int d\log k \Delta_\chi(k), \quad \Delta_\chi(k) \equiv \frac{k^3}{2\pi^2} \mathcal{P}_\chi, \quad \langle \chi_{\mathbf{k}} \chi_{\mathbf{k}'}^* \rangle \equiv (2\pi)^3 \mathcal{P}_\chi(k) \delta(\mathbf{k} - \mathbf{k}'). \quad (4.8)$$

We note that in Eq. (4.7) two dimensionless *resonance parameters* have emerged

$$q_3 \equiv \frac{\sigma \Phi_i}{m_\phi^2}, \quad \text{and} \quad q_\chi \equiv \frac{\lambda \Phi_i^2}{m_\phi^2}, \quad (4.9)$$

with Φ_i the initial amplitude of the homogeneous field at the end of inflation. These parameters play a fundamental role in the model because they control the intensity of the daughter field excitation. Taking a look at the mode frequency in Eq. (4.7), we can write

$$\tilde{\omega}_k^2 = m_\chi^2 + (\kappa/a)^2, \quad m_\chi^2 = \left(q_3 \tilde{\phi}(t) + 3q_\chi \langle \tilde{\chi}^2 \rangle\right). \quad (4.10)$$

The inflaton field oscillates in a quadratic potential as $\tilde{\phi}(t) \simeq \Phi(t) \sin(m_\phi t)$, with $\Phi(t) \propto \frac{1}{t} \propto \frac{1}{a^{3/2}}$ [212], we see that every half oscillation the inflaton amplitude becomes negative. Hence, in Eq. (4.10) we can see that every time that $\phi < 0$, $\tilde{\omega}_k^2$ becomes negative for sufficiently infrared momenta with κ below the cut-off $\kappa^2 < -m_\chi^2$ there is an exponential growth of those modes $\tilde{\chi}_k$. This is also at the same time an exponential growth of the variance $\langle \tilde{\chi}^2 \rangle$ (or equivalently into an exponential growth of the power spectrum $\mathcal{P}_\chi(k)$ of the IR modes). Such exponential growth must happen in a step-like manner, as every time the inflaton amplitude is negative we expect the variance to grow, whereas every time the inflaton is positive we expect the variance to oscillate (simply due to the mode oscillations when $\tilde{\omega}_k^2 > 0$ for $\phi > 0$). An overall exponential growth in steps is then expected.

In order to study the non-linear dynamics of the subsequent stages of the field excitation, we switch to using lattice simulations. In this work, we have considered the set of equations Eq. (4.3) and Eq. (4.4) and solved numerically a discrete version of them

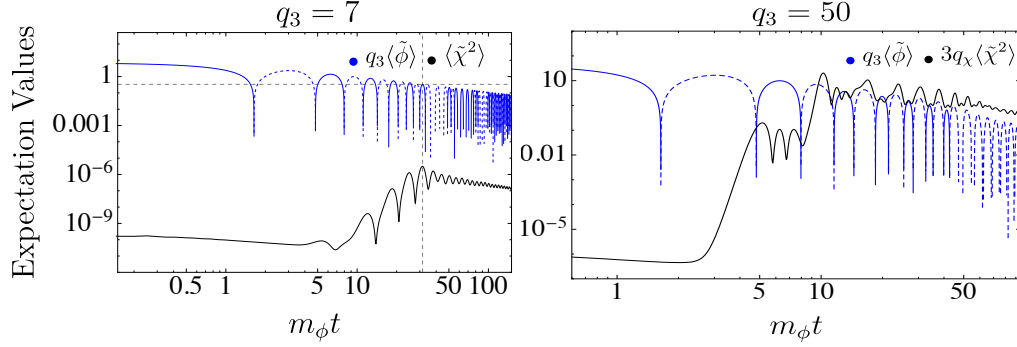


FIGURE 4.1: *Left:* Time evolution of the inflaton's volume average $\langle \tilde{\phi} \rangle$ times the resonance parameter q_3 ($q_{3\text{eff}} = q_3 \langle \tilde{\phi} \rangle$), and the variance of the daughter field $\langle \tilde{\chi}^2 \rangle$ for $q_3 = 7$. The horizontal dashed line corresponds to $q_{3\text{eff}} = 1/3$, whereas the vertical dashed line represents the time at which the maximum value of the variance is reached, $\langle \tilde{\chi}^2 \rangle_{\text{max}}$. *Right:* Time evolution of $q_3 \langle \tilde{\phi} \rangle$ and the variance of the daughter field $\langle \tilde{\chi}^2 \rangle$ times $3q_\chi$ for $q_3 = 50$. We see that once the term $3q_\chi \langle \tilde{\chi}^2 \rangle$ overtakes $q_3 \langle \tilde{\phi} \rangle$, the growth of the variance stops. For both panels, the solid (dashed) blue lines correspond to the positive (negative) part of the inflaton mean value.

using `CosmoLattice` [165, 213]. The initial conditions for the homogeneous part of the inflaton field during inflation is obtained by solving the Friedmann equation $H^2 \equiv (\frac{\dot{a}}{a})^2 = \frac{1}{3m_p^2} \left(\frac{1}{2} \dot{\phi}^2 + V(\phi) \right)$, together with Eq. (4.3) in the absence of the χ field (as well as in the absence of the $\nabla^2 \phi$ term). We obtain the initial amplitude and velocity from the breaking of slow roll condition, $\epsilon_H \equiv -\frac{\dot{H}}{H^2} = 1$, which leads to the homogeneous initial condition

$$\Phi_i \equiv \phi(t_i) = 1.00738 m_p \quad \text{and} \quad \dot{\Phi}_i \equiv \dot{\phi}(t_i) = -0.712295 m_\phi m_p. \quad (4.11)$$

In the non-linear regime, the inflaton field will develop inhomogeneities $\tilde{\phi}(\mathbf{x}, t)$, therefore if we want to extract its homogeneous part we will compute volume average $\langle \tilde{\phi} \rangle$. The initial conditions for the χ field are such that the homogeneous amplitude is set to zero, and standard quantum vacuum fluctuations are added on top, for which we follow initial conditions as introduced in Sec. 2.2.4

We performed a series of simulations with $\mathcal{N} = 512$ number of points per dimension and a infrared momentum $\kappa_{\text{IR}} \in [0.2 : 0.75]$, such that we make sure all relevant scales of the problem are well captured. Simulations were run for several values of q_3 while keeping the relation $q_\chi = q_3^2$ for $q_3 \in [1 : 10^4]$. Such relation between the resonance parameters simply implies that we are using $\lambda = 2\lambda_c$, in order to be safe from run-away solutions in the numerical dynamics. Simulations were forced to end at time $m_\phi t_f = 400$, as by this time in all cases (independently of the coupling σ), the system has reached a steady state with no further energy exchange between fields.

We proceed to discuss and explain the dynamics we identify in our lattice simulations. There is a initial stage is dominated by the inflaton oscillations, which stays as a

homogeneous field whenever the backreaction of the χ field is not relevant. In this regime, the solution to the equation of motion of the inflaton is $\phi(t) \simeq \sqrt{\frac{8}{3} \frac{m_p}{m_\phi t}} \cdot \sin(m_\phi t)$ [30]. The effective frequency of oscillation of the modes χ_k , as given in Eq. (4.7), is negative every time the inflaton is negative, which induces an exponential growth of the modes below the cut-off

$$\kappa < \sqrt{q_3 |\tilde{\phi}|} a. \quad (4.12)$$

Looking at the mode frequency $\tilde{\omega}_k^2$ in Eq. (4.10), we identify an effective tachyonic resonance parameter as $q_{3,\text{eff}} = q_3 |\tilde{\phi}|$, with q_3 the initial resonance parameter defined in Eq. (4.9). Due to the expansion of the universe, the inflaton amplitude $\Phi(t)$ decays with time, and the effective resonance parameter vanishes with time following the same decay as $q_{3,\text{eff}} \propto |\Phi| \propto \frac{1}{a^{3/2}}$. Once the value of q_{eff} drops below $1/3$, the system is switched off to a narrow resonance regime [41]. In this regime, the tachyonic excitation is not effective. These suggests a parametric dependence on the efficiency of the mechanism. Hence, depending on the value of the trilinear coupling, two regimes can be identified: *i*) For 'small' values of q_3 , $q_{3,\text{eff}}$ drops below $1/3$ before or soon enough after the first zero-crossing of the inflaton, and hence the growth of the variance of χ shuts down at a maximum value $\langle \tilde{\chi}^2 \rangle_{\text{max}}$ determined by the moment when $q_{3,\text{eff}} = 1/3$, as shown in the *Left* panel of Fig. 4.1. *ii*) For 'large' values of q_3 , the variance $\langle \tilde{\chi}^2 \rangle$ grows exponentially in steps (as many as negative semi-cycles of the inflaton) up to a new larger maximum value $\langle \tilde{\chi}^2 \rangle_{\text{max}}$, which is reached before $q_{3,\text{eff}}$ drops below $1/3$. We refer to this new maximum as the *critical* value of the variance, $\langle \tilde{\chi}^2 \rangle_{\text{max}} = \langle \tilde{\chi}^2 \rangle_{\text{crit}}$, see *Right* panel of Fig. 4.1. In this second case, just about when the variance reaches $\langle \tilde{\chi}^2 \rangle_{\text{crit}}$, the backreaction of the daughter field into the inflaton cannot be ignored any longer. In particular, the growth of $\langle \tilde{\chi}^2 \rangle$ towards $\langle \tilde{\chi}^2 \rangle_{\text{crit}}$ forces the position of the minimum to be shifted, and the inflaton is effectively 'locked' to oscillate around a negative value after the variance of χ reaches $\langle \tilde{\chi}^2 \rangle_{\text{crit}}$.

In our lattice simulations, we discover the distinction between the two regimes to be at around $q_3 \simeq 10$, i.e. the regime *i*) of 'small' q_3 values corresponds to $q_3 < 10$, whereas the regime *ii*) of 'large' q_3 values corresponds to $q_3 > 10$. Although the maximum value $\langle \tilde{\chi}^2 \rangle_{\text{max}}$ corresponds in both regimes to the end of the tachyonic resonance, in the regime *i*) such value is reached when the condition $q_{3,\text{eff}} \leq 1/3$ is satisfied, whereas in the regime *ii*) the maximum value corresponds to the moment when the χ self-interaction overtakes the trilinear interaction for the first time in χ 's effective mass $m_\chi^2 = q_3 \langle \tilde{\phi} \rangle + 3q_\chi \langle \tilde{\chi}^2 \rangle$. By canceling out the two terms with each other (when $\tilde{\phi} < 0$), we can obtain an analytical estimation for the critical value $\langle \tilde{\chi}^2 \rangle_{\text{crit}}$ in the regime *ii*) as

$$\langle \tilde{\chi}^2 \rangle_{\text{crit}} \simeq \frac{q_3 |\langle \tilde{\phi} \rangle|}{3q_\chi}, \quad (q_3 \gtrsim 10). \quad (4.13)$$

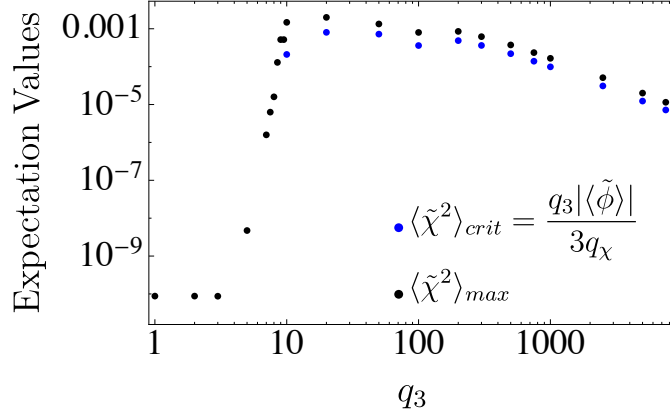


FIGURE 4.2: Lattice results of $\langle \tilde{\chi}^2 \rangle_{max}$ for different values of q_3 (black dots). It is clearly observed that for $q_3 \lesssim 10$ the amplitude of $\langle \tilde{\chi}^2 \rangle_{max}$ is much smaller than for $q_3 \gtrsim 10$, as the tachyonic growth of $\langle \tilde{\chi}^2 \rangle$ is simply stopped by the condition $q_{3,eff} = 1/3$. We choose $\langle \tilde{\chi}^2 \rangle_{max}$ as the value of $\langle \tilde{\chi}^2 \rangle$ averaged over one oscillation when it reaches its maximum value. For comparison, we also show the corresponding theoretical prediction for $q_3 \gtrsim 10$ in Eq. (4.13) (blue dots).

In Fig. 4.1, we show the time evolution of the mean value of the inflaton times the resonance parameter, $q_3 \langle \tilde{\phi} \rangle$, and the variance of the daughter field $\langle \tilde{\chi}^2 \rangle$, for $q_3 = 7$ (*Left* panel) and $q_3 = 50$ (*Right* panel), and for q_χ values that satisfy the relation $q_\chi = q_3^2$. Using the definitions of q_3 and q_χ in Eqs. (4.9), we can easily see that the critical relation between σ and λ in Eq. (4.2) translates into $q_\chi = \frac{q_3^2}{2}$. Thus, the relation $q_\chi = q_3^2$ used represents a conservative choice to avoid a runaway unstable solution of the dynamics. We can observe that $\langle \tilde{\chi}^2 \rangle$ grows whenever the inflaton is negative while it oscillates during whenever the inflaton is positive. The variance stops growing when it reaches its maximum value $\langle \tilde{\chi}^2 \rangle_{max}$.

In Fig. 4.2, we show $\langle \tilde{\chi}^2 \rangle_{max}$ from our lattice simulations as a function of the resonance parameter q_3 , and compare them against the theoretical estimation for the critical value $\langle \tilde{\chi}^2 \rangle_{crit}$ in the regime *ii*) with $q_3 \gtrsim 10$, c.f. Eq. (4.13). Recalling the relation $q_\chi = q_3^2$ we used in our simulations, we observe that the critical variance for $q_3 \gtrsim 10$ decreases inversely proportional to q_3 , as $\langle \tilde{\chi}^2 \rangle_{crit} \simeq \langle \tilde{\phi} \rangle / (3q_3)$, hence inversely proportional to the coupling constant σ . Although our analytical prediction Eq. (4.13) only captures roughly the real amplitude $\langle \tilde{\chi}^2 \rangle_{max}$ ³, it captures exactly the dependence on q_3 .

4.2.2 Gravitational wave production

In this section, we study GW production during preheating. We follow the production of GWs, h_{ij} , given by Eq. (1.86) and sourced by $\Pi_{ij} = \{\partial_i \phi \partial_j \phi + \partial_i \chi \partial_j \chi\}$. We characterize them by its energy density power spectrum of GWs normalized by the critical energy

³We estimate the maximum variance $\langle \tilde{\chi}^2 \rangle_{max}$ as the average over one period of oscillation of the variance $\langle \tilde{\chi}^2 \rangle$, once it has reached its maximum value.

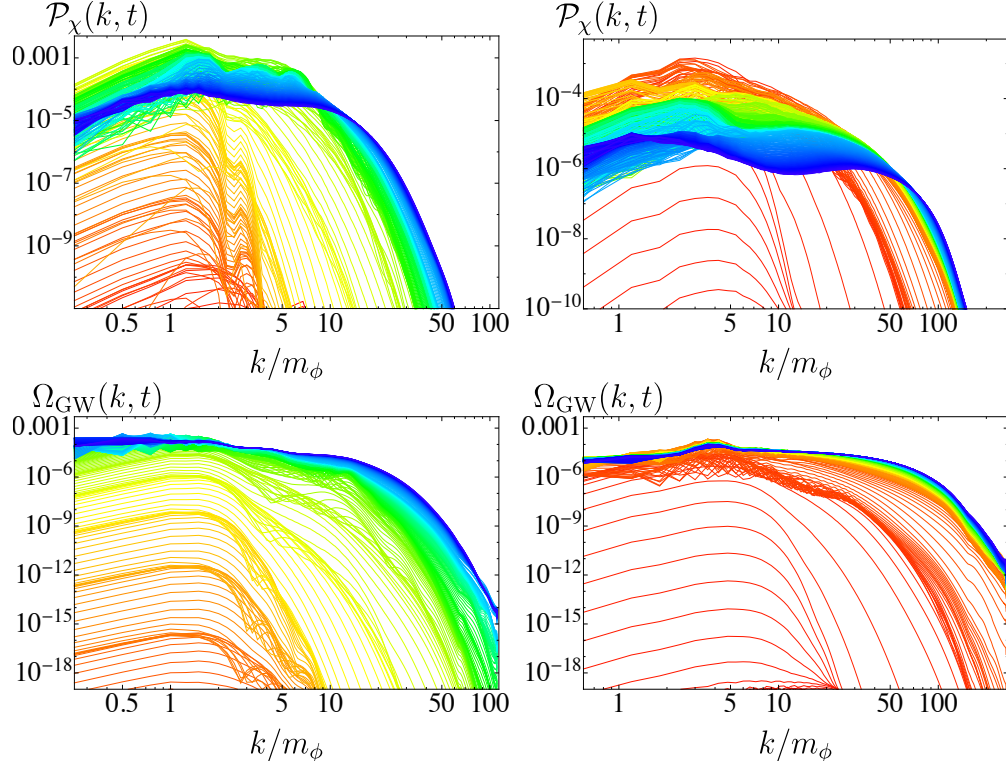


FIGURE 4.3: *Top* panels: Matter power spectrum of the daughter field χ , $\mathcal{P}_\chi(k)$, as a function of the momentum k/m_ϕ for $q_3 = 10$ (*Left*) and for $q_3 = 100$ (*Right*). *Bottom* panels: GW energy density power spectrum $\Omega_{\text{GW}}(k)$, for $q_3 = 10$ (*Left*) and for $q_3 = 100$ (*Right*). In all panels, colors run from red (early times) to blue (late times). Each spectra is measured every $\Delta(m_\phi t) = 0.2$.

density of the universe given by Eq. (1.98). Lattice simulations follow the procedure described in Sec. 1.4 and Sec. 2.3 where we solve a discrete version of Eq. (1.86). In order to compute the GW energy density power spectrum we make use of a discrete version of Eq. (1.98). To obtain the total energy density of GWs, we integrate Eq. (1.98) over momentum k ,

$$\Omega_{\text{GW}}^{\text{tot}}(t) = \int d \log k \frac{1}{\rho_c} \frac{d\rho_{\text{GW}}}{d \log k}(\mathbf{k}, t). \quad (4.14)$$

In Fig. 4.3, we show in the *Top* panels the matter power spectrum of the daughter field, $\mathcal{P}_{\tilde{\chi}}$, whereas in the *Bottom* panels we show the normalized GW energy density power spectrum Ω_{GW} , as a function of $\kappa = k/m_\phi$. Spectra are measured every $\Delta(m_\phi t) = 0.2$ steps up to a final time $m_\phi t_f = 400$. In this way, we make sure that the system has reached a steady state and the transfer of energy between ϕ and χ is shut down. The GWs are produced in pulses coinciding with the periods of tachyonic instability of χ , with step-like jumps visible in both the matter spectra $\mathcal{P}_{\tilde{\chi}}(k, t)$ and in $\Omega_{\text{GW}}(k, t)$. The production of GWs finishes when the variance of the χ field stops growing due to the end of tachyonic resonance (from that moment the χ_k modes are no longer excited, therefore the spatial gradients of $\tilde{\chi}$ do not grow). In Fig. 4.4, we show the evolution of the total GW energy density as a function of time. We observe its exponential increase until it reaches a threshold value; this saturation occurs when $\langle \tilde{\chi}^2 \rangle$ reaches its maximum

value $\langle \tilde{\chi}^2 \rangle_{\max}$. For instance, we see that GW production for $q_3 = 50$ stops around $m_\phi t \sim 15$, and looking into the *Right* panel of Fig. 4.1, we observe that $\langle \tilde{\chi}^2 \rangle$ stopped increasing around the same time. At the moment the GW spectrum saturates, the GWB redshifts freely. Since the maximum value of $\langle \tilde{\chi}^2 \rangle$ depends on q_3 , the same dependence is inherited in the maximum value of the GW energy density, see Fig. 4.4. Another feature worth noticing is that for $q_3 \gtrsim 10$, the final GW energy density ratio $\Omega_{\text{GW}}^{\text{tot}}(t)$ reaches a maximum fixed amplitude, which can be as high as $\Omega_{\text{GW}}^{\text{tot}}(t_f) \simeq 5 \cdot 10^{-4}$ for $q_3 \simeq 10$, while it decreases smoothly for larger values of q_3 [see Eq. (4.19) for a parametrization of the q_3 dependence of the peak amplitude of the GWB].

The GWB produced during preheating is redshifted once its amplitude saturates. Today's amplitude and peak frequency of the GWB depend on the expansion history of the universe, so in order to estimate them, we need to redshift the background from the final time of our simulations t_f , till the present time t_0 . This requires to know the expansion history from t_f till the moment t_{RD} when radiation domination (RD) is fully established. We follow redshift procedure introduced in Sec. 1.4.2, in this case we parametrize the era between end of production and onset of RD with an effective equation of state $\bar{w} = p/\rho$, we can write the redshifted frequency and amplitude today of the GWB as

$$f_{\text{GW}} \simeq 4 \times 10^{10} \epsilon_f^{1/4} \frac{k_{\text{GW}}}{a_f H_f} \left(\frac{H_f}{m_p} \right)^{1/2} \text{ Hz}, \quad (4.15)$$

$$h_0^2 \Omega_{\text{GW}}^{(0)} \simeq 1.6 \times 10^{-5} \epsilon_f \Omega_{\text{GW}}^{(f)}, \quad (4.16)$$

where ϵ_f is defined as

$$\epsilon_f \equiv \left(\frac{a_f}{a_{\text{RD}}} \right)^{1-3\bar{w}}, \quad (4.17)$$

with $a_f \equiv a(t_f)$ and $a_{\text{RD}} \equiv a(t_{\text{RD}})$. In order to estimate $h_0^2 \Omega_{\text{GW}}^{(0)}$ and f_{GW} we have characterized the position κ_p and the amplitude $\Omega_{\text{GW}}^{(f)}$ of the peak in the GW spectra at t_f . The position of the peak is characterized by the initial instability tachyonic band given in Eq. (4.12), and the final amplitude of the peak $\Omega_{\text{GW}}^{(f)}$ decreases smoothly with the resonance parameter. We have also measured the values of a_f and H_f in our simulations. We have observed that for $q_3 \gtrsim 10$ the contribution that dominates to the total energy of the system is the one of the daughter field, which behaves as radiation. Therefore, the system at t_f is already quite close to RD, with an equation of state $\bar{w}$ slightly smaller than (but very close to) $\bar{w} \simeq 1/3$. This implies that we can estimate ϵ_f to be $\lesssim 1$. Putting everything together and taking $\epsilon_f = 1$ for simplicity, we find the

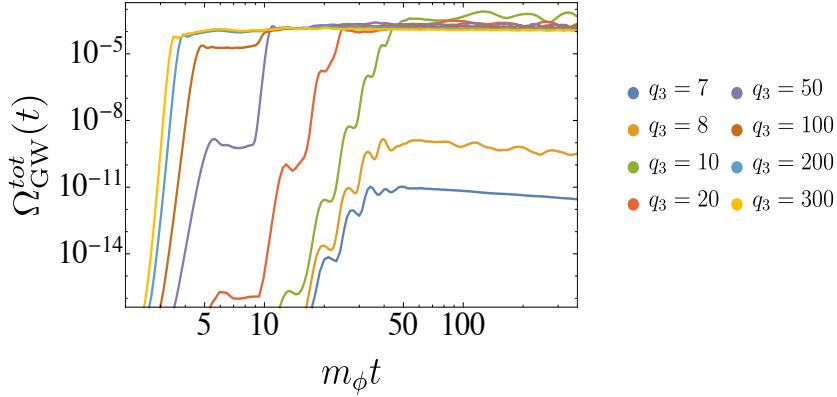


FIGURE 4.4: Time evolution of the GW total energy density for different values of q_3 .

following parametrization for the GWB peak today (valid only for $q_3 \gtrsim 9$)

$$f_{\text{GW}}^{(p,0)}(q_3) \simeq (1.4 \pm 0.1) \times 10^8 \left(\frac{q_3}{10}\right)^{0.53 \pm 0.03} \text{ Hz}, \quad (4.18)$$

$$h_0^2 \Omega_{\text{GW}}^{(p,0)}(q_3) \simeq (4.9 \pm 1.6) \times 10^{-9} \left(\frac{q_3}{10}\right)^{-0.63 \pm 0.14}. \quad (4.19)$$

We observe that the produced GWB has a large amplitude but is peaked at high frequencies. We also note that the exponent of our frequency characterization coincides with that in Eq. (4.12). As previously discussed, a high-frequency detection program at $\sim$ MHz frequencies and above, has been recently put forward [163], this is yet in its infancy. The reality of the situation is that neither current nor planned GW detection experiments are able to detect these high frequency GWBs. Direct detection is therefore challenging, as it is also usually the case in $\phi^2\chi^2$ preheating scenarios with monomial inflaton potentials as seen in previous chapter.

For this case, the GW production is in any case very significant. We may look for additional ways to probe these backgrounds, for instance using indirect constraining means. The GWs produced during preheating are created at sub-horizon scales at the time of production, they contribute to the energy density of the universe as radiation [164]. It is possible to set bounds on the amount of GW radiation that can be produced. For this regard, we can resort to cosmological constraints on the total radiation density species allowed in the universe. This is typically parameterized by an excess of the effective number of relativistic degrees of freedom beyond the Standard Model $\Delta N_{\text{eff}} = N_{\text{eff}} - N_{\text{eff,SM}}$, with $N_{\text{eff,SM}} = 3.0440$ [214]. Assuming that only the produced gravitational radiation contributes to the extra radiation energy density at the time of CMB decoupling, we can set an upper threshold on the extra radiation degrees of freedom ΔN_{eff} which sets a constraint on $h_0^2 \Omega_{\text{GW}}^{(0)}$ as [164]

$$\frac{h_0^2 \Omega_{\text{GW}}^{(0)}}{h_0^2 \Omega_{\gamma}^{(0)}} = \frac{7}{8} \left(\frac{4}{11}\right)^{4/3} \Delta N_{\text{eff}}, \quad (4.20)$$

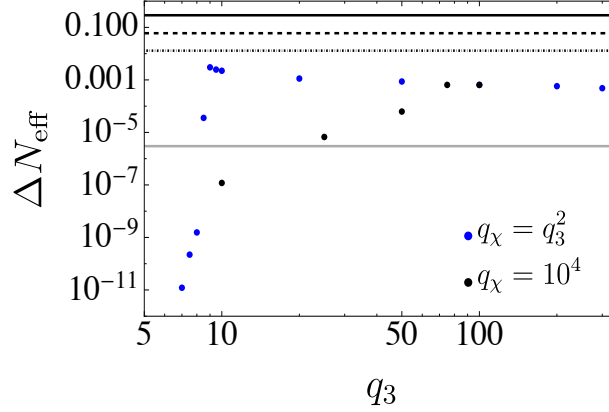


FIGURE 4.5: Number of relativistic degrees of freedom beyond Standard Model for different values of q_3 , for the relation $q_\chi = q_3^2$ and fixed $q_\chi = 10^4$. The solid top line corresponds to the current Planck bound $\Delta N_{\text{eff}} \lesssim 0.29$, whereas the dashed line below represents the future CMB-S4 $\Delta N_{\text{eff}} \lesssim 0.06$ upper bound at 2σ . The dash-dotted line corresponds to the upper bound imposed by next generation satellite missions at 2σ , $\Delta N_{\text{eff}} \lesssim 0.013$. The gray bottom solid line corresponds to a hypothetical cosmic variance limited (CVL) CMB polarization experiment upper bound $\Delta N_{\text{eff}} \lesssim 3 \cdot 10^{-6}$.

where the present energy density of photons is $h_0^2 \Omega_\gamma^{(0)} = 2.47 \times 10^{-5}$. The Planck observations set a limit $|\Delta N_{\text{eff}}| \lesssim 0.29$ at 95% C.L. [206, 207], whereas the next-generation CMB-S4 experiments will probe $\Delta N_{\text{eff}} \lesssim 0.06$ at 2σ [208]. Additionally, the next generation of satellite missions like CORe [209] and Euclid [210] will impose bounds at 2σ on $\Delta N_{\text{eff}} \lesssim 0.013$. A hypothetical cosmic variance limited (CVL) CMB polarization experiment could presumably get down to $\Delta N_{\text{eff}} \lesssim 3 \cdot 10^{-6}$ [215], though this does not seem experimentally realistic. In Fig. 4.5, we show ΔN_{eff} for different values of q_3 considering two cases: $q_\chi = q_3^2$ (blue dots) and $q_\chi = 10^4$ (black dots). We found that the coupling that produces the largest amount of GWs is $q_3 = 9$, yielding $\Delta N_{\text{eff}} \simeq 0.003$, which is above the bound expected in a hypothetical CVL experiment, but still below the sensitivity of actually planned realistic experiments. We can safely conclude that both direct or indirect detection methods cannot probe the GWBs predicted in the scenario studied in this section. The reason are either the peak of excitation in frequency is placed at regions far from our current detectors in the high frequency domain. On the other hand, the amount of GWs produced is just insufficient to be constrained by ΔN_{eff} . We can speculate on how a scenario could be observable, for which we should focus on two circumstances that could be sought. First, if the energy scale of inflation is decreased, the peak of excitation could be shifted to lower frequencies domains. Second, if the amount of GWs produced is increased relative to the field energy budget of the universe it could be constrained by N_{eff} bound. In the next section, we introduce an inflationary scenario driven by polynomial function for the inflaton which, *a priori*, seems to allow to satisfy both conditions. We therefore explore next the extent to which GW production during preheating after polynomial inflation can improve the detectability of the GWB

due to a trilinear interaction.

4.3 Polynomial potential

In this section, we consider the most general renormalizable single-field model of inflation, where the inflaton potential is a polynomial of degree four [216]

$$\begin{aligned} V_{\text{inf}}(\phi) &= \frac{1}{2}m_\phi^2\phi^2 + \sigma_\phi\phi^3 + \lambda_\phi\phi^4 \\ &= \lambda_\phi \left(2\phi_0^2\phi^2 - \frac{8}{3}(1-\beta)\phi_0\phi^3 + \phi^4 \right), \end{aligned} \quad (4.21)$$

where we have parametrized the mass and trilinear coupling of the inflaton as $m_\phi \equiv 2\sqrt{\lambda_\phi}\phi_0$ and $\sigma_\phi \equiv -\frac{8}{3}\lambda_\phi(1-\beta)\phi_0$. The linear term in ϕ is absent in the potential of Eq. (4.21) since it can be absorbed by shifting the field, whereas a constant term is ignored since we set it to equal to the value of the cosmological constant, which is negligible compared to the energy scale of inflation [216]. The three terms in Eq. (4.21) are used to match the observational data, and the existence of a (near) inflection-point at ϕ_0 guarantees the flatness of the polynomial potential of quartic order. In this setup, λ_ϕ is the inflaton self-coupling, and β is a parameter that controls the flatness configuration around ϕ_0 . Notice that, if $\beta < 0$, there is a false vacuum for $\phi > \phi_0$, and the inflaton can get stuck in this second minimum [217]. Hence, we focus on the case where $0 < \beta \ll 1$, and on small inflaton amplitudes, so that the inflaton field value is sub-Planckian at the time when the curvature perturbations observed at the CMB are generated, i.e., we will assume $\phi_0 \leq m_p$ [216, 217]. Interestingly, the inflection-point ϕ_0 constitutes the only free parameter in this inflationary model, since all the couplings in the potential Eq. (4.21) can be expressed at its expense. In particular, by considering the CMB measurements from Planck data [25] on the scalar power spectrum $A_s = (2.1 \pm 0.1) \times 10^{-9}$, and on the spectral index $n_s = 0.9649 \pm 0.0042$, and assuming these were generated $N_e = 65$ e-folds before the end of inflation, the quantities λ_ϕ , β and m_ϕ^2 , can be fixed as

$$\lambda_\phi \simeq 6.61 \times 10^{-16} \left(\frac{\phi_0}{m_p} \right)^2, \quad (4.22)$$

$$\beta \simeq 9.73 \times 10^{-7} \left(\frac{\phi_0}{m_p} \right)^4, \quad (4.23)$$

$$m_\phi = 2\sqrt{\lambda_\phi}\phi_0 \simeq 5 \times 10^{-8} \left(\frac{\phi_0}{m_p} \right)^2 m_p. \quad (4.24)$$

As a lower bound $\phi_0 \geq 3 \times 10^{-5}m_p$ emerges naturally by demanding the stability of the inflationary scenario against radiative corrections [216, 217], the inflection point ϕ_0

ranges in practice within the interval $3 \times 10^{-5} m_p < \phi_0 \leq m_p$. The inflaton mass lies therefore in our modeling within the range $10^2 \text{ GeV} < m_\phi \leq 10^{11} \text{ GeV}$.

The full potential of the system inflaton-daughter field is

$$V(\phi, \chi) = V_{\text{inf}}(\phi) + \frac{\sigma}{2} \phi \chi^2 + \frac{\lambda}{4} \chi^4. \quad (4.25)$$

To make sure that this potential is bounded from below, we require $\lambda_\phi, \lambda > 0$. Identifying a critical value for the self-coupling of χ as

$$\lambda_c(\sigma, \lambda_\phi, \phi_0) \equiv \frac{\sigma^2}{8 \lambda_\phi \phi_0^2}. \quad (4.26)$$

We observe that for $0 < \lambda < \lambda_c$, the potential (4.25) has two minima at non-vanishing values of χ and ϕ , where the potential is negative, $V(\phi, \chi) < 0$. For $\lambda \geq \lambda_c$ instead, there is only one minimum at $\phi = \chi = 0$ with $V(\phi, \chi) = 0$. In the following analysis we use the limiting case $\lambda = \lambda_c$, as this value corresponds to the case for which the production of GWs is maximal.

4.3.1 Preheating dynamics. Lattice simulations

In a similar manner to the case of the monomial potential, we can perform a change of variables such that we are able to write the equations of motion with dimensionless quantities. We choose conveniently

$$d\tilde{x}^\mu = \sqrt{\lambda_\phi} \phi_0 dx^\mu, \quad \tilde{\phi} = \frac{\phi}{\phi_0}, \quad \tilde{\chi} = \frac{\chi}{\phi_0}, \quad \tilde{H} = \frac{H}{\sqrt{\lambda_\phi} \phi_0}, \quad (4.27)$$

such that the equations of motion for ϕ and χ become

$$\tilde{\phi}'' + 3\tilde{H}\tilde{\phi}' + 4\tilde{\phi} - 8(1-\beta)\tilde{\phi}^2 + 4\tilde{\phi}^3 + \frac{q_3}{2}\tilde{\chi}^2 - \frac{\tilde{\nabla}^2 \tilde{\phi}}{a^2} = 0, \quad (4.28)$$

$$\tilde{\chi}'' + 3\tilde{H}\tilde{\chi}' + q_3\tilde{\chi}\tilde{\phi} + q_\chi\tilde{\chi}^3 - \frac{\tilde{\nabla}^2 \tilde{\chi}}{a^2} = 0, \quad (4.29)$$

where $'$ denotes derivatives with respect to the new time variable. The resonance parameters emerging now in the problem are defined as

$$q_3 \equiv \frac{\sigma}{\lambda_\phi \phi_0} \quad \text{and} \quad q_\chi \equiv \frac{\lambda}{\lambda_\phi}. \quad (4.30)$$

A relation between q_χ and q_3 , equivalent to $\lambda \geq \lambda_c$, so that there is only one minimum at $\tilde{\phi} = \tilde{\chi} = 0$ where $V(\tilde{\phi}, \tilde{\chi}) = 0$, can be obtained as

$$q_\chi > \frac{q_3^2}{8}. \quad (4.31)$$

Introducing the field decomposition from Eq. (4.6) into Eq. (4.29), and using again the Hartree approximation $\tilde{\chi}^3 \rightarrow 3\tilde{\chi} \langle \tilde{\chi}^2 \rangle$, we find that the mode functions of $\tilde{\chi}$ obey

$$\tilde{\chi}_k'' + 3\tilde{H}\tilde{\chi}_k' + \left(3q_\chi \langle \tilde{\chi}^2 \rangle + q_3 \tilde{\phi}(t) + \frac{\kappa^2}{a^2}\right) \tilde{\chi}_k = 0, \quad \text{where } \kappa \equiv \frac{k}{\sqrt{\lambda_\phi \phi_0}} \quad (4.32)$$

The χ field has therefore an effective time-dependent mass $m_\chi^2 \approx (q_3 \tilde{\phi}(t) + 3q_\chi \langle \tilde{\chi}^2 \rangle)$, which oscillates between negative and positive values due to the oscillations of the inflaton. Whenever $m_\chi^2 < 0$ and $\kappa^2/a^2 < -m_\chi^2$, the mode frequency $\tilde{\omega}_k^2 \equiv \kappa^2/a^2 + m_\chi^2$ will become negative. This effect, once again, leads to an exponential growth of the corresponding $\tilde{\chi}_k$ modes. The power spectrum of the excited modes $\mathcal{P}_{\tilde{\chi}}(k, t)$, and consequently the variance of the daughter field $\langle \tilde{\chi}^2 \rangle$ will grow exponentially in a step-like manner for each moment the inflaton amplitude becomes negative.

The full dynamics of tachyonic resonance in this scenario, similarly as in Sec. 4.2.1, will lead to the backreaction of the preheat field onto the inflaton background. We will consider Eqs. (4.28) and (4.29) and solve them numerically. We obtain the initial amplitude and velocity of the inflaton from the breaking of the slow-roll condition when $\epsilon_H = 1$, for different values of ϕ_0 , see Table. 4.1. The initial condition for the χ field is again set to an initially zero homogeneous amplitude, on top of which we add quantum vacuum fluctuations, following the standard recipe for initial lattice power spectra as explained in Sec. 2.2.4. We have performed a series of lattice simulations with $\mathcal{N} = 512$

ϕ_0	$\tilde{\phi}(t_i)$	$\tilde{\phi}'(t_i)$
m_p	0.368	-0.395
$10^{-1} m_p$	0.473	-0.464
$10^{-2} m_p$	0.485	-0.470
$10^{-3} m_p$	0.486	-0.471

TABLE 4.1: The initial amplitude and velocity of the inflaton for different values of ϕ_0 .

sites per dimension and a minimum infrared momentum $\kappa_{\text{IR}} \in [0.5 : 6]$, making sure we capture well all relevant scales. Simulations were run for different values of q_3 considering the relation $q_\chi = q_3^2/8$ for $q_3 \in [25 : 5 \times 10^4]$, finishing at $\sqrt{\lambda_\phi} \phi_0 t_f = 400$, as this guarantees that the system has reached a steady state.

The dynamics of preheating in the polynomial scenario can be described as follows. In an initial stage, the inflaton oscillates around the minimum of the potential, similar to the monomial scenario, except that in this case the potential of Eq. (4.21) makes the first oscillations to be an-harmonic. As the amplitude decays with the expansion of the universe, the oscillations become gradually more harmonic in the sense it only probes the minimum of the potential which is essentially quadratic. Given that for each oscillation the inflaton becomes negative during half of it, the effective mass $m_{\tilde{\chi}}$ of the $\tilde{\chi}_k$ modes becomes negative for $\kappa < \sqrt{\sigma|\phi|}a$. This induces an exponential growth of those IR modes during the semi oscillations, which can be observed in Fig. 4.6. As in the previous scenario we identify two regimes depending on the value of q_3 :

- For $q_3 \lesssim 100$, tachyonic resonance ends due to the expansion of the universe, since the effective resonance parameter $q_{3,\text{eff}} = q_3|\langle\tilde{\phi}\rangle|$ decreases with the scale factor. When $q_{3,\text{eff}} < 1$ is achieved, the system enters into a narrow regime where tachyonic resonance is not efficient. This can be observed in the *Left* panel of Fig. 4.6.
- For $q_3 \gtrsim 100$, tachyonic resonance rather ends (before it becomes narrow) due to the presence of the self-interaction of the χ field. This happens when the χ self-interaction term in $m_{\tilde{\chi}}^2$ overtakes the trilinear interaction term. In that moment the variance $\langle\tilde{\chi}^2\rangle$ reaches a maximal critical value, which can be expressed as

$$\langle\tilde{\chi}^2\rangle_{\text{crit}} \simeq \frac{q_3|\langle\tilde{\phi}\rangle|}{3q_\chi}. \quad (4.33)$$

The end of tachyonic resonance when $\langle\tilde{\chi}^2\rangle$ reaches its critical value (4.33) can be observed in the *Right* panel of Fig. 4.6.

In Fig. 4.7, it is shown the theoretical prediction for $q_3 \gtrsim 100$ of the maximum value of $\langle\tilde{\chi}^2\rangle$ given by Eq. (4.33) (blue dots), together with the values obtained in our lattice simulations (black dots). We observe that for $q_3 \lesssim 100$ the variance $\langle\tilde{\chi}^2\rangle$ grows with q_3 , while for $q_3 \gtrsim 100$ it decays inversely proportional to q_3 , in correspondence with the theoretical description $\langle\tilde{\chi}^2\rangle_{\text{crit}} \simeq \frac{q_3\langle\tilde{\phi}\rangle}{3q_\chi} = \frac{8\langle\tilde{\phi}\rangle}{3q_3}$ [in the last identity we have introduced the saturation relation $q_\chi = \frac{q_3^2}{8}$ from Eq. (4.31)]. While our theoretical estimation in Eq. (4.33) captures only approximately the truly measured maximum variance $\langle\tilde{\chi}^2\rangle_{\text{max}}$ for $q_3 \gtrsim 100$, it describes nonetheless exactly its dependence on q_3 . In such regime, after the variance reaches the critical value $\langle\tilde{\chi}^2\rangle_{\text{max}} \simeq \langle\tilde{\chi}^2\rangle_{\text{crit}}$, the system dynamics become fully non-linear in the following stages, due to the backreaction of the χ field onto the inflaton. In a similar manner as in the scenario with the monomial potential, due to the non-zero value acquired by the variance of χ , the inflaton is forced to oscillate around a new minimum developed in its effective potential, a minimum that corresponds to a negative value of ϕ .

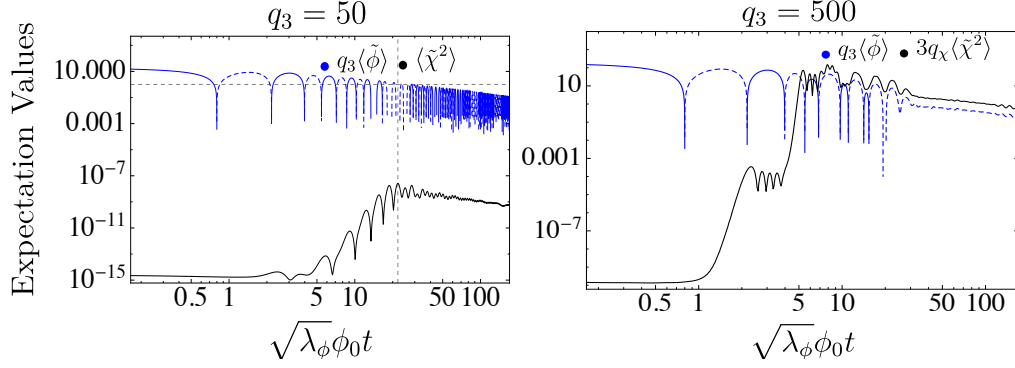


FIGURE 4.6: *Left:* Time evolution of the inflaton's volume average $\langle \tilde{\phi} \rangle$ times the resonance parameter q_3 ($q_{3,\text{eff}}$), and the variance of the daughter field $\langle \tilde{\chi}^2 \rangle$ for $q_3 = 50$. The horizontal dashed line represents $q_{3,\text{eff}} = 1$, whereas the vertical dashed line is the time at which the maximum value of the variance, $\langle \tilde{\chi}^2 \rangle_{\text{max}}$, is reached. *Right:* Time evolution of $q_3 \langle \tilde{\phi} \rangle$ and the variance of the daughter field $\langle \tilde{\chi}^2 \rangle$ times $3q_\chi$ for $q_3 = 500$. We see that once the term $3q_\chi \langle \tilde{\chi}^2 \rangle$ overtakes $q_3 \langle \tilde{\phi} \rangle$ the growth of the variance stops. For both panels, the solid (dashed) blue lines are the positive (negative) part of the inflaton mean value.

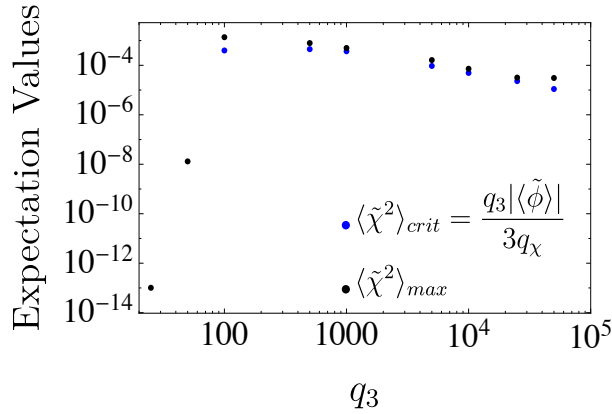


FIGURE 4.7: Lattice results of $\langle \tilde{\chi}^2 \rangle_{\text{max}}$ for different values of q_3 (black dots). It is clearly observed that for $q_3 \lesssim 100$ the amplitude of $\langle \tilde{\chi}^2 \rangle_{\text{max}}$ is much smaller than for $q_3 \gtrsim 100$, as the tachyonic growth of $\langle \tilde{\chi}^2 \rangle$ is simply stopped by the condition $q_{3,\text{eff}} = 1$. We choose $\langle \tilde{\chi}^2 \rangle_{\text{max}}$ as the value of $\langle \tilde{\chi}^2 \rangle$ averaged over one oscillation when it reaches its maximum value. For comparison we also show the corresponding theoretical prediction for $q_3 \gtrsim 100$ in Eq. (4.33) (blue dots).

4.3.2 Gravitational wave production

In this section, we study the production of GWs during preheating after polynomial inflation. We follow the same procedure explained in Sec. 4.2.2, tracking the real time dynamics of the GWs and measuring their energy density power spectrum $\Omega_{\text{GW}}(k, t)$. In Fig. 4.8, we show $\Omega_{\text{GW}}(k, t)$ and the matter power spectrum of the daughter field $\mathcal{P}_{\tilde{\chi}}(k)$. Spectra are measured every $\Delta(\sqrt{\lambda_\phi} \phi_0 t) = 0.2$ intervals, up to a final time $\sqrt{\lambda_\phi} \phi_0 t_f = 400$. We see that GWs are produced in a step-like manner coinciding with the growth of the variance $\langle \tilde{\chi}^2 \rangle$ of the daughter field. The production of GWs in

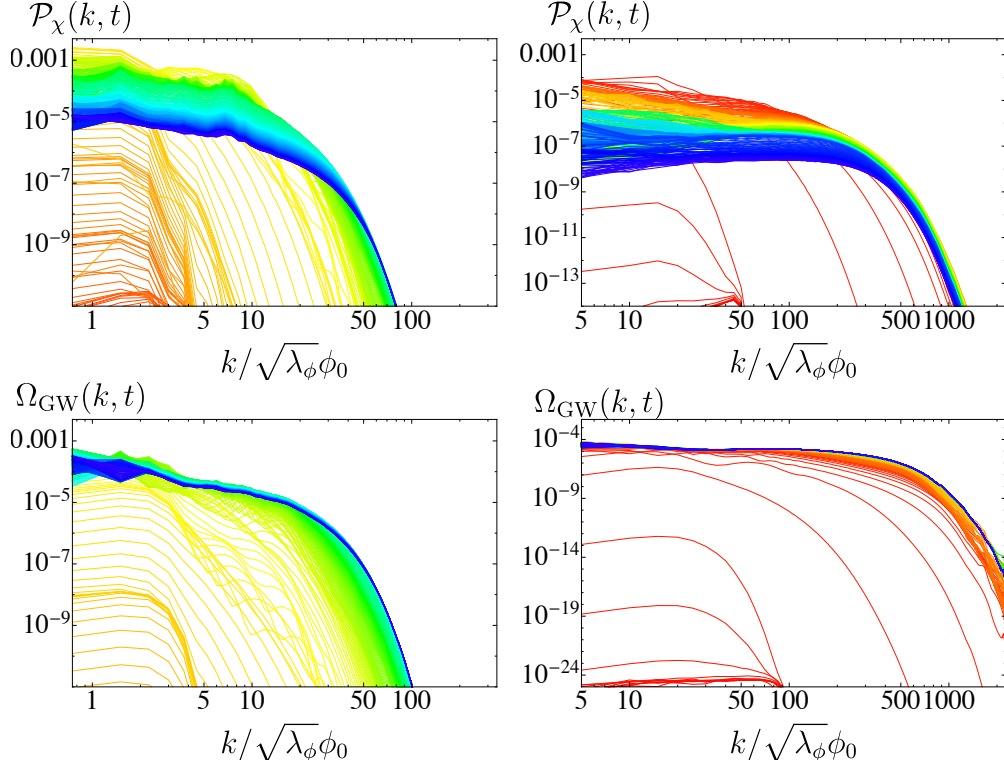


FIGURE 4.8: *Top panels:* Matter power spectrum of the daughter field $\mathcal{P}_\chi(k, t)$, as a function of the momentum k/m_ϕ for $q_3 = 100$ (*Left*) and for $q_3 = 10^4$ (*Right*). *Bottom panels:* GW energy density power spectrum $\Omega_{\text{GW}}(k, t)$, for $q_3 = 100$ (*Left*) and for $q_3 = 10^4$ (*Right*). In all panels, colors run from red in early times to blue in late times. Spectra are measured every $\Delta(\sqrt{\lambda_\phi}\phi_0 t) = 0.2$ until saturation.

the regime $q_3 \gtrsim 100$, stops around the time when $\langle \tilde{\chi}^2 \rangle$ reaches its maximum critical value $\langle \tilde{\chi}^2 \rangle_{\text{crit}}$. At this moment, the growth of $\Omega_{\text{GW}}(k, t)$ ceases and the non-linearities developed in the dynamics of the inflaton-daughter system leads to a transfer of power into more ultraviolet (UV) modes in the GW spectrum.

We fix the value $q_\chi = q_3^2/8$ according to the saturation relation and we have characterized the amplitude and the peak of excitation of the GWB as a function of ϕ_0 and q_3 . In the *Left* panel of Fig. 4.9, we show the evolution of the GW total energy density for different values of ϕ_0 , and observe that the total GW energy density decreases with ϕ_0 . On the other hand, making ϕ_0 smaller implies decreasing the energy scale of inflation, and hence also the peak of excitation of the signal today $f_{\text{GW}}^{(p,0)}$ shifts to smaller frequencies. In the *Right* panel of Fig. 4.9, we plot the evolution of the GW total energy density for different values of q_3 , for which we can observe the final amplitude decrease only mildly when increases the parameter q_3 . The dependence of $f_{\text{GW}}^{(p,0)}$ on the resonance parameter q_3 is determined by the initial instability band, $\kappa^2 < q_3 |\tilde{\phi}| a$. Regarding the expansion history of the universe, which is relevant for the determination the ϵ_f parameter to redshift the signal, our lattice simulations indicate that for sufficiently large couplings $q_3 \gtrsim 10^3$ the

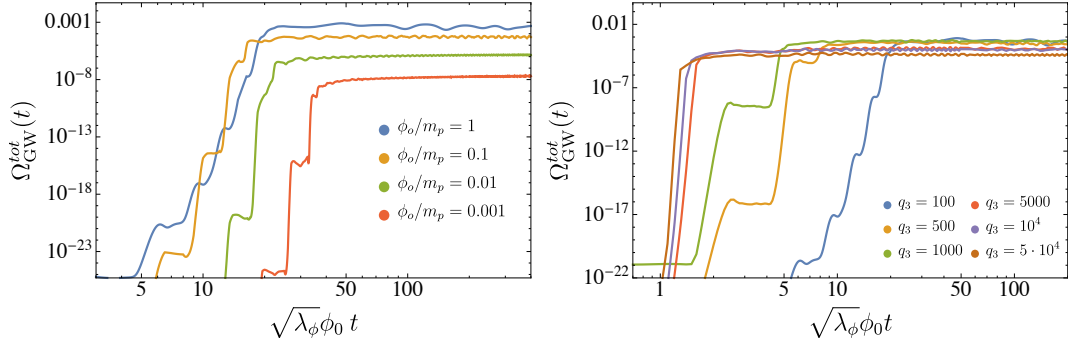


FIGURE 4.9: Time evolution of GWs total energy density: *Left*: for different values of ϕ_0 , with $q_3 = 100$ and $q_\chi = q_3^2/8$. We observe that the largest amount of GWs is produced by $\phi_0 = m_p$. *Right*: For different values of q_3 , with $q_\chi = q_3^2/8$ and $\phi_0 = m_p$.

energy budget of the system is dominated by kinetic and gradient energy of the daughter field, which behave as radiation. The universe is therefore in RD by the time we end our simulations, and therefore we take $\epsilon_f = 1$. Using Eqs. (4.15) and (4.16), we obtain a parametrization of the GWB's peak position and amplitude, as function of ϕ_0 and q_3 , valid for $q_3 \gtrsim 10^3$ and $3 \times 10^{-5} m_p \leq \phi_0 \leq m_p$, as

$$f_{\text{GW}}^{(p,0)}(\phi_0, q_3) \simeq (5.05 \pm 2.03) \times 10^6 \left(\frac{\phi_0}{m_p} \right)^{0.31 \pm 0.1} \left(\frac{q_3}{100} \right)^{0.4 \pm 0.06} \text{ Hz}, \quad (4.34)$$

$$h_0^2 \Omega_{\text{GW}}^{(p,0)}(\phi_0, q_3) \simeq (4.5 \pm 4) \times 10^{-9} \left(\frac{\phi_0}{m_p} \right)^{1.41 \pm 0.13} \left(\frac{q_3}{100} \right)^{-0.75 \pm 0.12}. \quad (4.35)$$

From the parametrization we observe that the maximum amplitude for the GWB is obtained for $\phi_0 \simeq m_p$, but in that case the background is peaked at very high frequencies $f_{\text{GW}}^{(p,0)} \gtrsim 5 \cdot 10^6$ Hz. While decreasing ϕ_0 allow us to move the frequency window to smaller values, down to values $f_{\text{GW}}(\phi_0 = 3 \cdot 10^{-5} m_p) \sim 10^5$ Hz, this can only be done the expense of suppressing the amplitude of the GWB spectrum, which then becomes $h_0^2 \Omega_{\text{GW}}^{(0)}(\phi_0 = 10^{-5} m_p) \lesssim 10^{-15}$. Furthermore, the dependence on the resonance parameter q_3 just follows a similar behavior as in parametric resonance [107], so that increasing q_3 parametrically shifts the signal to higher frequencies while decreasing its amplitude.

In conclusion, despite the fact that in the polynomial inflation scenario allowed for lower energy scales. It only achieves marginally our intention of moving to smaller frequencies the peak of the GWB, which can only be done at the expense of suppressing the amplitude. In order to have a large amplitude background we are forced to have it peaked, as usual, at very high frequencies, where direct detection experiments are yet far from the desired capabilities of detection. Given the inability to detect the GWB directly also in this model, we can still see whether constraints on a maximum trilinear interaction coupling could be placed using current and planned constraints on

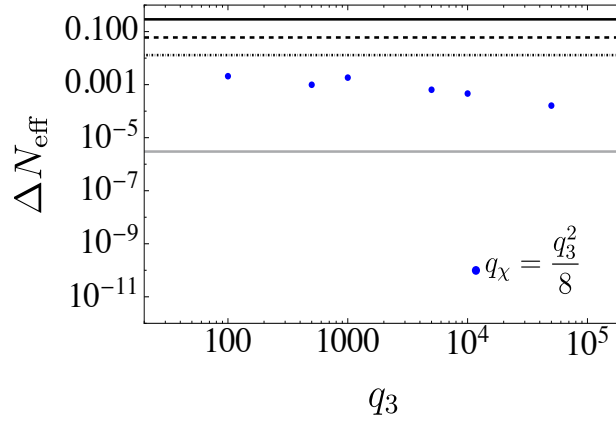


FIGURE 4.10: Number of relativistic degrees of freedom beyond Standard Model, ΔN_{eff} , for different values of q_3 with $q_\chi = q_3^2/8$. The top solid black line corresponds to the current bound coming from Planck data $\Delta N_{\text{eff}} \lesssim 0.29$, the dashed line below represents the future CMB-S4 $\Delta N_{\text{eff}} \lesssim 0.06$ upper bound at 2σ . The dot-dashed black line corresponds the projected bounds at 2σ on $\Delta N_{\text{eff}} \lesssim 0.013$ from next-generation satellite missions, whereas the bottom gray solid line corresponds to a hypothetical cosmic variance limited (CVL) CMB polarization experiment upper bound $\Delta N_{\text{eff}} \lesssim 3 \cdot 10^{-6}$.

ΔN_{eff} . Recalling Eq. (4.20), we translate the energy density of the GW background into ΔN_{eff} and compare with the present/projected bounds on ΔN_{eff} , as we can see in Fig. 4.10. We find that for $q_3 = 100$ and $\phi_0 = m_p$, we obtain a maximum value as $\Delta N_{\text{eff}} \simeq 0.0021$, which is well above a futuristic CVL experiment, but still below the more realistic expected bounds in near future experiments. We conclude therefore that it is not possible to probe directly or indirectly the predicted signal with upcoming experiments.

4.4 Discussion

In this work, we have studied the production of GWs during preheating induced by a trilinear interaction $\phi\chi^2$, involving a singlet inflaton ϕ coupled to a scalar daughter field χ . We considered a post-inflationary scenario where the inflaton oscillates in a quadratic potential, $V(\phi) = \frac{1}{2}m_\phi^2\phi^2$, and included a quartic self-interaction term for the daughter field, $\frac{1}{4}\lambda\chi^4$, in order to ensure the stability of the potential. This introduces a consistency condition between the resonance parameters q_3 and q_χ , which respectively control the strength of the trilinear and quartic couplings. Only configurations satisfying $q_\chi > \frac{q_3^2}{2}$ avoid runaway growth of χ . We used *CosmoLattice* to perform lattice simulations, for which we find that for $q_3 \lesssim 10$, the variance of the daughter field $\langle \tilde{\chi}^2 \rangle$ stops growing early due to the redshifting of modes as the system enters a narrow resonance regime. On the other hand, for $q_3 > 10$, the variance grows until it reaches a critical value $\langle \tilde{\chi}^2 \rangle_{\text{crit}}$,

defined by the moment when the quartic self-interaction becomes dominant over the trilinear coupling. Gravitational wave production follows the step-like amplification of the daughter field power spectrum and ceases once $\langle \tilde{\chi}^2 \rangle$ saturates, see Figs. 4.1 and 4.4. Although the resulting stochastic GWB can reach amplitudes as high as $h_0^2 \Omega_{\text{GW}}^{(0)} \simeq 10^{-9}$, the signal is peaked at very high frequencies $f_{\text{GW}} \gtrsim 10^8$ Hz, well beyond the reach of current and near-future GW observatories. Nonetheless, this large GW production translates into an effective contribution to the number of relativistic species as high as $\Delta N_{\text{eff}} \sim 0.003$ (for $q_3 = 9$), though still below the projected sensitivity of next-generation CMB experiments.

We have also studied the case in which the inflaton evolves in a polynomial potential (both during inflation and preheating), whose shape is controlled by a single parameter: the location of a near-inflection point ϕ_0 in the potential, varying within the range $3 \times 10^{-5} m_p < \phi_0 < m_p$. This scenario offers potential improvements in two directions:

- lowering the inflationary energy scale, thereby shifting the peak of the GWB to smaller frequencies
- enhancing the fraction of GW energy density relative to the total field energy budget.

In this context, runaway solutions for the daughter field χ are avoided if the resonance parameters satisfy the condition $q_\chi > \frac{q_3^2}{8}$. We used *CosmoLattice* to perform lattice simulations, for which we find that for $q_3 \lesssim 100$, tachyonic resonance terminates shortly after the onset of inflaton oscillations, as the system enters narrow resonance due to the expansion of the universe. For $q_3 \gtrsim 100$, however, resonance ends when the variance of the daughter field reaches a critical value $\langle \tilde{\chi}^2 \rangle_{\text{crit}}$, marking the moment when the quartic self-interaction overtakes the trilinear coupling.

Regarding gravitational wave production, we find that the largest amplitude of the GWB is obtained by choosing the highest possible value of the inflection point, $\phi_0 = m_p$, yielding a signal as strong as $\Omega_{\text{GW}}^{(0)} h_0^2 \simeq 5 \cdot 10^{-9}$. Nevertheless, the peak of the spectrum lies at high frequencies, $f_{\text{GW}} \simeq 5 \cdot 10^6$ Hz, which remain well beyond the sensitivity range of upcoming GW detectors. Although it is possible to shift the peak down to frequencies around $\sim 10^5$ Hz by lowering ϕ_0 to its minimal value consistent with CMB observations, this comes at the cost of a significant suppression in amplitude, down to $h^2 \Omega_{\text{GW}}^{(p,0)} \simeq 10^{-15}$. In terms of the effective number of relativistic degrees of freedom contributed by the GWB, we find a maximum value of $\Delta N_{\text{eff}} \simeq 0.0021$, which still lies below the detection threshold of next-generation CMB experiments.

Chapter 5

Geometric reheating of the universe

In this chapter, we study the post-inflationary transfer of energy from the inflaton field (ϕ) into a scalar field (χ) non-minimally coupled to gravity through a term $\xi R|\chi|^2$, considering models where the inflaton potential behaves as $V_{\text{inf}} \approx |\phi|^p$ around $\phi = 0$. We extend the scope of preheating scenarios beyond the standard case of direct particle couplings, by exploring this indirect coupling mechanism. This setup corresponds to the paradigm of geometric preheating, which we now investigate in the non-linear regime by means of lattice simulations. Using α -attractor T-model potentials as representative cases, we analyze the viability of efficient reheating for $p = 2, 4, 6$, and determine whether radiation domination (RD), characterized by the energetic dominance of χ over ϕ , can be achieved. We identify the conditions under which preheating becomes efficient and successfully completes. This chapter is based on Ref. [3].

5.1 Introduction

The possibilities for reheating depend largely upon the manner in which the inflaton interacts with the *daughter* fields. As we have seen in the previous two chapters, direct couplings allow for (non-)perturbative decays of the inflaton into the daughter species [9, 30, 31, 35–40, 50–67, 110, 111, 183, 184, 197–202, 218–226]. On the other hand, there exist scenarios in which only gravity mediates the interaction between the inflaton and the daughter sectors, see e.g. [45–48, 73–75, 78–90]. In both types of scenarios, non-perturbative *preheating* effects can arise, leading to an efficient and often exponential transfer of energy into the daughter fields.

We focus on (p)reheating scenarios that are gravity-mediated, which we refer to as *gravitational (p)reheating*. We consider a single daughter scalar field χ non-minimally coupled to gravity through an interaction of the form $\xi|\chi|^2 R$, where R is the Ricci scalar and ξ is a dimensionless coupling constant. The inclusion of such a term is required by the renormalizability of the theory in curved spacetime [227, 228], with ξ acting as a running parameter that cannot be set to zero across all energy scales. We refer to χ as a non-minimally coupled (NMC) field. An early formulation of gravitational reheating [78] was based on two key ingredients: the excitation of a NMC scalar field towards the end of inflation [78, 79, 229], and the presence of a period of *kination* domination (KD) immediately following inflation. Under these circumstances, the initially negligible energy density of the NMC field, assumed to behave as radiation, can eventually dominate the energy content of the Universe. This mechanism, first introduced in Ref. [78] and later extended in [79, 230], plays a central role in *Quintessential inflation* scenarios [229, 231–238]. However, this original proposal has been shown to be inconsistent with CMB and BBN constraints [164, 239], due to an overproduction of gravitational waves (GWs) accumulated during the KD phase [114, 116, 240, 241], as demonstrated in [115]. Furthermore, Ref. [82] showed that the NMC field does not behave as radiation during KD, as it was assumed in [78, 79, 229], but rather experiences a tachyonic instability. This instability arises from the sign change in the Ricci scalar, $R \propto (1 - 3w)$, which occurs for a stiff equation of state $w > 1/3$. This crucial effect constitutes the mechanism of what is known as *Ricci reheating* [45, 82–85], where the NMC scalar is exponentially excited during KD, leading to successful reheating and suppressing the otherwise excessive production of GWs.

Gravitational reheating can be realized through the excitation of a NMC scalar during a sustained period of KD, as described above. However, a NMC scalar can also be naturally excited immediately after inflation in a broader class of scenarios. In particular, if the inflaton potential governing the post-inflationary dynamics has a monomial form, inflaton oscillations begin as soon as inflation ends. These oscillations induce an oscillatory behavior in the Ricci scalar R , which becomes negative during parts of each oscillation. This effect, similar to the trilinear interaction, leads to a tachyonic excitation of the NMC scalar. As a result, the energy density of the NMC field grows exponentially during these tachyonic phases. This mechanism, first studied and coined as *geometric preheating* in Ref. [47] (see also Ref. [48]), has since been explored in the context of higher-order curvature inflationary models [49, 80], and extended to multi-field inflationary scenarios [73–75, 87, 88].

In this chapter, we focus on geometric preheating effects in single-field inflationary

scenarios, selecting an inflaton potential with a monomial shape during the post-inflationary phase. In this context, α -attractor T-models [180] provide particularly useful frameworks, as their potentials asymptote to a plateau at large field values while reducing to a monomial form, $V_{\text{inf}} \propto |\phi|^p$, at small field values. A notable feature of these models is their capacity to tune the inflationary energy scale across a wide range of values. In this work, we exploit this property to investigate geometric preheating-like effects at various energy scales, including values well below the current upper bound set by CMB observations. We perform lattice simulations to study the dynamics of the system once it enters the non-linear regime,¹ determining whether proper *reheating*, i.e., the onset of radiation domination (RD) driven by the energetic dominance of χ over ϕ , can be realized. In this way, we extend the concept of geometric preheating into that of *geometric reheating*, identifying the conditions under which reheating can be successfully achieved.

Using α -attractor T-model potentials as a proxy, we have studied the non-linear dynamics of the system for $p = 2, 4, 6$, as a function of the energy scale Λ . We find that in high-energy scale scenarios, reheating is never achieved for $p = 2$, can be partially achieved for $p = 4$, and becomes highly efficient for $p = 6$. However, if the NMC field features self-interactions of the form $V_{\text{NMC}} \propto \lambda \chi^4$, reheating is frustrated even in the $p = 6$ case, unless the coupling is extremely small, $\lambda \ll 1$. When the energy scale of inflation Λ falls below a certain threshold, the Universe enters radiation domination (RD) very rapidly, triggered by efficient inflaton fragmentation after inflation. In this case, there is no significant energy transfer into the NMC field, and reheating (into χ) never completes. We have characterized the relevant energy and time scales for RD onset, whether due to proper reheating or inflaton fragmentation, as functions of both Λ and the non-minimal coupling ξ .

We emphasize that while the initial tachyonic growth of χ takes place in the linear regime of the system's dynamics, the completion of reheating, or the onset of any frustration mechanism, is governed by the emergence of non-linear effects. These effects are either due to self-interactions or to inflaton fragmentation. Specifically, the exponential energy transfer into χ is eventually controlled by a regularization of the tachyonic mass. This can occur due to:

- *Gravitational backreaction*: when the NMC field becomes the dominant energy component in the universe and proper reheating is achieved².

¹We note that the Ricci scalar R in our setup is treated as a background quantity, sourced by the volume averages of the matter fields. This is a valid approximation as long as local gravitational perturbations remain small relative to the background metric.

²We note that the name gravitational backreaction here does not refer to backreaction of the gravitational perturbations onto the field dynamics, but rather to the fact that when χ starts dominating the energy budget, the Ricci scalar tends to zero, thus switching off the energy transfer process itself.

- *Self-interaction backreaction*: when self-interactions of the NMC field are present and they block eventually the energy transfer into χ .
- *Inflaton fragmentation*: when inflaton backreacts onto itself and fragmentation becomes efficient, preventing further energy transfer into χ .

5.2 Inflationary model

We consider a family of inflationary potentials based on α -attractor T-models [180]

$$V_{\text{inf}}(\phi) = \frac{\Lambda^4}{p} \tanh^p \left(\frac{|\phi|}{M} \right) \simeq \begin{cases} \frac{\Lambda^4}{p} & , |\phi| \gg M \\ \frac{\Lambda^4}{p} \left(\frac{|\phi|}{M} \right)^p & , |\phi| \ll M \end{cases} \quad (5.1)$$

with p a dimensionless parameter, and Λ and M dimension one parameters which control the height and the width of the potential, respectively. We shall work with $p \geq 2$, so that $\partial V_{\text{inf}}/\partial\phi$ and $\partial^2 V_{\text{inf}}/\partial\phi^2$ are well defined when $\phi = 0$. Potentials (5.1) exhibit the characteristic property that for large field values, $|\phi| \gg M$, they approach a plateau as $V_{\text{inf}}(\phi) \rightarrow \Lambda^4/p$, whereas for small field values, $|\phi| \ll M$, they adopt a monomial shape as $V_{\text{inf}}(\phi) \propto |\phi|^p$.

As a first approach to understand the inflationary dynamics, we obtain the *potential slow-roll* parameters, which read

$$\epsilon_V \equiv \frac{m_p^2}{2} \left(\frac{V'_{\text{inf}}(\phi)}{V_{\text{inf}}(\phi)} \right)^2 = \frac{2p^2 \text{csch}^2(\frac{2\phi}{M})}{(M/m_p)^2}, \quad \eta_V \equiv m_p^2 \frac{V''_{\text{inf}}(\phi)}{V_{\text{inf}}(\phi)} = \frac{4p(p - \cosh(\frac{2\phi}{M})) \text{csch}^2(\frac{2\phi}{M})}{(M/m_p)^2}. \quad (5.2)$$

From the condition $\epsilon_V = 1$, we immediately obtain the inflaton amplitude ϕ_{end} at the end of inflation³. Indicating with a sub-index k the time when a scale k crossed the Hubble radius $-N_k$ efolds before the end of inflation, from the expression $N_k \simeq -\int_{\phi_{\text{end}}}^{\phi_k} \frac{d\phi}{\sqrt{2\epsilon_V}}$ we can obtain the inflaton amplitude ϕ_k as

$$\frac{\phi_k(M, p)}{m_p} \simeq \frac{\mathcal{M}}{2} \text{arccosh} \left\{ \frac{\mathcal{M} \sqrt{2p^2 + \mathcal{M}^2} - 4pN_k}{\mathcal{M}^2} \right\}, \quad (5.3)$$

³Such value is only approximated, as the end of inflation should be rather identified with the moment when $\epsilon_H \equiv -d \log H / d \log N = 1$, with H the Hubble rate and N the number of efoldings. The condition $\epsilon_V = 1$ is simply an approximation to this.

where, without loss of generality, we have chosen $N_k < 0$ and the positive branch $\phi > 0$ during inflation. For later convenience we have also introduced the dimensionless mass-parameter

$$\mathcal{M} \equiv \frac{M}{m_p}. \quad (5.4)$$

The slow roll framework relates the amplitude A_s and spectral index n_s of the scalar perturbations produced during inflation, as well as the amplitude of inflationary tensor perturbations, parameterized through the tensor-to-scalar ratio $r \equiv A_t/A_s$, with the inflaton potential and its derivatives, via the expressions [24]

$$A_s \simeq \frac{1}{24\pi^2\epsilon_{V_k}} \frac{V_{\text{inf}}(\phi_k)}{m_p^4}, \quad n_s - 1 \simeq 2\eta_{V_k} - 6\epsilon_{V_k}, \quad r \simeq 16\epsilon_{V_k}. \quad (5.5)$$

where V_k is the potential evaluated at $V_k \equiv V(\phi_k)$. Introducing the value ϕ_k from (5.3) into expressions (5.5), leads to

$$A_s(\Lambda, M, p) = \frac{\Lambda^4}{m_p^4} \times \frac{\left[8pN_k^2 + \mathcal{M}^2 \left(p - 4N_k \sqrt{1 + \frac{2p^2}{\mathcal{M}^2}} \right) \right] \left[\mathcal{M}^2 \left(\sqrt{1 + \frac{2p^2}{\mathcal{M}^2}} - 1 \right) - 4pN_k \right]^p}{24\pi^2 p^2 \mathcal{M}^2 \left[\mathcal{M}^2 \left(\sqrt{1 + \frac{2p^2}{\mathcal{M}^2}} + 1 \right) - 4pN_k \right]^p}, \quad (5.6)$$

$$n_s(M, p) = \frac{8p(N_k + 2)N_k - \mathcal{M}^2 \left(4(N_k + 1) \sqrt{1 + \frac{2p^2}{\mathcal{M}^2}} + p \right)}{8pN_k^2 + \mathcal{M}^2 \left(p - 4N_k \sqrt{1 + \frac{2p^2}{\mathcal{M}^2}} \right)}, \quad (5.7)$$

$$r(M, p) = \frac{16p\mathcal{M}^2}{8pN_k^2 + \mathcal{M}^2 \left(p - 4N_k \sqrt{1 + \frac{2p^2}{\mathcal{M}^2}} \right)}. \quad (5.8)$$

Though not very illuminating by themselves, expressions (5.6)-(5.8) can be used readily to constrain the parameters $\{\Lambda, M, p\}$ that characterize the potential (5.1), by confronting them against CMB observations. In particular, CMB data constrain scalar and tensor perturbations at the pivot scale $k_{\text{CMB}} = 0.05 \text{ Mpc}^{-1}$ as [25–27]

$$A_s = 2.099_{-0.292}^{+0.296} \cdot 10^{-9} \text{ (68\% CL)}, \quad (5.9)$$

$$n_s = 0.9649 \pm 0.0042 \text{ (68\% CL)}, \quad (5.10)$$

$$r < 0.032 \text{ (95\% CL)}. \quad (5.11)$$

As $r(M, p)$ in Eq. (5.8) is an increasing function of M for fixed p , the upper bound on the tensor-to-scalar ratio (5.11) sets an upper limit $M \leq M_{\text{max}}^{(p)}$ for a given p , independently of Λ . Saturating the bound (5.11) and setting $-N_k = -N_{\text{CMB}} = 60$, leads to $M_{\text{max}}^{(2)} \simeq 8.79 m_p$, $M_{\text{max}}^{(4)} \simeq 8.22 m_p$, and $M_{\text{max}}^{(6)} \simeq 8.08 m_p$ for $p = 2, 4, 6$, respectively. Furthermore, we observe that $n_s(M, p)$, cf. Eq. (5.7), is always well within the allowed range indicated

in Eq. (5.10) for $M \leq M_{\max}^{(p)}$, for any p , and independently of Λ . In other words, our inflationary modeling is, by construction, compatible with CMB constraints on the spectral index. Finally, the value of Λ , which controls the energy scale of inflation, can be determined from the expression of the scalar amplitude in Eq. (5.6), allowing us to express Λ as a function of $\{M, p\}$, for fixed A_s .

$$\frac{\Lambda^4(M, p; A_s)}{m_p^4} = \frac{24\pi^2 A_s \mathcal{M}^2 p^2 \left[\mathcal{M}^2 \left(\sqrt{1 + \frac{2p^2}{\mathcal{M}^2}} + 1 \right) - 4pN_k \right]^p}{\left[8pN_k^2 + \mathcal{M}^2 \left(p - 4N_k \sqrt{1 + \frac{2p^2}{\mathcal{M}^2}} \right) \right] \left[\mathcal{M}^2 \left(\sqrt{1 + \frac{2p^2}{\mathcal{M}^2}} - 1 \right) - 4pN_k \right]^p} \quad (5.12)$$

We can inspect Eq. (5.12) for which we observe that $\Lambda(M, p; A_s)$ is an increasing function of M , for given values of p and A_s . Therefore, fixing A_s , say to the central value in Eq. (5.9), and taking $M = M_{\max}^{(p)}$, leads to an upper bound on the scale of inflation as $\Lambda \leq \Lambda_{\max}^{(p)}$, with $\Lambda_{\max}^{(2)} \simeq 7.16 \cdot 10^{-3} m_p$, $\Lambda_{\max}^{(4)} \simeq 8.47 \cdot 10^{-3} m_p$, $\Lambda_{\max}^{(6)} \simeq 9.37 \cdot 10^{-3} m_p$, for $p = 2, 4, 6$, respectively. In the same manner, we can also obtain an approximate expression for the Hubble rate H at the Hubble crossing time of CMB scales. Evaluating (5.3) at $N_k = N_{\text{CMB}} = -60$, using (5.12), and the fact that deep inside inflation the potential energy dominates over the kinetic, i.e. $H_{\text{inf}}^2 \simeq \frac{1}{3m_p^2} V_{\text{inf}}(\phi_{\text{CMB}})$, leads to

$$\frac{H_{\text{inf}}^2(M, p; A_s)}{m_p^2} \simeq \frac{8p \mathcal{M}^2 A_s \pi^2}{28800 p + \mathcal{M}^2 \left(p + 240 \sqrt{1 + \frac{2p^2}{\mathcal{M}^2}} \right)}, \quad (5.13)$$

in correspondence with the idea that decreasing M reduces the energy scale of inflation. We will see next that decreasing the value of M does not only lower the energy scale of inflation, but affects drastically the way the inflaton oscillates around the minimum after inflation.

Summarizing, although the exact details depend on the choice of p , CMB constraints imply a maximum mass scale approximately given by $M \leq M_{\max} \sim 8 m_p$, and a corresponding maximum energy scale $\Lambda \leq \Lambda_{\max} \sim 0.01 m_p$, or equivalently $H_{\text{inf}} \leq H_{\max} \sim 5 \cdot 10^{13} \text{ GeV}$. Interestingly, there is nothing that prevent us to consider the regime $M \ll M_{\max}$ (or equivalently $\Lambda \ll \Lambda_{\max}$). One of the advantages of the chosen model is precisely that it allows us to lower the energy scale of inflation well below its current observational upper bound, while remaining fully consistent with all CMB constraints. If we wish to reduce the energy scale of inflation, it suffices to decrease the value of M . The *small-scale* limit $M \ll m_p$ can be readily obtained from Eqs. (5.6)–(5.8)

and (5.13), as

$$n_s(M, p) \xrightarrow{\frac{M}{m_p} \ll 1} \frac{29}{30} \simeq 0.967, \quad (5.14)$$

$$r(M, p) \xrightarrow{\frac{M}{m_p} \ll 1} \frac{(M/m_p)^2}{1800} = 5.56 \cdot 10^{-4} \left(\frac{M}{m_p} \right)^2, \quad (5.15)$$

$$\frac{\Lambda(M, p)}{m_p} \xrightarrow{\frac{M}{m_p} \ll 1} \left(\frac{24\pi^2 p A_s}{28800} \right)^{1/4} \sqrt{\frac{M}{m_p}} \simeq 2.04 \cdot 10^{-3} p^{1/4} \sqrt{\frac{M}{m_p}}, \quad (5.16)$$

$$H_{\text{inf}}(M, p) \xrightarrow{\frac{M}{m_p} \ll 1} 5.8 \times 10^{12} \left(\frac{M}{m_p} \right) \text{ GeV}. \quad (5.17)$$

We can notice that for $M = m_p$, the exact expressions of $n_s(M, p)$, $r(M, p)$ and $\Lambda(M, p; A_s)$, [c.f. Eqs. (5.7), (5.8) and (5.12), respectively], approach the asymptotic expressions in the *rhs* of Eqs. (5.14)-(5.16) to better than 0.1%, 0.2% and 2%, while for $M = 0.1m_p$, they approach the asymptotic behavior to better than 0.002%, 0.03% and 0.1%, respectively.

The expressions (5.3), (5.6)-(5.8) and (5.12)-(5.13), as well as their asymptotic expressions (5.14)-(5.17), are all based on the *slow-roll* description. Actually, for a chosen p , once we determine Λ as a function of M with the *slow-roll* approximation, i.e. via Eq. (5.12), we can compute the dynamics of the homogeneous inflaton by solving its Klein-Gordon equation (for each M), and determine when inflation really ends according to the condition $\epsilon_H \equiv -d \log H / d \log N = 1$, with H the Hubble rate and N the number of efoldings. This allows us to determine the correct value ϕ_{CMB} of the inflaton 60 e-folds before the end of inflation [instead of using approximation (5.3)], for given $\{p, M, \Lambda\}$. Using Eq. (5.5) and the observed value (5.9) of the scalar perturbation amplitude A_s , leads to correct the value of Λ . We repeat this process iteratively, till there is no appreciable change in ϕ_{CMB} and Λ . As a result of this procedure, the value of Λ is corrected with respect to the SR prediction (5.12) by $\sim 1\%$ in a first iteration, and by $\sim 0.001\%$ in a second iteration, for all values of M used. We conclude therefore that two iterations are more than enough.

5.2.1 Inflaton and Ricci curvature oscillations

Towards the end of inflation, the energy budget of the Universe is initially dominated by the inflaton, which can be safely assumed to be homogeneous.⁴ Following the end of

⁴We will assume the inflaton to remain homogeneous throughout Sec. 5.2.1. While this is a valid approximation for sufficiently large energy scales, in Sec. 5.2.2 we will see that inflaton self-interactions eventually induce inhomogeneities (the smaller the mass scale M , the earlier this occurs). For sufficiently low scales, the analysis in Sec. 5.2.1 will ultimately become invalid. We will revisit this issue in Sec. 5.2.2 and at the end of Sec. 5.4.

inflation, the homogeneous inflaton amplitude begins oscillating around the minimum of the potential (5.1), located at $\phi = 0$. The nature of these oscillations depends crucially on the value of M

$$\frac{\phi_\star}{M} = \operatorname{arcsinh} \left\{ \sqrt{\frac{p-1}{2}} \right\}. \quad (5.18)$$

The potential is thus divided into two field regions: $\phi < \phi_\star$, where the curvature is positive, $V''_{\text{inf}} > 0$, and $\phi > \phi_\star$, where the curvature is negative, $V''_{\text{inf}} < 0$. In *small-scale* scenarios with $M \lesssim m_p$, it holds that $\phi_{\text{end}} > \phi_\star$, whereas in *large-scale* scenarios with $m_p \lesssim M \leq M_{\text{max}}$ (i.e. those close to saturating the r bound), the opposite occurs: $\phi_{\text{end}} < \phi_\star$. The boundary between small and large scales is always of the order $M \sim m_p$, although the precise value depends on the choice of p . For this reason, we will loosely distinguish between large- and small-scale scenarios by simply indicating $M \gtrsim m_p$ and $M \lesssim m_p$, respectively. In Fig. 5.1, we plot, as an illustrative example, the inflaton potential for $p = 4$, marking ϕ_{end} and $\phi_\star$ with vertical lines. In the left panel, corresponding to a large-scale model with $M = 5 m_p$, we see that inflation ends below the inflection point, i.e. $\phi_{\text{end}} < \phi_\star$. In the right panel, corresponding to a small-scale model with $M = 10^{-3} m_p$, we observe that inflation ends above the inflection point, $\phi_{\text{end}} > \phi_\star$.

In large-scale scenarios, the inflaton amplitude oscillates within the positive-curvature region of the potential, i.e. $|\phi| < \phi_\star$. Taking the small-field limit $\phi < M$ in Eq. (5.1), and using the fact that the inflaton amplitude at the onset of oscillations is $\sim m_p$, we infer that the initial oscillation frequency is of the order $\Omega_{\text{osc}}(\phi_{\text{end}}) \sim (\Lambda/m_p)^2 (M/m_p)^{-p/2} m_p$. Due to the expansion of the Universe, the homogeneous field amplitude decays as $\propto a^{-6/(p+2)}$, and consequently, the oscillation frequency is damped as $\Omega_{\text{osc}}(\phi) \propto a^{-3(p-2)/(p+2)}$ [183, 184]. However, since the Hubble rate also decays as $H \propto a^{-3p/(2+p)}$, their ratio evolves as $\Omega_{\text{osc}}/H \propto a^{6/(2+p)}$, indicating that the inflaton completes an increasing number of oscillations per Hubble time as the Universe expands.

In small-scale scenarios, the inflaton amplitude probes both positive ($|\phi| < \phi_\star$) and negative ($\phi_\star < |\phi| < \phi_{\text{end}}$) branches of the potential. The smaller the scale M , the more time the field spends in the negative region, as compared to the positive region. The fraction of time spent within one oscillation in the negative curvature decreases however in time, as the inflaton amplitude decays due to the expansion of the universe. In addition, the effective frequency of oscillation could be approximated as $\Omega_{\text{eff}} \sim V'(\phi)/\phi$, which for $\phi \sim M$, it approaches a value independent of M . This means that the smaller the value of M , the faster the oscillations are compared to the expansion of the

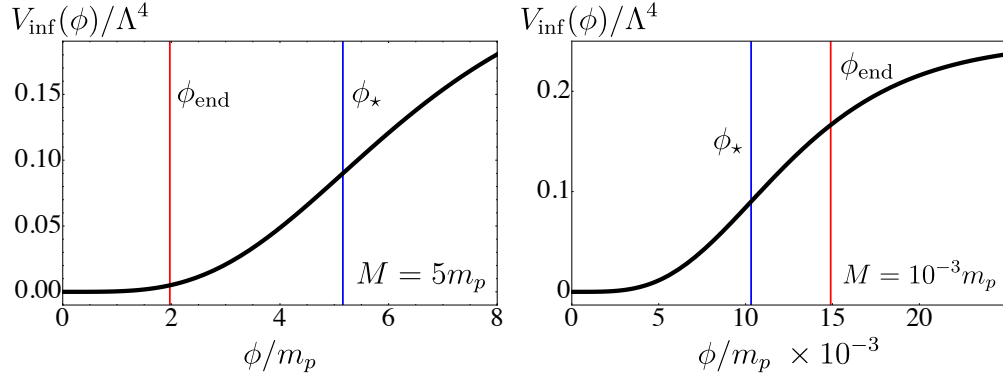


FIGURE 5.1: Inflaton potential profile for $p = 4$, *Left*: $M = 5m_p$ and *Right*: $M = 10^{-3}m_p$. The choice of $M = 5m_p$ shows that the field value at the end of inflation (ϕ_{end}) is below its inflection point (ϕ_*), this means that the inflaton will only probe the monomial potential of the form $\propto \phi^p$ when it oscillates. Alternatively, for $M = 10^{-3}m_p$ the inflaton ends inflation at a value higher than the inflection point meaning that the first oscillations of the inflaton will probe the change of inflection in the potential. This explains why the first oscillations are potential dominated.

universe, as it holds that $\Omega_{\text{eff}}/H_{\text{inf}} \propto 1/M$. Independently of whether we are in large or small-scale oscillatory regimes, we can always define an *oscillation-averaged* version of the (homogeneous) pressure and energy densities of the inflaton, $\bar{p}_\phi$ and $\bar{\rho}_\phi$, where the overline - denotes averaging over one or more inflaton oscillations. This allows us to define an *effective equation of state* (*eos*) as $w_{\text{osc}} = \bar{p}_\phi/\bar{\rho}_\phi$. Although $\bar{p}_\phi$ and $\bar{\rho}_\phi$ are generally time-dependent, in large-scale scenarios the *eos* takes on a simple constant value, $w_{\text{osc}} \equiv (p-2)/(p+2)$ [212], reflecting the fact that only the positive-curvature region of the potential, with $V_{\text{inf}} \propto |\phi|^p$, is sampled during oscillations. In small-scale scenarios, the inflaton initially probes both sides of the potential, including the negative-curvature region. However, as time progresses, the fraction of each oscillation spent in the negative-curvature region decreases, and w_{osc} gradually increases. Eventually, after a number of oscillations, the inflaton amplitude falls below the inflection point ϕ_* , and from that moment onward, oscillations are confined entirely within the positive-curvature region. As a result, w_{osc} asymptotically approaches the same p -dependent constant value as in the large-scale regime. In summary, we can write

$$w_{\text{osc}} \equiv \frac{\bar{p}_\phi}{\bar{\rho}_\phi} \simeq \begin{cases} \frac{p-2}{p+2} & (\text{as } |\phi| < \phi_* \text{ always}) \quad , \quad M \gtrsim m_p \\ \left(\begin{array}{c} \text{increasing} \\ \text{function} \end{array} \right) & \xrightarrow{\text{(Once } |\phi| < \phi_*)} \frac{p-2}{p+2} \quad , \quad M \lesssim m_p \end{cases} \quad (5.19)$$

A consequence of the inflaton's oscillations is that the Ricci scalar R must also oscillate. This has important implications for the scenarios studied later in Sec. 5.3 and 5.4. To understand the oscillatory behavior of R , we note that in a (flat) FLRW background

metric the Ricci scalar can be written as [46]

$$R = \frac{6}{a^{2\alpha}} \left[\frac{a''}{a} - (\alpha - 1) \left(\frac{a'}{a} \right)^2 \right], \quad (5.20)$$

where $a(\eta)$ is the scale factor, η is the α -time variable defined before (which corresponds, for example, to cosmic time for $\alpha = 0$, or to conformal time for $\alpha = 1$), and $'$ denotes derivatives with respect to η . At early times, the Ricci scalar is determined solely by the inflaton field, so one should be able to express R directly in terms of the inflaton's energy components. To make this explicit, we first write the Friedmann equations sourced by the inflaton:

$$\frac{a''}{a} = \frac{a^{2\alpha}}{3m_p^2} \left\langle (\alpha - 2)K(\phi) + \alpha G(\phi) + (\alpha + 1)V_{\text{inf}}(\phi) \right\rangle, \quad (5.21)$$

$$\left(\frac{a'}{a} \right)^2 = \frac{a^{2\alpha}}{3m_p^2} \left\langle K(\phi) + G(\phi) + V_{\text{inf}}(\phi) \right\rangle, \quad (5.22)$$

where $\langle \dots \rangle$ denotes volume averaging, and where $K(\phi) \equiv \frac{1}{2a^{2\alpha}}\phi'^2$, $G(\phi) \equiv \frac{1}{2a^2}(\vec{\nabla}\phi)^2$, and $V_{\text{inf}}(\phi)$ are the inflaton's kinetic, gradient, and potential energy densities, respectively. Plugging Eqs. (5.21) and (5.22) into Eq. (5.20) leads to

$$R = \frac{2}{m_p^2} \left\langle 2V_{\text{inf}}(\phi) + G(\phi) - K(\phi) \right\rangle \xrightarrow{\vec{\nabla}\phi \rightarrow 0} \frac{2}{m_p^2} \left(2V_{\text{inf}}(\phi) - K(\phi) \right), \quad (5.23)$$

where in the expression on the right-hand side, we have used the fact that $\langle V_{\text{inf}}(\phi) \rangle = V_{\text{inf}}(\phi)$, $\langle K(\phi) \rangle = K(\phi)$, and $\langle G(\phi) \rangle = 0$ as long as the inflaton remains homogeneous. From Eq. (5.23), we see that the oscillations of the homogeneous inflaton are imprinted in R through the competition between twice the potential energy, $2V_{\text{inf}}(\phi)$, and the kinetic energy, $K(\phi)$. In order to understand the behavior of the curvature as a function of M , we notice an exact relation between R , the Hubble rate $H^2 (= (\dot{a}/a)^2$ in cosmic time), and the background *eos* w , using Eq. (5.20) and the Friedmann equations, as

$$\frac{R}{H^2} = 3(1 - 3w). \quad (5.24)$$

Aforementioned expression (5.24) is valid for arbitrary matter content in a FLRW expanding background, i.e. it does not require inflaton domination. We can specialize it to our initial case of a homogeneous inflaton, and furthermore take an oscillation-average on both sides. This leads to

$$\frac{\bar{R}}{\bar{H}^2} = 3(1 - 3w_{\text{osc}}) \simeq \begin{cases} 6 \frac{(4-p)}{(2+p)} & (\text{as } |\phi| < \phi_\star \text{ always}) \quad , \quad M \gtrsim m_p \\ \left(\begin{array}{c} \text{decreasing} \\ \text{function} \end{array} \right) & \xrightarrow{(\text{Once } |\phi| < \phi_\star)} 6 \frac{(4-p)}{(2+p)} \quad , \quad M \lesssim m_p \end{cases} \quad (5.25)$$

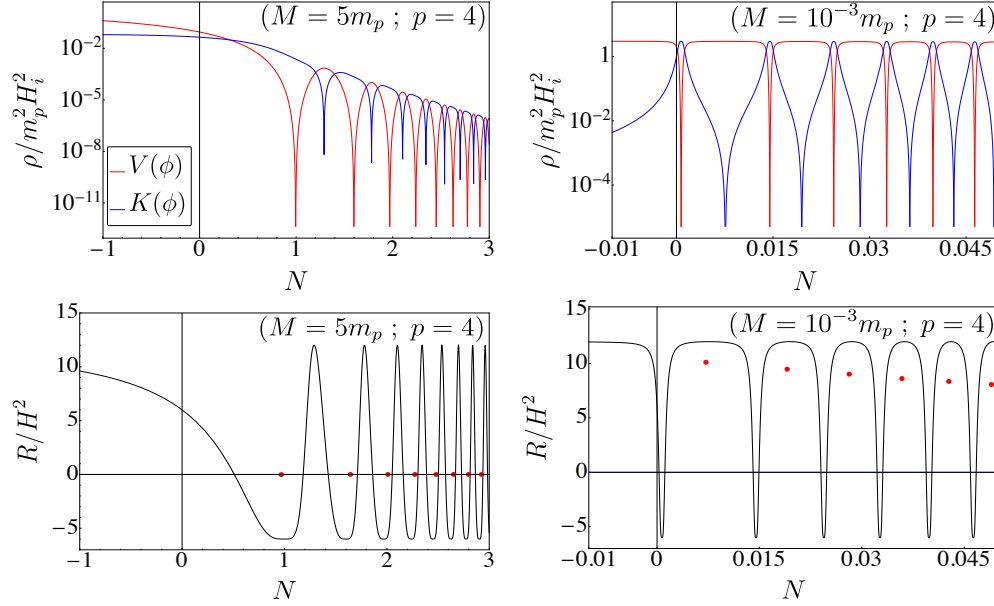


FIGURE 5.2: Time evolution of the kinetic and potential energy of the inflaton and R/H^2 for $p = 4$, for $M = 5m_p$ (Left Panel) and $M = 10^{-3}m_p$ (Right Panel). We can see that whenever the kinetic energy dominates over the potential, the Ricci scalar becomes negative. Red dots indicate the ratio of the oscillation averaged quantities, $\overline{R}/\overline{H^2}$, c.f. Eq. (5.25).

with $\overline{H^2} \equiv 3m_p^2 \bar{\rho}_\phi$ and w_{osc} given by Eq. (5.19). Eq (5.25) is very informative, as it indicates that the condition to obtain an oscillation-averaged negative curvature, $\overline{R} < 0$, relies on having a *stiff* effective *eos* as $w_{\text{osc}} > 1/3$. For the oscillatory regime that samples only the positive curvature region $V'' > 0$ of the potential, the *eos* is given by $w_{\text{osc}} = (p - 2)/(p + 2)$, c.f. Eq. (5.19), which implies that $\overline{R} < 0$ requires $p > 4$.

In summary, potentials of the form $\propto \tanh^p(\phi/M)$ with $p > 4$, irrespective of the value of M , eventually yield an oscillation-averaged negative curvature, $\overline{R} < 0$. In high-scale models, the condition $\overline{R} < 0$ is established immediately upon the onset of oscillations. In contrast, low-scale models exhibit an initial phase of oscillations during which $\overline{R}$ is positive but decays in time. Eventually, once the inflaton amplitude drops below the inflection point, the Hubble-normalized oscillation-averaged curvature settles to a negative constant value, given by $\overline{R}/\overline{H^2} = 6(4 - p)/(2 + p) < 0$, cf. Eq. (5.25). As we will see in Sec. 5.3 and 5.4, this interesting *delay* in the onset of $\overline{R} < 0$, characteristic of low-scale scenarios with $p > 4$, will have important implications for enabling reheating in our setup. Although less relevant for our later purposes of reheating, we also note that potentials of the form $\propto \tanh^p(\phi/M)$ with $p \leq 4$ always yield, in large-scale models, an oscillation-averaged curvature of the form $\overline{R}/\overline{H^2} = 6(4 - p)/(2 + p)$, which is positive ($\overline{R} > 0$) for $p < 4$, and vanishing⁵ ($\overline{R} = 0$) for $p = 4$. In the case of low-scale models with

⁵The fact that $\overline{R} = 0$ for $p = 4$ in large-scale models can be anticipated from the observation that $V \propto \phi^4$ is a scale-free potential. Oscillations in such a potential lead to a radiation-dominated background, for which one expects $\overline{R} = 0$.

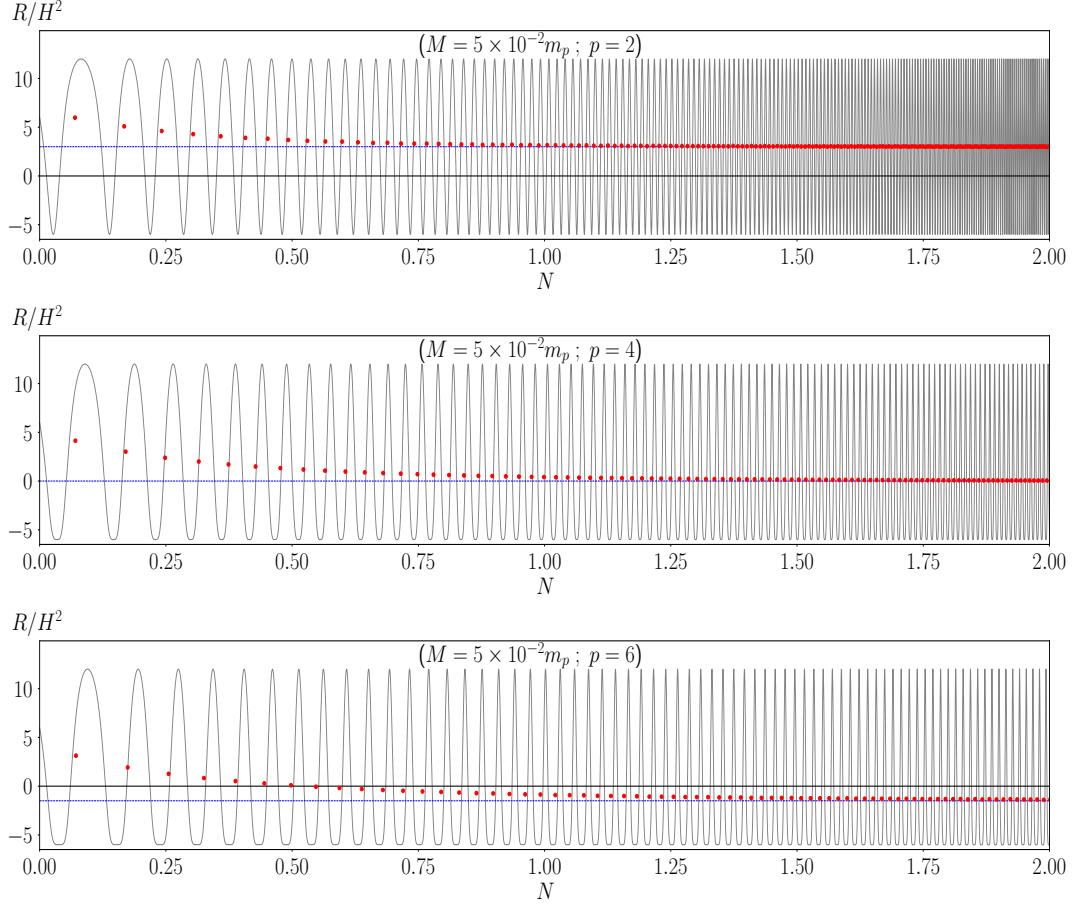


FIGURE 5.3: Time evolution of R/H^2 for low scale scenarios with $M = 5 \cdot 10^{-2} m_p$, for $p = 2, 4, 6$ (top to bottom). Red dots indicate the ratio of the oscillation averaged quantities, $\bar{R}/\bar{H}^2$, whereas blue dashed lines represent the asymptotic value (5.25).

$p \leq 4$, we expect that $\bar{R}$ is initially positive and decreases over time. Eventually, once the condition $|\phi| < \phi_\star$ is satisfied, the averaged curvature settles to the same asymptotic expression as in large-scale models, namely $\bar{R}/\bar{H}^2 = 6(4 - p)/(2 + p)$, which is again positive for $p < 4$, and vanishing for $p = 4$.

As an example, in Fig. 5.2 we show the potential and kinetic energy of the homogeneous inflaton during the first few oscillations after inflation, for scenarios with $p = 4$. We choose representative cases of large-scale and low-scale scenarios, with $M = 5 m_p$ (left panels) and $M = 10^{-3} m_p$ (right panels). In the large-scale scenario, we observe that the kinetic and potential energies are balanced within each oscillation, resulting in an oscillation-averaged curvature $\bar{R} = 0$. In the low-scale scenario, by contrast, the potential energy dominates over the kinetic energy within each oscillation, leading to $\bar{R} > 0$. However, as noted earlier, the inflaton amplitude in the low-scale case eventually decays below the inflection point, leading to an asymptotic approach toward $\bar{R} = 0$. While this feature is not apparent in the bottom-right panel of Fig. 5.2, it becomes clearly visible in the middle panel of Fig. 5.3, where the behavior of the system in a low-scale scenario with $p = 4$ is depicted at much later times. The corresponding behavior of $\bar{R}$

for low-scale scenarios with $p = 2$ and $p = 6$ can also be observed in the top and bottom panels of Fig. 5.3, respectively.

5.2.2 Inflaton fragmentation

In the previous Sec. 5.2.1, we assumed that the inflaton remained homogeneous at all times. In this section, we show that this assumption does not always hold, as the inflaton condensate may fragment into inhomogeneous fluctuations due to its own self-interactions; see [221, 223]. To understand inflaton fragmentation, we begin by writing the equation of motion (*com*) for the inflaton as

$$\phi'' + (3 - \alpha) \left(\frac{a'}{a} \right) \phi' - a^{-2(1-\alpha)} \nabla^2 \phi = -a^{2\alpha} V_{,\phi}. \quad (5.26)$$

Setting $\alpha = 3(p-2)/(p+2)$, we introduce the following variables:

$$\varphi = \frac{1}{m_p} a^{\frac{6}{p+2}} \phi, \quad d\tilde{\eta} = H_* a^{\frac{3(2-p)}{p+2}} d\eta, \quad d\tilde{x}_i = H_* dx_i, \quad (5.27)$$

with H_* the Hubble rate deep inside the *slow-roll* regime of inflation. This allows us to rewrite the inflaton's equation of motion as

$$\varphi'' - a^{-\frac{16-4p}{2+p}} \tilde{\nabla}^2 \varphi + \left(\tilde{V}_{,\varphi} + F(a) \right) = 0, \quad (5.28)$$

where

$$F(a) = \frac{6}{p+2} \left(\frac{p-4}{p+2} \left(\frac{a'}{a} \right)^2 - \frac{a''}{a} \right), \quad (5.29)$$

and where $\tilde{\nabla}_i \equiv \partial/\partial\tilde{x}_i$, and $\tilde{V} \equiv V_{\text{inf}}(\phi(\varphi))/(m_p^2 H_*^2)$. We now decompose the field into homogeneous and inhomogeneous components, $\bar{\varphi}(\tilde{\eta})$ and $\delta\varphi(\tilde{\eta}, \tilde{x}_i)$, respectively:

$$\varphi(\tilde{\eta}, \tilde{x}_i) \equiv \bar{\varphi}(\tilde{\eta}) + \delta\varphi(\tilde{\eta}, \tilde{x}_i), \quad (5.30)$$

and treat the inhomogeneous part as a perturbation over the background, i.e. $|\delta\varphi(\tilde{\eta}, \tilde{x}_i)| \ll |\bar{\varphi}(\tilde{\eta})|$, at least initially.

We can distinguish two regimes. For $M \gtrsim m_p$, inflaton oscillations sample only the positive-curvature region of the potential, which can be written in dimensionless form as $\tilde{V}(\varphi) = \frac{1}{p} \mu^2 \varphi^p$, with $\mu \equiv (\Lambda^2 m_p^{p/2-1})/(M^{p/2} H_*)$ acting as an effective dimensionless mass parameter for the inflaton. The equation of motion for the inflaton fluctuations in

momentum space then reads

$$\delta\varphi_k'' + \omega_k^2 \delta\varphi_k = 0, \quad \omega_k^2 = \kappa(a)^2 + (p-1)\mu^2 \bar{\varphi}^{p-2} + F(a), \quad (5.31)$$

where

$$\kappa(a) = \frac{k}{H_*} a^{\frac{2p-8}{2+p}}, \quad (5.32)$$

with k the comoving momentum. In this case, the oscillations of the inflaton induce parametric resonance of the inflaton fluctuations shortly after the onset of oscillations, since $F(a)$ becomes negligible rather quickly. Eq. (5.31) admits solutions of the form $\varphi_k \sim e^{\mu_k \tilde{\eta}}$, and a Floquet chart analysis shows that the growth exponent peaks at $\mu_k \sim 0.036$ for $p = 3.6$ [184]. However, this mechanism is not particularly efficient in fragmenting the inflaton condensate, as the typical timescale for fragmentation—i.e., for the fluctuations to backreact on the inflaton background—is generally $N_{\text{frag}} > 10$ e-folds; see e.g. [183, 184, 221, 223].

In the case of low-scale scenarios, $M \lesssim m_p$, the initial oscillations are dominated by the negative-curvature branch of the potential, as the inflaton ends inflation above the inflection point. In this situation, the potential cannot be approximated by a monomial; however, we can still gain insight by analyzing a more general form of the equation of motion for the fluctuations in this regime:

$$\delta\varphi_k'' + \omega_k^2 \delta\varphi_k = 0, \quad \omega_k^2 = \kappa(a)^2 + m_{\delta\varphi}^2 + F(a), \quad m_{\delta\varphi}^2 \equiv \frac{\partial^2 \tilde{V}}{\partial \varphi^2}. \quad (5.33)$$

It is useful to express quantities in the small-scale limit $M \ll m_p$. To begin with, we consider H_* to be dominated by the potential energy, which for $\phi > M$ approximates to the asymptotic value $V_{\text{inf}} \approx \Lambda^4/p$, such that $H_*^2 \approx \Lambda^4/(3pm_p^2)$. The dimensionless potential then reads $\tilde{V}(\varphi) \sim 3 \tanh^p(|\varphi|/\mathcal{M})$, where we have set the initial scale factor to $a = 1$ and introduced $\mathcal{M} = M/m_p$.

Our aim is to understand how the initial effective mass of the inflaton perturbations behaves. We derive the following expression for the effective mass:

$$m_{\delta\varphi}^2 = \frac{\partial^2 \tilde{V}}{\partial \varphi^2} = \frac{12p}{\mathcal{M}^2} \frac{\left(p - \cosh\left(\frac{2\varphi}{\mathcal{M}}\right)\right) \tanh^p\left(\frac{\varphi}{\mathcal{M}}\right)}{\sinh^2\left(\frac{2\varphi}{\mathcal{M}}\right)} \xrightarrow{\mathcal{M} \ll 1} -\frac{\sqrt{72}}{\mathcal{M}}, \quad (5.34)$$

where the asymptotic limit corresponds to $\mathcal{M} \ll 1$, with the hyperbolic functions evaluated at the background field amplitude satisfying the condition $\epsilon_V(\varphi_{\text{end}}) = 1$. We emphasize the negative sign, as expected from the fact that the oscillations begin above the inflection point. This implies that modes with $\kappa^2 \lesssim \sqrt{72} \mathcal{M}^{-1}$ will have

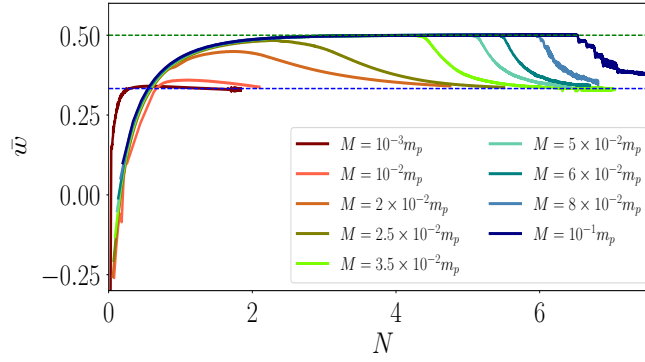


FIGURE 5.4: Time evolution of oscillation averaged equation of state $\bar{w}$ for low scale scenarios with $M \in [10^{-3}, 10^{-1}]m_p$, for $p = 6$. Green dashed lines represent the asymptotic value (5.25), Blue dashed line corresponds to radiation-domination eos .

$\omega_k^2 < 0$, and hence will undergo tachyonic growth. From this linearized analysis, we conclude that inflaton modes begin growing immediately after inflation ends, and the strength of their tachyonic growth increases with decreasing $\mathcal{M}$. This may ultimately lead to fragmentation of the inflaton condensate, provided that the fluctuations grow sufficiently to become comparable to the background amplitude.

To properly characterize the timescale of inflaton fragmentation, however, we must go beyond the linear regime. This requires the use of lattice simulations capable of fully capturing the non-linear dynamics of the system. We have simulated and measured the fragmentation timescale of the inflaton⁶. Simulations were performed on lattices with $\mathcal{N} = 385$ sites per spatial dimension, for different values of the mass scale ranging within $\mathcal{M} \in [10^{-1}, 10^{-3}]$. Each simulation was initialized at the end of inflation, defined by the condition $\epsilon_H = 1$, and evolved until the system reached radiation domination (RD) as a result of full inflaton fragmentation. To characterize the timescale of fragmentation, we followed the evolution of the background equation of state of the inflaton:

$$w = \frac{\langle p \rangle}{\langle \rho \rangle} = \frac{\langle K(\phi) - \frac{1}{3}G(\phi) - V(\phi) \rangle}{\langle K(\phi) + G(\phi) + V(\phi) \rangle}, \quad (5.35)$$

where $\langle \dots \rangle$ denotes volume averaging. We define the fragmentation timescale N_{frag} following [221], as the number of e -folds elapsed since the end of inflation when the envelope of the highly oscillatory evolution of w drops below 95 % of its initial maximum value.

As shown in Fig. 5.4, the fragmentation time decreases significantly for smaller values of M , consistent with the linear analysis previously discussed: tachyonic growth, which eventually drives fragmentation, is stronger for lower values of $\mathcal{M}$. The cases depicted in

⁶We have characterized the inflaton fragmentation time only for $p = 6$, as this is the most promising case to yield proper reheating at low energy scales; see Sec. 5.4.

the figure that show delayed fragmentation correspond to scenarios where inflaton self-resonance is inefficient. In these situations, the initial tachyonic growth is insufficient to fragment the inflaton condensate before the background field begins oscillating entirely within the positive-curvature region (i.e., below the inflection point). From that point onward, only parametric resonance operates, resulting in a slower fragmentation process.

In the upcoming Sec. 5.3 and 5.4, we will study the efficiency of geometric reheating of a NMC scalar field, under the assumption that the inflaton remains homogeneous during its oscillations. At the end of Sec. 5.4, we will revisit the concept of inflaton fragmentation and examine how it impacts the realization of proper reheating, depending on the value of M .

5.3 Geometric preheating: linear regime

Our starting point is the action

$$S = \int d^4x \sqrt{-g} \left(\frac{1}{2} m_p^2 R - \frac{1}{2} \xi R \chi^2 - \frac{1}{2} g^{\mu\nu} \partial_\mu \chi \partial_\nu \chi - \frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi, \chi) \right), \quad (5.36)$$

which describes the dynamics of the inflaton ϕ and a scalar spectator field χ with non-minimal coupling to gravity of the form $\xi \chi^2 R$, where R is the *Ricci* scalar and ξ is a dimensionless coupling constant. From now on, we will refer to χ as the non-minimally coupled (NMC) scalar field. The total potential is given by $V = V_{\text{inf}}(\phi) + V_{\text{NMC}}(\chi, \dots)$, where $V_{\text{inf}}(\phi)$ is the inflaton potential defined in Eq. (5.1), and $V_{\text{NMC}}(\chi, \dots)$ encodes interactions involving χ . The equation of motion for the NMC field in α -time η can be straightforwardly obtained by varying the action (5.36) with respect to χ :

$$\chi'' + (3 - \alpha) \mathcal{H} \chi' - a^{-2(1-\alpha)} \nabla^2 \chi + a^{2\alpha} (\xi R \chi + V_{,\chi}) = 0, \quad (5.37)$$

where $\mathcal{H} \equiv a'/a$, and $'$ denotes differentiation with respect to η .

In order to understand the dynamics of the NMC field during the inflaton oscillations following the end of inflation, we first canonically quantize χ in Eq. (5.37):

$$\chi = \int \frac{d^3k}{2\pi^3} \left[\chi_k \hat{a}_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{x}} + \chi_k^* \hat{a}_{\mathbf{k}}^\dagger e^{-i\mathbf{k}\cdot\mathbf{x}} \right], \quad (5.38)$$

where $\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{k}}^\dagger$ are standard annihilation and creation operators satisfying $[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{k}'}^\dagger] = (2\pi)^3 \delta(\mathbf{k} - \mathbf{k}')$, and the field modes are initially normalized such that $\chi_k \chi_k'^* - \chi_k' \chi_k^* = i$. To study the early-time behavior of the modes $\chi_k(\eta)$, we choose to work in conformal time τ and introduce a conformally rescaled field $\tilde{\chi}(\tau) = a(\tau) \chi$. The modes $\tilde{\chi}_k$ satisfy

the equation

$$\tilde{\chi}_k'' + \omega_k^2 \tilde{\chi}_k = 0, \quad \text{with} \quad \omega_k^2 \equiv k^2 + a^2 \left(\xi - \frac{1}{6} \right) R, \quad (5.39)$$

where we have neglected interactions of χ (i.e. $\partial_\chi V = 0$), and used the relation $a^3 R = 6a''$ valid in conformal time. Initially, backreaction from χ into the Ricci scalar is negligible, so we treat R as a time-dependent background quantity sourced solely by the homogeneous inflaton. Under these assumptions, Eq. (5.39) is a linear equation for $\tilde{\chi}_k$. We observe that for $\xi > 1/6$, whenever R becomes negative, $\tilde{\chi}_k$ undergoes a *tachyonic* instability for modes with $k \lesssim k_*$, where

$$k_* = \sqrt{(\xi - 1/6) |R| a^2}. \quad (5.40)$$

As a result, infrared modes with $k < k_*$ experience exponential growth, $\tilde{\chi}_k \sim e^{|\omega_k| \tau}$, due to $\omega_k^2 < 0$.

Since Eq. (5.24) indicates that the most negative value of the Ricci scalar normalized by the Hubble rate squared is bounded⁷ from below as $R/H^2 > -6$, we can normalize Eq. (5.40) by aH , yielding

$$\frac{k_*}{aH} < \sqrt{6\xi - 1}. \quad (5.41)$$

Using the relation $\overline{R}/\overline{H^2} = 6(4 - p)/(2 + p)$ from Eq. (5.24), valid when the inflaton oscillates entirely within the positive-curvature region of the potential, we can further estimate the cutoff scale as

$$\frac{k_*}{aH} \simeq \sqrt{6\xi - 1} \sqrt{\frac{|4 - p|}{2 + p}}, \quad (5.42)$$

which satisfies Eq. (5.41) for all p . In other words, as long as $\xi \gg 0.1$, it approximately holds that $k_* \lesssim \sqrt{6\xi} aH$. While for $1/6 < \xi \lesssim \mathcal{O}(1)$ only super-horizon or near-horizon modes are excited, for $\xi \gg 1$, sub-horizon modes are significantly amplified, up to a physical scale $\sim 1/\sqrt{6\xi}$ times smaller than the Hubble radius.

In order to determine the evolution of the modes $\chi_k \equiv \tilde{\chi}_k/a$, we follow the prescription of Ref. [46]. This consists of adopting the *Bunch–Davies* (BD) vacuum as the initial condition for modes deep inside the Hubble radius during inflation. This vacuum state is obtained by solving Eq. (5.39) in the limit $k^2 \gg a^2 R$, yielding

$$\tilde{\chi}_k(-k\tau \gg 1) \approx \frac{1}{\sqrt{2k}} e^{-ik\tau} \approx \frac{1}{\sqrt{2k}} e^{\frac{ik}{aH}}, \quad (5.43)$$

⁷The fact that the equation of state of the Universe is bounded from above as $w \leq 1$ emerges from the physical requirement that the sound speed $c_s^2 = \partial p / \partial \rho \equiv w$ of a fluid does not exceed the speed of light.

where, in the second equality, we used the fact that deep inside the slow-roll regime of inflation, the conformal time satisfies $\tau \approx -1/(aH)$. To ensure that the initial condition of each mode is accurately represented by the BD condition (5.43), we begin evolving them during inflation when they are at least a factor $\beta = 10^3$ inside the Hubble radius.⁸ We consider a range of modes spanning three decades in momentum space, and adjust the initial time such that the most ultraviolet mode in this range (initially satisfying $k/(aH) = \beta = 10^6$) exits the Hubble radius precisely at the end of inflation. This implies that the scale factor must grow by a factor $\sim 10^6$ during inflation, which corresponds to $-N_{\text{init}} \approx \ln(10^6) \approx 14$ e -folds before the end of inflation. In practice, we evolve the system over a total range of $\Delta N_{\text{tot}} = -N_{\text{init}} + N_{\text{final}}$ e -folds, where $N_{\text{init}} \simeq -14$ marks the starting point deep inside inflation, $N = 0$ corresponds to the end of inflation, and $N_{\text{final}} = 2 - 4$ e -folds accounts for post-inflationary evolution. This allows us to capture the effects of the evolution of R on the dynamics of the χ field after inflation.

In the left panels of Fig. 5.5, we show the oscillations of R , normalized to the Hubble rate as a function of the number of e -folds after inflation. In the right panels of the same figure, to illustrate the growth of the tachyonic modes of χ , we plot the power spectrum $\Delta_\chi(k, N)$, defined via the variance of the χ field as

$$\langle \chi^2 \rangle = \langle 0 | \hat{\chi}(N, x) \hat{\chi}(N, x) | 0 \rangle = \int \frac{dk}{k} \Delta_\chi(k, N). \quad (5.44)$$

The first feature worth highlighting is the marked difference in the oscillatory regimes of the Ricci scalar depending on the value of p , as shown in the left panels of Fig. 5.5. For the chosen large-scale example with $M = 5 m_p$, the duration of a semi-oscillation with $R < 0$ is longer for $p = 6$ than for $p = 4$, and significantly longer than for $p = 2$. This implies that the tachyonic growth of the modes is more effective for higher values of p . This behavior is reflected in the more rapid growth of the power spectrum for $p = 6$ compared to $p = 4$ and $p = 2$, as can be seen in the right panels of Fig. 5.5. In Fig. 5.6, we plot the growth of χ modes for low-scale scenarios with $M = 0.05 m_p$ (left panels) and $M = 0.001 m_p$ (right panels), for $p = 2, 4$, and 6 (top to bottom panels). The case $M = 0.05 m_p$ corresponds to the scenarios considered in Fig. 5.3, where we previously analyzed the behavior of the ratio $\bar{R}/\bar{H}^2$. As explained in Sec. 5.2 and illustrated in Fig. 5.3, the oscillation-averaged Ricci scalar $\bar{R}$ in these scenarios remains a positive, decaying function while ϕ samples the potential above the inflection point $\phi_\star$ (indicated by red squares). However, the ratio $\bar{R}/\bar{H}^2$ eventually decays to its asymptotic value given by Eq. (5.25). In Fig. 5.6, we observe that χ modes are not significantly amplified when $M = 0.05 m_p$ for $p = 2$ and $p = 4$, as $\bar{R}$ remains essentially non-negative (see top and middle panels of Fig. 5.3). For $p = 6$, although the modes do not grow initially

⁸In practice, $\beta \gtrsim 100$ is sufficient, but we choose $\beta = 10^3$ to remain on the conservative side.

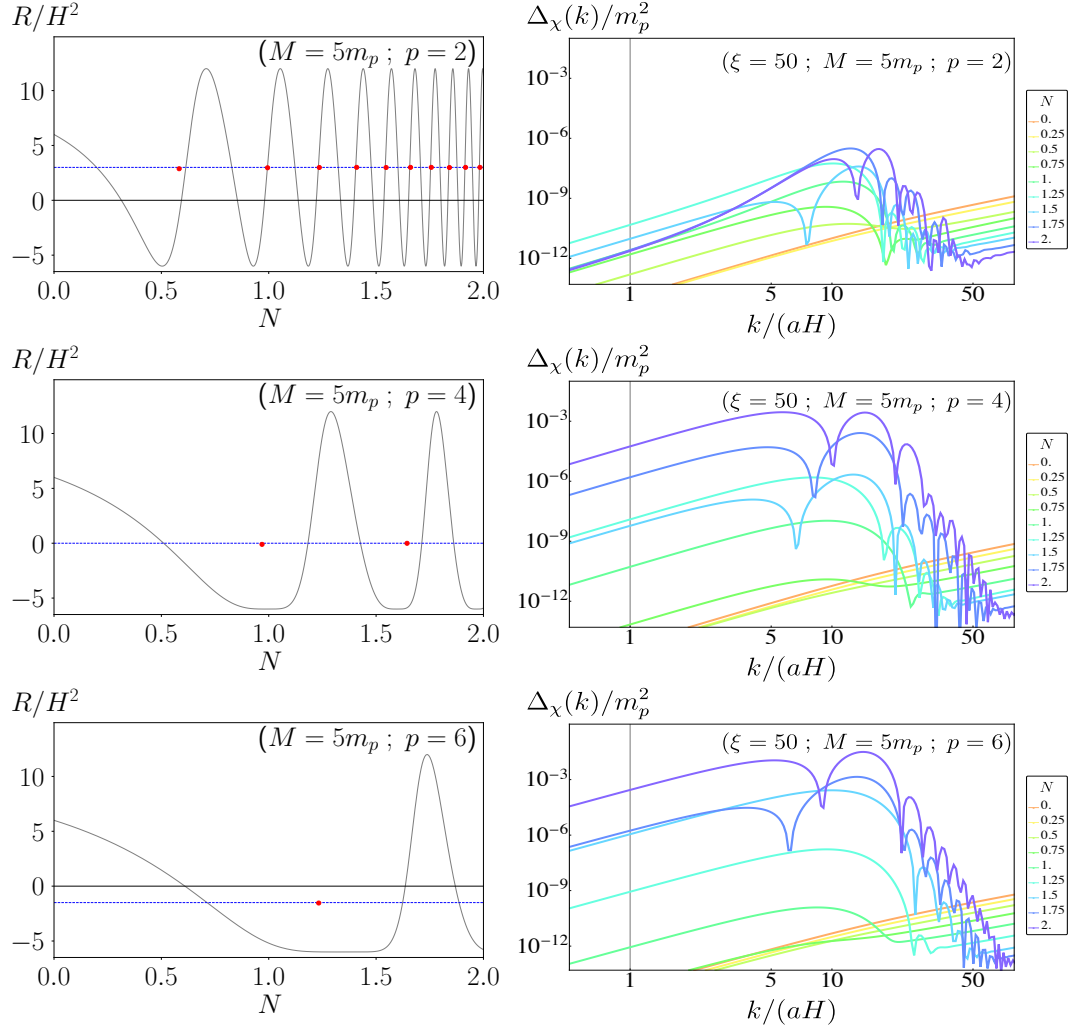


FIGURE 5.5: *Left-Panels:* Time Evolution of R/H^2 for $M = 5m_p$ and $p = \{2, 4, 6\}$. Red dots indicate R/H^2 average per oscillation and blue dashed line is given by the asymptotic value of Eq. (5.25). *Right-Panels:* Time evolution of the matter power spectrum of the NMC field $\Delta_\chi(k, N)/m_p^2$ for each of the cases with $\xi = 50$.

while R remains positive, once the averaged Ricci scalar becomes negative (see bottom panel of Fig. 5.3), the tachyonic instability leads to exponential growth of the χ modes. By comparing the left and right panels of Fig. 5.6, we further observe how the efficiency of mode amplification depends sensitively on the choice of M . Decreasing M suppresses the tachyonic instability. For $p = 2$ and $p = 4$, this suppression becomes so severe that the modes are not excited at all. In the case of $p = 6$, the onset of the tachyonic instability is delayed for smaller values of M , implying that, for a fixed time after the end of inflation, the χ modes grow less when M is smaller. We have analyzed the presence of a tachyonic instability in the dynamics of the NMC field, induced by the oscillations of R . A relevant aspect to address is the duration of this instability. Clearly, the tachyonic growth cannot be sustained indefinitely, as it is eventually suppressed by one of two mechanisms:

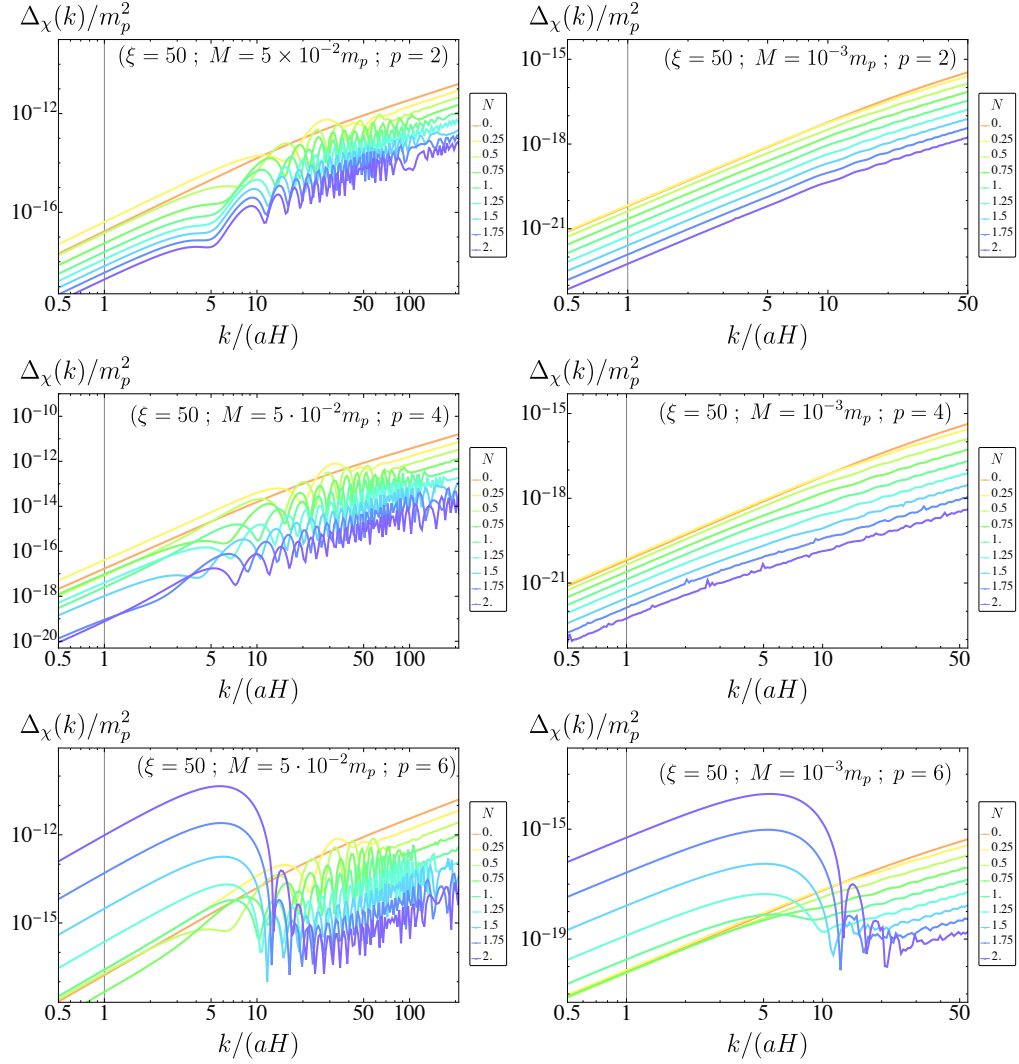


FIGURE 5.6: Time evolution of the power spectrum of the NMC field $\Delta_\chi(k, N)/m_p^2$.

- the instability gradually shuts off during the linear regime due to the expansion of the universe.
- the energy transferred into the NMC field becomes sufficiently large to drive the system into a non-linear regime, thereby also terminating the instability.

While the case in which the tachyonic instability is switched off during the linear regime is less interest than the transition to non-linearities, it is nevertheless easier to understand, and we discuss it first. As long as the Ricci scalar is sourced solely by the homogeneous inflaton, the dynamics of χ remain in the linear regime. In particular, inspecting Eq. (5.39) and assuming $\xi \gg 1/6$, we find that the instability for a given mode k will cease whenever $a^2|R| \lesssim k^2/\xi$. Considering the oscillation-averaged Ricci curvature (5.25), we estimate $a^2|\bar{R}| \sim \mathcal{H}^2 \sim 1/\tau^2$, where $\mathcal{H} \equiv a'/a \sim 1/\tau$ is the conformal Hubble rate. This implies that, within a conformal time scale of order $\tau_k \sim \sqrt{\xi}/k$, a mode k typically exits the tachyonic regime. Consequently, within a

time $\tau_{\max} \sim \sqrt{\xi}/\mathcal{H}$, all sub-horizon modes with $k \gtrsim \mathcal{H}$ cease to be tachyonic, effectively terminating the instability. Depending on the value of p and the strength of ξ , this shutdown of the instability may occur before the energy transferred into χ becomes significant relative to the energy in ϕ . This behavior is illustrated, for instance, in the case $p = 2$ in the left-middle panels of Fig. 5.7, where even for large couplings such as $\xi = 500$, the energy transfer into the NMC field stops around $N_{\max} \lesssim 1$ e-foldings after the end of inflation, both for large ($M = 5m_p$, left panel) and small ($M = 0.1m_p$, right panel) scale scenarios. Similarly, in the $p = 4$ case, we observe in the top panels of Fig. 5.10 that even with $\xi = 300$, the energy density of the NMC field stops growing around $N_{\max} \lesssim 1$ e-foldings after inflation. The precise timing of the instability switch-off, as well as the amount of energy transferred at that moment, depends on the value of M . In particular, we find that the instability ends well within the linear regime for small ($M = 0.1m_p$, right panel) and intermediate ($M = m_p$, middle panel) scales, while it is marginally quenched at the onset of the non-linear regime for large scale scenarios ($M = 5m_p$, left panel).

The case in which the tachyonic instability is switched off due to the onset of non-linear dynamics is more complex to describe, and will therefore be discussed in detail in Sec. 5.4. We anticipate here, however, that in certain scenarios the transfer of energy into the NMC field becomes so efficient that χ starts to backreact on the background dynamics, driving the coupled inflaton-NMC system into a fully non-linear regime. This is, in fact, the most interesting situation, as it allows us to investigate whether proper *reheating* of the Universe can be achieved, i.e. whether the energy transfer results in the energetic dominance of the NMC scalar χ over the inflaton ϕ . In the next section, we will explore the two-field system using lattice simulations, which will enable us to probe the full non-linear regime and determine under which conditions reheating can occur. Before that, however, we can still gain preliminary insight into the upcoming non-linear dynamics by analyzing the total energy-momentum tensor of the system, which can be split into the individual contributions of the inflaton and the NMC field, $T_{\mu\nu} \equiv T_{\mu\nu}^\chi + T_{\mu\nu}^\phi$. The contribution from the NMC field reads [46]

$$T_{\mu\nu}^\chi = \partial_\mu \chi \partial_\nu \chi - g^{\mu\nu} \left(\frac{1}{2} g^{\rho\sigma} \partial_\rho \chi \partial_\sigma \chi + V_{\text{NMC}} \right) + \xi \left(G_{\mu\nu} + g_{\mu\nu} g^{\alpha\beta} \nabla_\alpha \nabla_\beta - \nabla_\mu \nabla_\nu \right) \chi^2, \quad (5.45)$$

which, upon taking the trace with respect to the background metric, gives

$$T_\chi \equiv g^{\mu\nu} T_{\mu\nu}^\chi = (6\xi - 1)(\partial^\mu \chi \partial_\mu \chi + \xi R \chi^2) + 6\xi \chi \partial_\chi V_{\text{NMC}} - 4V_{\text{NMC}}(\chi). \quad (5.46)$$

Following [46], the Ricci scalar can be expressed from the trace of Einstein's equations as $m_p^2 R = -(T_\chi + T_\phi)$. Using $T_\phi = \partial^\mu \phi \partial_\mu \phi - 4V_{\text{inf}}(\phi)$ and Eq. (5.46), we find

$$R = \frac{(1 - 6\xi)\langle \partial^\mu \chi \partial_\mu \chi \rangle + 4\langle V \rangle - 6\xi\langle \chi V_{,\chi} \rangle + \langle \partial^\mu \phi \partial_\mu \phi \rangle}{m_p^2 + (6\xi - 1)\xi\langle \chi^2 \rangle}, \quad (5.47)$$

where $\langle \dots \rangle$ denotes volume averaging, and $V = V_{\text{inf}}(\phi) + V_{\text{NMC}}(\chi)$. Equation (5.47) shows that as long as all the NMC contributions ($\{(1 - 6\xi)\langle \partial^\mu \chi \partial_\mu \chi \rangle, 6\xi\langle \chi V_{,\chi} \rangle, (6\xi - 1)\xi\langle \chi^2 \rangle\}$) remain small, the Ricci scalar will be well approximated by the inflaton-dominated expression in Eq. (5.23). However, for sufficiently large values of ξ , it is possible for one or more of these terms to become significant, indicating the onset of the non-linear regime. We will characterize this transition in the following section.

5.4 Geometric reheating: non-linear regime

In order to study the non-linear evolution of the system, including the backreaction of the NMC field on the Ricci scalar, and hence on the expansion of the background and the inflaton dynamics, it is necessary to make use of lattice simulations. We adopt the lattice formulation for NMC dynamics introduced in [46], and study the efficiency of preheating, as well as the conditions under which proper reheating is achieved, for $p = 2$, $p = 4$, and $p = 6$, as a function of the non-minimal coupling ξ and the mass scale M . To initialize the χ field on the lattice at a time N_{sim} after the end of inflation, we set a vanishing zero-mode and introduce initial fluctuations with a power spectrum $\Delta_\chi(k, N_{\text{sim}})$ obtained from the linear analysis. The e-fold N_{sim} is chosen such that a peak has clearly emerged in the χ spectrum due to tachyonic excitation during the linear regime, while the NMC field remains energetically subdominant compared to the inflaton. Simultaneously, we initialize the inflaton by providing both its homogeneous component (amplitude and velocity), read from the solution of its Klein-Gordon equation during the linear regime, and a set of fluctuations on top. These fluctuations are added using a spectrum that mimics quantum vacuum fluctuations, see Sec. 2.2.4 for details. We then evolve the system using the lattice implementation of the coupled differential equations presented and make use of Runge-Kutta evolver, see Sec. 2.2.2,

$$\chi'' + (3 - \alpha) \left(\frac{a'}{a} \right) \chi' - a^{-2(1-\alpha)} \nabla^2 \chi = -a^{2\alpha} (\xi R \chi + V_{,\chi}), \quad (5.48)$$

$$\phi'' + (3 - \alpha) \left(\frac{a'}{a} \right) \phi' - a^{-2(1-\alpha)} \nabla^2 \phi = -a^{2\alpha} V_{,\phi}, \quad (5.49)$$

$$\frac{a''}{a} + (1 - \alpha) \left(\frac{a'}{a} \right)^2 = \frac{a^{2\alpha}}{6} R, \quad (5.50)$$

with R computed from volume averages $\langle \dots \rangle$ as

$$R = \frac{\left[(6\xi - 1) \left(\frac{\langle \chi'^2 \rangle}{a^{2\alpha}} - \frac{\langle (\nabla \chi)^2 \rangle}{a^2} \right) - 6\xi \langle \chi V_{,\chi} \rangle + 4 \langle V \rangle - \frac{\langle \phi'^2 \rangle}{a^{2\alpha}} + \frac{\langle (\nabla \phi)^2 \rangle}{a^2} \right]}{m_p^2 + (6\xi - 1)\xi \langle \chi^2 \rangle}, \quad (5.51)$$

and where the total potential is given by $V = V_{\text{inf}}(\phi) + V_{\text{NMC}}(\chi)$. In the following we set $V_{\text{NMC}} = 0$, unless stated otherwise (only in Sec. 5.4.3.1 we will restore $V_{\text{NMC}} \neq 0$).

We note that the Hubble rate can be used as a constraint equation, as it is sourced from the energy density of both the inflaton and the NMC field,

$$\mathcal{H}^2 \equiv \left(\frac{a'}{a} \right)^2 = \frac{a^{2\alpha}}{3m_p^2} [E_\phi + E_\chi], \quad (5.52)$$

$$E_\phi = \frac{1}{2a^{2\alpha}} \langle \phi'^2 \rangle + \frac{1}{2a^2} \langle (\nabla \phi)^2 \rangle + \langle V_{\text{inf}}(\phi) \rangle, \quad (5.53)$$

$$E_\chi = \frac{1}{2a^{2\alpha}} \langle \chi'^2 \rangle + \frac{1}{2a^2} \langle (\nabla \chi)^2 \rangle + \langle V_{\text{NMC}}(\chi) \rangle + \frac{3\xi}{a^{2\alpha}} \mathcal{H}^2 \langle \chi^2 \rangle + \frac{6\xi}{a^{2\alpha}} \mathcal{H} \langle \chi \chi' \rangle \quad (5.54)$$

so that during evolution, we make sure that the left hand side of Eq. (5.52), $3m_p^2(a'/a)^2$, is well balanced by the right hand side, $a^{2\alpha}(E_\phi + E_\chi)$, with the solution for $\{a(\eta), E_\phi, E_\chi\}$ obtained from solving Eqs. (5.48)-(5.50).

Before discussing our results, we first describe how the initial power spectrum obtained from the linear analysis in a 1-D momentum grid is translated into initial conditions for the lattice simulations. In order to set the initial conditions of the non-minimally coupled (NMC) field, we solve Eq. (5.39) in cosmic time. We consider a set of modes that begin well inside the Hubble radius during inflation, such that the curvature contribution is negligible and the modes can be safely initialized in the *Bunch-Davies* vacuum.

$$\tilde{\chi}_k(k\tau \rightarrow \infty) = \frac{1}{\sqrt{2k}} e^{-ik\tau}, \quad (5.55)$$

where τ is the conformal time. `CosmoLattice` requires as initial condition the power spectrum of the field and its time derivative, both as functions of the comoving momenta. The values of the comoving momenta depend on the normalization of the scale factor. This is relevant because in the linear analysis we set $a = 1$ at $t = 0$, whereas `CosmoLattice` sets $a = 1$ at the beginning of the simulation. Therefore, it is necessary to match both normalizations. This can be achieved by equating scale-factor invariant quantities, such as the physical momenta. $p = k/a$ and the power spectrum Δ_χ

$$\langle \tilde{\chi}^2 \rangle = \langle 0 | \hat{\tilde{\chi}}(\tau, x) \hat{\tilde{\chi}}(\tau, x) | 0 \rangle = \int \frac{dk}{k} \Delta_{\tilde{\chi}}(k, \tau). \quad (5.56)$$

We can relate the power spectrum of the original field by

$$\Delta_\chi(k, \tau) = \frac{1}{a^2} \Delta_{\tilde{\chi}} = \frac{k^3}{2\pi^2 a^2} |\tilde{\chi}_k(\tau)|^2 \equiv \frac{k^3}{2\pi^2} |\chi_k(\tau)|^2, \quad (5.57)$$

by equating the physical momenta and the power spectrum we can convert the solution of the linear analysis to the initial conditions for the lattice simulation

$$\begin{aligned} p_{\text{CL}} &= p_{\text{LA}} \implies \frac{k_{\text{CL}}}{a_{\text{CL}}} = \frac{k_{\text{LA}}}{a_{\text{LA}}} \rightarrow k_{\text{CL}} = \frac{k_{\text{LA}}}{a_{\text{LA}}}, \\ \Delta_\chi(k, \tau) &= \frac{1}{a^2} \Delta_{\tilde{\chi}}(k, \tau) \rightarrow \frac{k_{\text{CL}}^3}{2\pi^2} |\chi_{k,\text{CL}}|^2 = \frac{k_{\text{LA}}^3}{2\pi^2 a_{\text{LA}}^2} |\tilde{\chi}_{k,\text{LA}}(\tau)|^2, \\ &\implies |\chi_{k,\text{CL}}|^2 = a_{\text{LA}} |\tilde{\chi}_{k,\text{LA}}(\tau)|^2. \end{aligned} \quad (5.58)$$

Subscript LA stands for linear analysis and CL for *CosmoLattice*. Meanwhile for the time derivative we have

$$|\chi'_{k,\text{CL}}|^2 = a_{\text{LA}} |\tilde{\chi}'_{k,\text{LA}}(\tau) - \mathcal{H} \tilde{\chi}_{k,\text{LA}}(\tau)|^2, \quad (5.60)$$

where $\mathcal{H} = a'/a$. Finally, *CosmoLattice* requires the initial conditions to be provided in units of GeV for the momenta, $(\text{GeV})^{-1}$ for the squared amplitude of the mode functions, and $(\text{GeV})^{-1}$ for time. Initially, when solving Eq. (5.39), we chose to express time and momenta in units of H_i (the Hubble rate at the beginning of the linear analysis). Therefore, the input values for *CosmoLattice* are given as

$$k_{\text{CL}} = \frac{k_{\text{LA}}}{a_{\text{LA}}} H_i, \quad (5.61)$$

$$|\chi_{k,\text{CL}}|^2 = \frac{a_{\text{LA}}}{H_i} |\tilde{\chi}_{k,\text{LA}}(\tau)|^2, \quad (5.62)$$

$$|\chi'_{k,\text{CL}}|^2 = a_{\text{LA}} H_i |\tilde{\chi}'_{k,\text{LA}}(\tau) - \mathcal{H} \tilde{\chi}_{k,\text{LA}}(\tau)|^2. \quad (5.63)$$

Simulations were run on lattices with $\mathcal{N} = [240, 1000]$ sites per dimension, depending on the case, always ensuring that the ultraviolet cutoff satisfied $k_{\text{UV}} > k_*$, where $k_*/aH \simeq \sqrt{\xi|R|/H^2}$ denotes the threshold scale separating tachyonically excited modes ($k \lesssim k_*$) from non-excited ones ($k \gtrsim k_*$). In the following, we present our results from lattice simulations for $p = 2$ with $M/m_p = \{5, 0.1\}$ and $\xi = 500$; for $p = 4$ with $M/m_p = \{5, 1, 0.1\}$ and $\xi = \{25, \dots, 500\}$; and for $p = 6$ with $M/m_p = \{5, 1, 10^{-1}, 10^{-2}, 10^{-3}\}$ and $\xi = \{25, \dots, 500\}$. We choose to plot the evolution of physical quantities in terms of the number of e-folds $N > 0$ after the end of inflation, defined as $N(\eta) \equiv \log(a(\eta)/a_{\text{end}})$.

5.4.1 Lattice results for $p = 2$

Here we review the relevant results on geometric (p)reheating for $p = 2$, i.e. when the inflaton potential is quadratic around $\phi = 0$. From the linear analysis in the previous section, we have already shown that inflaton oscillations in this case lead to a Ricci scalar R that remains mostly positive. Since the tachyonic instability requires $R < 0$ to be effective, we expect the energy transfer to the χ field to be highly inefficient in this scenario. This is indeed observed, for instance, in the top-right panel of Fig. 5.5, or in the top panels of Fig. 5.6, where the power spectra of χ display only mild growth compared to the cases with higher p . Nevertheless, we have performed lattice simulations for large values of the coupling ξ , in order to test whether the strength of the coupling can compensate for the short duration of the $R < 0$ stages.

In Fig. 5.7, we show the time evolution of the NMC field variance $\langle \chi^2 \rangle$ and the Ricci scalar R/H^2 for low-scale ($M = 0.1m_p$) and large-scale ($M = 5m_p$) scenarios with $\xi = 500$. We observe that $\xi \langle \chi^2 \rangle / m_p^2$ grows at most to a value $\mathcal{O}(10^{-1})$ for $M = 5m_p$, and only to $\mathcal{O}(10^{-3})$ for $M = 0.1m_p$, before beginning to decay. A corresponding inspection of the energy densities reveals that the NMC energy reaches at most a value $\mathcal{O}(10^{-1})$ times smaller than the inflaton energy in the large-scale case, and no more than $\mathcal{O}(10^{-3})$ in the low-scale case. Once the NMC energy peaks, indicating the end of the tachyonic instability in the linear regime, it begins to decay faster than the inflaton energy. This can be understood from the fact that, for $p = 2$, the inflaton behaves as non-relativistic matter, whereas the NMC field behaves as radiation. We also plot the equation of state (*eos*) w , and observe that in the case $M = 5m_p$, the effective value obtained from the ratio of oscillation-averaged quantities (see Eq. 5.19) approaches $w_{\text{osc}} = 1/3$, reflecting the fact that the NMC field has grown enough to backreact on the Ricci scalar, dragging it toward zero. In contrast, for $M = 0.1m_p$, the oscillations of w are only mildly affected. Eventually, w_{osc} resumes oscillations with increasing amplitude. In the case $M = 5m_p$, it does not recover the typical matter-dominated oscillations within the simulated time, whereas for $M = 0.1m_p$, it does. We therefore conclude that the $p = 2$ case leads to an inefficient energy transfer, even for large-scale inflation and strong couplings up to $\xi \lesssim 500$.

It is perhaps interesting to note that, had we chosen larger couplings, it is likely that, at least in large-scale inflationary models (i.e. $1 \lesssim M/m_p \lesssim 8$), the energy of the χ field might have reached the energy density of the inflaton, seemingly achieving reheating of the Universe. However, this would correspond only to a temporary reheating stage: since the energy densities of the NMC and inflaton fields redshift respectively as radiation and non-relativistic matter, the inflaton would eventually dominate again, making χ increasingly subdominant. In the case of small-scale inflationary models, the initial

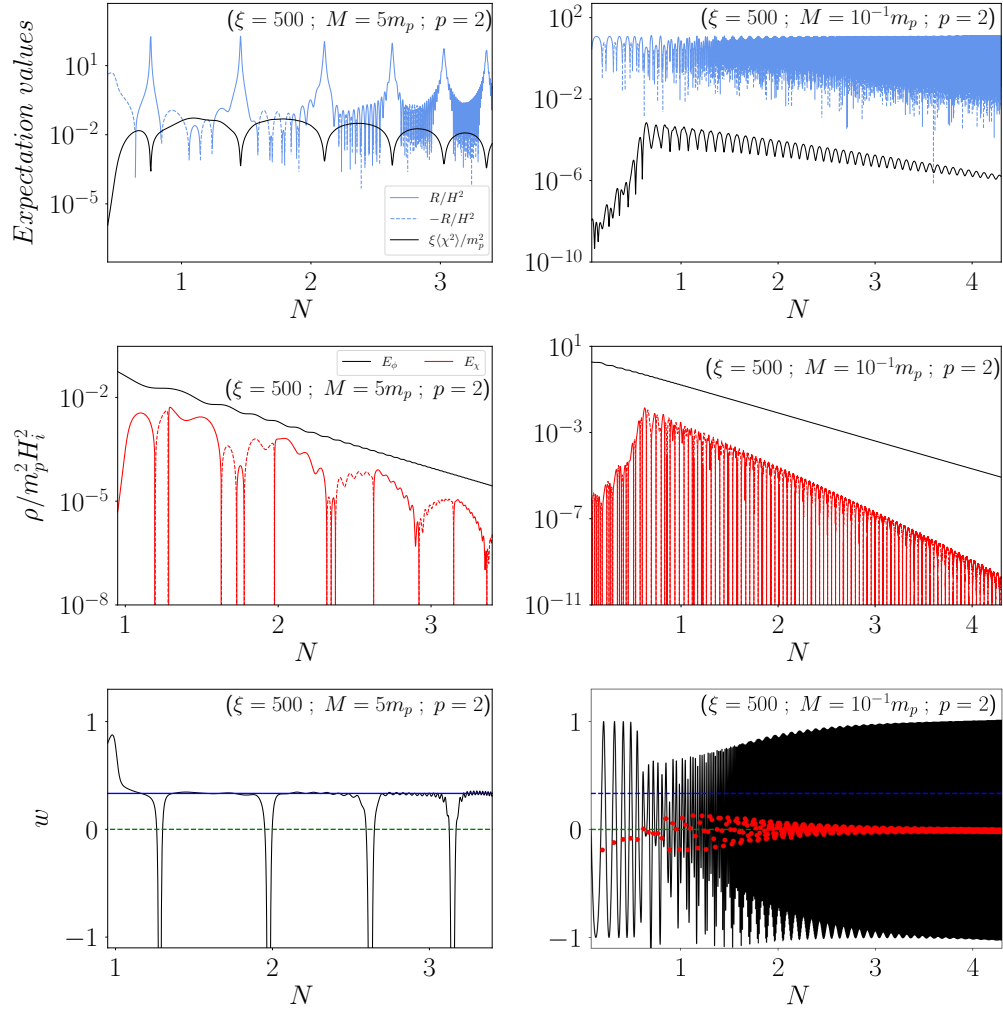


FIGURE 5.7: Time evolution of various quantities for $p = 2$, $\xi = 500$, and $M = 5m_p$ (left panels) and $M = 10^{-1}m_p$ (right panels). *Top Panel:* Evolution of the NMC field variance $\xi\langle\chi^2\rangle/m_p^2$ and R/H^2 . *Middle Panel:* Evolution of the total energy of the inflaton and of the NMC field. *Bottom Panel:* Evolution of the equation of state (w) computed from Eq. (5.24), with the green dashed line at $w = 0$ indicating matter domination, and red dots depicting w_{osc} per oscillation, c.f. Eq. (5.25). We note that in logarithmic plots solid lines stand for positive values and dashed lines for negative values.

energy transfer during the linear regime becomes increasingly inefficient as the scale M is reduced, preventing even the temporary realization of such a reheating stage.

5.4.2 Lattice results for $p = 4$

Here we review the relevant results on geometric (p)reheating for $p = 4$, i.e. when the inflaton potential is quartic around $\phi = 0$. In the top panels of Fig. 5.8, we show the evolution of the variance of the NMC field and the normalized Ricci scalar R/H^2 . In the top-left panel, we observe how the variance of the NMC field grows whenever $R < 0$, while it oscillates when $R > 0$. For a large-scale model ($M = 5m_p$), we see that for

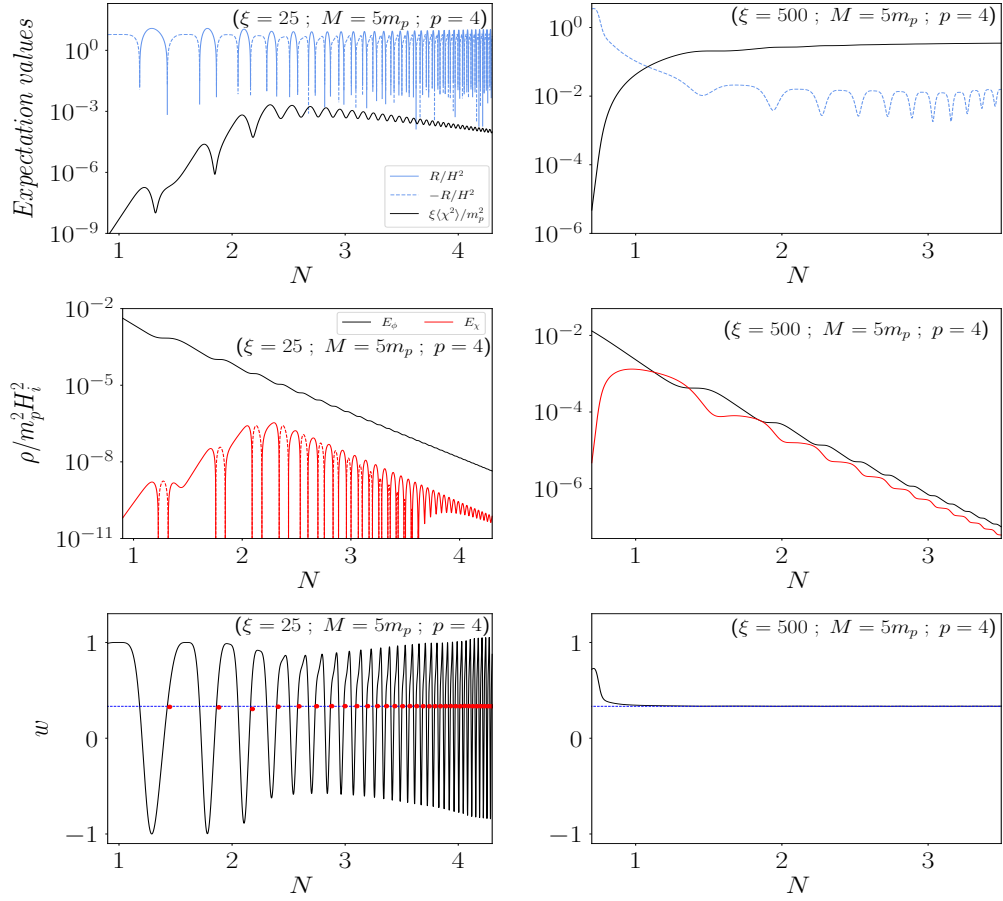


FIGURE 5.8: Time evolution of various quantities for $p = 4$, $M = 5m_p$, $\xi = 25$ and (left panels) and $\xi = 500$ (right panels). *Top Panel:* Evolution of the NMC field variance $\xi\langle\chi^2\rangle/m_p^2$ and R/H^2 . *Middle Panel:* Evolution of the total energy of the inflaton and of the NMC field. *Bottom Panel:* Evolution of the equation of state (w) computed from Eq. 5.24, blue line is at $w = 1/3$ (Radiation Domination), and red dots depicting w_{osc} per oscillation, c.f. Eq. (5.25). We note that in logarithmic plots solid lines stand for positive values and dashed lines for negative values.

$\xi = 25$, the growth of $\langle\chi^2\rangle$ stops at $N_{\text{max}} \simeq 2.2$ e-folds after inflation, during the linear regime. This occurs because, during a negative semi-oscillation of R , the expansion of the universe dilutes the field faster than the tachyonic resonance can amplify it, a behavior expected for small couplings. However, for larger couplings such as $\xi = 500$, the variance rapidly grows during the first negative semi-oscillation of R and saturates at $\xi\langle\chi^2\rangle/m_p^2 \simeq 1$. At this point, the backreaction of the NMC field onto the background becomes significant, driving R/H^2 close to zero. This can be understood from the expression of R in terms of field quantities, c.f. Eq. (5.51). The growth of $\langle\chi^2\rangle$ in the denominator leads to the suppression of R , while the terms in the numerator determine its sign. In the top-left panel of Fig. 5.8, we see R/H^2 decreasing as $\langle\chi^2\rangle$ increases, eventually stabilizing at a negative value due to the NMC field's gradient energy dominating over its kinetic energy.

In the middle panels, we show the evolution of the field energies. For $\xi = 25$, the energy

of χ grows alongside its variance. The NMC energy becomes temporarily negative when $R > 0$, corresponding to a rapid flip in the sign of the NMC velocity. During this transient stage, the energy is dominated by the $\langle\chi\chi'\rangle$ term, but quickly resumes its regular oscillatory behavior. In contrast, for $\xi = 500$ (see right panels), the energy of χ grows to a maximum within the first negative semi-oscillation of R , at which point nearly half of the total energy is stored in the NMC field. As R decays due to backreaction, the NMC field effectively becomes a massless free field, hence behaving as radiation. This explains why both the inflaton and NMC field energy densities decay at the same rate.⁹ In the bottom panels, we show the equation of state (*eos*) as computed from Eq. (5.24). For $\xi = 25$, the *eos* initially oscillates between $w = 1$ and $w = -1$, with an average of $1/3$. Eventually, backreaction squeezes the oscillation envelope while maintaining this average. However, it is not strong enough to fully suppress R , and once tachyonic resonance becomes inefficient, the *eos* returns to its oscillatory behavior. For $\xi = 500$, backreaction occurs so rapidly, within the first semi-oscillation of R , that w transitions directly to the fixed value $w = 1/3$, without noticeable oscillations.

In Fig. 5.9, we show the power spectra for a large scale scenario with $M = 5m_p$, $p = 4$, and two different couplings, $\xi = 25$ and $\xi = 500$. For $\xi = 25$, we observe that the initially growing spectrum of χ peaks at a fixed scale $k_*/(aH) \lesssim 8$. Whenever R/H^2 reaches its most negative values, modes below the cutoff scale $k \lesssim k_*$ experience exponential growth, while for $R/H^2 > 0$, these modes simply oscillate. The power spectrum grows until the expansion of the universe shuts down the tachyonic resonance during the linear regime. From that point onward, the range of excited modes simply redshifts and the amplitude of the spectrum decays. For the larger coupling case $\xi = 500$, the situation is qualitatively different. The power spectrum of χ grows while peaking at a larger scale $k_*/(aH) \lesssim 30$. However, in contrast to the weak coupling case, the expansion of the universe does not shut down the resonance. Instead, the resonance is interrupted by the backreaction of the NMC field on the Ricci scalar. This backreaction occurs rapidly, within the first semi-oscillation of R , damping the Ricci scalar to small values. As a consequence, the cutoff scale k_* decreases, forcing a shift in the power spectrum of χ toward larger physical wavelengths. This effect is visible in the bottom panel of Fig. 5.9, where we observe the drift of the peak towards the Hubble scale. This behavior is expected, as the threshold scale for tachyonic excitation satisfies $k_* \propto \sqrt{|R|}$. Once the smallest excited mode reaches the infrared boundary of the lattice, the simulation can no longer be trusted, so we typically stop the evolution slightly before this happens, when the infrared tail of the spectrum still extends sufficiently far from the peak. We focus on the dependence of the energy transfer efficiency on the mass scale M . To quantify

⁹Recall that for $p = 4$, the inflaton energy density also redshifts as radiation, once it oscillates within the positive-curvature region of its potential.

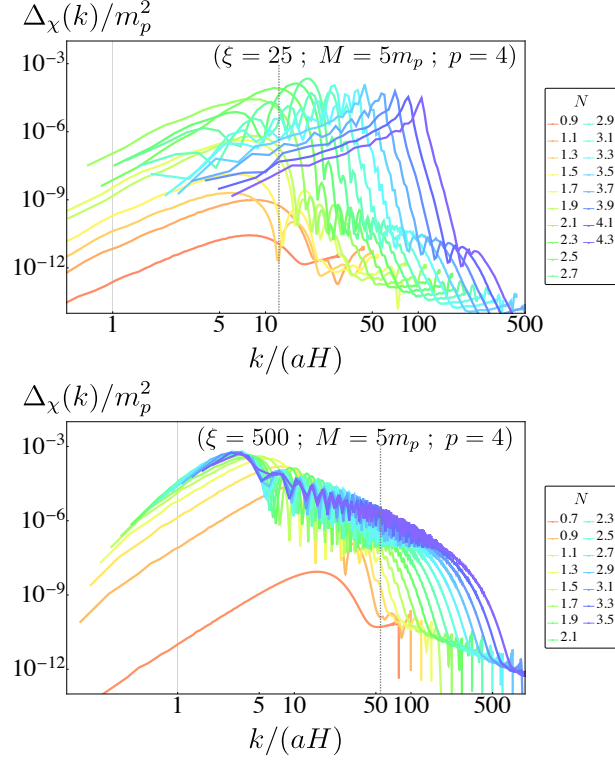


FIGURE 5.9: Power spectrum of NMC field versus k/aH for $M = 5m_p$ and $p = 4$. *Left:* $\xi = 25$ and *Right:* $\xi = 500$. Dashed line shows threshold scale $k_*/aH = \sqrt{6\xi}$

the how effective is reheating, we consider two key variables: the fraction of energy transferred into the NMC field, and the time scale required to reach an equation of state $w \simeq 1/3$, corresponding to radiation domination. We define reheating as successfully achieved whenever the energy of the NMC field accounts for at least 90% of the total energy of the two-field system. In Fig. 5.10, we show a series of panels displaying the evolution of the energy components and the equation of state (*eos*) for a fixed coupling $\xi = 300$, and decreasing values of M . For the large scale scenario $M = 5m_p$ (left panels), nearly half of the inflaton's energy is transferred to the NMC field by $N_{\max} \lesssim 1$, and since both fields behave effectively as radiation in this case, the *eos* rapidly settles to $w = 1/3$. In the middle panels, we analyze the intermediate scale case $M = m_p$. Here, reheating is not fully achieved, as the NMC field never accumulates more than approximately 5% of the total energy. Although the initial dynamics of χ induce oscillations in the *eos*, it ultimately converges to $w = 1/3$, dictated by the radiation-like scaling of the dominant inflaton. As the NMC field remains subdominant and continues interacting with R , it does not behave as pure radiation; hence, its energy density decays faster than $1/a^4$. Finally, in the right panels we show the small scale scenario with $M = 10^{-1}m_p$. In this case, reheating is never achieved. The tachyonic resonance is terminated within the linear regime, with the NMC field contributing less than 1% to the total energy. The effective *eos* initially starts with $w_{\text{osc}} < 1/3$, which is characteristic of low scale models, as the first inflaton oscillations are dominated by the potential and sample values above

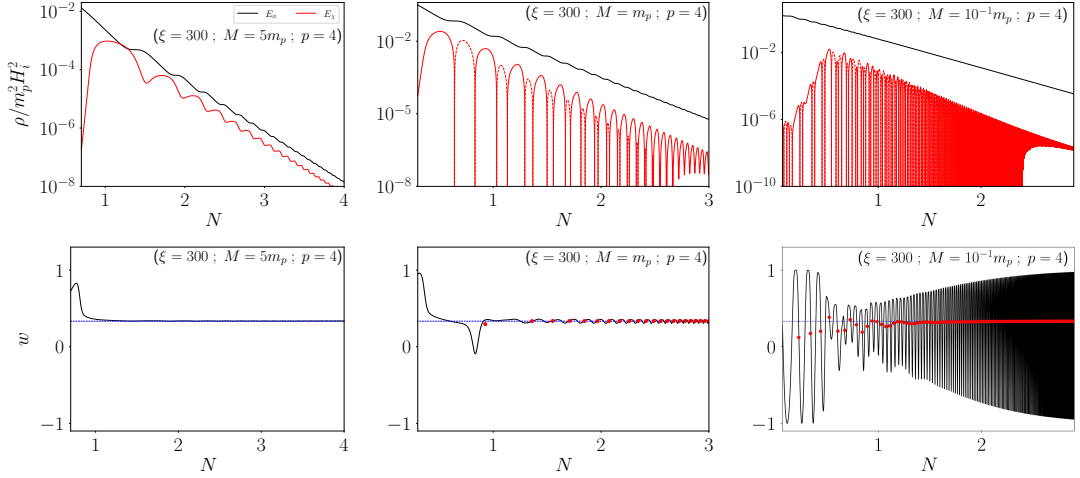


FIGURE 5.10: Energy densities and equation of state (*eos*) for $p = 4$ and fixed coupling $\xi = 300$, with $M = 5m_p$ (left panels), $M = m_p$ (central panels) and $M = 0.1m_p$ (right panels). *Top*: Evolution of the total energy density of the inflaton and of the NMC field. Solid lines denote positive values and dashed lines negative values. *Bottom*: Evolution of the *eos* w computed from Eq. (5.24), where the blue solid line at $w = 1/3$ denotes RD, and the red dots depict the effective value w_{osc} computed from Eq. (5.25).

the inflection point. Eventually, the *eos* saturates to $w_{\text{osc}} = 1/3$, as the oscillations become confined to the positive curvature region of the potential, where $V_{\text{inf}} \propto \phi^4$.

The efficiency of the energy transfer for $p = 4$ decreases as the scale M is lowered. This reduction in efficiency can be attributed to two main effects:

- Lowering the scale M forces the inflaton to end inflation above the inflection point of the potential. As a consequence, the first oscillations are dominated by potential energy, resulting in a Ricci scalar that remains mostly positive.
- Lowering the scale M causes the inflaton to oscillate faster relative to the Hubble rate. This leads to shorter durations of the negative semi-oscillations of R , making the tachyonic growth less effective when compared to the dilution due to cosmic expansion.

In Fig. 5.11, we plot the fraction of energy stored in the NMC field relative to the total energy. The dependence of this energy fraction on the parameters clearly illustrates the two effects described above. Increasing the coupling ξ enhances the efficiency of the energy transfer, and for sufficiently large couplings, energy can eventually be distributed almost equally between both fields, thus achieving partial reheating of the Universe. On the other hand, decreasing the mass scale M reduces the preheating efficiency, such that even for the largest coupling considered, $\xi = 500$, and a moderately small mass scale $M = 10^{-1}m_p$, the energy fraction of the NMC field accounts for only about 0.1% of the total energy.

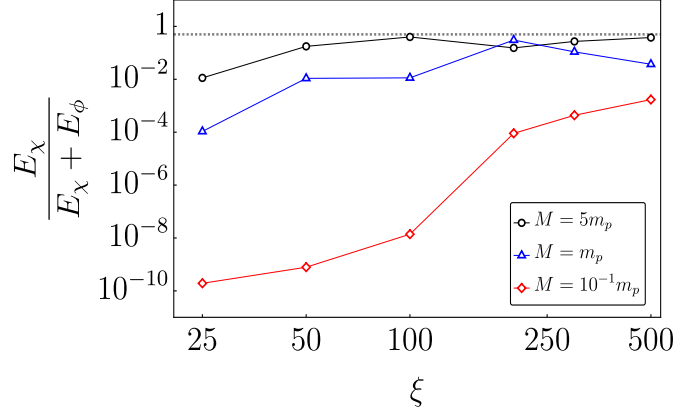


FIGURE 5.11: Final energy fraction of the NMC field compared to the total energy, for $p = 4$. The horizontal grey line denotes $E_\chi = E_\phi$.

5.4.3 Lattice results for $p = 6$

In this section, we review the most relevant results on geometric (p)reheating for the potential (5.1) with $p = 6$. In Fig. 5.12, we show the time evolution of the NMC field variance $\xi\langle\chi^2\rangle/m_p^2$ for different values of M , alongside the Hubble-normalized Ricci scalar R/H^2 . We observe that, for a given coupling, the variance starts growing at different moments depending on M , and eventually saturates to a value $\xi\langle\chi^2\rangle/m_p^2 \lesssim 1$. The backreaction onto the Ricci scalar becomes noticeable in all cases, as R/H^2 begins to decrease once the variance of χ grows sufficiently. Remarkably, this effect occurs for all values of M , i.e. regardless of how fast R oscillates relative to the expansion rate of the universe. The delay in the initial growth of $\langle\chi^2\rangle$ depends on the inflationary energy scale, becoming longer for smaller values of M , as shown in the bottom panels of Fig. 5.12 for $M/m_p = \{10^{-2}, 10^{-3}\}$. This delay arises because the initial inflaton oscillations sample the negative curvature region of the potential, leading to an initially positive oscillation-averaged Ricci curvature $\bar{R} > 0$ (see Sec. 5.2). The smaller the value of M , the longer it takes for the inflaton to transition into an oscillatory regime where only the positive curvature part of the potential is explored. Once this regime is established, the effective equation of state settles to $\bar{w} = 0.5$, c.f. Eq. (5.19) for $p = 6$, leading to an oscillation-averaged Ricci scalar $\bar{R}/\bar{H}^2 = -1.5$, c.f. Eq. (5.24).

The novelty for $p = 6$ is that, contrary to the cases with $p \leq 4$, the oscillation-averaged Ricci curvature eventually becomes negative. As a result, the tachyonic instability is guaranteed to be triggered at some point, the onset being delayed for smaller values of the energy scale M , as long as the inflaton condensate remains homogeneous¹⁰. Moreover, once the condition $\xi\langle\chi^2\rangle/m_p^2 \lesssim 1$ is reached, the energy of the χ field decays

¹⁰For the time being, we are ignoring inflaton fragmentation; see Sec. 5.2.2. Once this effect is taken into account, homogeneity of the inflaton is broken very rapidly, particularly for small mass scales M . For clarity, we postpone this discussion and later quantify how the achievement of proper reheating is affected.

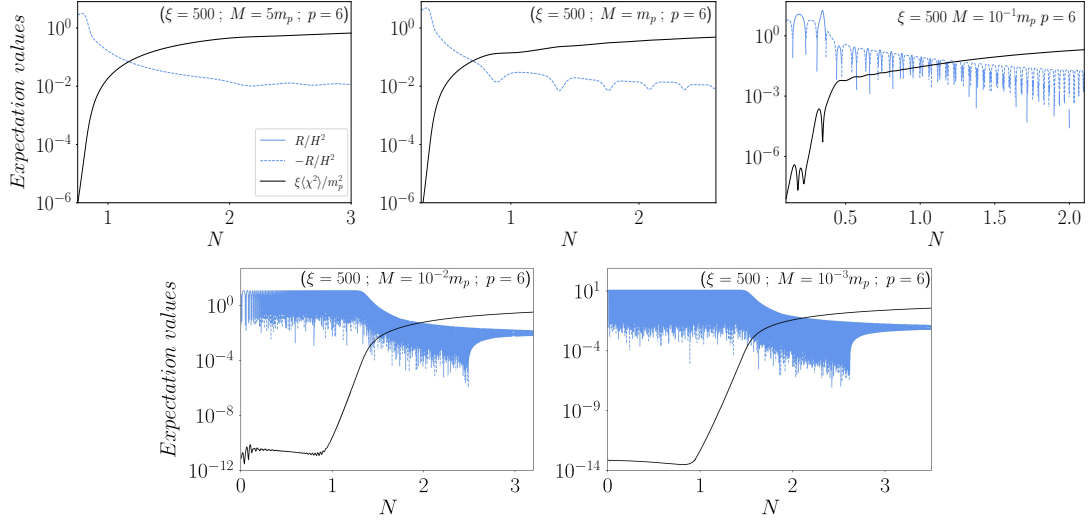


FIGURE 5.12: Time evolution of the variance of the NMC field times the coupling, $\xi\langle\chi^2\rangle/m_p^2$, and R/H^2 for $\xi = 500$ and (from left to right) $M = 5m_p$, $M = m_p$, $M = 10^{-1}m_p$, $M = 10^{-2}m_p$, $M = 10^{-3}m_p$. Solid lines denote positive values and dashed lines negative values.

as radiation, while the inflaton energy decays faster than radiation for $p = 6$. Therefore, χ will eventually dominate the energy budget, with the inflaton contribution becoming gradually negligible. We consider that proper reheating has been achieved when the energy densities become equal, $\rho_\chi = \rho_\phi$, with the understanding that the system will subsequently evolve naturally into a state with $\rho_\chi \gg \rho_\phi$. The evolution of the equation of state for $p = 6$ is shown in Fig. 5.13. After an initial stage with the *eos* satisfying $w > 0$, oscillatory, or negative, for large ($M \geq m_p$), intermediate ($M = 0.1m_p$) and small ($M = 0.01m_p, 0.001m_p$) scale models, respectively, backreaction eventually kicks in (the later the smaller the value of M), so that R/H^2 settles close to zero, driving the *eos* down to $w \simeq 1/3$ in all cases. The evolution of the energy components in this scenario is also worth highlighting. To begin with, we observe that once the tachyonic instability is triggered, the total energy of the NMC field grows at the same rate as the variance of χ , as it is the term $\propto \langle\chi^2\rangle$ in Eq. (5.54) that dominates the energy of the NMC scalar during this stage. In Fig. 5.14, we show the evolution of the total energy densities of the inflaton and of the NMC field, once the tachyonic instability has been established. We see that the energy of χ grows exponentially fast due to the instability, and eventually backreacts onto the Ricci scalar, switching off the instability and thus ceasing the exponential growth. Nonetheless, the NMC energy continues to grow relative to that of the inflaton, since the inflaton energy decays faster than radiation for $p = 6$, while the NMC field, once R has been suppressed due to backreaction, behaves effectively as a free massless field, with its energy decaying as radiation. As a result, the NMC field inevitably comes to dominate the energy budget, leading to a proper reheating of the Universe. In Fig. 5.15 we show the evolution of the power spectrum $\Delta_\chi(k, N)$ of

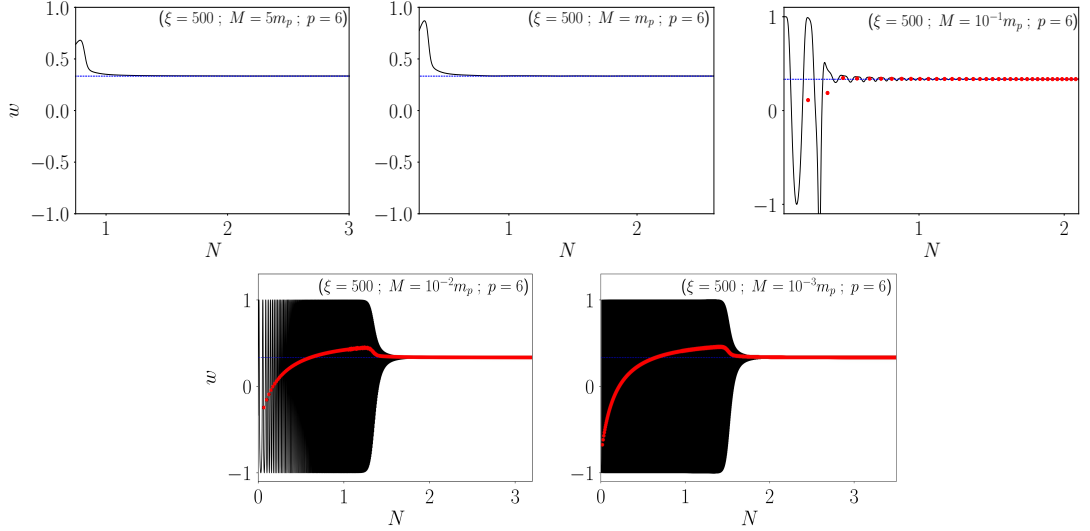


FIGURE 5.13: Time evolution of the equation of state (w) as computed from Eq. 5.24, blue line is at $w = 1/3$ (Radiation Domination), and red dots depict average value of w per oscillation for $\xi = 500$ and (from left to right) $M = 5m_p$, $M = m_p$, $M = 10^{-1}m_p$, $M = 10^{-2}m_p$, $M = 10^{-3}m_p$.

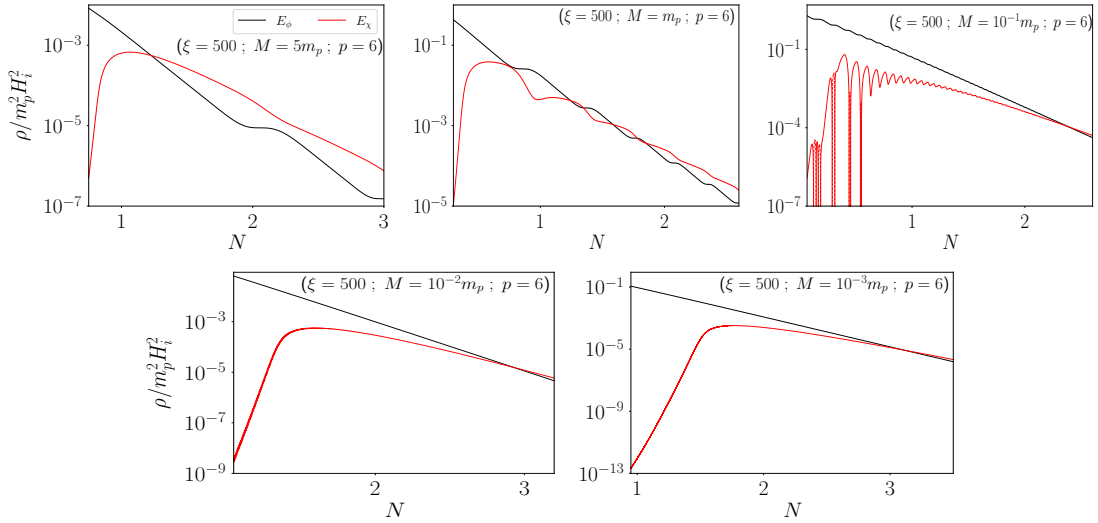


FIGURE 5.14: Time evolution of the total energy of the inflaton and the NMC field for $\xi = 500$ and (from left to right) $M = 5m_p$, $M = m_p$, $M = 10^{-1}m_p$, $M = 10^{-2}m_p$, $M = 10^{-3}m_p$.

the NMC field for different choices of M . The growth and saturation of the spectra in a large-scale scenario (e.g. top panel of Fig. 5.15 for $M = 5m_p$), occurs within the first negative semi-oscillations of R . The initial growth takes place up to a threshold momentum $k_*/(aH) \lesssim \sqrt{\xi}|\bar{R}|^{1/2}/H \simeq 50$. Once χ begins to backreact onto the Ricci scalar, R/H^2 is driven to smaller values, and consequently the threshold scale k_* that separates tachyonic ($k \lesssim k_*$) from non-tachyonic ($k \gtrsim k_*$) modes, shifts to smaller values, gradually approaching $k_*/(aH) \simeq 1$. In a small-scale scenario, as shown for instance in the middle panel of Fig. 5.15 for $M = 10^{-1}m_p$, the growth of the χ spectrum proceeds in two stages. Initially, modes up to a maximum cutoff $k_*/(aH) \sim \sqrt{6\xi}$ are amplified every time R becomes negative. Over time, the magnitude of the negative curvature

during each semi-oscillation becomes insufficient to overcome the dilution due to the expansion of the Universe. However, a net growth of the modes is still observed once the oscillation-averaged curvature $\bar{R}$ becomes negative, when the inflaton oscillations become dominated by kinetic energy. The tachyonic amplification is eventually suppressed as R/H^2 is sufficiently damped due to backreaction, shifting the peak of the χ power spectrum toward larger length scales, such that $k_*/(aH) \sim 1$. In even smaller-scale scenarios (see e.g. the bottom panel of Fig. 5.15 for $M = 10^{-3}m_p$), unstable modes are not excited even during a single negative semi-oscillation initially, as dilution from expansion dominates. However, once the regime with $\bar{R} < 0$ is entered, the averaged curvature tends to $\bar{R}/\bar{H}^2 = -3/2$, triggering the tachyonic amplification of the infrared modes of χ with $k \lesssim k_*$. The threshold scale at that moment can be estimated from Eq. (5.42) as $k_*/(aH) \lesssim \sqrt{6\xi} \sqrt{|4-p|/(p+2)} \simeq 27$. Eventually, the exponential growth of the variance ceases and saturates to a value $\xi\langle\chi^2\rangle/m_p^2 \lesssim 1$, displacing again the peak of the NMC field's power spectrum towards horizon scales, with $k_*/(aH) \simeq 1$. In order to parameterize the reheating dynamics as a function of M and ξ , we have characterized both the time scale when the NMC field χ backreacts on the Ricci scalar R , denoted N_{BR} , and the time when χ becomes the dominant component of the total energy density, signaling that proper reheating has been achieved, denoted N_{RH} . We define N_{BR} as the time when the envelope of the effective equation of state w_{osc} decays below 90% of its maximum value. This marks the transition of the Universe from a kination-dominated (KD) to a radiation-dominated (RD) phase, see Fig. 5.13. In the top panel of Fig. 5.16, we show the dependence of the backreaction time N_{BR} on the parameters M and ξ . As expected, increasing the coupling ξ reduces the time needed for backreaction to occur, as the growth rate of tachyonic modes increases with ξ . Conversely, decreasing M delays the onset of backreaction. This delay arises partly because the effective equation of state w_{eff} takes longer to reach the value $w_{\text{eff}} = 0.5$, and also because the initial value of w_{eff} is below $1/3$ for smaller M , implying that the averaged Ricci scalar $\bar{R}$ remains positive during the initial inflaton oscillations.

The time scale for the NMC field to become the dominant energy component, N_{RH} , marks the moment when the Universe is properly reheated. For practical purposes, we define N_{RH} as the time when the energy density of χ becomes equal to that of the inflaton, $E_\chi = E_\phi$. In the bottom panel of Fig. 5.16, we show the dependence of N_{RH} on M and ξ . As anticipated, N_{RH} is always larger than N_{BR} , since the latter measures the time when χ starts affecting the Ricci scalar R , while the former tracks when χ overtakes ϕ as the dominant energy component of the system. Our analysis so far has ignored the possibility that the inflaton condensate may fragment. However, Fig. 5.4 shows that for scales $M < 10^{-1}m_p$, inflaton fragmentation occurs before the backreaction of the NMC field χ becomes relevant. This invalidates the assumption of a homogeneous inflaton

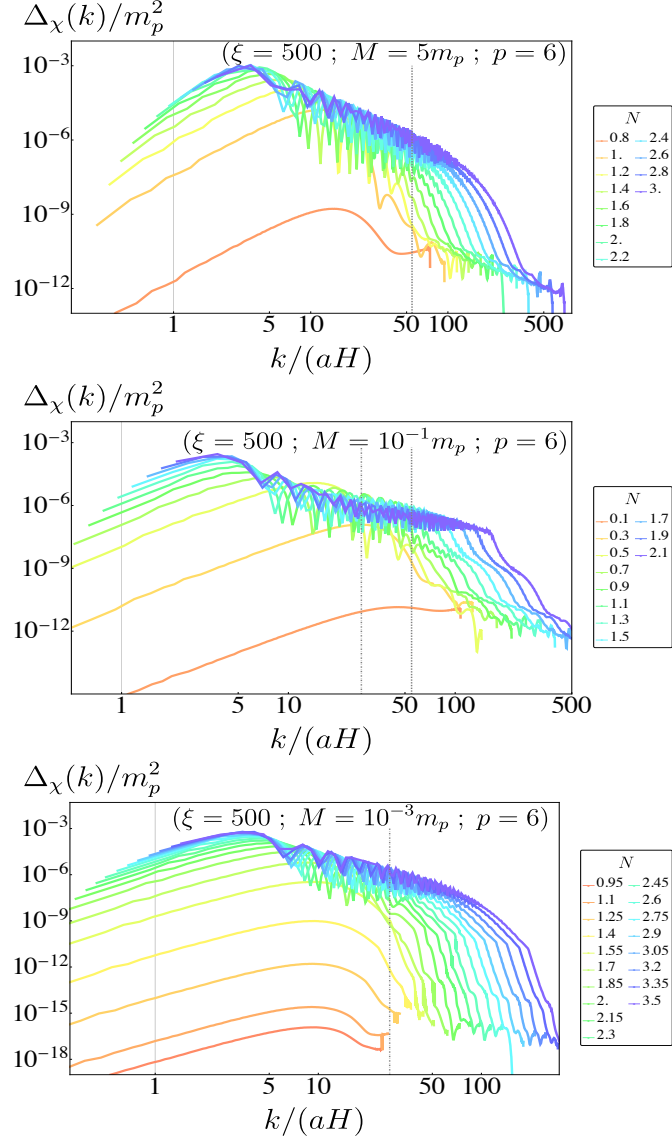


FIGURE 5.15: Power spectrum of NMC field versus k/aH for $p = 6$ and $\xi = 500$ for *Top*: $M = 5m_p$, *Mid*: $M = 10^{-1}m_p$ and *Bottom*: $M = 10^{-3}m_p$. Dashed line shows threshold scale $k_*/aH = \sqrt{6\xi}$, while dot-dashed line shows threshold scale $k_*/aH = \sqrt{3/2\xi}$

condensate underpinning our previous analysis. The region of parameter space where this occurs is marked by the blue shaded area in the top panel of Fig. 5.16. Similarly, in the bottom panel of the same figure, the blue region identifies the parameter space where our mechanism of proper reheating into χ no longer applies. Interestingly, in this case, the Universe still transitions into radiation domination (RD), as inflaton fragmentation becomes efficient. That is, for very small inflationary scales, $M \ll 0.1 m_p$, the Universe enters the RD era almost immediately after the end of inflation, typically within less than one e-fold, see bottom panel of Fig. 5.16 for $M < 0.01 m_p$. Nevertheless, proper reheating in these cases would require the inflaton to decay into other relativistic species through direct couplings.

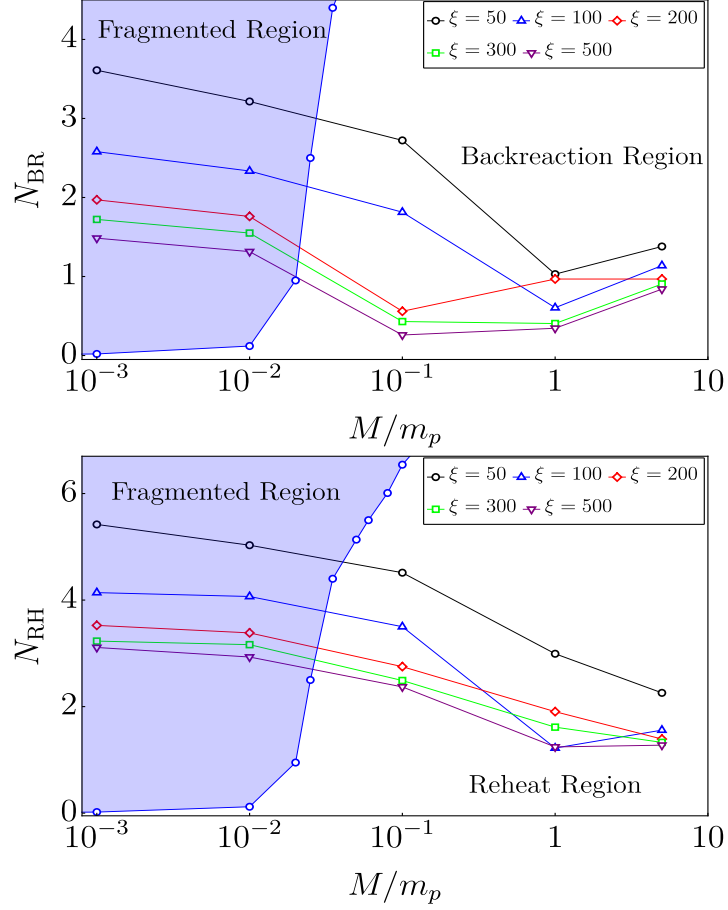


FIGURE 5.16: *Top*: Time scale to develop backreaction (top panel) and to achieve reheating (bottom panel), as a function of the scale M and the coupling ξ . Blue regions indicate the parameter space ruled out due to inflaton fragmentation.

In summary, due to the onset of inflaton fragmentation, only relatively high-scale inflationary models with $M \gtrsim 0.1 m_p$ can yield proper reheating in our scenario. This conclusion is summarized in Table 5.1, where we list the time scale and reheating temperature for the cases where proper reheating is achieved. The temperature T_{RH} is computed from the relation $\rho_{\text{NMC}}(N_{\text{RH}}) = \frac{\pi^2}{30} g_*(T) T_{\text{RH}}^4$. In order to obtain concrete numerical values, we have assumed $g_*(T) = 106.75$, corresponding to the Standard Model.

5.4.3.1 Including NMC field self-interactions

In Sec. 5.4.1 and 5.4.3, we have assumed that interactions of the NMC field are negligible, simply setting $V_{\text{NMC}}(\chi) = 0$. In this section, we relax this assumption and consider self-interactions of the NMC field of the form $V_{\text{NMC}}(\chi) = \frac{\lambda}{4} \chi^4$, where λ is a dimensionless coupling. If the NMC field possesses sufficiently strong self-interactions, the system's dynamics may also develop non-linearities, effectively regularizing the tachyonic instability. This occurs due to the field's self-backreaction, as during the

	$M = 10^{-1} m_p$		$M = m_p$		$M = 5 m_p$	
	N_{RH}	$T_{\text{RH}} [\text{GeV}]$	N_{RH}	$T_{\text{RH}} [\text{GeV}]$	N_{RH}	$T_{\text{RH}} [\text{GeV}]$
$\xi = 50$	4.51	5.46×10^{12}	2.99	6.84×10^{13}	2.26	1.76×10^{14}
$\xi = 100$	3.50	1.71×10^{13}	1.22	4.69×10^{14}	1.56	3.76×10^{14}
$\xi = 200$	2.75	3.95×10^{13}	1.91	2.25×10^{14}	1.39	4.84×10^{14}
$\xi = 300$	2.49	5.29×10^{13}	1.61	3.08×10^{14}	1.33	5.32×10^{14}
$\xi = 500$	2.37	6.04×10^{13}	1.24	4.66×10^{14}	1.28	5.75×10^{14}

TABLE 5.1: Efolding time scale and temperature of reheating for representative cases of the parameter space where inflaton fragmentation does not spoil proper reheating, see bottom panel of Fig. 5.16.

tachyonic growth, the NMC field acquires a positive (squared) interaction mass $m_{\text{int}}^2 = 3\lambda\langle\chi^2\rangle > 0$. If λ is large enough, this term can grow to the point of screening the tachyonic gravitational mass $m_{\text{grav}}^2 = \xi\bar{R} < 0$, such that the total effective mass of the NMC field, $m_{\text{tot}}^2 = m_{\text{int}}^2 + \xi\bar{R}$, flips from negative to positive, thereby ending the tachyonic instability. The condition for this ‘self-interaction backreaction’ to occur is simple: it requires that $m_{\text{tot}}^2 = 0$. This allows us to determine the field amplitude at that moment as

$$\chi_{\text{br}} \equiv \sqrt{\langle\chi^2\rangle}\Big|_{m_{\text{tot}}^2=0} \simeq \sqrt{\frac{\xi}{3\lambda}}\sqrt{\bar{R}}. \quad (5.64)$$

While self-interaction backreaction can, in principle, occur for any exponent p , from the perspective of reheating this phenomenon is only relevant for $p > 4$, i.e. for $p = 6$ in our case. The reason is straightforward: once the tachyonic mass is screened, the dynamics of the NMC field are governed by its self-interactions, which for a quartic potential implies an energy density scaling as radiation¹¹ [181]. This implies that, after the tachyonic growth is regularized, the NMC energy density decreases (typically after a short time) as $\rho_\chi \propto a^{-4}$, while the inflaton energy density continues scaling as $\rho_\phi \propto a^{-6p/(p+2)}$. Consequently, the energy ratio evolves as

$$\frac{\rho_\chi}{\rho_\phi} \propto a^{2(p-4)/(p+2)}, \quad (5.65)$$

and will increase, remain constant, or decrease with time depending on whether $p > 4$, $p = 4$, or $p < 4$, respectively. Since at the onset of self-interaction backreaction the

¹¹This statement is strictly valid once the gravitational mass $|m_{\text{grav}}^2| = \xi|\bar{R}|$ becomes negligible compared to the self-interaction mass $m_{\text{int}}^2 = 3\lambda\langle\chi^2\rangle$, i.e. when $\xi\bar{R} \ll 3\lambda\langle\chi^2\rangle$. This is expected to occur, as the expansion of the Universe eventually drives the inflaton to oscillate only within the positive-curvature part of its potential, such that $\bar{R}/\langle\chi^2\rangle$ becomes a decreasing function of time.

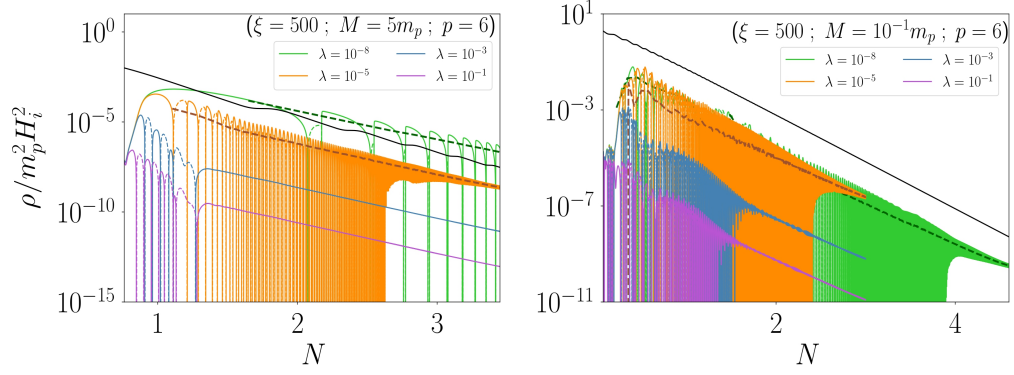


FIGURE 5.17: Energy density time evolution for different choices of λ and M with fixed $\xi = 500$ value. Dashed lines depict average per oscillation values.

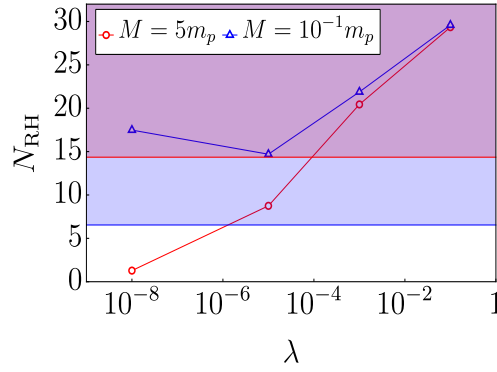


FIGURE 5.18: Time scale for reheating as a function of λ and M for fixed $\xi = 500$ value. Colored regions depict excluded region for reheating due to inflaton fragmentation, each for the corresponding M case.

NMC field energy is subdominant, $\rho_\chi/\rho_\phi \ll 1$, this ratio can only grow in time, allowing for proper reheating ($\rho_\chi \geq \rho_\phi$) only if $p > 4$. To characterize the impact of λ , together with ξ and M , on the reheating time scale for $p = 6$, we have simulated the case of non-vanishing self-interaction of the NMC field, $\lambda \neq 0$. We have measured the number of e-folds N_{RH} required for the energy of the NMC field to equal that of the inflaton. When necessary, we have extrapolated N_{RH} beyond the simulation time by extending the scaling behavior of both the inflaton and NMC energy densities, as measured from our lattice simulations. We find that for low-energy scenarios, even the smallest chosen self-interaction delays reheating so significantly that inflaton fragmentation occurs before reheating is achieved. For higher scales, proper reheating is only possible for the largest mass scales (e.g. $M = 5m_p$ in the figure), and only for very small self-coupling constants, $\lambda \lesssim 10^{-6}$. In conclusion, introducing self-interactions for the NMC field effectively spoils the possibility of proper reheating, unless λ is extremely small. Our numerical findings are summarized in Fig. 5.18 and in the set of allowed cases listed in Table 5.2.

	$M = 5 m_p$	
	N_{RH}	$T_{\text{RH}} [\text{GeV}]$
$\lambda = 10^{-8}$	1.281	6.511×10^{14}
$\lambda = 10^{-5}$	8.747	3.382×10^{11}

TABLE 5.2: Time scale and temperature of reheating for the allowed cases in the parameter space Fig. 5.18

5.5 Convergence tests

Here we present the impact of the lattice resolution on the results shown in Fig. 5.9, corresponding to the parameter set $p = 4$, $M = 5m_p$, and $\xi = \{25, 500\}$. For each case, we provide lattice simulation results using four different lattice sizes characterized by $\mathcal{N}$ points per spatial dimension. In Fig. 5.19, we present four simulations for a fixed IR scale $k_{\text{IR}}/H_i = 0.0055$ and increasing $\mathcal{N}$, which corresponds to an increasing UV coverage.

In this particular case, the peak of excitation is well resolved for the chosen lattice parameters. Moreover, the excitation of the NMC field only mildly affects the volume-averaged quantities. To demonstrate this, we monitor the time evolution of the Ricci scalar. This confirms that any choice of lattice size that resolves the peak of excitation leads to an identical evolution of the volume-averaged quantities. In the right panel of Fig. 5.19, we display the power spectra of the NMC field for different lattice sizes. As seen, for simulations with $\mathcal{N} \geq 240$, the spectra are nearly identical on the relevant scales and essentially overlap. The spectrum corresponding to $\mathcal{N} = 120$ shows distortions at the most UV scales, where lattice artifacts become apparent. Nevertheless, these effects do not significantly impact the peak of excitation, which explains why the Ricci scalar evolves identically across simulations. These tests therefore validate the robustness of our results for these cases. In Fig. 5.20, we present four lattice simulations for a fixed UV cutoff $k_{\text{UV}}/H_i = 5.19615$ and increasing $\mathcal{N}$, which corresponds to enhanced IR coverage. In this case, the peak of excitation follows the expression $k_*/(aH) = \sqrt{(\xi - 1/6)R/H^2}$, as given by Eq. (5.40). This implies that as R decreases due to the exponential excitation of the NMC field, the peak of excitation progressively shifts toward near-horizon scales. Consequently, the excited comoving momenta are redshifted to sub-horizon values, eventually causing the peak of excitation to exit the range of momenta resolved by the lattice. For instance, in the left panel of Fig. 5.20, we observe that for simulations where the peak is well resolved throughout the evolution ($\mathcal{N} = \{500, 1000\}$), the value of R/H^2 settles close to its asymptotic value near zero. In contrast, for simulations with insufficient resolution ($\mathcal{N} = \{125, 250\}$), the value of

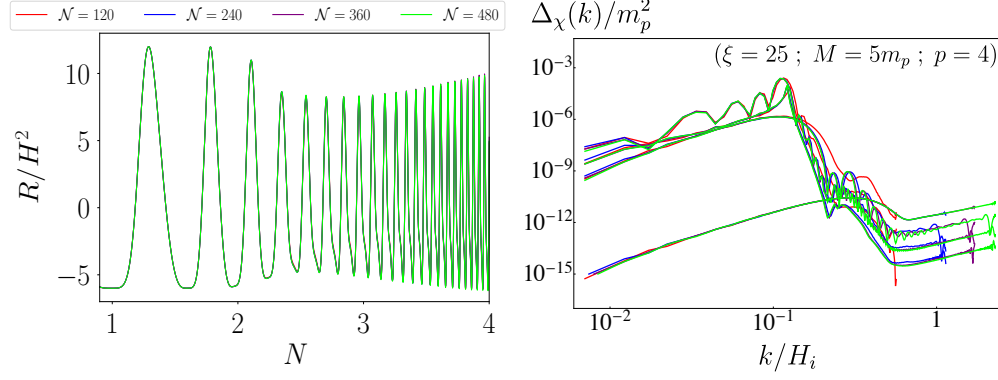


FIGURE 5.19: Convergence tests for the case $p = 4$, $M = 5m_p$ and $\xi = 25$ and $\mathcal{N} = \{120, 240, 360, 480\}$. *Left*: Time evolution of Ricci scalar normalized by the Hubble squared. *Right*: Power spectrum of the NMC field vs. comoving momenta normalized by the initial Hubble rate k/H_i measured at $N = \{0.9, 1.93, 2.96, 4\}$

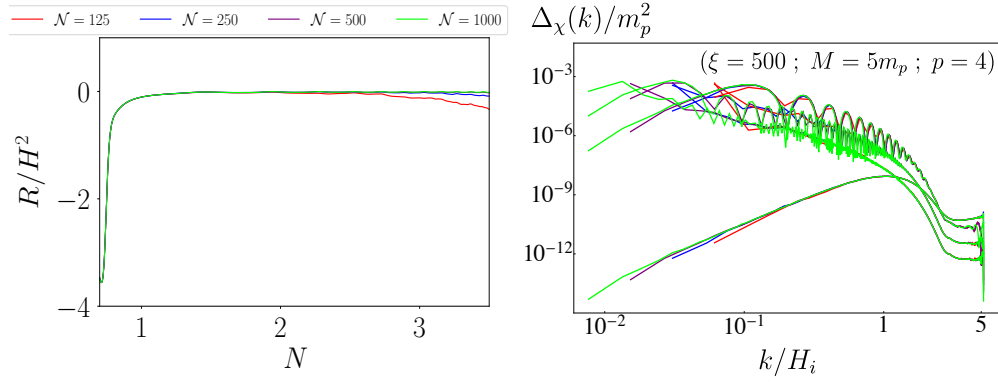


FIGURE 5.20: Convergence tests for the case $p = 4$, $M = 5m_p$ and $\xi = 500$ and $\mathcal{N} = \{125, 250, 500, 1000\}$. *Left*: Time evolution of Ricci scalar normalized by the Hubble squared. *Right*: Power spectrum of the NMC field vs. comoving momenta normalized by the initial Hubble rate k/H_i measured at $N = \{0.7, 1.63, 2.56, 3.5\}$

R/H^2 deviates significantly from its expected behavior, as the lattice fails to capture the peak accurately. For this reason, we ensure that our simulations are performed with lattice parameters that adequately resolve the excitation peak during the relevant stages of the dynamics, thus validating the reliability of our results.

5.6 Conclusions

We have analyzed the feasibility of geometric reheating into a non-minimally coupled (NMC) scalar field χ , within the framework of α -attractor T-models characterized by a tunable mass scale M . This parameter controls both the inflationary energy scale Λ and the width of the monomial shape of the inflaton potential around $\phi = 0$, $V_{\text{inf}} \propto \phi^p$.

Choosing small values of M , significantly below the critical value M_{max} that saturates current bounds on the tensor-to-scalar ratio from the CMB, impacts the post-inflationary dynamics of the inflaton in several ways. We find that

1. the inflaton amplitude at the end of inflation exceeds the inflection point of the potential once M becomes a factor $\sim 1/8$ smaller than M_{max} .
2. as long as the inflaton remains homogeneous, its oscillations after inflation become increasingly fast compared to the Hubble rate, with the frequency scaling inversely with M .

Both effects reduce the efficiency of preheating by suppressing the occurrence of negative values of the Ricci scalar R , thereby limiting the energy transfer into χ . Furthermore, for sufficiently small values of M , we also find that:

3. the inflaton condensate fragments rapidly after the onset of oscillations, breaking its spatial coherence and thereby deactivating the geometric preheating mechanism. Notably, the time scale for fragmentation decreases drastically as M is reduced.

Given that inflaton fragmentation and geometric (p)reheating are independent mechanisms, we have analyzed their dynamics and time scales separately, identifying the regions of parameter space that support viable reheating, and discarding those where inflaton fragmentation occurs before gravitational backreaction becomes relevant. We have studied the cases $p = 2$, $p = 4$, and $p = 6$, which exhibit qualitatively different behaviors. For $p = 2$, geometric preheating is inefficient regardless of the value of the non-minimal coupling ξ . This is due to the fact that inflaton oscillations yield a Ricci scalar that remains positive on average, thereby suppressing the tachyonic instability. For $p = 4$, we find that preheating becomes inefficient for $M \lesssim 10^{-1}m_p$, as both effects 1) and 2) discussed previously, the inflaton ending inflation above the inflection point and oscillating faster than the Hubble rate, conspire to suppress the excitation of χ particles. The case $p = 6$ is different: energy transfer into the NMC field can be successfully achieved even in low-scale scenarios, due to the stiff effective equation of state of the inflaton, which drives the Ricci scalar to negative average values, $\bar{R} < 0$. This is sufficient to trigger a tachyonic instability and produce χ particles efficiently. However, for scenarios with $M \lesssim 10^{-2}m_p$, inflaton fragmentation proceeds so rapidly that there is no time to transfer sufficient energy into χ , thereby frustrating proper reheating. Additionally, for $p = 6$, we have explored the impact of self-interactions of the NMC field. We find that self-couplings effectively suppress the energy transfer

into χ by introducing a positive mass term that can screen the tachyonic instability. Nonetheless, for certain cases, large-scale scenarios with very small NMC self-couplings, proper reheating remains feasible before inflaton fragmentation occurs. The parameter space that allows for successful reheating is summarized in Tables 5.1 and 5.2.

Chapter 6

The non-linear dynamics of axion inflation

In this chapter, we present a study of the fully inhomogeneous nonlinear dynamics of axion inflation. We argue the necessity to use lattice simulations in order to go beyond the analytical and semi-analytical methods used under severe assumptions. We demonstrate that those methods fail to accurately describe the truly local dynamics of the case when strong backreaction is non-negligible. We use lattice techniques that explicitly preserve gauge invariance and shift symmetry, and which we validate against other computational methods of the linear dynamics and of the homogeneous backreaction regime. We investigate the convergence of simulations of local backreaction, determining the requirements to achieve an accurate description of the dynamics, and providing useful parameterizations of the delay of the end of inflation. We identify key features emerging from a proper local treatment of strong backreaction: the dominance of magnetic energy against the electric counterpart, the excitation of the longitudinal mode, and the generation of a scale-dependent chiral (im)balance. This chapter is based on article [4].

6.1 Introduction

Several implementations of axion-driven inflation have been proposed [28, 29, 242–246]. In this work, we focus on scenarios where the operator of lowest dimensionality that couples the inflaton to an Abelian gauge sector is present

$$\mathcal{L}_{\text{int}} = -\frac{1}{4\Lambda}\phi F_{\mu\nu}\tilde{F}^{\mu\nu}, \quad (6.1)$$

with Λ an energy scale, and $F_{\mu\nu}$ ($\tilde{F}_{\mu\nu}$) the (dual) field strength tensor of a gauge field A_μ . These scenarios, referred to as *axion inflation*, are particularly interesting from an observational perspective due to the rich variety of phenomena they can generate. A distinctive feature is the exponential production of a helical gauge field during inflation [247–253], which can lead to several interesting cosmological consequences. In the present analysis, we assume that ϕ couples only to a hidden Abelian gauge sector, *i.e.* A_μ is a *dark photon*. For a given inflaton potential $V(\phi)$, the excitation of the gauge field is then fully determined by the strength of the coupling scale Λ . In practice, the dynamics of gauge field production are controlled by the dimensionless parameter $\xi \equiv |\dot{\phi}|/(2\Lambda H)$, where $\dot{\phi}$ is the inflaton velocity and H is the Hubble rate during inflation.

During inflation, the excited gauge field sources curvature perturbations through inflaton fluctuations generated by the inverse decay process $A + A \rightarrow \delta\phi$. The resulting scalar perturbations are non-Gaussian and of the equilateral type, with a non-linear parameter given by $f_{\text{NL}}^{\text{eq}} \simeq 7 \cdot 10^5 \times (H^6/|\dot{\phi}|^3)(e^{6\pi\xi}/\xi^9)$ [251]. Current CMB constraints on equilateral non-Gaussianities [254], namely $f_{\text{NL}}^{\text{eq}} = -26 \pm 47$, place a stringent bound on the value of ξ at CMB scales. For the specific case of a quadratic inflaton potential, $V(\phi) = \frac{1}{2}m^2\phi^2$, this constraint reads $\xi_{\text{CMB}} \lesssim 2.5$ at 95% C.L. [251, 255–257], which is equivalent to $\Lambda^{-1} \lesssim 35 m_p^{-1}$, where $m_p \simeq 2.435 \cdot 10^{18}$ GeV is the reduced Planck mass.

The quantity ξ , often referred to as the particle production parameter, controls the excitation of the gauge field A_μ . Typically, ξ increases as the inflaton rolls down its potential, since H decreases and $|\dot{\phi}|$ increases. Therefore, an initially small value $\xi \lesssim 2.5$ at CMB scales is expected to grow at much shorter scales. For a given inflaton potential $V(\phi)$, the evolution of ξ , especially towards the end of inflation, depends on the level of backreaction from the gauge field A_μ , both on the inflationary expansion and on the inflaton dynamics. A growing ξ can lead to large overdensities in the matter perturbations, which may collapse into primordial black holes (PBHs) upon horizon re-entry during radiation domination (RD) [258]. To accurately quantify this effect, one must track the evolution of large curvature perturbations in the strong backreaction regime [255]. Taking this into account, Ref. [258] showed that in order to avoid overproduction of dark matter in the form of PBHs, a tighter bound $\xi_{\text{CMB}} \lesssim 1.5$ must be imposed at CMB scales. For a quadratic inflaton potential, this translates into an upper limit on the axion-gauge coupling as $\Lambda^{-1} \lesssim 23 m_p^{-1}$ [258]. While the PBH bound is more stringent than the non-Gaussianity limit, it is based on approximations within the strong backreaction regime.

Tensor metric perturbations are also copiously sourced by the helical gauge field [251]. Once these tensor modes re-enter the horizon after the end of inflation, they contribute to

a gravitational wave background (GWB) with a distinctive phenomenological signature: a pronounced chirality, where one polarization mode dominates over the other. The amplitude of such a parity-violating background could be potentially observable by future space- and ground-based interferometers, such as the Laser Interferometer Space Antenna (LISA) and the Einstein Telescope (ET) [252, 255, 259–263]. A realistic assessment of the detectability of this background, however, requires a careful treatment of the impact of strong backreaction effects on the production of gravitational waves. Furthermore, a high-frequency gravitational wave background (GWB) is also expected to emerge from the preheating dynamics following axion inflation [67, 111, 264]. Current limits on the effective number of relativistic degrees of freedom translate into a constraint on the axion-gauge coupling. For the potential $V(\phi) = \frac{1}{2}m^2\phi^2$, this yields $\Lambda^{-1} \lesssim 14 m_p^{-1}$ [67, 111]. While the preheating GWB constraint is stronger than those derived from non-Gaussianity and PBH overproduction, it is sensitive to the detailed dynamics of the final stages of inflation and the possible occurrence of an early PBH-dominated phase after inflation [265]. Since the strong backreaction effects explored in this work could modify these dynamics, we leave open the possibility of exploring axion-gauge couplings beyond current preheating bounds, with the aim of refining observational constraints on axion inflation.

It is possible to follow the dynamics of axion inflation during gauge field backreaction using semi-analytical techniques that have been developed in recent years. After initial studies neglecting backreaction and solving the linear dynamics of A_μ [266, 267], it was soon realized that gauge field backreaction is unavoidable for sufficiently large couplings [253, 265, 268–273]. Assuming a homogeneous inflaton background, the current state-of-the-art approach to model the backreaction dynamics in axion inflation is the *gradient expansion formalism* (GEF). This method involves an infinite tower of coupled partial differential equations for the expectation values of gauge field correlators. Phenomenological results based on this formalism have been studied in [274–276]. While computationally efficient, this approach neglects the local structure of the $F\tilde{F}$ term and the coupling between inflaton gradients and the gauge field A_μ . A perturbative extension of the GEF, which incorporates inflaton gradient effects, has also been recently proposed [277].

Lattice approaches, on the other hand, capture the full locality of the problem, and have been used to reassess various phenomenological effects [67, 111, 264, 278–282]. It was initially shown in [280] that a proper lattice implementation can reproduce previous homogeneous backreaction results when axion gradients are switched off. More importantly, it demonstrated the necessity of consistently capturing the inhomogeneity of the system. In this chapter, we extend the analysis showing significant deviations in the dynamics when axion gradients are included. While lattice simulations accurately

resolve the full non-linear problem, including the local features of strong backreaction, they remain computationally very demanding.

In summary, axion inflation can lead to a rich variety of phenomenological consequences. Besides the production of non-Gaussian curvature perturbations [251, 255, 255, 257, 283, 284] and chiral gravitational waves (GWs) [252, 255, 259, 260, 285, 286], it may also give rise to successful magnetogenesis [220, 248, 249, 274], mechanisms for baryon asymmetry generation [287–292], efficient (p)reheating [65, 66, 264], and significant post-inflationary GW production [67, 110, 111]. The specific outcome depends on the form of the inflaton potential $V(\phi)$, the coupling scale Λ , and the field content of the theory. These dynamics can be rather intricate, potentially requiring the inclusion of fermion production [293–297] and thermal effects [298, 299]. Scenarios where the gauge sector is represented by a non-Abelian $SU(2)$ group have also been extensively studied [300–317]. A major goal of this chapter is to demonstrate that, even in the simplest case where the inflaton couples only to a $U(1)$ dark photon, the non-linear dynamics induced by strong backreaction are highly complex. As such, any phenomenological predictions relying on this regime should not be trusted unless the backreaction effects are fully and reliably controlled.

In the following, we present the details of a consistent lattice formulation for the interaction between a shift-symmetric scalar field and a gauge field, including scenarios where the expansion of the universe is driven by these fields. This formalism constitutes a natural extension of the formulations introduced in [66, 224], and has already been employed in [280]. In this manuscript, we elaborate on the technical challenges encountered when implementing this lattice formulation. In particular, we discuss the numerical difficulties that arise in axion inflation scenarios where the inflaton is coupled to a dark photon, and the potential takes the form $V(\phi) = \frac{1}{2}m^2\phi^2$. Although a quadratic potential is in tension with current CMB observations [26, 27], we adopt this choice to facilitate comparison with previous results in the literature. Nevertheless, our methodology is general and can be readily applied to arbitrary inflaton potentials.

6.2 Axion inflation

In this section we provide an overview of the axion inflation model. We consider an action $S_{\text{tot}} = S_g + S_m$, with $S_g \equiv \int d^4x \sqrt{-g} \frac{1}{2} m_p^2 R$ the standard Hilbert-Einstein term for gravity, and a matter action given by

$$S_m = - \int d^4x \sqrt{-g} \left[\frac{1}{2} \partial_\mu \phi \partial^\mu \phi + V(\phi) + \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{\alpha_\Lambda}{4} \frac{\phi}{m_p} F_{\mu\nu} \tilde{F}^{\mu\nu} \right],$$

with ϕ the inflaton, and A_μ the gauge field of a hidden $U(1)$ gauge sector. We indicate the coupling strength of the axion-gauge interaction by $\alpha_\Lambda \equiv m_p/\Lambda$, and define the field strength of A_μ and its dual, as usual, by

$$F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu, \quad \tilde{F}_{\mu\nu} \equiv \frac{1}{2}\epsilon_{\mu\nu\rho\sigma}F^{\rho\sigma}, \quad (6.2)$$

where $\epsilon_{\mu\nu\rho\sigma}$ is the completely antisymmetric Levi-Civita tensor in a curved space-time, with $\epsilon_{0123} = 1/\sqrt{-g}$.

The action is invariant under local $U(1)$ transformations, $A_\mu \rightarrow A_\mu + \partial_\mu \alpha$, with $\alpha(x)$ an arbitrary real function. Except for the potential term $V(\phi)$, the action displays a shift symmetry $\phi \rightarrow \phi + c$, with $c \in \mathbb{R}$. The potential $V(\phi)$ is assumed to be generated by an external mechanism and breaks the shift symmetry explicitly. The axion-inflation model includes the lowest dimensional interaction compatible with a shift-symmetric inflaton, *i.e.* the five-dimensional axial coupling $\phi F \tilde{F}$ in Eq. (6.1). While our approach is applicable to any potential, we choose a quadratic one

$$V(\phi) = \frac{1}{2}m^2\phi^2, \quad (6.3)$$

to compare with previous results from the literature. The inflaton mass is set to $m \simeq 6.16 \cdot 10^{-6}m_p$, as dictated by fitting the observed temperature anisotropies of the CMB [25].

We assume, once again, a spatially flat Friedmann-Lemaître-Robertson-Walker (FLRW) background. Varying the action leads to the equations of motion (*eom*)

$$\ddot{\phi} = -3H\dot{\phi} + \frac{1}{a^2}\nabla^2\phi - m^2\phi + \frac{\alpha_\Lambda}{a^3m_p}\vec{E} \cdot \vec{B}, \quad (6.4)$$

$$\dot{\vec{E}} = -H\vec{E} - \frac{1}{a^2}\vec{\nabla} \times \vec{B} - \frac{\alpha_\Lambda}{am_p}\left(\dot{\phi}\vec{B} - \vec{\nabla}\phi \times \vec{E}\right), \quad (6.5)$$

$$\vec{\nabla} \cdot \vec{E} = -\frac{\alpha_\Lambda}{am_p}\vec{\nabla}\phi \cdot \vec{B}, \quad (6.6)$$

where $a(t)$ is the scale factor, and $H = \dot{a}/a$ is the Hubble rate. We define the electric field $\vec{E}$ in the temporal gauge $A_0 = 0$, and the magnetic field $\vec{B}$, as

$$E_i \equiv \dot{A}_i, \quad B_i \equiv \epsilon_{ijk}\partial_j A_k, \quad (6.7)$$

where ϵ_{ijk} is the Levi-Civita symbol with $\epsilon_{123} = 1$. We note that Eq. (6.6) represents the Gauss constraint.

The expansion of the Universe is governed by the Friedmann equations

$$\ddot{a} = -\frac{a}{3m_p^2}(2\rho_K - \rho_V + \rho_{\text{EM}}), \quad (6.8)$$

$$H^2 = \frac{1}{3m_p^2}(\rho_K + \rho_G + \rho_V + \rho_{\text{EM}}), \quad (6.9)$$

where the different homogeneous energy density contributions are given by

$$\begin{aligned} \rho_K &\equiv \frac{1}{2}\langle\dot{\phi}^2\rangle, & \rho_G &\equiv \frac{1}{2a^2}\langle(\vec{\nabla}\phi)^2\rangle, \\ \rho_V &\equiv \frac{1}{2}m^2\langle\phi^2\rangle, & \rho_{\text{EM}} &\equiv \frac{1}{2a^4}\langle a^2\vec{E}^2 + \vec{B}^2\rangle, \end{aligned} \quad (6.10)$$

with $\langle\ldots\rangle$ indicating volume averaging. Labels K, G, V denote the kinetic, gradient and potential energy densities of the inflaton, respectively, while EM denotes the electromagnetic energy density associated to A_μ . We note that Eq. (6.9) represents the Hubble constraint.

The treatment on the dynamics of the system of Eqs. (6.4)-(6.6) and (6.8)-(6.9) can be done at various levels of approximation, which we review in the following.

6.2.1 Linear regime: chiral instability

In the linear regime the expansion of the universe is just dictated by the inflaton, which is considered to be homogeneous, *i.e.* $\vec{\nabla}\phi = 0$. The contributions of the gauge field to the expanding background or to the inflaton dynamics are neglected. In this regime the *eoms* read

$$\ddot{\phi} = -3H\dot{\phi} - m^2\phi, \quad (6.11)$$

$$\dot{\vec{E}} = -H\vec{E} - \frac{1}{a^2}\vec{\nabla} \times \vec{B} - \frac{\alpha_\Lambda}{am_p}\dot{\phi}\vec{B}, \quad (6.12)$$

$$\ddot{a} = -\frac{a}{3m_p^2}(2\rho_K - \rho_V), \quad (6.13)$$

while the constraint equations are reduced to

$$\vec{\nabla} \cdot \vec{E} = 0, \quad (6.14)$$

$$H^2 = \frac{1}{3m_p^2}(\rho_K + \rho_V). \quad (6.15)$$

In order to understand the gauge field dynamics we first write Eq. (6.12) in terms of $\vec{A}(t, \mathbf{x})$ and using the conformal time $d\tau \equiv dt/a(t)$,

$$\left(\frac{d^2}{d\tau^2} - \nabla^2 + \frac{\alpha_\Lambda}{m_p} \frac{d\phi}{d\tau} \vec{\nabla} \times \right) \vec{A}(\tau, \mathbf{x}) = 0. \quad (6.16)$$

Next, we Fourier transform $\vec{A}$, decomposing its Fourier modes into a polarization vector basis,

$$\vec{A}(\tau, \mathbf{x}) = \sum_{\lambda} \int \frac{d^3k}{(2\pi)^3} A^{\lambda}(\tau, \mathbf{k}) \vec{\varepsilon}^{\lambda}(\hat{\mathbf{k}}) e^{i\mathbf{k}\cdot\mathbf{x}}. \quad (6.17)$$

We choose a chiral vector basis $\{\vec{\varepsilon}^{+}(\hat{\mathbf{k}}), \vec{\varepsilon}^{-}(\hat{\mathbf{k}})\}$, which satisfies the properties

$$\hat{\mathbf{k}} \cdot \vec{\varepsilon}^{\lambda}(\hat{\mathbf{k}}) = 0, \quad \hat{\mathbf{k}} \times \vec{\varepsilon}^{\lambda}(\hat{\mathbf{k}}) = -i\lambda \vec{\varepsilon}^{\lambda}(\hat{\mathbf{k}}), \quad (6.18)$$

$$\varepsilon_i^{\lambda}(\hat{\mathbf{k}})^* = \varepsilon_i^{\lambda}(-\hat{\mathbf{k}}), \quad \vec{\varepsilon}^{\lambda'}(\hat{\mathbf{k}}) \cdot \vec{\varepsilon}^{\lambda}(\hat{\mathbf{k}})^* = \delta_{\lambda\lambda'}. \quad (6.19)$$

We promote the Fourier amplitude into a quantum operator $A^{\lambda}(\tau, \mathbf{k}) \rightarrow \hat{A}^{\lambda}(\tau, \mathbf{k}) \equiv \hat{a}_{\mathbf{k}} \mathcal{A}^{\lambda}(\tau, \mathbf{k}) + \hat{a}_{-\mathbf{k}}^{\dagger} \mathcal{A}^{\lambda}(\tau, -\mathbf{k})^*$ by means of standard annihilation and creation operators that verify $[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{k}'}^{\dagger}] \equiv \delta^{(3)}(\mathbf{k} - \mathbf{k}')$. Using the slow-roll condition $\tau \simeq -1/(aH)$, the mode functions $\mathcal{A}^{\pm}(\tau, \mathbf{k})$ follow the equation

$$\left(\partial_{\tau}^2 + k^2 \pm \text{sign}(\dot{\phi}) \frac{2k\xi}{|\tau|} \right) \mathcal{A}^{\pm}(\tau, \mathbf{k}) = 0, \quad (6.20)$$

$$\text{with} \quad \xi \equiv \frac{|\dot{\phi}|}{2H\Lambda}. \quad (6.21)$$

For values $\xi \gtrsim \mathcal{O}(1)$, one of the chiral modes develops an exponential growth, while the other remains in vacuum. For $\dot{\phi} < 0$ (> 0), it is A^{+} (A^{-}) the one that experiences the instability. Considering ξ to be approximately constant, as expected deep inside the slow-roll dynamics, a general solution for Eq. (6.20) is given by

$$\mathcal{A}^{\pm}(\tau, \mathbf{k}) = \frac{1}{\sqrt{2k}} [iF_0(\pm\xi, -k\tau) + G_0(\pm\xi, -k\tau)], \quad (6.22)$$

with F_0 and G_0 the regular and irregular Coulomb wave functions [249, 250]. The coefficients have been chosen so that for scales deep inside the Hubble radius, $-k\tau \gg 1$, both chiralities are in the so-called Bunch-Davies (BD) vacuum,

$$\mathcal{A}^{\pm}(\tau, \mathbf{k}) \xrightarrow{-k\tau \gg 1} \frac{1}{\sqrt{2k}} e^{-ik\tau}. \quad (6.23)$$

Without loss of generality, we choose $\dot{\phi} < 0$, so that the excited mode is $\mathcal{A}^{+}$. In the limit $-k\tau \ll 2\xi$, the solution (6.22) for this mode can be approximated by

$$\mathcal{A}^{+}(\tau, \mathbf{k}) \underset{k|\tau| \ll 2\xi}{\simeq} \frac{1}{\sqrt{2k}} \left(\frac{k}{2\xi aH} \right)^{1/4} e^{\pi\xi - 2\sqrt{2\xi k/(aH)}}, \quad (6.24)$$

which exhibits very clearly the exponential nature of the chiral instability.

Depending on the evolution of ξ , the exponential growth of the unstable gauge mode

may become sufficiently large for its contribution to the expansion of the universe (in the form of electromagnetic energy density) to become increasingly relevant. Moreover, the term $\vec{E} \cdot \vec{B}$ on the right-hand side of Eq. (6.4) may also start playing a significant role, thereby affecting the dynamics of the inflaton. In other words, for sufficiently large coupling α_Λ , there always comes a point during inflation when the backreaction-less approximation breaks down. We must therefore refine our treatment of the dynamics, as we do next, to consistently incorporate the backreaction of the gauge field.

6.2.2 Homogeneous backreaction

In order to account for the gauge field backreaction in the system, it is essential to include both the electromagnetic energy in Eq. (6.8), as well as the term $\propto \vec{E} \cdot \vec{B}$ in Eq. (6.4). The homogeneous backreaction approach incorporates the latter through the expectation value $\langle \vec{E} \cdot \vec{B} \rangle$, thereby ensuring that the inflaton dynamics remains homogeneous. This leads to the following set of *eoms*:

$$\ddot{\phi} = -3H\dot{\phi} - m^2\phi + \frac{\alpha_\Lambda}{a^3 m_p} \langle \vec{E} \cdot \vec{B} \rangle, \quad (6.25)$$

$$\dot{\vec{E}} = -H\vec{E} - \frac{1}{a^2} \vec{\nabla} \times \vec{B} - \frac{\alpha_\Lambda}{am_p} \dot{\phi} \vec{B}, \quad (6.26)$$

$$\ddot{a} = -\frac{a}{3m_p^2} (2\rho_K - \rho_V + \rho_{\text{EM}}), \quad (6.27)$$

while the constraint equations read now

$$\vec{\nabla} \cdot \vec{E} = 0, \quad (6.28)$$

$$H^2 = \frac{1}{3m_p^2} (\rho_K + \rho_V + \rho_{\text{EM}}). \quad (6.29)$$

Different approaches have been developed to solve these equations. A straightforward method is to evaluate the expectation value $\langle \vec{E} \cdot \vec{B} \rangle$ at every time step, as an integral over the gauge field mode functions, and then plug this back into the equations of motion, repeating the procedure iteratively [253, 265, 268, 269]. An alternative approach is to reformulate the equations of motion entirely in terms of expectation values. This is the basis of the so-called *Gradient Expansion Formalism* (GEF), which results in an infinite tower of coupled ordinary differential equations [270, 274–276, 318]. These new equations are solved iteratively up to a truncated order that guarantees convergence of the solution when compared to the mode-by-mode method.

To a first approximation, the most significant impact of homogeneous backreaction is on the inflaton dynamics. The gauge field excitation acts as a form of ‘friction’ that reduces the inflaton’s velocity, thereby delaying the end of inflation. Additionally, an oscillatory

behavior emerges in $\dot{\phi}$, and consequently in ξ , as a result of the delayed response of the inflaton's velocity to the gauge field excitation [253, 265, 268, 269]. Remarkably, in the homogeneous backreaction regime the system remains fully chiral. That is, only the exponentially enhanced gauge mode (A^+ in our case) is excited, while the opposite chirality (A^- in our convention) remains in vacuum, or at most becomes marginally excited if the inflaton velocity changes sign due to a strong oscillation.

A more refined approach has been recently developed to incorporate the gradients of the inflaton field into the GEF framework [277]. This extended formalism goes beyond the standard GEF by including not only 2-point functions of the gauge field, but also 3-point functions involving the inflaton gradient. The results reproduce earlier findings from the standard GEF, such as the delayed end of inflation and the oscillatory behavior of ξ . While this method effectively captures the onset of genuinely local backreaction, it ultimately fails once the inflaton gradients surpass a certain threshold, a regime where fully non-linear lattice simulations become necessary.

6.2.3 Local backreaction

In order to accurately capture the dynamics of backreaction, it is necessary to allow for inhomogeneities in both the inflaton and the gauge field, consistently and without restrictions. The truly local dynamics of the system are governed by Eqs. (6.4), (6.5) and (6.8), for which no analytical solution exists. Due to the non-linear nature of these equations, approximation schemes such as the previously discussed homogeneous approaches fail to correctly track the backreaction dynamics. Therefore, a lattice approach is required to faithfully resolve the local dynamics. Before we proceed to the technical details of our lattice formulation, we introduce the form of the exact local equations of motion in the continuum, adapted to the inflationary dynamics. To do so, we express them in terms of the number of e-folds N , defined by

$$dN = H dt. \quad (6.30)$$

Keeping the previous definition of the electric field, $E_i \equiv \dot{A}_i \equiv H A'_i$, with $' \equiv d/dN$ and $H \equiv \dot{a}/a$, and introducing the inflaton conjugate momentum $\pi_\phi \equiv \dot{\phi} = H\phi'$, we rewrite the *eom* of the system as

$$H(\pi'_\phi + 3\pi_\phi) = \frac{1}{a^2} \nabla^2 \phi - m^2 \phi + \frac{\alpha_\Lambda}{a^3 m_p} \vec{E} \cdot \vec{B}, \quad (6.31)$$

$$H(\vec{E}' + \vec{E}) = \frac{1}{a^2} \vec{\nabla} \times \vec{B} - \frac{\alpha_\Lambda}{a m_p} (\pi_\phi \vec{B} - \vec{\nabla} \phi \times \vec{E}), \quad (6.32)$$

$$-3HH' = \frac{1}{m_p^2} (3\rho_K + \rho_G + 2\rho_{\text{EM}}), \quad (6.33)$$

while the constraint equations

$$\vec{\nabla} \cdot \vec{E} = -\frac{\alpha_\Lambda}{am_p} \vec{\nabla} \phi \cdot \vec{B}, \quad (6.34)$$

$$H^2 = \frac{1}{3m_p^2} (\rho_K + \rho_G + \rho_V + \rho_{EM}), \quad (6.35)$$

remain intact. We note that the scale factor can be written now explicitly as $a \equiv a_* e^{(N-N_*)}$, with a_* and N_* of arbitrary choice, while the homogeneous energy densities ρ_K , ρ_G , ρ_V and ρ_{EM} are still given by Eq. (6.10).

The key difference of this system of equations with respect to the homogeneous approaches is the inclusion of inflaton inhomogeneities. This implies that the backreaction of the gauge field on the inflaton is local, via the term $\vec{E} \cdot \vec{B}$, and not through its expectation value $\langle \vec{E} \cdot \vec{B} \rangle$. It also implies that, for consistency, we must retain the Laplacian term $\propto \nabla^2 \phi$ in the inflaton's *eom*. Furthermore, the term $\pi_\phi \vec{B}$ in the gauge field's *eom* can no longer be interpreted as the magnetic field times a homogeneous inflaton velocity, as the latter also becomes spatially dependent.

6.2.4 Chiral Projection

We notice that the chirality decomposition of the gauge field is performed in Fourier space. However, we aim to solve the non-linear dynamics of the system in real space, where the decomposition into the two linearly independent chiralities is not explicit. It is therefore convenient to establish a procedure to translate back and forth between the Cartesian and chiral bases. We begin by recalling the property of the chirality vectors $\{\vec{\varepsilon}^+, \vec{\varepsilon}^-\}$, namely $\epsilon_{ijl} \hat{k}_j \varepsilon_l^\lambda = -i\lambda \varepsilon_i^\lambda$, which can be conveniently rewritten as

$$(-i\epsilon_{ijl} \hat{k}_l) \varepsilon_j^\lambda = \lambda \varepsilon_i^\lambda. \quad (6.36)$$

This allows us to define the chirality operator [319]:

$$\Sigma_{ij}(\hat{\mathbf{k}}) \equiv -i\epsilon_{ijl} \hat{k}_l, \quad (6.37)$$

which, by construction, is antisymmetric, $\Sigma_{ij} = -\Sigma_{ji}$, and has $\vec{\varepsilon}^\pm$ as eigenvectors, with eigenvalues ± 1 , *i.e.* $\Sigma_{ij} \varepsilon_j^\lambda = \lambda \varepsilon_i^\lambda$. Extending the basis to a full orthonormal one $\{\vec{\varepsilon}^+, \vec{\varepsilon}^-, \hat{k}\}$, and relabeling the vectors as $\vec{\varepsilon}^{(-1)} \equiv \vec{\varepsilon}^-$, $\vec{\varepsilon}^{(0)} \equiv \hat{k}$, $\vec{\varepsilon}^{(+1)} \equiv \vec{\varepsilon}^+$, we see that $\vec{\varepsilon}^{(0)}$ is also an eigenvector of Σ_{ij} with vanishing eigenvalue. We can therefore write

$$\Sigma_{ij} \varepsilon_j^{(\sigma)} = \sigma \varepsilon_i^{(\sigma)}, \quad \sigma = -1, 0, +1. \quad (6.38)$$

Any vector $\vec{A}(\mathbf{k})$ that lives in Fourier space can always be expressed as a linear combination of the cartesian basis vectors, $\vec{A} \equiv A_1 \hat{e}_1 + A_2 \hat{e}_2 + A_3 \hat{e}_3$, or as a linear combination of the chiral basis as $\vec{A} \equiv A^{(-1)} \vec{\varepsilon}^{(-1)} + A^{(0)} \vec{\varepsilon}^{(0)} + A^{(+1)} \vec{\varepsilon}^{(+1)}$, with $A^{(+1)} \equiv A^+$ the positive chiral component, $A^{(-1)} \equiv A^-$ the negative chiral component, and $A^{(0)} \equiv A^\parallel$ the longitudinal component. We can think of $\vec{A}(\mathbf{k})$ as the Fourier transform of the gauge field $\vec{A}(N, \mathbf{x})$ at an arbitrary time N . Then, it holds that

$$\Sigma_{ij}(\hat{\mathbf{k}}) A_j^{(\sigma)}(\mathbf{k}) = \sigma A_i^{(\sigma)}(\mathbf{k}), \quad \sigma = -1, 0, +1, \quad (6.39)$$

where $A_i^{(\sigma)} \equiv A^{(\sigma)} \varepsilon_i^{(\sigma)}$ are the chiral components of $\vec{A}$. This property can be used to construct a proper chirality projector as

$$\begin{aligned} \Pi_{ij}^\pm(\hat{\mathbf{k}}) &\equiv \frac{1}{2} \left[(\Sigma^2(\hat{\mathbf{k}}))_{ij} \pm \Sigma_{ij}(\hat{\mathbf{k}}) \right] \\ &= \frac{1}{2} \left(P_{ij}(\hat{\mathbf{k}}) \pm \Sigma_{ij}(\hat{\mathbf{k}}) \right), \end{aligned} \quad (6.40)$$

where in the second line we have used $(\Sigma^2(\hat{\mathbf{k}}))_{ij} \equiv \Sigma_{il}(\hat{\mathbf{k}}) \Sigma_{lj}(\hat{\mathbf{k}}) = P_{ij}(\hat{\mathbf{k}})$, with $P_{ij}(\hat{\mathbf{k}}) \equiv \delta_{ij} - \hat{k}_i \hat{k}_j$ the transverse projector. It then follows that $\Pi_{ij}^\lambda(\hat{\mathbf{k}}) A_j^\parallel(\mathbf{k}) = 0$ and $\Pi_{ij}^\lambda(\hat{\mathbf{k}}) A_j^{\lambda'}(\mathbf{k}) = \frac{1}{2} (1 + \lambda \lambda') A_i^{\lambda'}(\mathbf{k}) \equiv A_i^\lambda(\mathbf{k}) \delta_{\lambda \lambda'}$. From here we arrive at the desired property for a chiral projector: given the cartesian components of a vector, say the gauge field $\vec{A} = (A_1, A_2, A_3)$ or the electric field $\vec{E} = (E_1, E_2, E_3)$, we can obtain their chiral components by means of the projection operation

$$A_i^\pm(\mathbf{k}) \equiv \Pi_{ij}^\pm(\hat{\mathbf{k}}) A_j(\mathbf{k}), \quad E_i^\pm(\mathbf{k}) \equiv \Pi_{ij}^\pm(\hat{\mathbf{k}}) E_j(\mathbf{k}). \quad (6.41)$$

We can also obtain the cartesian components of a vector, given its chiral components. This requires to write explicitly the cartesian form of the chiral vectors, which can be constructed as $\vec{\varepsilon}^\pm(\hat{\mathbf{k}}) \equiv [\vec{u}(\hat{\mathbf{k}}) \pm i \vec{v}(\hat{\mathbf{k}})] / \sqrt{2}$, with $\vec{v}(\hat{\mathbf{k}}) \equiv (\hat{e}_3 \times \hat{\mathbf{k}}) / |\hat{e}_3 \times \hat{\mathbf{k}}| = (-\sin \varphi, \cos \varphi, 0)$ and $\vec{u}(\hat{\mathbf{k}}) \equiv \vec{v} \times \hat{\mathbf{k}} = (\cos \theta \cos \varphi, \cos \theta \sin \varphi, -\sin \theta)$, given that $\hat{\mathbf{k}} = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta)$, where θ and φ are the spherical polar and azimuthal angles. In figure 6.1, we show how the chirality vector are build around the position vector $\vec{k}$ in reciprocal space. We then obtain

$$A_i = \varepsilon_i^+ A^+ + \varepsilon_i^- A^- + \hat{k}_i A^\parallel, \quad i = 1, 2, 3 \quad (6.42)$$

where $\varepsilon_i^\pm(\theta, \varphi) \equiv (u_i(\theta, \varphi) \pm i v_i(\theta, \varphi)) / \sqrt{2}$.

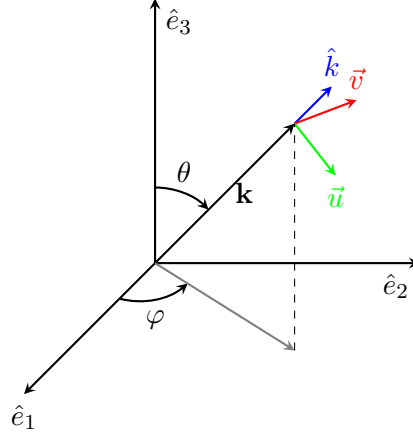


FIGURE 6.1: Definition of the orthonormal basis around the direction of the vector $\hat{\mathbf{k}}$ in the reciprocal space used to define the polarization vectors $\vec{\epsilon}^{\pm}$.

6.3 Lattice formulation

In this section, we present a consistent lattice formulation of an axion coupling in a self-consistently expanding background. This includes a suitable spatial discretization scheme for the Eqs. (6.31)–(6.35), designed to respect all relevant symmetries of the system. We also introduce our implementation of a lattice version of the chiral projector operator in Sec. 6.3.1. It is important to emphasize that this formulation is also applicable to other scenarios in which the axion coupling plays a role. For instance, see its implementation in the study of chiral charge dynamics in Abelian gauge theories at finite temperature [226, 320].

A consistent lattice formulation of the *eoms* of a system with axion-gauge interaction $\phi F\tilde{F}$ was originally introduced in [224] for Abelian gauge sectors (see also [321, 322]), and later on adapted in [66] to account for self-consistent evolution in an expanding background. There, an appropriate lattice gauge invariant version of $\phi F\tilde{F}$ was derived, so that the Bianchi identities and the shift symmetry of the system are preserved exactly at the lattice level, see [66, 224] for details. The lattice equations presented in those works required, however, an implicit evolution scheme for symplectic time integrators, due to the fact that the evolution of the electric field depends on itself at other lattice sites, as discussed in section 2.2.2. Here, we present a modified version of such lattice equations, suitable for non-symplectic time integrators like Runge-Kutta 2, see Sec. 2.2.2. This evades the need to use implicit schemes, while still reproducing the continuum equations to order $\mathcal{O}(\delta t^2, \delta x^2)$.

We consider that ϕ lives on the lattice sites $\mathbf{n}$, A_i 's on the links (*i.e.* halfway in between lattice sites, at $\mathbf{n} + \hat{i}/2$), and we define spatial lattice derivatives as usual, using forward/backward operators $\Delta_i^{\pm} \varphi \equiv \pm \frac{1}{\delta x} (\varphi_{\pm i} - \varphi)$, where δx is the lattice spacing. We

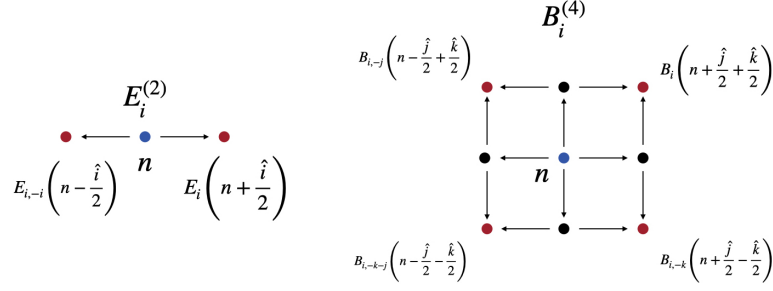


FIGURE 6.2: Definition of the electric field ($E_i^{(2)}(n)$) and the magnetic field ($B_i^{(4)}(n)$) living at lattice sites n . The field $E_i^{(2)}$ is obtained by averaging the field amplitude E at two neighboring points along the direction $\hat{i}$, while $B_i^{(4)}$ is obtained by averaging the field amplitude B at four lattice sites located at the centers of the *plaquettes* along directions $\hat{j}$ and $\hat{k}$.

define electric and magnetic fields on the lattice as

$$E_i(\mathbf{n} + \hat{i}/2) \equiv \dot{A}_i, \quad B_i(\mathbf{n} + \hat{j}/2 + \hat{k}/2) \equiv \sum_{j,k} \epsilon_{ijk} \Delta_j^+ A_k, \quad (6.43)$$

and introduce the definition of the Electric and Magnetic fields living on the lattice sites $\mathbf{n}$, see Fig. 6.2, as

$$\begin{aligned} E_i^{(2)}(\mathbf{n}) &\equiv \frac{1}{2} (E_i + E_{i,-i}), \\ B_i^{(4)}(\mathbf{n}) &\equiv \frac{1}{4} (B_i + B_{i,-j} + B_{i,-k} + B_{i,-j-k}). \end{aligned} \quad (6.44)$$

The definitions of $E_i^{(2)}$ and $B_i^{(4)}$ are such that the object representing the term $\vec{E} \cdot \vec{B}$ at the lattice level lives at lattice sites n and is equivalent up to order $\mathcal{O}(\delta x^2)$. This is required to satisfy the Bianchi identities at the lattice level; see [224] for a detailed derivation of the action. Following [66, 224], we use the above ingredients to build an action which preserves gauge invariance and the shift symmetry of ϕ . The lattice action that preserves all symmetries at lattice level then reads as

$$\begin{aligned} S_m^L = \int dt \sum_{\vec{n}} \delta x^3 &\left[\frac{1}{2} a^3 \pi_\phi^2 - \sum_i a \frac{1}{2} (\Delta_i^+ \phi)^2 - V(\phi) \right. \\ &\left. + \sum_i \frac{1}{2} a (E_i^2 - a^{-2} B_i^2) + \frac{\alpha_\Lambda \phi}{m_p} \sum_i E_i^{(2)} B_i^{(4)} \right], \end{aligned} \quad (6.45)$$

where $\pi_\phi = \dot{\phi}$. We choose to show a hybrid version of the action with time derivatives respect to N e-folds is kept at the continuum level, but the spatial derivatives are discretized. The variation of this hybrid action yields to the discrete *eoms* for the matter

sector:

$$\dot{\pi}_\phi = -3H\pi_\phi + \frac{1}{a^2} \sum_i \Delta_i^- \Delta_i^+ \phi - \frac{dV(\phi)}{d\phi} + \frac{\alpha_\Lambda}{a^3 m_p} \sum_i E_i^{(2)} B_i^{(4)}, \quad (6.46)$$

$$\begin{aligned} \dot{E}_i &= -HE_i - \frac{1}{a^2} \sum_{j,k} \epsilon_{ijk} \nabla_j^- B_k - \frac{\alpha_\Lambda}{2am_p} \left(\pi_\phi B_i^{(4)} + \pi_{\phi,+i} B_{i,+i}^{(4)} \right) \\ &\quad + \frac{\alpha_\Lambda}{4am_p} \sum_{\pm} \sum_{j,k} \epsilon_{ijk} \left\{ \left[(\Delta_j^\pm \phi) E_{k,\pm j}^{(2)} \right]_{+i} + \left[(\Delta_j^\pm \phi) E_{k,\pm j}^{(2)} \right] \right\}, \end{aligned} \quad (6.47)$$

with the discrete Gauss constraint

$$\sum_i \Delta_i^- E_i = -\frac{\alpha_\Lambda}{2am_p} \sum_{\pm} \sum_i (\Delta_i^\pm \phi) B_{i,\pm i}^{(4)}. \quad (6.48)$$

The lattice counterparts of the Friedmann equations, Eq. (6.8) and (6.9), are

$$-3HH' = \frac{1}{m_p^2} (3\rho_K^L + \rho_G^L + 2\rho_{EM}^L), \quad (6.49)$$

$$3H^2 = \frac{1}{m_p^2} (\rho_K^L + \rho_G^L + \rho_V^L + \rho_{EM}^L), \quad (6.50)$$

where the different energy density contributions (6.10) are computed in the lattice as

$$\begin{aligned} \rho_K^L &\equiv \frac{1}{2} \langle \pi_\phi^2 \rangle_V, \quad \rho_G^L \equiv \frac{1}{2a^2} \langle (\vec{\Delta}^+ \phi)^2 \rangle_V, \\ \rho_V^L &\equiv \frac{1}{2} m^2 \langle \phi^2 \rangle_V, \quad \rho_{EM}^L \equiv \frac{1}{2a^4} \langle a^2 \vec{E}^2 + \vec{B}^2 \rangle_V, \end{aligned} \quad (6.51)$$

with $\langle \dots \rangle_V \equiv \frac{1}{\mathcal{N}^3} \sum_{\mathbf{n}} (\dots)$ denoting volume averaging over a lattice of $\mathcal{N}$ points per dimension. As long as $\mathcal{N}$ is sufficiently large to guarantee that the comoving lattice length $L = \mathcal{N}\delta x$ is much bigger than the comoving wavelength $\lambda_* \sim 2\pi/k_*$ of the excited fields. This procedure leads to a well-defined notion of a homogeneous and isotropic expanding background, within the given comoving volume $V = L^3$.

The prescription is not restrictive, given that time is not discretized. In this sense, it does not force us to use only symplectic time integrators, such as leapfrog or velocity-Verlet schemes. This issue is particularly relevant in axion inflation, where the gauge conjugate momentum appears in its *com*, making symplectic integrators suitable only for implicit solving schemes [66]. Therefore, the prescription we present here is naturally compatible with non-symplectic integrators, which allow for explicit schemes. We choose a 2nd-order Runge-Kutta (RK2) integrator, with which the continuous equations are satisfied up to $\mathcal{O}(\delta t^2, \delta x^2)$, or $\mathcal{O}(\delta N^2, \delta x^2)$ if e-folds are used as the time variable.

6.3.1 The reciprocal lattice

For the specific purpose of axion inflation we choose a specific lattice derivative. In our case, we use backward/forward operators which, if interpreted as centered halfway of the lattice sites, at $\mathbf{n} + \hat{i}/2$, lead to the lattice momentum (both for forward and backward operations)

$$k_{L,i} = 2 \frac{\sin(\pi \tilde{n}_i / \mathcal{N})}{\delta x} . \quad (6.52)$$

As expected, Eq. (6.52) tends to the linear momentum $k_{L,i}(\tilde{\mathbf{n}}) \simeq 2\pi \tilde{n}_i / (\delta x \mathcal{N}) \equiv k_{\text{IR}} \tilde{n}_i$, at the IR scales of the lattice ($|\tilde{\mathbf{n}}| \ll \mathcal{N}/2$).

We use Eq. (6.52) to build the lattice version of the chiral projector (6.40) as

$$\Pi_{ij}^{L,\lambda}(\tilde{\mathbf{n}}) = \frac{1}{2} \left(\delta_{ij} - \frac{k_{L,i} k_{L,j}}{k_L^2} - \lambda \frac{i}{k_L} \epsilon_{ijk} k_{L,k} \right) , \quad (6.53)$$

where $\lambda = \pm$ and $k_L = (k_{L,1}^2 + k_{L,2}^2 + k_{L,3}^2)^{1/2}$. Given the Cartesian components of a vector on the lattice, $A_i(\tilde{\mathbf{n}})$, we will obtain its chiral components $\lambda = \pm$ as

$$A_j^\lambda(\tilde{\mathbf{n}}) = \Pi_{ij}^{L,\lambda}(\tilde{\mathbf{n}}) A_j(\tilde{\mathbf{n}}) . \quad (6.54)$$

The Fourier transform of these $A_i^\lambda(\tilde{\mathbf{n}}) \rightarrow A_i^\lambda(\mathbf{n})$ will be transverse only with respect to the forward/background derivative operators, *i.e.* $\Delta_i^\pm A_i^\lambda(\mathbf{n}) = 0$ ¹.

The lattice version of Eq. (6.42),

$$A_i = \varepsilon_{L,i}^+ A^+ + \varepsilon_{L,i}^- A^- + \hat{k}_{L,i} A^\parallel , \quad i = 1, 2, 3 \quad (6.55)$$

allow us to obtain the Cartesian components of a vector, given its chiral components. The difference between Eq. (6.42) and Eq. (6.55) is that, in the latter, we use a lattice version of the chiral vectors, which we construct using the azimuthal and polar angles θ_L, φ_L of the lattice momentum $\hat{\mathbf{k}}_L = (\sin \theta_L \cos \varphi_L, \sin \theta_L \sin \varphi_L, \cos \theta_L)$, with

$$\vec{\varepsilon}_L^\pm(\hat{\mathbf{n}}) \equiv [\vec{u}(\hat{\mathbf{k}}_L) \pm i \vec{v}(\hat{\mathbf{k}}_L)] / \sqrt{2} , \quad (6.56)$$

$$\vec{v}(\hat{\mathbf{k}}_L) = (-\sin \varphi_L, \cos \varphi_L, 0) , \quad (6.57)$$

$$\vec{u}(\hat{\mathbf{k}}_L) = (\cos \theta_L \cos \varphi_L, \cos \theta_L \sin \varphi_L, -\sin \theta_L) . \quad (6.58)$$

We explicitly construct the chiral basis in the lattice. Our choice of forward/background lattice derivative interpreted as living at mid points gives a lattice momentum given by

¹Note that $\pm$ in $\Delta_i^\pm$ refers to forward (+) / backward (-) derivatives, unrelated to the chiral component λ , which are also either + or - (independently of the choice of derivative).

Eq. (6.52). We build the chiral orthonormal basis using the lattice momenta $\hat{\mathbf{k}}_L$ where the angles defining Eqs. (6.57)-(6.58) in the orthonormal basis $\{\vec{u}_L, \vec{v}_L, \hat{\mathbf{k}}_L\}$ are given by the following expressions:

$$\sin \theta_L = \frac{\sqrt{k_{L,1}^2 + k_{L,2}^2}}{\sqrt{k_{L,1}^2 + k_{L,2}^2 + k_{L,3}^2}}, \quad (6.59)$$

$$\sin \varphi_L = \frac{k_{L,2}}{\sqrt{k_{L,1}^2 + k_{L,2}^2}}, \quad (6.60)$$

$$\cos \theta_L = \frac{k_{L,3}}{\sqrt{k_{L,1}^2 + k_{L,2}^2 + k_{L,3}^2}}, \quad (6.61)$$

$$\cos \varphi_L = \frac{k_{L,1}}{\sqrt{k_{L,1}^2 + k_{L,2}^2}}. \quad (6.62)$$

We note that for the vector $\hat{\mathbf{k}}_L$ pointing in the $\hat{e}_3$ direction, $\sin \varphi_L$ and $\cos \varphi_L$ are ill-defined. Nevertheless, this is just a consequence of choosing $\hat{e}_3$ as our reference direction to define the chiral basis, therefore we can just choose any value for φ_L as long as $\{\vec{u}_L, \vec{v}_L, \hat{\mathbf{k}}_L\}$ are orthogonal among them. We can just replace all the continuous definitions of the chiral vectors and the projector by its lattice counterpart to obtain the desired properties. Similarly, as described in Sec. 6.3, we can construct the chiral projector on the lattice using the appropriate lattice momentum, Eq. (6.53), for which similar properties to those of the chiral vectors are exactly satisfied on the lattice:

$$k_{L,i} \Pi_{ij}^{L,\lambda}(\hat{\mathbf{k}}_L) = 0, \quad (6.63)$$

$$\epsilon_{ijl} \hat{k}_{L,j} \Pi_{lm}^{L,\lambda}(\hat{\mathbf{k}}_L) = -i\lambda \Pi_{ij}^{L,\lambda}(\hat{\mathbf{k}}_L), \quad (6.64)$$

$$\Pi_{ij}^{L,\lambda}(\hat{\mathbf{k}}_L) \Pi_{ij}^{L,\lambda'}(\hat{\mathbf{k}}_L) = \delta_{\lambda\lambda'}, \quad (6.65)$$

$$\Pi_{ij}^{L,\lambda}(\hat{\mathbf{k}}_L) = \Pi_{ij}^{L,\lambda}(-\hat{\mathbf{k}}_L) = \Pi_{ij}^{L,\lambda}(\hat{\mathbf{k}}_L)^* = \Pi_{ji}^{L,\lambda}(\hat{\mathbf{k}}_L). \quad (6.66)$$

In addition, we make sure that the Fourier transforms/anti-transforms are consistent with our lattice prescription. In our formulation, we use backward/forward spatial derivatives that live at the intermediate points relative to the position of the field they act upon. This implies that if the scalar field ϕ lives at $\mathbf{n}$, $\Delta_i^\pm \phi$ lives at $\mathbf{n} \pm \hat{i}/2$, so does the i th component of the gauge field in order to preserve gauge invariance. We make this explicit in the transformation of the gauge fields

$$A_i(\mathbf{n} + \hat{i}/2) \equiv \frac{1}{N^3} \sum_{\tilde{\mathbf{n}}} e^{-i\frac{2\pi}{N} \tilde{\mathbf{n}}(\mathbf{n} + \hat{i}/2)} A_i(\tilde{\mathbf{n}}), \quad (6.67)$$

$$A_i(\tilde{\mathbf{n}}) \equiv \sum_{\mathbf{n}} e^{+i\frac{2\pi}{N} (\mathbf{n} + \hat{i}/2) \tilde{\mathbf{n}}} A_i(\mathbf{n} + \hat{i}/2), \quad (6.68)$$

and similarly for the electric field. With this redefinitions we can naturally use (6.52) as the lattice momentum to construct the chiral vectors (6.56) through the angles (6.59)-(6.62), and the lattice chirality projector (6.53). Furthermore, we ensure that Gauss's law is satisfied to machine precision during the initialization. A similar procedure was used in [323] to make explicit the locations of gauge fields for Fourier transforms. Finally, we note that this procedure is equivalent to that shown in [165] for pure gauge theories, where a complex lattice momentum is used for gauge initialization. However, the complex nature of the chiral vectors and the chiral projector prevent the use of complex lattice momenta.

Finally, we discuss the different observables we use to study the dynamics of the system. We compute the power spectrum of the gauge, electric and magnetic field amplitudes, as well as the chiral components of the gauge and electric fields. As explained in Sec. 2.2, there are different approaches to define the power spectrum on a lattice. For this regard, we use **Type I** power spectra and canonical binning. In practice, we compute power spectra in our simulations for $\phi, \vec{A}^\pm, \vec{A}, \vec{E}^\pm, \vec{E}, \vec{B}$ as $\Delta_\phi, \Delta_A^{(\pm)}, \Delta_A, \Delta_E^{(\pm)}, \Delta_E, \Delta_B$ via Eq. (2.44) with $|f(\tilde{\mathbf{n}})|^2 = |\phi(\tilde{\mathbf{n}})|^2, |\vec{A}^\pm(\tilde{\mathbf{n}})|^2, |\vec{A}(\tilde{\mathbf{n}})|^2, |\vec{E}^\pm(\tilde{\mathbf{n}})|^2, |\vec{E}(\tilde{\mathbf{n}})|^2$, and $|\vec{B}(\tilde{\mathbf{n}})|^2$, respectively, with the electric and magnetic fields given by Eq. (6.43).

6.4 Lattice simulations

Our simulations must capture the evolution of all relevant modes involved in the dynamics, starting from the initial vacuum configuration of the gauge field, progressing through the initial linear evolution, and eventually reaching the non-linear regime. As we will see, the range of modes $k_{\text{IR}} \leq k \leq k_{\text{UV}}$ that are dynamically relevant depends strongly on the strength of the axion-gauge interaction. This range becomes wider as the coupling α_Λ increases, making simulations with the largest couplings computationally highly demanding. Moreover, in simulations requiring such a wide range of modes, the integrated contribution from the initial vacuum tail of the gauge field may dominate over the inflaton contribution. This would artificially affect the initial dynamics, which should be solely inflaton-driven during the linear regime. In Sec. 6.4.1, we present our method for setting up initial conditions on the lattice when the gauge field is in either vacuum or excited configurations, with the latter being the one used in our results. In Sec. 6.4.2, we introduce a simplified version of the lattice *eom* suitable for simulating only the linear regime.

6.4.1 Initial conditions on the lattice

In this section, we introduce a procedure to set proper initial conditions for the fields on the lattice. In the case of the gauge field, we know that the Bunch-Davies (BD) vacuum (6.23) provides the correct initial condition for each chirality component, for wavelengths well inside the Hubble radius. This condition characterizes an initial quantum vacuum state $|0\rangle$ (for which $\hat{a}_{\mathbf{k}}|0\rangle = 0$), whereas solving Eqs. (6.31)-(6.33) implicitly assumes the use of classical fields. Given the quantum expectation value $s_A^2(\tau, \mathbf{k}) \equiv \langle 0 | \hat{A}^\lambda(\tau, \mathbf{k}) \hat{A}^{\lambda\dagger}(\tau, \mathbf{k}) | 0 \rangle \equiv |\mathcal{A}^\lambda(\tau, \mathbf{k})|^2$ for the chirality λ , the BD condition implies $s_A^2 \rightarrow \frac{1}{2k}$. This corresponds to initially vanishing occupation numbers, $n_{\mathbf{k}} = 0$. However, we know that one of the gauge field chiralities (specifically $\lambda = +$ in our convention) will grow exponentially during the linear regime, leading to exponentially large occupation numbers $n_{\mathbf{k}} \gg 1$. This allows us to approximate the vacuum expectation value of products of field operators by ensemble averages of classical random fields, see [324]. This corresponds to the classical limit of the excited field(s).

In practice, at the initial time when wavelengths are still deep inside the Hubble radius, we create a random realization of the corresponding Fourier amplitudes for each chirality. Initially, such a classical stochastic configuration cannot, of course, fully capture the genuinely quantum nature of the gauge field, as the commutator between the field amplitude and its conjugate momentum vanishes. However, as the gauge field chirality grows exponentially during the linear regime, the field becomes effectively classical, so that quantum expectation values become well approximated by ensemble averages over classical field realizations on the lattice. Once the quantum nature is no longer relevant, the quantum field commutator becomes negligible compared to the field amplitude, and we find that $|A(\tau, \mathbf{k})|^2 \approx |A^+(\tau, \mathbf{k})|^2 \gg 2 \times \frac{1}{2k}$ within the finite range of modes where the excitation is supported.

In order to set the gauge field initial values, we create a random realization of the chirality components as described above. In practice, we first write the BD condition (6.23) as a function of the e-folding time variable N . Making use of the conformal time relation $\tau \simeq -1/(aH)$ deep inside inflation, we write the gauge and the electric field chiral amplitudes as

$$A^\pm(N, \tilde{\mathbf{n}}) \simeq f_k^\pm e^{ik/(aH)}, \quad (6.69)$$

$$E^\pm(N, \tilde{\mathbf{n}}) \simeq -\frac{i}{a} k g_k^\pm e^{ik/(aH)}, \quad (6.70)$$

where $f_k^+, f_k^-, g_k^+, g_k^-$ are the independent amplitude realizations of a Gaussian random field with zero mean and root-mean-square $1/\sqrt{2k}$. We note that N appears only

implicitly throughout $a = a(N)$ and $H = H(N)$. We notice that that k in Eqs. (6.69)-(6.70) corresponds to the modulus of the linear momentum, $k = |\mathbf{k}| = k_{\text{IR}}|\tilde{\mathbf{n}}|$, to mimic the continuum spectra. Using Eq. (6.55), we finally convert the chiral fluctuations (6.69)-(6.70) into Cartesian components of the gauge and the electric field, $A_i(N, \mathbf{n})$ and $E_i(N, \mathbf{n})$, where we highlight that the angles θ_L, φ_L in Eq. (6.55) correspond to the polar and azimuthal angles of the lattice momentum $\mathbf{k}_L(\tilde{\mathbf{n}})$, the Cartesian components of which are given in Eq. (6.52).

In order to obtain the field and conjugate momentum amplitudes, we solve the mode equations of the gauge field in the linear regime on a 1D grid, for modes well inside the Hubble radius, $k \gg aH$, and for different initial conditions. By inspecting the Bunch-Davies (BD) solution, we observe that the relative phase between the gauge field amplitude and the electric field should be $\pi/2$. If this condition is not satisfied, the subsequent evolution may lead to anomalous oscillations in the power spectrum. In our simulations, we aim to start at the onset of the non-linear regime (to be explained later), and we set the infrared cutoff as $k_{\text{IR}} = 10 a(N_{\text{start}})H(N_{\text{start}})$. We find that the positive chirality is excited for a range of momenta $k \lesssim aH$, up to a transition scale k_{tr} , above which the fields remain in vacuum. From these results, we observe that the relative phase of the positive chirality component differs in the excited infrared region (super-Hubble modes) compared to the non-excited region (sub-Hubble modes), where it remains at $\pi/2$. In fact, the exponential excitation appears to eliminate any initial phase difference in the excited region. For the negative mode, by contrast, a global phase difference of approximately $\pi/2$ remains. Taking this effect into account, we incorporate it in the initial conditions in the following way:

$$A^+(\mathcal{N}, \tilde{\mathbf{n}}) \simeq A_{\text{PS}(\mathbf{k})}^+ e^{ik/(aH)}, \quad (6.71)$$

$$A^-(\mathcal{N}, \tilde{\mathbf{n}}) \simeq A_{\text{PS}(\mathbf{k})}^- e^{ik/(aH)}, \quad (6.72)$$

$$E^+(\mathcal{N}, \tilde{\mathbf{n}}) \simeq E_{\text{PS}(\mathbf{k})}^+ e^{i[k/(aH) - \Theta(k, k_{\text{tr}}, \nu)\pi/2]}, \quad (6.73)$$

$$E^-(\mathcal{N}, \tilde{\mathbf{n}}) \simeq E_{\text{PS}(\mathbf{k})}^- e^{i[k/(aH) - \pi/2]}, \quad (6.74)$$

where $k = |\mathbf{k}| = k_{\text{IR}}|\tilde{\mathbf{n}}|$, and $A_{\text{PS}(\mathbf{k})}^\pm$ and $E_{\text{PS}(\mathbf{k})}^\pm$ correspond to realizations of the spectra extracted from the mode by mode linear solution at the onset of non-linearities. The function $\Theta(k, k_{\text{tr}}, \nu)$ accounts for the smooth transition of the relative phase in the excited mode, which we model by

$$\Theta(k, k_{\text{tr}}, \nu) = \frac{1}{2} \left[\tanh \left(\nu \log \frac{k}{k_{\text{tr}}} \right) + 1 \right], \quad (6.75)$$

with k_{tr} and ν free parameters that control the position and velocity of the transition between the two phases.

Finally, the initialization of the inflaton and of the expansion rate is, on the other hand, more straightforward compared to the gauge field. We start our simulations when the inflationary dynamics are still well dominated by the homogeneous inflaton. We set up a spatially independent initial amplitude ϕ and conjugate momentum according to the slow-roll condition $\pi_\phi \simeq -m^2\phi/(3H)$. The initial Hubble rate at that moment is dictated by the kinetic and potential contributions, ρ_K^L and ρ_V^L . Technically, we should also include fluctuations on top of the homogeneous field, such as those corresponding to a spectrum of quantum vacuum fluctuations in the Bunch-Davies state. However, in practice, we can safely set the inflaton fluctuations to zero during the linear regime, as they would not be classically well balanced in Eq. (6.48), given the gauge field initialization described above. Furthermore, once the dynamics enter the non-linear regime, the inflaton naturally develops classical inhomogeneities due to the local backreaction of the gauge field, which rapidly become much larger than those expected from quantum vacuum fluctuations.

6.4.2 Linear regime on the lattice

If we start with all modes deep inside the Hubble radius, the initial dynamics captured in the lattice should only correspond to the linear regime. We therefore start our simulations evolving first a lattice version of the *eom* that capture only the linear physics,

$$\pi'_\phi = -3\pi_\phi - \frac{1}{H}m^2\phi, \quad (6.76)$$

$$E'_i = -E_i - \frac{1}{H} \left[\frac{1}{a^2} \sum_{j,k} \epsilon_{ijk} \Delta_j^- B_k + \frac{\alpha_\Lambda}{2am_p} \left(\pi_\phi B_i^{(4)} + \pi_{\phi,+i} B_{i,+i}^{(4)} \right) \right], \quad (6.77)$$

$$-HH' = \frac{1}{m_p^2} \rho_K^L, \quad (6.78)$$

and the corresponding constraints,

$$\sum_i \Delta_i^- E_i = 0, \quad (6.79)$$

$$3H^2 = \frac{1}{m_p^2} (\rho_K^L + \rho_V^L), \quad (6.80)$$

which correspond to Eqs. (6.11)-(6.13) and the constraints (6.14)-(6.15). From now on we set $N = 0$ to be the moment at which slow-roll inflation ends, hence for cases where inflation end is delayed it will happen for $N > 0$. In Fig. 6.3, we show the results of lattice simulations in the linear regime for the case $\alpha_\Lambda = 18$, together with the results from solving the mode equations on a 1D grid in momentum space. The purpose of these results is to provide a first test of the accuracy of lattice simulations in describing the dynamics of the full non-linear regime. As can be observed in the figure, the power spectrum computed on the lattice lacks power in the most infrared bins. The lack of

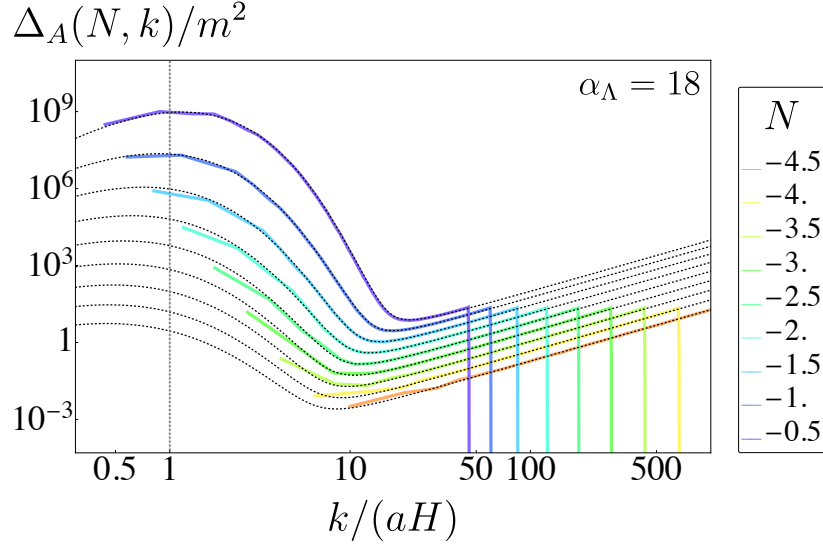


FIGURE 6.3: Power spectrum of gauge field amplitude as a function of comoving momenta per horizon size. In colors, results of lattice simulation in the linear regime starting at $N = -4.5$ e-folds and ending at $N = -0.5$ e-folds. In black dashed lines result of solution to the modes equations for 1D array of momenta k .

statistics in the first bins, which contain the fewest modes per bin, causes the largest deviations from the continuum approximation. In order to obtain a proper lattice-based description that corresponds to the continuum, we must ensure that the peak of excitation is located at intermediate scales, $k_{\text{IR}} < k_{\text{exc}} < k_{\text{UV}}$. From the linear mode *eom*, we can infer that the peak of excitation approximately follows the horizon as $k_{\text{exc}}/(aH) < 2\xi = \dot{\phi}/(2H\Lambda)$; recall that $\alpha_\Lambda \equiv m_p/\Lambda$. This behavior can be observed in the 1D grid solution, where the peak of excitation always remains inside the horizon. It is only when the simulation, due to the expansion of the universe, reaches the time when the intermediate scales of the lattice are at some penetration factor that both descriptions differ only marginally.

We can run these linear *eoms* until the contribution of the terms that drive the dynamics into the non-linear regime becomes relevant. At that moment, we switch to solving the full system of *eoms* (6.46)-(6.47) and (6.49), along with their constraints (6.48) and (6.50). To determine when the linear regime ceases to be valid, we refer again to Fig. 6.3. As mentioned before, the 1D grid mode solution and the lattice simulation agree only when the peak of excitation lies within the intermediate scales of the lattice. This ensures that we avoid the large variance present in the first bins, which causes significant deviations between continuum and lattice simulations. The time at which this occurs is denoted as N_{start} . For each simulation with a different coupling α_Λ , an appropriate value of N_{start} is chosen to ensure that the system starts with excited modes but with the backreaction of the gauge fields still negligible. This means simultaneously

several things, before N_{start} we have the electromagnetic energy to be subdominant in the energy budget $\rho_{\text{EM}} \ll \rho_K, \rho_V$. Furthermore, the terms in the inflaton equation of motion describing the backreaction are subdominant, $\alpha_\Lambda \langle \vec{E} \cdot \vec{B} \rangle \ll 3H\dot{\phi}, m^2\phi$. Only if this conditions are met, we can safely trust that the initial condition of our simulation is both a good representation of the continuum result and the gauge fields are still negligible.

6.5 Phenomenology of the Non-Linear Regime

In this section, we discuss the main results obtained from a series of lattice simulations summarized in Table 6.1. Each of these simulations was performed using the aforementioned lattice setup, with excited initial conditions, and was run until energy conservation was sufficiently maintained. For all simulations, we reach the end of inflation; however, radiation domination is not achieved in every case. In the following, we will discuss the phenomenology of the non-linear regime and present the main findings, which are: a delayed end of inflation, an electromagnetic slow-roll phase, and a scale-dependent chiral imbalance.

6.5.1 Delayed End of Inflation

We already anticipated a key characteristic of gauge field backreaction onto the inflaton dynamics provided by the homogeneous backreaction formalism, this is a delayed end of inflation due to the energy dominance of the gauge fields over the kinetic energy. In the homogeneous backreaction, an additional feature observed was an oscillatory pattern in the particle production parameter ξ . In light of our results from lattice simulations where the backreaction happens at local level, we have argued that the oscillatory features in the homogeneous backreaction regime is due to a delayed response on the gauge field growth. In what it regards to us, the particle production parameter ξ ceases to be a good observable in the non-linear regime. In this regime, first the inflaton velocity does not follow the *canonical* trajectory and it is clearly driven by the gauge field backreaction, hence it is neither a good indicator of the growth of gauge fields in this regime. In order to quantify the dynamics we will follow the evolution of the slow roll parameter ϵ_H , which it can be seen also as a measure of the different energy contributions

$$\epsilon_H \equiv -\frac{\dot{H}}{H^2} = 1 + \frac{2\rho_K - \rho_V + \rho_{\text{EM}}}{\rho_{\text{tot}}} . \quad (6.81)$$

We can immediately anticipate two conditions to end inflation, $\epsilon_H(N_{\text{end}}) = 1$, if the gauge fields do not grow enough inflation will end when twice the kinetic energy equals

α_Λ	$\mathcal{N}$	k_{IR}/m	k_{UV}/m	N_{start}
10	320	0.1932	53.54	-1.1
11	320	0.1932	53.54	-1.1
12	320	0.1932	53.54	-1.1
13	320	0.1932	53.54	-1.1
14	480	0.1932	80.31	-1.1
15	640	0.1932	107.08	-1.1
16	1152	0.1932	192.75	-1.1
17	2048	0.1932	342.66	-1.4
18	3072	0.1932	514.00	-1.7
19	3072	0.1544	410.81	-2.0
20	3072	0.1233	327.95	-2.4

TABLE 6.1: Lattice simulations performed for different α_Λ , $\mathcal{N}$ number of lattice sites per dimension, k_{IR} infrared momentum, k_{UV} ultraviolet momentum, N_{start} the efold at which we start each simulation.

the potential, $2\rho_K = \rho_V$. On the other hand, if the gauge fields grow enough they will finish inflation at the moment it equals the potential energy $\rho_{\text{EM}} = \rho_V$. In Fig. 6.4, we show the time evolution of ϵ_H for different coupling strengths α_Λ from which we identify three different levels of backreaction depending whether inflation's end is delayed or not. The regimes are:

- **Weak backreaction:** For couplings smaller than $\alpha_\Lambda \lesssim 13$, the gauge fields do not grow enough during inflation. Hence, inflation's end is not delayed and it only affects the dynamics later on during preheating. If the coupling strength is enough electromagnetic fields will become the main contribution to the total energy density. For very low couplings $\alpha_\Lambda \lesssim 8$ gauge fields backreaction are completely negligible [66].
- **Mild backreaction:** For couplings $13 \lesssim \alpha_\Lambda < 15$, backreaction becomes relevant during inflation nevertheless it still ends as given by the slow roll trajectory. Later on the system re-enters inflation as the EM energy keeps growing and later on

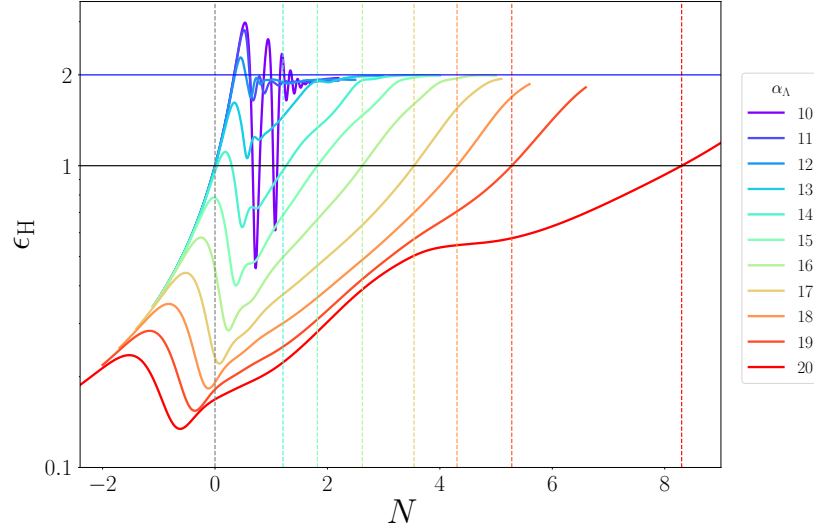


FIGURE 6.4: Time evolution of the slow roll parameter ϵ_H for the full set of lattice simulations summarized in Table 6.1. Vertical dashed lines correspond to the moment when $\epsilon_H = 1$ meaning end of inflation for each case.

effectively inflation ends. In Fig. 6.4, we can observe this behavior for the case $\alpha_\Lambda = 14$.

- **Strong backreaction:** For couplings $\alpha_\Lambda \geq 15$, gauge fields backreaction happen early enough such that it delays the end of inflation and actually drives it during the last e-folds. For this case, inflation ends when the electromagnetic energy equals the potential one, leaving the kinetic energy very subdominant. In a state where the inflaton is effectively stopped.

We now focus on how the different energy contributions interplay in each of the regimes described. To this end, we choose three benchmark values of $\alpha_\Lambda = \{12, 14, 18\}$, representative of each regime. In Fig. 6.5, top-left panel, we show the time evolution of the energy fraction of each component for a *weak backreaction* case. As expected from the ϵ_H plot, we observe that electromagnetic energy grows only after the end of slow-roll inflation. Only during the preheating phase do the gauge fields overtake the energy budget of the universe, ultimately resulting in a radiation-dominated stage with $\epsilon_H = 2$, a well-known result from axion-gauge preheating. In the top-right panel of Fig. 6.5, we show the evolution of the energy fractions for a *mild backreaction* case. In this scenario, the gauge fields are excited during inflation, but not enough to affect the timing of the end of inflation. Nevertheless, the electromagnetic energy continues to grow afterward, eventually driving the system back into an inflationary phase with $\epsilon_H < 1$, which ends shortly after, once the EM energy becomes comparable to the potential energy. The system ends up with most of the energy in the form of EM fields, effectively reheating

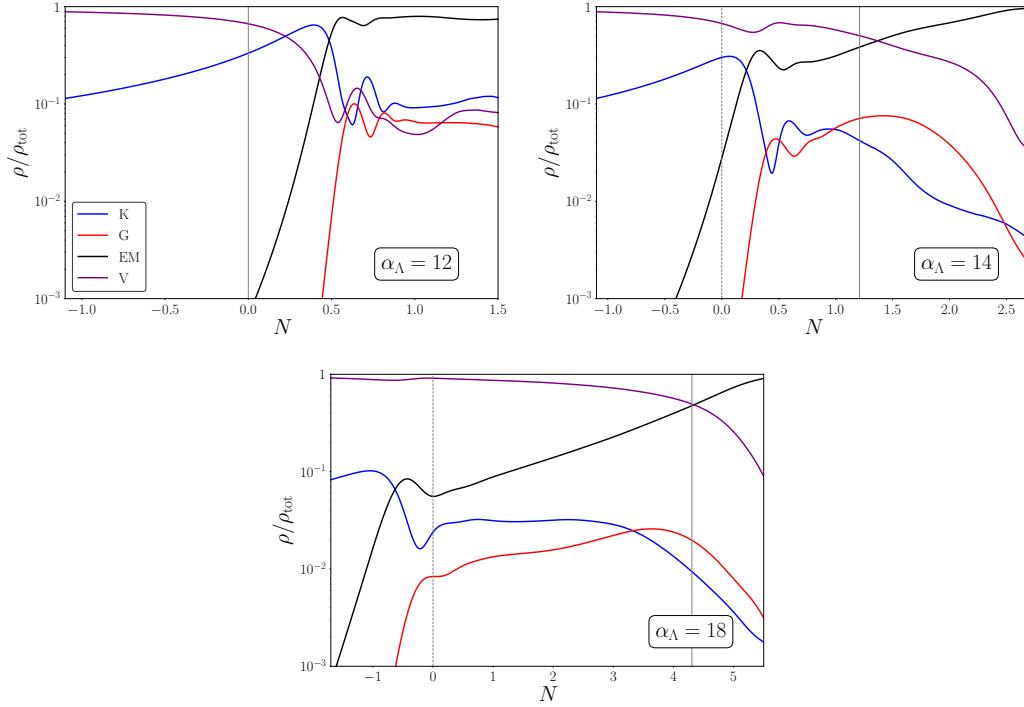


FIGURE 6.5: Time evolution of the energy components for the three regimes represented by the cases $\alpha_\Lambda = 12, 15, 18$. Dashed vertical line shows end of inflation for the slow roll case. Dashed vertical shows end of inflation when it is delayed due to backreaction.

the universe with a radiation-dominated equation of state. Finally, in the bottom panel of Fig. 6.5, we observe how the EM energy in the *strong backreaction* regime surpasses the kinetic energy even before the end of slow-roll inflation. It enters a phase where it grows steadily while the kinetic energy remains approximately constant. This continues until the EM energy reaches the level of the potential energy, and inflation ends with most of the energy in the form of EM fields. In this case, the lattice simulations could not follow the evolution all the way to radiation domination, as they failed earlier. However, all indications suggest that the system would eventually reach a radiation-dominated equation of state soon after the end of inflation. In all cases, we observe that at the end of inflation, the gradient energy becomes comparable to the kinetic energy, illustrating how the inflaton fluctuations become significant due to the system's non-linearities.

The mild and strong backreaction regimes clearly show a parametric dependence on the delayed inflation's end as a function of the coupling α_Λ , we will call this delay ΔN_{br} where br stands for backreaction. As shown in Fig. 6.4, we have measured how much late inflation ends for each of the simulations and it can be seen approximately that it follows a linear increase on it as a function of the coupling. In Fig. 6.6, we show the measured ΔN_{br} for each α_Λ and a fit to the points in order to extract a parametrization

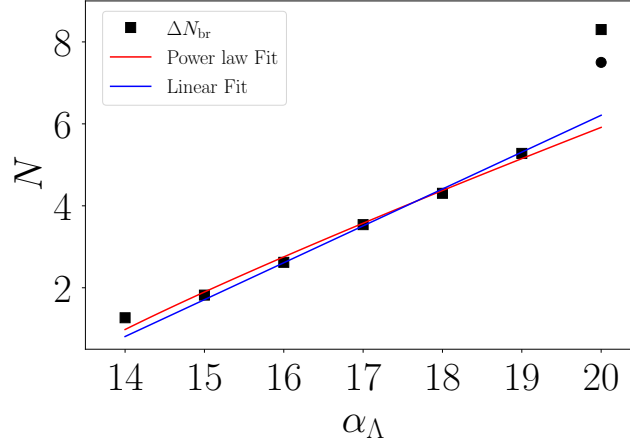


FIGURE 6.6: Delay on inflation's end (ΔN_{br}) as a function of the coupling α_Λ . Black squares are results obtained from lattice simulations, blue solid line shows linear fit and red solid line shows power law fit given by Eqs. 6.82. Black circle located at ($\alpha_\Lambda = 20$, $\Delta N_{\text{br}} \sim 7.5$) represents a failed simulation with better UV resolution ($\mathcal{N} = 3648$) whose extrapolation establishes $\Delta N_{\text{br}} \sim 7.5$, see Sec. 6.6.

that could help us predict the delay on inflation's end for couplings that are inaccessible by simulations. We obtained the following parametrization, observing for instance that at coupling $\alpha_\Lambda = 13$ there is not delayed on the end of inflation and excluding $\alpha_\Lambda = 20$ data point

$$\begin{cases} \Delta N_{\text{br}} = (0.90 \pm 0.02)[\alpha_\Lambda - 13.1], & \text{Linear Fit,} \\ \Delta N_{\text{br}} = (1.1 \pm 0.1)[\alpha_\Lambda - 13.1]^{0.88 \pm 0.06}, & \text{Power law Fit.} \end{cases} \quad (6.82)$$

where $\alpha_\Lambda = 13.1$ is the coupling at which the system enters the mild backreaction regime; see [4]. We can observe outlier data points in the plot corresponding to the $\alpha_\Lambda = 20$ case: the black square represents a simulation with $\mathcal{N} = 3072$, while the black circle corresponds to a failed simulation with $\mathcal{N} = 3648$, for which we extrapolated the results in order to estimate ΔN_{br} ; see Fig. 6.16. However, it can be seen that data point corresponding to $\alpha_\Lambda = 20$ completely falls out of the trend. The lattice simulation for this case did not reach the desired convergence criteria, most likely due to the lack of UV coverage in this particular setup. In Section 6.6, we will discuss the implications of UV coverage and how we set the convergence conditions for a series of simulations. Additionally, we can notice that the fitting formulas in Eq. (6.82) fall within the error bands of those provided in Eqs. (77) and (78) of Ref. [4], which were obtained from a different set of simulations but for an equivalent set of results

Useful information about the dynamics of the system can be obtained by examining the power spectrum. In Fig. 6.7, we show the power spectrum of the gauge field

amplitude plotted against the comoving momenta k , normalized by the Hubble horizon $1/(aH)$. Several features of the power spectrum can be highlighted, particularly those that illustrate why obtaining reliable results for large couplings is technically challenging. Let us start with the top panel, corresponding to the case $\alpha_\Lambda = 12$. We observe how the power spectrum grows in the linear regime while inflation is still ongoing, causing the comoving set of momenta to cross the horizon, i.e., $k/(aH) > 1 \rightarrow k/(aH) < 1$. In this case, inflation ends shortly after simulation begins, and the peak of excitation remains sub-horizon, closely tracking the horizon scale. We also observe that the peak is well resolved within the chosen window, exhibiting an exponential fall-off in the UV and an approximate fall-off in the IR. In the middle panel, corresponding to the case $\alpha_\Lambda = 14$, we observe that the growth of the power spectrum during inflation resembles that of the linear regime. Eventually, modes stop moving to super-horizon scales and later begin to move again, indicating the reentry into an inflationary phase. Once again, the peak of excitation remains at sub-horizon scales, approximately tracking the horizon. Finally, the bottom panel, depicting the case $\alpha_\Lambda = 18$ (the strong backreaction regime) displays several features that explain why this regime is so difficult to capture accurately on the lattice. First, the modes excited during inflation are exponentially redshifted from the UV to the IR, which is a typical feature of inflation. Since the same gauge modes that are excited and redshifted out of the horizon are the ones driving inflation, these modes freeze once they become sufficiently super-horizon. The most UV modes, which have been excited and pulled to near-horizon scales, become dominant. Again, the peak of excitation is located near the horizon. However, for this to occur, many decades of modes excited during the non-linear regime, originally deep inside the horizon, must become super-horizon. This exponential influx of modes from the UV to the IR needs a large separation of scales to properly describe the dynamics. In Section 6.6, we will describe how the dynamics of the gauge fields are altered and potentially misrepresented if the size of the lattice box is not properly chosen.

Finally, let us discuss the power spectrum of the inflaton field. As previously explained, the inflaton is initially completely homogeneous from the simulation's point of view. However, these fluctuations rapidly grow due to the local backreaction of the gauge fields. In Fig. 6.8, we show the time evolution of the inflaton power spectrum as a function of the comoving momenta normalized by the Hubble horizon. It can be seen that shortly after the initial time, the power spectrum grows following a shape similar to that of the gauge field amplitude, this occurs during the linear stage at the beginning of the simulation. However, once non-linearities begin to dominate the dynamics, a shift toward UV modes occurs and power is transferred to them. This is another clear indication of the system's non-linear nature and highlights the necessity of accounting for the local backreaction in order to properly follow the dynamics.

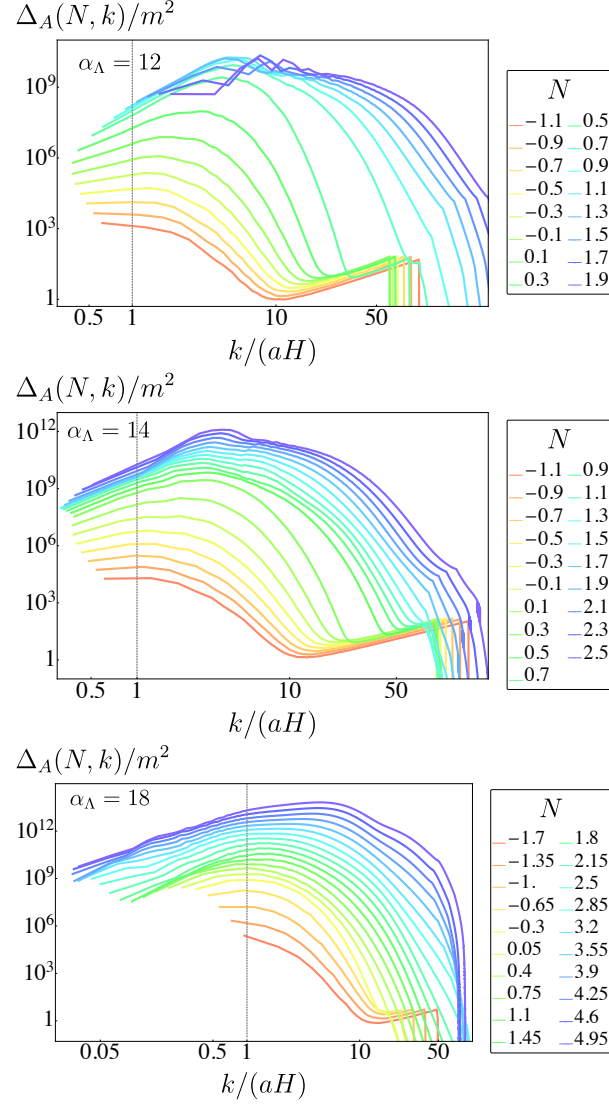


FIGURE 6.7: Power spectrum of the gauge field amplitude $\Delta_A(N, k)$ as a function of the comoving momentum normalized per the Hubble horizon, $k/(aH)$. Time is shown from warm to colder colors. Solid vertical line shows the position of the Hubble horizon.

6.5.2 Electromagnetic slow roll

In this section, we analyze the role played by the electric and magnetic fields in the electromagnetic (EM) energy budget. In Fig. 6.9, we show the time evolution of the individual energy fractions in the form of electric and magnetic fields, as well as the total EM energy. In the top-left panel, we display the evolution of EM energy for the weak backreaction regime ($\alpha_\Lambda = 12$). As anticipated, the backreaction occurs only after the end of inflation and is primarily driven by the electric field. However, the magnetic energy eventually takes over and becomes the dominant component of the EM energy. In this case, we observe an interesting phase: once EM energy dominates the total energy density of the universe, i.e. the universe has reheated, the electric

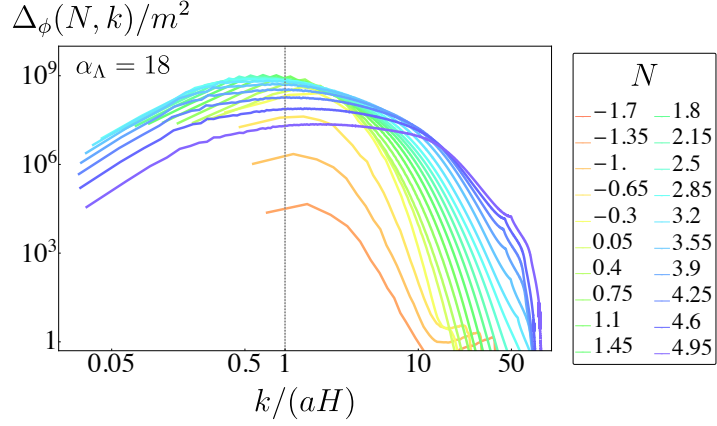


FIGURE 6.8: Power spectrum of the inflaton amplitude $\Delta_\phi(N, k)$ as a function of the comoving momentum normalized per the Hubble horizon, $k/(aH)$. Time is shown from warm to colder colors. Solid vertical line shows the position of the Hubble horizon.

and magnetic energies reach equipartition. This feature appears in any scenario where gauge reheating has been achieved. In the top-right panel, we present the case of mild backreaction ($\alpha_\Lambda = 14$). Here, the gauge fields grow to a certain level during inflation, which nevertheless ends, only to later reenter an inflationary phase. By the time inflation ends for good, the magnetic energy dominates, comprising nearly all the EM energy. After inflation ends, the EM energy continues to grow until, once again, the universe is effectively reheated and the electric and magnetic energies become equal. In the bottom panel, representing the strong backreaction regime ($\alpha_\Lambda = 18$), we observe the same trend. Initially, the electric energy dominates during the linear regime. As backreaction sets in and begins to drive the dynamics, the electric energy dominates over the kinetic energy in the initial stage. Inflation is still ongoing as the potential energy dominates the energy budget. Because of this, we call the phase *electromagnetic slow-roll*, in contrast to single-field slow-roll. The backreaction onto the background dynamics makes the inflaton perturbations grow, as can be seen in Fig. 6.10, where the root-mean-square of the inflaton grows and the volume-averaged inflaton trajectory momentarily seems to stop. The growth of the inflaton perturbations can also be observed in Fig. 6.8, where the power spectrum grows and then flattens due to the non-linear interactions with the gauge field. However, it is actually the magnetic energy that dominates the energy budget during this phase, reaching its maximum precisely when inflation ends. At the same time, the electric energy remains nearly stationary and begins to grow only toward the end, as the universe approaches radiation domination. Although the simulation does not reach this phase completely, a sudden increase in electric energy toward the end can still be seen. The dominance of magnetic over electric energy is further proof of the system's non-linear nature and reinforces the necessity of a true local backreaction description to properly capture the dynamics. We can also examine the power spectrum

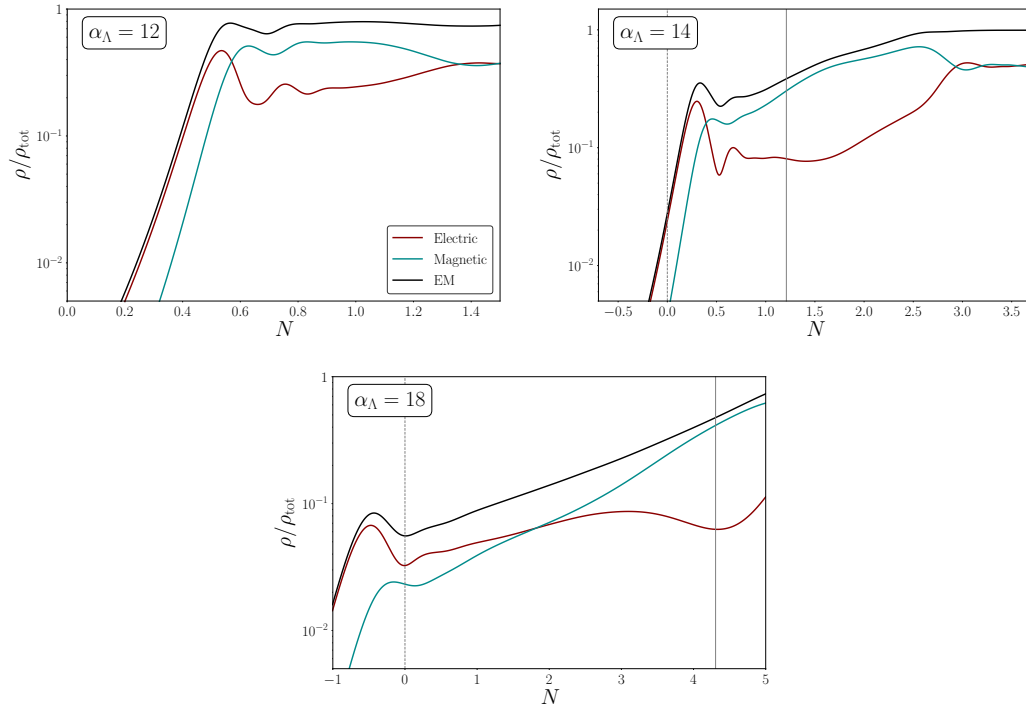


FIGURE 6.9: Time evolution of the electromagnetic energy components for the three regimes represented by the cases $\alpha_\Lambda = 12, 15, 18$. Dashed vertical line shows end of inflation for the slow roll case. Dashed vertical shows end of inflation when it is delayed due to backreaction.

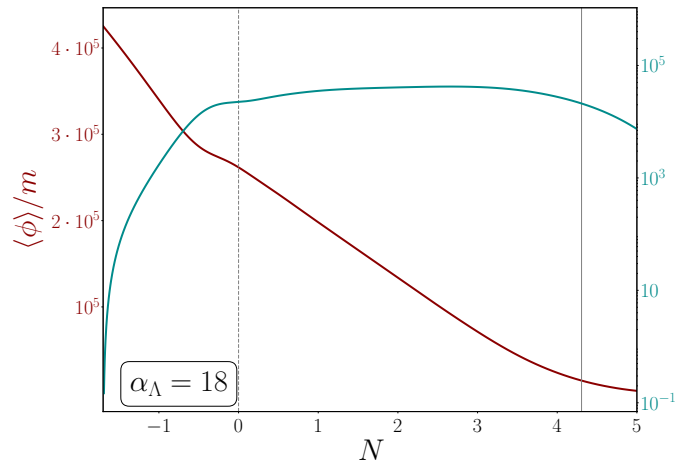


FIGURE 6.10: Time evolution of the inflaton volume average ($\langle\phi\rangle$) and the inflaton root mean square (ϕ_{rms}) with $\phi_{\text{rms}} = \sqrt{\langle\phi^2\rangle - \langle\phi\rangle^2}$ for the case $\alpha_\Lambda = 18$. Dashed vertical line shows end of inflation for the slow roll case. Dashed vertical shows end of inflation when it is delayed due to backreaction.

of each contribution. In Fig. 6.11, we show in solid lines the power spectrum of the electric field, $a^2\Delta_E(N, k)$, multiplied by the square of the scale factor to account for the redshifting due to the expansion of the universe. In dashed lines, we show the power spectrum of the magnetic field, $\Delta_B(N, k)$. On the right, we also plot a measure of how much the magnetic field dominates over the electric one as a function of the comoving momenta normalized per the Hubble horizon:

$$\delta_{\text{EM}}(N, k) \equiv \frac{\Delta_B(N, k) - a^2\Delta_E(N, k)}{\Delta_B(N, k) + a^2\Delta_E(N, k)}, \quad (6.83)$$

which becomes maximal, $\delta_{\text{EM}} = 1$, when the magnetic field dominates, and minimal, $\delta_{\text{EM}} = -1$, when the electric field dominates. In the top panels, for $\alpha_\Lambda = 12$, we see that initially both the electric and magnetic fields are present, and only at smaller scales (larger k) does a peak appear, this is just a feature of the initialization. As time advances, the peak of excitation emerges in the spectra, and there is a slight dominance of the magnetic field over sub-horizon scales that flattens over a range of comoving momenta. Eventually, due to the reheating of the universe, the electric and magnetic fields become comparable in the IR modes, while there remains a slight $\sim 50\%$ dominance of the magnetic field in the UV modes. In the middle panel, for $\alpha_\Lambda = 14$, the initial features of both electric and magnetic fields are again present. A clear distinction emerges: modes that become super-horizon and spend some time there become dominated by the electric field, whereas modes excited during the non-linear regime, which remain sub-horizon and peak at a certain scale, are dominated by the magnetic field and show a flattening over a range of UV momenta. Finally, in the bottom panel, for $\alpha_\Lambda = 18$, the same features described above become more pronounced over a wider range of momenta. Super-horizon scales are dominated by the electric field, while sub-horizon scales are dominated by the magnetic field. This also explains why the magnetic energy dominates over the electric energy, the integral of the power spectrum receives its largest contribution from the UV region rather than the IR.

6.5.3 Chiral imbalance

Historically, a key feature of the phenomenology of axion inflation has been the prediction of fully chiral excitation of gauge modes. This means that only gauge modes of a certain chirality are excited, while the other remains in the vacuum. In this case the excited chirality is the positive one. This property is also crucial for gravitational wave production, as the prediction states the emergence of a fully chiral gravitational wave background (GWB), which could serve as a *smoking gun* signature of axion inflation. However, all of these predictions were made within the context of the linear regime or under the assumption of homogeneous backreaction. In this section, we explore whether

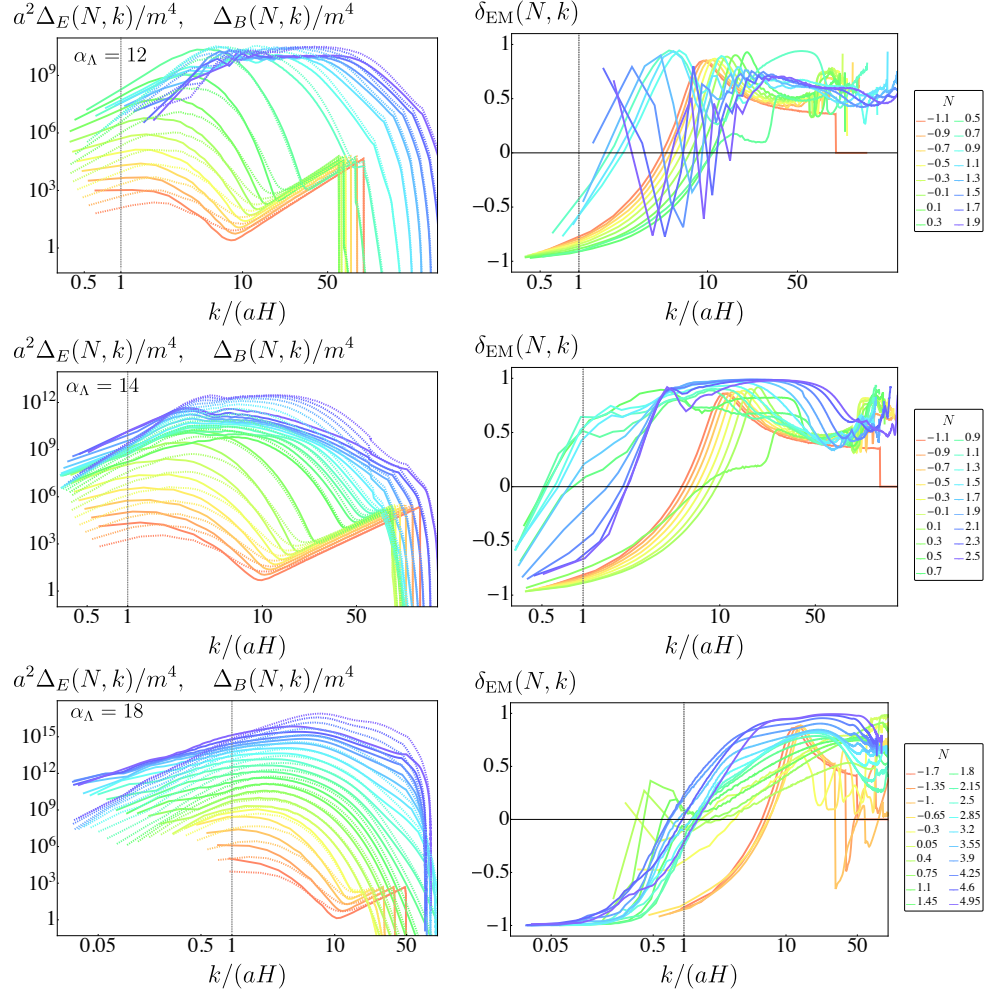


FIGURE 6.11: In the Left, power spectrum of the electric field amplitude $a^2 \Delta_E(N, k)$ corrected to account for the expansion of the universe and the magnetic field amplitude $\Delta_B(N, k)$ as a function of the comoving momentum normalized per the Hubble horizon, $k/(aH)$. In the right, fraction of electric or magnetic dominance $\delta_{EM}(N, k)$, where $\delta_{EM} = 1$ means magnetic dominance and $\delta_{EM} = -1$ is electric dominance. Time is shown from warm to colder colors. Solid vertical line shows the position of the Hubble horizon.

this situation persists or if a local treatment of the backreaction changes it. For this purpose, we examine the power spectrum of both chiral components of the gauge field amplitude. In the left panels of Fig. 6.12, we show the time evolution of the power spectra of the chiral components: the positive component $\Delta_A^+(N, k)$ is depicted in solid lines, and the negative component $\Delta_A^-(N, k)$ in dashed lines. We also define an observable that quantifies the level of chirality, or chiral imbalance, of the gauge field amplitude as a function of comoving momentum:

$$\mathcal{H}(N, k) \equiv \frac{\Delta_A^+(N, k) - \Delta_A^-(N, k)}{\Delta_A^+(N, k) + \Delta_A^-(N, k)}. \quad (6.84)$$

In the top panel, corresponding to the case $\alpha_\Lambda = 12$, we observe some key features in the power spectra of each chirality, which are expected from the initial conditions of the simulations. The positive chirality starts with an excited power spectrum, while the negative chirality begins in the vacuum state. We see that the negative chirality eventually grows, but at a slower rate than the positive one. The peak of excitation is dominated by the positive chirality, while toward the UV the net chirality is reduced, and $\mathcal{H}$ approaches zero. In the middle panel, for $\alpha_\Lambda = 14$, we again observe that initially $\mathcal{H} = 1$ for the excited range of the power spectrum. This net positive chirality is sustained for sub-horizon modes, particularly those around the peak of excitation, while modes that are super-horizon show a chirality imbalance of about $\sim 50\%$. Finally, in the bottom panel for the case $\alpha_\Lambda = 18$, the same pattern remains: modes at sub-horizon scales around the peak of excitation remain nearly fully chiral, while super-horizon modes are less chiral, with the chirality imbalance reduced to about $\sim 30\%$.

The emergence of negative chirality gauge fields can be understood only through the non-linear nature of the system. In Fig. 6.13, we show the time evolution of the integrated power spectrum, $\langle A_\sigma^2 \rangle$, for each chirality $\sigma = \{+, -, ||\}$, normalized by the total integrated power spectrum of the gauge field amplitude. The contribution of the longitudinal component is obtained by subtracting the positive and negative integrated power spectra from the full spectrum: $\langle A_{||}^2 \rangle = \langle A^2 \rangle - \langle A_+^2 \rangle - \langle A_-^2 \rangle$. For the case $\alpha_\Lambda = 18$, we observe, as expected, that the positive chirality, which grows exponentially, initially accounts for the totality of the gauge spectrum. Once non-linearities set in, both the longitudinal and negative modes begin to grow, with the longitudinal component dominating over the negative one. The longitudinal mode eventually constitutes nearly $\sim 10\%$ of the total gauge amplitude.

In order to illustrate this, we write the inflaton-gauge interaction terms from Eq. (6.5) in Fourier space,

$$(\pi_\phi \vec{B})_{\vec{k}} = -i \sum_{\lambda=\pm} \lambda \int d^3q \, \phi_{(\vec{k}-\vec{q})}^* q A_{\vec{q}}^\lambda \vec{\varepsilon}_{\vec{q}}^\lambda, \quad (6.85)$$

$$(\vec{\nabla} \phi \times \vec{E})_{\vec{k}} = i \sum_{\sigma=\pm, ||} \int d^3q \, \phi_{(\vec{k}-\vec{q})}^* \dot{A}_{\vec{q}}^\sigma \left((\vec{k} - \vec{q}) \times \vec{\varepsilon}_{\vec{q}}^\sigma \right), \quad (6.86)$$

where we explicitly show that $\vec{E}$ contains projections along all three components of the chiral basis ($\sigma = -, +, ||$), whereas $\vec{B}$ is transverse ($\lambda = +, -$), as $\vec{\nabla} \cdot \vec{B} = 0$. Given that the integrals in the convolutions span all momenta $\vec{q}$, we observe that for $\vec{q} \not\propto \vec{k}$, the projections $\vec{k} \cdot \vec{\varepsilon}_{\vec{q}}^\lambda \neq 0$ and $\vec{\varepsilon}_{\vec{k}}^\lambda \cdot \vec{\varepsilon}_{\vec{q}}^\lambda \neq 0$ are generally non-zero, since they depend on the angle between the external momentum $\vec{k}$ and the internal modes $\vec{q}$, defined via $\cos \theta \equiv (\vec{k} \cdot \vec{q}) / (|\vec{k}| |\vec{q}|)$, which spans values from -1 to $+1$. This implies that there will

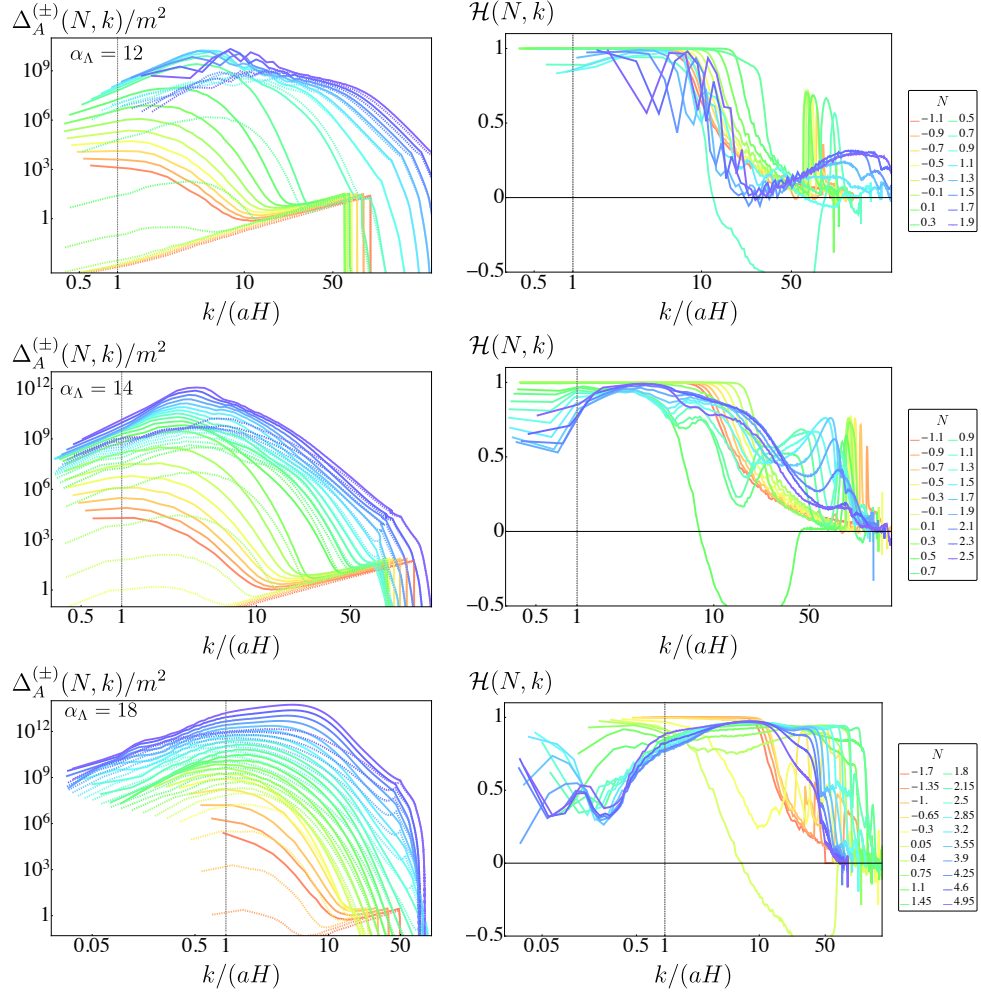


FIGURE 6.12: In the Left, power spectrum of the chiral components of gauge field amplitude $\Delta_A^{\pm}(N, k)$ as a function of the comoving momentum normalized per the Hubble horizon, $k/(aH)$. In the right, fraction of chiral imbalance $\mathcal{H}(N, k)$, where $\mathcal{H} = 1$ means positive chirality dominance and $\mathcal{H} = -1$ is negative dominance. Time is shown from warm to colder colors. Solid vertical line shows the position of the Hubble horizon.

always be non-zero projections of $(\pi_\phi \vec{B})_{\vec{k}}$ and $(\vec{\nabla} \phi \times \vec{E})_{\vec{k}}$ onto both chiralities $\pm$ and onto the longitudinal mode.

For example, in the *com* for the gauge field, the negative chirality $A_{\vec{k}}^-$ will now be sourced by the component $(\pi_\phi \vec{B})_{\vec{k}}^- \equiv \vec{\varepsilon}_{\vec{k}}^+ \cdot (\pi_\phi \vec{B})_{\vec{k}}$, which is given by

$$\begin{aligned}
 (\pi_\phi \vec{B})_{\vec{k}}^- &\equiv -i \int d^3 \vec{q} \, \phi_{(\vec{k}-\vec{q})}^* q \left(A_{\vec{q}}^+ (\vec{\varepsilon}_{\vec{k}}^+ \cdot \vec{\varepsilon}_{\vec{q}}^+) - A_{\vec{q}}^- (\vec{\varepsilon}_{\vec{k}}^+ \cdot \vec{\varepsilon}_{\vec{q}}^-) \right), \\
 &\simeq -\frac{i}{2} \int d^2 \hat{q} \, dq \, q^3 \, \phi_{(\vec{k}-\vec{q})}^* (1 - \cos \theta) A_{\vec{q}}^+, \tag{6.87}
 \end{aligned}$$

where we used $\vec{\varepsilon}_{\vec{k}}^+ \cdot \vec{\varepsilon}_{\vec{q}}^\lambda = (1 - \lambda \cos \theta)/2$, and in the second line we assumed $|A_{\vec{q}}^+| \gg |A_{\vec{q}}^-|$, which holds during the initial stages of non-linear dynamics over the region of excitation.

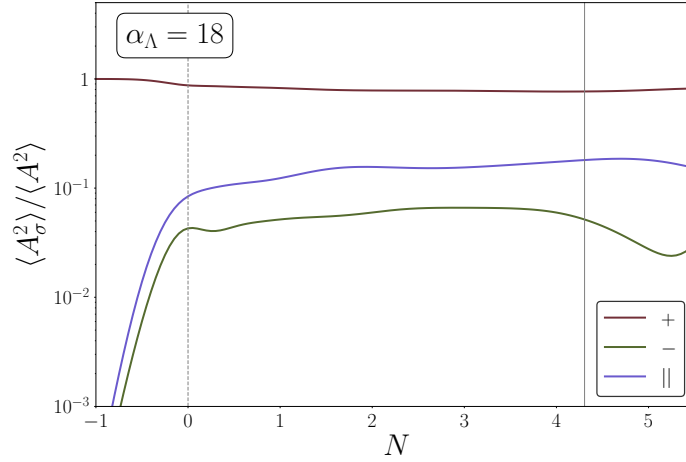


FIGURE 6.13: Time evolution of the integrated power spectrum of the chiral components of the gauge field amplitude normalized per total gauge field amplitude. Dashed vertical lines shows end of slow roll inflation, vertical solid line shows delayed end of inflation for the corresponding case.

Similarly, we can obtain analogous expressions for $(\pi_\phi \vec{B})_{\vec{k}}^\parallel \equiv \hat{k} \cdot (\pi_\phi \vec{B})_{\vec{k}}$, which sources $A_{\vec{k}}^\parallel$, and also for $(\vec{\nabla} \phi \times \vec{E})_{\vec{k}}^- \equiv \vec{\varepsilon}_{\vec{k}}^+ \cdot (\vec{\nabla} \phi \times \vec{E})_{\vec{k}}$ and $(\vec{\nabla} \phi \times \vec{E})_{\vec{k}}^\parallel \equiv \hat{k} \cdot (\vec{\nabla} \phi \times \vec{E})_{\vec{k}}$, which further source $A_{\vec{k}}^-$ and $A_{\vec{k}}^\parallel$, respectively. Remarkably, this means that the negative and longitudinal components of the gauge field will be sourced to large amplitudes by the exponentially excited tachyonic modes $A_{\vec{q}}^+$, as explicitly shown by the expression in the second line of Eq. (6.87).

If the inflaton were to be homogeneous, $\vec{q}$ collapses to $\vec{k}$, and there is no power transfer between different components. In Eq. (6.87), this is understood as $\cos \theta \rightarrow 1$ in the second line. More generally, all components $(\pi_\phi \vec{B})_{\vec{k}}^-$, $(\pi_\phi \vec{B})_{\vec{k}}^\parallel$, $(\vec{\nabla} \phi \times \vec{E})_{\vec{k}}^-$, and $(\vec{\nabla} \phi \times \vec{E})_{\vec{k}}^\parallel$ vanish if the inflaton is homogeneous, because $\vec{\varepsilon}_{\vec{k}}^+ \cdot \vec{\varepsilon}_{\vec{k}}^+ = 0$ and $\hat{k} \cdot \vec{\varepsilon}_{\vec{k}}^\pm = 0$. In other words, only if the inflaton develops inhomogeneities will the inflaton-gauge interaction terms act as sources for the longitudinal and negative chirality components. The inhomogeneities generated during the local backreaction dynamics, especially those of the inflaton, are responsible for the excitation of the longitudinal and negative chirality components. This once again highlights that only the local description of backreaction can truly capture the inhomogeneous non-linear dynamics of axion inflation.

6.6 Convergence of the simulations and UV sensitivity

In order to validate our results and ensure they are independent of the lattice parameters, we perform several lattice simulations with different UV resolutions. In Fig. 6.14,

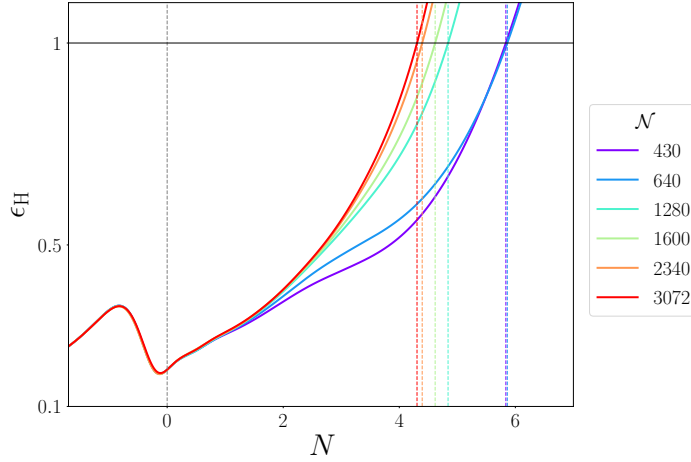


FIGURE 6.14: Time evolution of slow roll parameter ϵ_H for different UV resolutions $\mathcal{N}$ for the case $\alpha_\Lambda = 18$.

we show the time evolution of the slow-roll parameter ϵ_H for the case $\alpha_\Lambda = 18$ and various lattice resolutions. This means we fix the IR cutoff k_{IR} and increase the UV cutoff according to $k_{\text{UV}} = \frac{\sqrt{3}}{2}\mathcal{N}k_{\text{IR}}$. We observe that for the case with the poorest UV resolution, $\mathcal{N} = 430$, the end of inflation is delayed by approximately ~ 6 e-folds compared to the baseline slow-roll scenario. Inflation ends due to the energetic dominance of the gauge fields over the kinetic energy. However, with low UV resolution, many of the modes that should be excited are not captured, and the 'true' amplitude of the gauge fields is underestimated. This causes inflation to last significantly longer. Another consequence of insufficient resolution is that the peak of excitation appears near the horizon scale, as can be seen in Fig. 6.15. As the resolution improves, the peak of excitation shifts toward sub-horizon scales, and more modes are excited, contributing to a larger gauge field amplitude. This means that the better the UV resolution, the more accurately the true amplitude of the gauge field is captured. Consequently, the energetic dominance of the gauge fields over the kinetic energy and thus the end of inflation, which occurs when $\rho_{\text{EM}} = \rho_V$, happens earlier.

Increasing the UV resolution eventually leads to saturation toward a fixed value for the delayed end of inflation. We test the convergence of a series of simulations by tracking the delay at the end of inflation for the three largest runs. The observed delays are $\Delta N_{\text{br}} = [4.6177, 4.3944, 4.3054]$ for $\mathcal{N} = [1600, 2340, 3072]$. We compute

$$R = \frac{|\Delta N_{\text{br}}^{\text{L}} - \Delta N_{\text{br}}^{\text{M}}|}{|\Delta N_{\text{br}}^{\text{M}} - \Delta N_{\text{br}}^{\text{H}}|} \approx 2.5089, \quad (6.88)$$

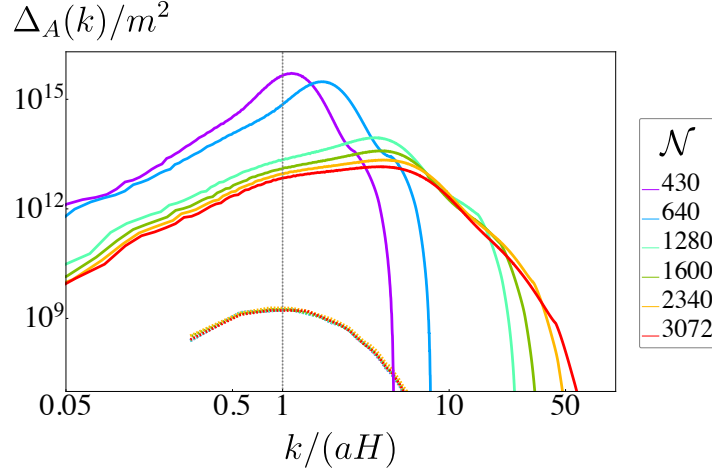


FIGURE 6.15: Power spectrum of the gauge field amplitude $\Delta_A(N, k)$ as a function of the comoving momentum normalized per the Hubble horizon, $k/(aH)$ for different UV resolutions $\mathcal{N}$. Dashed lines correspond to power spectrum measure at $N = 0$, solid lines correspond to power spectrum measured at end of inflation for each UV resolution. Solid vertical line shows the position of the Hubble horizon.

where L, M, H denote Low, Mid and High resolution, corresponding to $\mathcal{N}^L = 1600$, $\mathcal{N}^M = 2340$, $\mathcal{N}^H = 3072$. The order of convergence p is obtained by solving

$$R = \frac{|\mathcal{N}_L^{-p} - \mathcal{N}_M^{-p}|}{|\mathcal{N}_M^{-p} - \mathcal{N}_H^{-p}|}, \quad (6.89)$$

which yields $p \approx 1.77$, consistent with first-to-second order convergence.

In Fig. 6.15, we show the power spectrum of the gauge field at two different times. The dashed lines correspond to $N = 0$, where we see that all spectra overlap perfectly. This indicates that the UV resolution in each case is sufficient to support the initial exponential growth of the gauge fields. This behavior is also visible in Fig. 6.14, where the early-time evolution of ϵ_H overlaps across simulations. However, as time progresses, the trajectories for simulations with insufficient resolution begin to deviate and develop a plateau. We argue that the appearance of a plateau in the time evolution of ϵ_H is a clear indication of poor UV resolution. The other moment at which the power spectrum is plotted corresponds to the end of inflation for each resolution. Here, we observe that the spectrum becomes flatter and extends toward sub-horizon scales. Moreover, the power spectra of the two highest-resolution simulations nearly overlap. The small difference is due to the fact that they are shown at slightly different times. Additionally, we note that for a simulation with good UV resolution, no plateau appears in the evolution of ϵ_H . For this reason, we discard the result for the case $\alpha_\Lambda = 20$ in Fig. 6.4, because a plateau appears and is later corroborated by the fact that this result lies outside the linear trend. Nevertheless, we have seen that convergence of this case is close to be reached

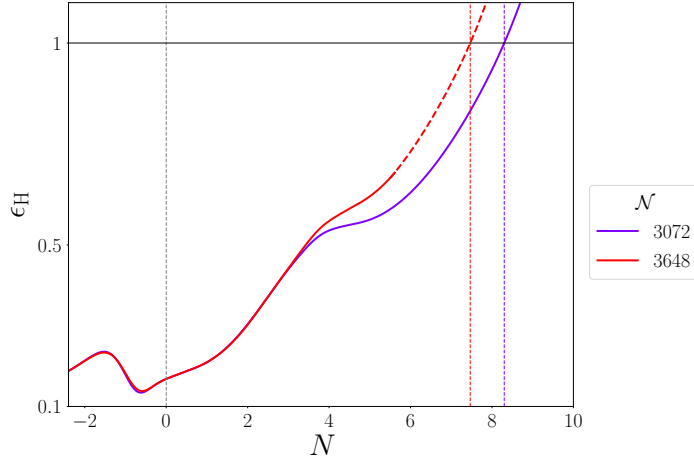


FIGURE 6.16: Time evolution of slow roll parameter ϵ_H for different UV resolutions $\mathcal{N}$ for the case $\alpha_\Lambda = 20$.

as a simulation with better UV resolution ($\mathcal{N} = 3648$) brings the predicted delay on the end of inflation down to $\Delta N_{\text{br}} \sim 7.5$, as seen in Fig. 6.16. Here, we show how the extrapolation (dashed lines) of a failed simulation points in the direction of convergence of simulations with better UV resolution. From a qualitative point of view, we expect to reach convergence for the case $\alpha_\Lambda = 20$ when the plateau disappears. However, a detailed convergence study with higher-resolution simulations is still required in order to discard the featured plateau as a nonphysical defect.

6.7 Discussion

Axion inflation provides a natural framework for a realistic particle physics realization of inflation. In particular, thanks to its shift symmetry, inflation is protected from quantum corrections. It is also phenomenologically very rich, offering a variety of unique observational signatures. In this respect, the accurate description of the dynamics is a necessary condition to correctly assess the observational predictions that can be used to probe this scenario. Previous approaches to study the dynamics of axion inflation, based on increasing levels of complexity, somehow fail to capture the true behavior of the system, in particular of the strong backreaction dynamics. In order to describe accurately the latter regime, it is necessary to resort to a fully local treatment of the dynamics, which only real time field theory lattice simulations can provide.

Here we have shown a proper lattice formulation of axion inflation that makes the symmetries of the theory explicit, and satisfies continuum constraints. Moreover, our lattice formulation agrees very well with the linear regime, which is exactly solvable.

Hence, it is validated in this way by these tests and sets the standard for performing lattice simulations for axion inflation. A series of simulations have been performed for a range of couplings, identifying three regimes: weak-, mild- and strong-backreaction, depending on the extension of inflation. For each coupling, except for the biggest coupling considered, we achieve convergence of simulations with increasingly better UV resolution. Our results show the difficulty in performing simulations for large couplings ($\alpha_\Lambda \geq 20$), as these require an increasingly larger separation of scales, which translates into increasingly expensive computational capabilities. In particular, our quantitative results, summarized *e.g.* in our fits in Eq. (6.82), are the only accurate parametrization of the delay of the end of inflation for large couplings that have reached convergence.

The truly local nature of strong backreaction has made us understand how the electromagnetic energy density eventually drives inflation towards an end, in a new regime that we coin “(electro)magnetic slow-roll”. We observe that the magnetic field actually contributes the most to the gauge field energy, way above the electric counterpart (especially around the peak of the spectra). The lattice study shows the need to resolve well the peak of excitation, which is very sensitive to the lattice UV cutoff. In this respect, we show the dependence of the end of inflation on the UV lattice resolution, as we obtain only convergence among simulations (for a given fixed coupling) if the UV scales simulated are sufficiently separated away from the initial IR scales required for the linear regime. We demonstrate in full detail how not resolving properly the necessary UV coverage leads to physically wrong results. The non-trivial evolution of the comoving Hubble scale is another feature that shows how the strong backreaction dynamics require a very large separation of IR-to-UV scales, in order to resolve properly the peak of excitation of the gauge field. We also find that the system does not remain completely chiral, as the non-tachyonic chirality and the longitudinal mode are generated during the non-linear dynamics of inflation. We show that this is effect is driven uniquely by the fact that the inflaton becomes inhomogeneous, as this leads to mode-couplings between the chiral and longitudinal components of the gauge field, that would not be present if the inflaton remained homogeneous. We observe that in the strong backreaction cases the system reached radiation domination soon after the end of inflation, with the gauge field dominating the total energy budget, presenting a non-trivial coupling- and scale-dependent partial balance between chiralities.

Chapter 7

Cosmological perturbations sourced by non-linear dynamics

We have emphasized the necessity of lattice simulations to accurately follow the dynamics of systems where particle production can backreact on the evolution of the background. We have also shown that these highly energetic phenomena are efficient sources of gravitational waves, a form of cosmological perturbation. Therefore, we expect that other types of perturbations may also be produced. In particular, the large excitation of scalar perturbations could lead to phenomenological consequences that serve as additional observational probes or constraints on these scenarios. For example as explained in Chapter 6, axion inflation sources scalar perturbations through the inverse decay of gauge quanta into inflaton fluctuations, a process that becomes relevant during the non-linear regime. These scalar perturbations could lead to the formation of primordial black holes or generate non-Gaussian signatures in the bispectrum of scalar perturbations. Although current observations of the CMB and large-scale structure (LSS) have not revealed evidence for non-Gaussianities, upcoming experiments with improved sensitivity will be able to better constrain, or potentially observe, such effects. Moreover, gravitational perturbations could themselves backreact on the dynamics, challenging the common assumption that matter drives the metric solely through its homogeneous component.

In this regard, we have developed a formulation to track the production of cosmological perturbations arising from the non-linear dynamics of interacting scalar fields (which could be easily extended to other bosonic degrees of freedom). In this framework, we consider linear metric perturbations on top of a FLRW background, while retaining the full non-linear interacting Lagrangian for the matter sector. In what follows, we introduce the formalism and describe how one can measure the spectrum, or even

the full Probability Density Function (PDF), of comoving curvature perturbations. Alternatively, one could consider full general relativity in cosmological simulations (see [325]), which introduces significant additional complexity but offers a treatment of the full non-linear dynamics of the gravitational degrees of freedom.

7.1 Linear perturbations of the metric

Let us consider a perturbed metric up to linear order. Notice, that we do not write it in Scalar-Vector-Tensor decomposition which will be understood later

$$ds^2 = -(1 + 2\delta N)dt^2 + a \delta N_i dx^i dt + a^2(\delta_{ij} + \gamma_{ij})dx^i dx^j . \quad (7.1)$$

We will write the quantities in terms of the background and perturbed components, for instance $g_{**} = \bar{g}_{**} + \delta g_{**}$ or $\Gamma_{**}^* = \bar{\Gamma}_{**}^* + \delta \Gamma_{**}^*$. We can write the background connection coefficients as

$$\bar{\Gamma}_{00}^0 = 0 , \quad \bar{\Gamma}_{i0}^0 = 0 , \quad (7.2)$$

$$\bar{\Gamma}_{00}^i = 0 , \quad \bar{\Gamma}_{ij}^0 = a\dot{a}\delta_{ij} , \quad (7.3)$$

$$\bar{\Gamma}_{j0}^i = H\delta_j^i , \quad \bar{\Gamma}_{jk}^i = 0 , \quad (7.4)$$

and the perturbed ones as

$$\delta\Gamma_{00}^0 = \delta\dot{N} , \quad (7.5)$$

$$\delta\Gamma_{i0}^0 = \partial_i\delta N + \dot{a}\delta N_i , \quad (7.6)$$

$$\delta\Gamma_{00}^i = a^{-2}(\partial^i\delta N) + a^{-1}H\delta N^i + a^{-1}\delta\dot{N}^i , \quad (7.7)$$

$$\delta\Gamma_{ij}^0 = -2a\dot{a}\delta N\delta_{ij} - \frac{a}{2}(\partial_i\delta N_j + \partial_j\delta N_i) + a\dot{a}\gamma_{ij} + \frac{a^2}{2}\dot{\gamma}_{ij} , \quad (7.8)$$

$$\delta\Gamma_{j0}^i = \frac{a^{-1}}{2}[\partial_j(\delta N^i) - \partial^i(\delta N_j)] + \frac{1}{2}\dot{\gamma}_j^i , \quad (7.9)$$

$$\delta\Gamma_{jk}^i = -\dot{a}\delta_{jk}\delta N^i + \frac{1}{2}(\partial_j\gamma_k^i + \partial_k\gamma_j^i - \partial^i\gamma_{jk}) . \quad (7.10)$$

Now we can write the background Ricci Tensor $\bar{R}_{\mu\nu}$

$$\bar{R}_{00} = -3(\dot{H} + H^2) , \quad (7.11)$$

$$\bar{R}_{0i} = 0 , \quad (7.12)$$

$$\bar{R}_{ij} = (2\dot{a}^2 + a\ddot{a})\delta_{ij} , \quad (7.13)$$

$$(7.14)$$

and the perturbed Ricci Tensor $\delta R_{\mu\nu}$

$$\delta R_{00} = a^{-2} \nabla^2 \delta N + 3H \delta \dot{N} + a^{-1} H (\partial_k \delta N_k) + a^{-1} (\partial_k \delta \dot{N}_k) - H \dot{\gamma} - \frac{1}{2} \ddot{\gamma} , \quad (7.15)$$

$$\begin{aligned} \delta R_{0i} = & 2H \partial_i (\delta N) - (2\ddot{a} + \dot{a}H) \delta N_i + \frac{a^{-1}}{2} [\partial_i (\partial_k \delta N_k) - \nabla^2 \delta N_i] \\ & + \frac{1}{2} \partial_j (\dot{\gamma}_{ij}) - \frac{1}{2} \partial_i (\dot{\gamma}) , \end{aligned} \quad (7.16)$$

$$\begin{aligned} \delta R_{ij} = & (-4\dot{a}^2 \delta N - a\dot{a}\delta \dot{N} - 2a\ddot{a}\delta N) \delta_{ij} - \partial_i \partial_j \delta N \\ & - \dot{a} (\partial_i \delta N_j + \partial_j \delta N_i) - \frac{a}{2} (\partial_i \delta \dot{N}_j + \partial_j \delta \dot{N}_i) - \dot{a} (\partial_k \delta N_k) \delta_{ij} \\ & + (2\dot{a}^2 + a\ddot{a}) \gamma_{ij} + \frac{a^2}{2} \ddot{\gamma}_{ij} + \frac{3}{2} a \dot{a} \dot{\gamma}_{ij} - \frac{\nabla^2}{2} \gamma_{ij} + \frac{1}{2} a \dot{a} \dot{\gamma} \delta_{ij} - \partial_i \partial_j \gamma \\ & + \frac{1}{2} (\partial_k \partial_i \gamma_{kj} + \partial_k \partial_j \gamma_{ki}) . \end{aligned} \quad (7.17)$$

The background Ricci Scalar $\bar{R} = \bar{g}^{\mu\nu} \bar{R}_{\mu\nu}$ is

$$\bar{R} = 6 \left(\left(\frac{\dot{a}}{a} \right)^2 + \frac{\ddot{a}}{a} \right) , \quad (7.18)$$

and the perturbed one δR is

$$\begin{aligned} \delta R = & -2a^{-1} \nabla^2 \delta N - 12 \left(\left(\frac{\dot{a}}{a} \right)^2 + \frac{\ddot{a}}{a} \right) \delta N - 6H \delta \dot{N} \\ & - 6a^{-1} H (\partial_k \delta N_k) - 2a^{-1} (\partial_k \delta \dot{N}_k) \\ & + 4H \dot{\gamma} + \ddot{\gamma} - a^{-2} \nabla^2 \gamma + a^{-2} (\partial_k \partial_m \gamma_{km}) . \end{aligned} \quad (7.19)$$

We can also write the background Einstein tensor for the 00, 0i, ij-components

$$\bar{G}_{00} = 3H^2 , \quad (7.20)$$

$$\bar{G}_{0i} = 0 , \quad (7.21)$$

$$\bar{G}_{ij} = -(\dot{a}^2 + 2a\ddot{a}) \delta_{ij} , \quad (7.22)$$

and the perturbed ones

$$\delta G_{00} = 2a^{-1} H (\partial_k \delta N_k) + H \dot{\gamma} - \frac{a^{-2}}{2} \nabla^2 \gamma + \frac{a^{-2}}{2} \partial_k \partial_l (\gamma_{kl}) , \quad (7.23)$$

$$\begin{aligned}
\delta G_{0i} = & 2H\partial_i(\delta N) - (2\ddot{a} + \dot{a}H)\delta N_i + \frac{a^{-1}}{2}[\partial_i(\partial_k\delta N_k) - \nabla^2\delta N_i] \\
& + \frac{1}{2}\partial_j(\dot{\gamma}_{ij}) - \frac{1}{2}\partial_i(\dot{\gamma}) ,
\end{aligned} \tag{7.24}$$

$$\begin{aligned}
\delta G_{ij} = & (2\dot{a}^2\delta N + 4a\ddot{a}\delta N + 2a\dot{a}\delta\dot{N} + \nabla^2\delta N)\delta_{ij} - \partial_i\partial_j(\delta N) \\
& + (2\dot{a}(\partial_k\delta N_k) + a(\partial_k\delta\dot{N}_k))\delta_{ij} - \frac{a}{2}(\partial_i\delta\dot{N}_j + \partial_j\delta\dot{N}_i) - \dot{a}(\partial_i\delta N_j + \partial_j\delta N_i) \\
& + \frac{a^2}{2}\ddot{\gamma}_{ij} + \frac{3}{2}a\dot{a}\dot{\gamma}_{ij} - \frac{\nabla^2}{2}\gamma_{ij} - (\dot{a}^2 + 2a\ddot{a})\gamma_{ij} \\
& - \left(\frac{a^2}{2}\ddot{\gamma} + \frac{3}{2}a\dot{a}\dot{\gamma} - \frac{\nabla^2}{2}\gamma\right)\delta_{ij} \\
& - \frac{1}{2}\partial_i\partial_j\gamma + \frac{1}{2}(\partial_k\partial_i\gamma_{kj} + \partial_k\partial_j\gamma_{ki}) - \frac{1}{2}(\partial_k\partial_m\gamma_{km})\delta_{ij} .
\end{aligned} \tag{7.25}$$

7.1.1 Einstein equations order by order

First, let us consider the action

$$S = \int d^4x \sqrt{-g} \left(-\frac{1}{2}g^{\mu\nu}\partial_\mu\phi_a\partial_\nu\phi_a - V(\{\phi_a\}) \right) , \tag{7.26}$$

where $T_{\mu\nu}$ reads as

$$T_{\mu\nu} = \partial_\mu\phi\partial_\nu\phi - g_{\mu\nu} \left(\frac{1}{2}g^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi + V(\{\phi\}) \right) . \tag{7.27}$$

The components of the energy-momentum tensor are

$$\begin{aligned}
T_{00} = & \frac{1}{2}\dot{\phi}^2 + \frac{1}{2a^2}(\vec{\nabla}\phi)^2 + V(\phi) \\
& + 2\delta N \left(-\frac{1}{2}\dot{\phi}^2 + \frac{1}{2}a^{-2}(\vec{\nabla}\phi)^2 + V(\phi) \right) \\
& + \delta N\dot{\phi}^2 + a^{-1}\dot{\phi}(\partial_k\phi)\delta N_k - \frac{1}{2}a^{-2}\gamma_{km}\partial_k\phi\partial_m\phi ,
\end{aligned} \tag{7.28}$$

$$\begin{aligned}
T_{0i} = & \dot{\phi}\partial_i\phi \\
& - a\delta N_i \left(-\frac{1}{2}\dot{\phi}^2 + \frac{1}{2}a^{-2}(\vec{\nabla}\phi)^2 + V(\phi) \right) ,
\end{aligned} \tag{7.29}$$

$$T_{ij} = \partial_i\phi\partial_j\phi - a^2\delta_{ij} \left(-\frac{1}{2}\dot{\phi}^2 + \frac{1}{2}a^{-2}(\vec{\nabla}\phi)^2 + V(\phi) \right) \tag{7.30}$$

$$\begin{aligned}
& - a^2\delta_{ij} \left(\delta N\dot{\phi}^2 + a^{-1}\dot{\phi}(\partial_k\phi)\delta N_k - \frac{1}{2}a^{-2}\gamma_{km}\partial_k\phi\partial_m\phi \right) \\
& - a^2\gamma_{ij} \left(-\frac{1}{2}\dot{\phi}^2 + \frac{1}{2}a^{-2}(\vec{\nabla}\phi)^2 + V(\phi) \right) .
\end{aligned} \tag{7.31}$$

We will obtain Einstein equations order by order. First, let us recall

$$G_{\mu\nu} = m_p^{-2} T_{\mu\nu} . \quad (7.32)$$

We can decompose the *lhs* of the equation into background and perturbed terms as $G_{**} = \bar{G}_{**} + \delta G_{**}$, where $\bar{G}$ is the Einstein tensor built from the background metric (in this case FLRW), and δG is the Einstein tensor built from the linear perturbations of the metric. We will not keep track of indices, as the equations will be presented schematically. We can then say $\bar{G}_{**} \sim \mathcal{O}(\epsilon^0)$ and $\delta G_{**} \sim \mathcal{O}(\epsilon)$, where $\epsilon \ll 1$,

$$\bar{G}_{**} + \delta G_{**} + \mathcal{O}(\epsilon^2) = m_p^{-2} T_{**} . \quad (7.33)$$

Now, let us assume a similar decomposition exists for the energy-momentum tensor T_{**}

$$T_{**} = \bar{T}_{**} + \delta T_{**} , \quad (7.34)$$

where $\bar{T}_{**}$ is the energy-momentum tensor obtained from the variation of the matter Lagrangian with respect to the background metric, and δT is the part of the decomposition proportional to both linear metric perturbations and matter perturbations, which could be of order $\mathcal{O}(\epsilon)$ or $\mathcal{O}(\epsilon^2)$. However, we can already notice that $\bar{T}_{**}$ is not of order $\mathcal{O}(\epsilon^0)$, as it contains the homogeneous mode of the matter sector and matter perturbations on top of it. We define the homogeneous part of the energy-momentum tensor as its volume average, denoted by $\langle \bar{T}_{**} \rangle$. Let us add and subtract $\langle \bar{T}_{**} \rangle$ from T_{**}

$$T_{**} = \bar{T}_{**}^m + \delta T_{**} + \langle \bar{T}_{**}^m \rangle - \langle \bar{T}_{**}^m \rangle . \quad (7.35)$$

We can now identify the term $\langle \bar{T}_{**} \rangle$ to be of order $\mathcal{O}(\epsilon^0)$, as it satisfies the symmetries of the zeroth-order Einstein equations; that is, it does not depend on the spatial coordinates. On the other hand, we have the term $\bar{T}_{**} - \langle \bar{T}_{**} \rangle$, which is the local fluctuation of the energy-momentum tensor of the matter sector and is of order $\mathcal{O}(\epsilon)$. We can then write Einstein's equations order by order

$$\bar{G}_{**} + \delta G_{**} + \mathcal{O}(\epsilon^2) = m_p^{-2} (\bar{T}_{**}^m + \delta T_{**} + \langle \bar{T}_{**}^m \rangle - \langle \bar{T}_{**}^m \rangle) , \quad (7.36)$$

At order $\mathcal{O}(\epsilon^0)$ we have:

$$\bar{G}_{**} = m_p^{-2} \langle \bar{T}_{**}^m \rangle , \quad (7.37)$$

while at order $\mathcal{O}(\epsilon)$ we have:

$$\delta G_{**} = m_p^{-2} (\bar{T}_{**}^m - \langle \bar{T}_{**}^m \rangle + \delta T_{**}) . \quad (7.38)$$

Finally, we focus on the term δT_{**} . This contains terms that are proportional to the linear metric perturbations; hence it includes terms of the form $\delta T_{**} \sim (\delta t[\phi, \dot{\phi}, \partial_i \phi] \cdot \delta g)_{**}$, where $\delta t[\phi, \dot{\phi}, \partial_i \phi]$ refers to terms in the energy-momentum tensor depending on the field amplitude and its temporal and spatial derivatives, and δg are terms proportional to the linear metric perturbations, which are of $\mathcal{O}(\epsilon)$. The $\delta t[\phi, \dot{\phi}, \partial_i \phi]$ terms can be further decomposed into local and volume average components of different orders, where $\langle \delta t[\phi, \dot{\phi}, \partial_i \phi] \rangle \sim \mathcal{O}(\epsilon^0)$ and $(\delta t[\phi, \dot{\phi}, \partial_i \phi] - \langle \delta t[\phi, \dot{\phi}, \partial_i \phi] \rangle) \sim \mathcal{O}(\epsilon)$. If we keep only terms up to $\mathcal{O}(\epsilon)$, we obtain:

$$\delta G_{**} = m_p^{-2} \left(\bar{T}_{**} - \langle \bar{T}_{**}^m \rangle + (\delta g \langle \delta t[\phi, \dot{\phi}, \partial_i \phi] \rangle)_{**} \right). \quad (7.39)$$

Now we can write down the equations order by order for the specific matter content of Eq. (7.28). The Einstein equations now reads for $\mathcal{O}(\epsilon^0)$ as

$$3H^2 = \frac{1}{m_p^2} \left\langle \frac{1}{2} \dot{\phi}^2 + \frac{1}{2a^2} (\vec{\nabla} \phi)^2 + V(\phi) \right\rangle = \frac{1}{m_p^2} \langle \rho \rangle, \quad (7.40)$$

$$-(\dot{a}^2 + 2a\ddot{a})\delta_{ij} = \frac{1}{m_p^2} \left[-a^2 \delta_{ij} \left\langle -\frac{1}{2} \dot{\phi}^2 + \frac{1}{6} a^{-2} (\vec{\nabla} \phi)^2 + V(\phi) \right\rangle \right] = \frac{a^2}{m_p^2} \delta_{ij} \langle p \rangle. \quad (7.41)$$

Subsequently, we can write down the equations at $\mathcal{O}(\epsilon)$ in the perturbations

$$\begin{aligned} -2a^{-1}H(\partial_k \delta N_k) + H\dot{\gamma} - \frac{a^{-2}}{2} \nabla^2 \gamma + \frac{a^{-2}}{2} \partial_k \partial_l (\gamma_{kl}) = \\ + \frac{1}{m_p^2} \left[\rho - \langle \rho \rangle + 2\delta N \left\langle -\frac{1}{2} \dot{\phi}^2 + \frac{1}{2a^2} (\vec{\nabla} \phi)^2 + V(\phi) \right\rangle \right] \end{aligned} \quad (7.42)$$

$$+ \frac{1}{m_p^2} \left[+\delta N \langle \dot{\phi}^2 \rangle + a^{-1} \langle \dot{\phi} \partial_k \phi \rangle \delta N_k - \frac{1}{2a^2} \gamma_{km} \langle \partial_k \phi \partial_m \phi \rangle \right], \quad (7.43)$$

$$\begin{aligned} 2H\partial_i(\delta N) - (2\ddot{a} + \dot{a}H)\delta N_i + \frac{a^{-1}}{2} [\partial_i(\partial_k \delta N_k) - \nabla^2 \delta N_i] - \frac{1}{2} \partial_j (\dot{\gamma}_{ij}) - \frac{1}{2} \partial_i (\dot{\gamma}) = \\ \frac{1}{m_p^2} \left[\dot{\phi} \partial_i \phi - \langle \dot{\phi} \partial_i \phi \rangle - a \delta N_i \left\langle -\frac{1}{2} \dot{\phi}^2 + \frac{1}{2} a^{-2} (\vec{\nabla} \phi)^2 + V(\phi) \right\rangle \right], \end{aligned} \quad (7.44)$$

$$\begin{aligned}
& (2\dot{a}^2\delta N + 4a\ddot{a}\delta N + 2a\dot{a}\delta\dot{N} + \nabla^2\delta N)\delta_{ij} - \partial_i\partial_j(\delta N) \\
& + (2\dot{a}(\partial_k\delta N_k) + a(\partial_k\delta\dot{N}_k))\delta_{ij} - \frac{a}{2}(\partial_i\delta\dot{N}_j + \partial_j\delta\dot{N}_i) - \dot{a}(\partial_i\delta N_j + \partial_j\delta N_i) \\
& \frac{a^2}{2}\ddot{\gamma}_{ij} + \frac{3}{2}a\dot{a}\dot{\gamma}_{ij} - \frac{\nabla^2}{2}\gamma_{ij} - (\dot{a}^2 + 2a\ddot{a})\gamma_{ij} - \left(\frac{a^2}{2}\ddot{\gamma} + \frac{3}{2}a\dot{a}\dot{\gamma} - \frac{\nabla^2}{2}\gamma\right)\delta_{ij} \\
& - \frac{1}{2}\partial_i\partial_j\gamma + \frac{1}{2}(\partial_k\partial_i\gamma_{kj} + \partial_k\partial_j\gamma_{ki}) - \frac{1}{2}(\partial_k\partial_m\gamma_{km})\delta_{ij} = \\
& \frac{1}{m_p^2} \left[\partial_i\phi\partial_j\phi - a^2\delta_{ij} \left(-\frac{1}{2}\dot{\phi}^2 + \frac{1}{2}a^{-2}(\vec{\nabla}\phi)^2 + V(\phi) \right) - a^2\delta_{ij}\langle p \rangle \right] \\
& - \frac{1}{m_p^2} \left[a^2\delta_{ij} \left(\delta N\langle \dot{\phi}^2 \rangle + a^{-1}\langle \dot{\phi}(\partial_k\phi) \rangle\delta N_k - \frac{1}{2}a^{-2}\gamma_{km}\langle \partial_k\phi\partial_m\phi \rangle \right) \right] \\
& - \frac{1}{m_p^2} \left[a^2\gamma_{ij} \left\langle -\frac{1}{2}\dot{\phi}^2 + \frac{1}{2}a^{-2}(\vec{\nabla}\phi)^2 + V(\phi) \right\rangle \right], \quad (7.45)
\end{aligned}$$

We can notice the following: on the *lhs*, we have the term $-(\dot{a}^2 + 2a\ddot{a})\gamma_{ij}$, and on the *rhs*, we have the term $\frac{a^2}{m_p^2}\gamma_{ij} \left\langle \frac{1}{2}\dot{\phi}^2 - \frac{1}{6}a^{-2}(\vec{\nabla}\phi)^2 - V(\phi) \right\rangle$. This means we can just cancel them, as this is basically Eq. (7.41). Now we should also write the *eom* for the scalar fields and how the perturbations backreact into their dynamics. We need to compute

$$\frac{1}{\sqrt{-g}}\partial_\mu(\sqrt{-g}g^{\mu\nu}\partial_\nu\phi) = \frac{\partial V}{\partial\phi}, \quad (7.46)$$

up to linear order in the metric perturbations. Considering $\sqrt{-g} = a^3(1 + \delta N + \gamma)$, the equation for the scalar field is

$$\begin{aligned}
& \ddot{\phi} + 3H\dot{\phi} - \frac{\nabla^2}{a^2}\phi(1 + 2\delta N) - (\delta\dot{N} - \dot{\gamma} + a^{-1}\partial_i\delta N_i)\dot{\phi} \\
& - 2a^{-1}\delta N_i\partial_i\dot{\phi} + a^{-1}H\delta N_i\partial_i\phi - a^{-1}\delta\dot{N}_i\partial_i\phi - \frac{\gamma_{ij}}{a^2}\partial_i\partial_j\phi = -(1 + 2\delta N)\frac{\partial V}{\partial\phi}. \quad (7.47)
\end{aligned}$$

7.1.2 Evolution scheme

We notice that the constraint equations contain at most first derivatives in time with respect to the scalar and vector perturbations δN and δN_i . Hence, they cannot serve as evolution equations for these *dofs*. We choose to write second-order time derivative equations for δN and δN_i by imposing the Harmonic gauge condition

$$\bar{g}^{\mu\nu}\delta\Gamma_{\mu\nu}^\rho = 0, \quad (7.48)$$

where $\bar{g}_{\mu\nu}$ is the background metric.

We can check how the perturbations transform under a change of coordinates $x_\mu \rightarrow x'_\mu = x_\mu + \xi_\mu(t, \vec{x})$, where $\xi_\mu = (\xi_0, \xi_i)$ is of the same order as the perturbations. Under

this transformation, the metric perturbations change as

$$\begin{aligned}\delta N' &= \delta N - \dot{\xi}_0 , \\ \delta N'_i &= \delta N_i - a\dot{\xi}_i + a^{-1}\partial_i\xi_0 , \\ \gamma'_{ij} &= \gamma_{ij} - 2H\xi_0\delta_{ij} - \partial_i\xi_j - \partial_j\xi_i .\end{aligned}\tag{7.49}$$

Now we can see how our gauge fixing condition changes under the transformation, which leads to the following equations

$$\begin{aligned}\ddot{\xi}_0 - \frac{\nabla^2}{a^2}\xi_0 - 2H\partial_i\xi_i &= 0 , \\ \ddot{\xi}_i - \frac{\nabla^2}{a^2}\xi_i + 2H\dot{\xi}_i + 3a^{-1}H\partial_i\xi_0 &= 0 ,\end{aligned}\tag{7.50}$$

which for static background $a = 1$, reduces to the known condition for the Harmonic Gauge $\square\xi_\mu = 0$. In order to get *eoms* for the scalar and vector *dofs*, we take a time derivative of Eq. (7.48) obtaining

$$\delta\ddot{N} + 6\dot{H}\delta\dot{N} + 6H\delta\dot{N} - a^{-2}\dot{a}(\partial_k\delta N_k) + a^{-1}(\partial_k\delta\dot{N}_k) - \dot{H}\gamma - H\dot{\gamma} - \frac{1}{2}\ddot{\gamma} = 0 ,\tag{7.51}$$

and

$$\begin{aligned}\delta\ddot{N}_i + 4\dot{H}\delta\dot{N}_i + 4H\delta\dot{N}_i - a^{-2}\dot{a}\partial_i\delta N + a^{-1}\partial_i\delta\dot{N} \\ + \frac{a^{-2}}{2}\dot{a}(2\partial_k(\gamma_{ki}) - \partial_i(\gamma)) - \frac{a^{-1}}{2}(2\partial_k(\dot{\gamma}_{ki}) - \partial_i(\dot{\gamma})) = 0 .\end{aligned}\tag{7.52}$$

Now that we have written down all the relevant evolution equations, we can express them in a Hamiltonian scheme. This will help us visualize the appropriate evolution strategy we want to use, which will be a Runge-Kutta type evolver. Let us start with the scalar perturbation

$$\begin{cases} \delta\dot{N} &= \pi_{\delta N} \\ \dot{\pi}_{\delta N} &= -6H\delta\dot{N} - 6\dot{H}\delta N + a^{-2}\dot{a}(\partial_k\delta N_k) - a^{-1}(\partial_k\delta\dot{N}_k) + H\dot{\gamma} + \dot{H}\gamma + \frac{1}{2}\ddot{\gamma} \end{cases},\tag{7.53}$$

while for the vector perturbation

$$\begin{cases} \delta\dot{N}_i &= \pi_{i,\delta N} \\ \dot{\pi}_{i,\delta N} &= -4H\delta\dot{N}_i - 4\dot{H}\delta N_i + a^{-2}\dot{a}\partial_i\delta N - a^{-1}\partial_i\delta\dot{N} - \frac{a^{-2}}{2}\dot{a}[2\partial_k(\gamma_{ki}) - \partial_i\gamma] \\ &\quad + \frac{a^{-1}}{2}[2\partial_k(\dot{\gamma}_{ki}) - \partial_i\dot{\gamma}] \end{cases}.$$

Finally, we can write the scheme for γ_{ij}

$$\begin{cases} \dot{\gamma}_{ij} &= \pi_{ij,\gamma} \\ \dot{\pi}_{ij,\gamma} &= -3H\dot{\gamma}_{ij} + \frac{\nabla^2}{a^2}\gamma_{ij} + \delta_{ij} \left(\ddot{\gamma} + 3H\dot{\gamma} - \frac{\nabla^2}{a^2}\gamma \right) \\ &+ \mathcal{T}_{ij}[\gamma, \dot{\gamma},] + \mathcal{S}_{ij}[\delta N, \delta \dot{N}] + \mathcal{V}_{ij}[\delta N_i, \delta \dot{N}_i] + \mathcal{M}_{ij}[\phi, \dot{\phi}] \end{cases},$$

with each of the contributions given by

$$\mathcal{T}_{ij}[\gamma, \dot{\gamma}] = a^{-2}\partial_i\partial_j\gamma - a^{-2}(\partial_k\partial_i\gamma_{kj} + \partial_k\partial_j\gamma_{ki}) + a^{-2}(\partial_k\partial_m\gamma_{km})\delta_{ij}, \quad (7.54)$$

$$\mathcal{S}_{ij}[\delta N, \delta \dot{N}] = -\delta_{ij} \left(4H^2\delta N + 8\frac{\ddot{a}}{a}\delta N + 4H\delta \dot{N} + 2\frac{\nabla^2}{a^2}\delta N \right) + \frac{2}{a^2}\partial_i\partial_j\delta N, \quad (7.55)$$

$$\begin{aligned} \mathcal{V}_{ij}[\delta N_i, \delta \dot{N}_i] &= -\delta_{ij}(4a^{-1}H(\partial_k\delta N_k) + 2a^{-1}(\partial_k\delta \dot{N}_k)) + \\ &a^{-1}(\partial_i\delta \dot{N}_j + \partial_j\delta \dot{N}_i) + 2a^{-1}H(\partial_i\delta N_j + \partial_j\delta N_i), \end{aligned} \quad (7.56)$$

$$\begin{aligned} \mathcal{M}_{ij}[\phi, \dot{\phi}] &= \frac{1}{m_p^2} \left[a^{-2} \left(\partial_i\phi\partial_j\phi - \frac{\delta_{ij}}{3}(\vec{\nabla}\phi)^2 \right) + \delta_{ij}\delta p \right] \\ &- \frac{1}{m_p^2} \left[\delta_{ij} \left(\delta N \langle \dot{\phi}^2 \rangle + a^{-1} \langle \dot{\phi}(\partial_k\phi) \rangle \delta N_k - \frac{1}{2}a^{-2}\gamma_{km} \langle \partial_k\phi\partial_m\phi \rangle \right) \right] \\ &- \frac{1}{m_p^2} \left[\gamma_{ij} \left\langle \frac{1}{3}a^{-2}(\vec{\nabla}\phi)^2 \right\rangle \right]. \end{aligned} \quad (7.57)$$

We can write the same for the matter fields, which in this case are scalar fields

$$\begin{cases} \dot{\phi} = a^{-3}\pi_\phi \\ \dot{\pi}_\phi = a^3\mathcal{K}[\phi, \dot{\phi}, \dots] \end{cases}, \quad (7.58)$$

where

$$\begin{aligned} \mathcal{K}[\phi, \dot{\phi}, \dots] &= \frac{\nabla^2}{a^2}\phi(1 + 2\delta N) + (\delta \dot{N} - \dot{\gamma} + a^{-1}\partial_i\delta N_i)\dot{\phi} + 2a^{-1}\delta N_i\partial_i\dot{\phi} \\ &- a^{-1}H\delta N_i\partial_i\phi + a^{-1}\delta \dot{N}_i\partial_i\phi + \frac{\gamma_{ij}}{a^2}\partial_i\partial_j\phi - (1 + 2\delta N)\frac{\partial V}{\partial \phi}. \end{aligned} \quad (7.59)$$

7.1.3 Initial Conditions

We will choose the initial conditions such that they satisfy the Hamiltonian and momentum constraints on the initial time slice, i.e., at $t_0 = 0$. This means we will set initial conditions for the dynamical degrees of freedom, namely ϕ and γ_{ij} , following the

procedure explained in Sec. 2.2.4 for the matter sector, and, for instance, setting $\gamma_{ij} = 0$ for the tensor perturbations. Once the dynamical *dofs* are initialized, we can use the constraints to initialize the scalar and vector perturbations δN and δN_i by solving the following set of equations in Fourier space

$$\begin{aligned} \delta N \frac{1}{m_p^2} \left\langle \frac{1}{2a^2} (\vec{\nabla} \phi)^2 + V(\phi) \right\rangle + H(\partial_k \delta N_k) + \frac{1}{2m_p^2} a^{-1} \langle \dot{\phi} \partial_k \phi \rangle \delta N_k = \\ \frac{a}{2} H \dot{\gamma} - \frac{a^{-1}}{4} \nabla^2 \gamma + \frac{a^{-1}}{4} \partial_k \partial_l (\gamma_{kl}) + \frac{a}{m_p^2} \delta \rho - \frac{1}{2a^2 m_p^2} \gamma_{km} \langle \partial_k \phi \partial_m \phi \rangle, \end{aligned} \quad (7.60)$$

$$\begin{aligned} 2H \partial_i (\delta N) - (2\ddot{a} + \dot{a}H) \delta N_i + \frac{a^{-1}}{2} [\partial_i (\partial_k \delta N_k) - \nabla^2 \delta N_i] \\ + \frac{2a}{m_p^2} \delta N_i \left\langle -\frac{1}{2} \dot{\phi}^2 + \frac{1}{2} a^{-2} (\vec{\nabla} \phi)^2 + V(\phi) \right\rangle = \\ \frac{1}{2} \partial_j (\dot{\gamma}_{ij}) + \frac{1}{2} \partial_i (\dot{\gamma}) + \frac{2}{m_p^2} \left(\dot{\phi} \partial_i \phi - \langle \dot{\phi} \partial_i \phi \rangle \right) \end{aligned} \quad (7.61)$$

Let us call $v = \left\langle -\frac{1}{2} \dot{\phi}^2 + \frac{1}{2} a^{-2} (\vec{\nabla} \phi)^2 + V(\phi) \right\rangle$ and $\mu = \left\langle \frac{1}{2} a^{-2} (\vec{\nabla} \phi)^2 + V(\phi) \right\rangle$. We will Fourier transform this using

$$\delta N(\mathbf{x}, t) = \int \frac{d^3 q}{(2\pi)^3} \delta N_q(\mathbf{q}, t) e^{-i\mathbf{q} \cdot \mathbf{x}}. \quad (7.62)$$

In Fourier space Eqs. (7.60) and (7.61) take the form of a matrix equation

$$\mathbf{C} \cdot \delta \vec{N} = \vec{F}, \quad (7.63)$$

with $\mathbf{C}$ being a matrix with elements

$$\begin{aligned} \mathbf{C}_{00} = \mu/m_p^2, \quad \mathbf{C}_{0i} = -iHk_i + \frac{1}{am_p^2} \langle \dot{\phi} \partial_i \phi \rangle, \quad \mathbf{C}_{i0} = -2iHk_i, \\ \mathbf{C}_{ij} = -(2\ddot{a} + \dot{a}H) \delta_{ij} + \frac{a^{-1}}{2} (k^2 \delta_{ij} - k_i k_j) + \frac{2av}{m_p^2} \delta_{ij}. \end{aligned} \quad (7.64)$$

Finally, in order to obtain the initial conditions for the perturbations δN and δN_i we solve the linear system of equations by inverting the matrix $\mathbf{C}$

$$\delta \vec{N} = \mathbf{C}^{-1} \cdot \vec{F}, \quad \text{with} \quad F_0 = \frac{a}{m_p^2} \mathcal{F}[\delta \rho], \quad F_i = \frac{1}{m_p^2} \mathcal{F}[\dot{\phi} \partial_i \phi - \langle \dot{\phi} \partial_i \phi \rangle], \quad (7.65)$$

where $\mathcal{F}[\dots]$ stands for the Fourier transform.

7.1.4 Gauge invariant quantities

One of the goals of this work is to extract the 3-D field configuration of the comoving curvature perturbation $\mathcal{R}(\mathbf{x}, t_*)$. The definition of $\mathcal{R}$ is [24]

$$\mathcal{R} \equiv \Psi - \frac{H}{\langle \rho \rangle + \langle p \rangle} \delta q , \quad (7.66)$$

where Ψ is the scalar part of the spatial metric γ_{ij} , and δq is the scalar part of the 3-momentum density, with $T_i^0 = \partial_i \delta q$.

In order to extract these from the cosmological perturbations and matter fields, we will consider the SVT decomposition of γ_{ij}

$$\gamma_{ij} = -2\Psi\delta_{ij} + E_{ij} , \quad (7.67)$$

where

$$E_{ij} = 2\partial_i\partial_j E + \partial_i F_j + \partial_j F_i + h_{ij} , \quad \text{with } \partial_i F_i = 0 , \quad h_{ii} = 0 , \quad \partial_i h_{ij} = 0 . \quad (7.68)$$

Hence, we can write

$$\nabla^2 \gamma - \partial_i \partial_j \gamma_{ij} = -4\nabla^2 \Psi , \quad (7.69)$$

which can be inverted “easily” in Fourier space.

On the other hand, we need to extract δq . In the regime where only linear terms in the perturbations are kept, we can write

$$T^0_i = -\dot{\phi} \partial_i \phi . \quad (7.70)$$

We can extract the scalar part in the following manner

$$\nabla^2 \delta q = -\partial_i \dot{\phi} \partial_i \phi - \dot{\phi} \nabla^2 \phi , \quad (7.71)$$

which again can be extracted in Fourier space.

This procedure can be applied whenever we need to measure the curvature perturbations. Had we decided to write down an *ecom* for $\mathcal{R}$ itself, we would have required a different gauge choice and, more importantly, a decomposition of the energy-momentum tensor from the matter sector in the Scalar-Vector-Tensor (SVT) formalism. This would require performing a Fourier transform at every time step, as the projection of the source into its SVT components is carried out in reciprocal space. Therefore, we follow an approach

similar to that used to evolve gravitational waves, with the difference that now we are able to evolve the full set of cosmological perturbations.

7.2 Lattice formulation of cosmological perturbations sourced by non-linear dynamics

In this section, we present a lattice formulation for the evolution of cosmological perturbations at linear order, sourced by the non-linear dynamics of the matter sector.

7.2.1 Equations of motion of cosmological perturbations

In order to implement one of the various evolution schemes, it is first necessary to rewrite the *eoms* such that the second derivatives of the *dofs* depend at most on first derivatives of the remaining *dofs*. For instance, Eqs. 7.53, 7.54, and 7.54 contain $\ddot{a}$, $\dot{H}$, and $\ddot{\gamma}$, which need to be expressed in terms of other *eoms* so that the resulting system contains only first derivatives. We begin by recalling the zeroth-order Einstein equation for the *ij*-component

$$\dot{a}^2 + 2a\ddot{a} = -\frac{a^2}{m_p^2} \left\langle \frac{1}{2}\dot{\phi}^2 - \frac{1}{6}(\vec{\nabla}\phi)^2 - V(\phi) \right\rangle = -\frac{a^2}{m_p^2} \langle p \rangle, \quad (7.72)$$

for which, we can write the following relations

$$\begin{aligned} \frac{\ddot{a}}{a} &= -\frac{1}{2m_p^2} \left(\langle p \rangle + \frac{1}{3} \langle \rho \rangle \right), \\ \dot{H} &= -\frac{1}{2m_p^2} (\langle p \rangle + \langle \rho \rangle). \end{aligned} \quad (7.73)$$

Finally, we can use Eqn. (7.54) in order to write down $\ddot{\gamma}$.

$$\begin{aligned} \frac{\ddot{\gamma}}{2} &= -\frac{3}{2}H\dot{\gamma} + \frac{\nabla^2}{4a^2}\gamma - \frac{1}{4a^2}\partial_k\partial_m\gamma_{km} + 3H^2\delta N - \frac{3}{4m_p^2}(\langle p \rangle + \frac{1}{3}\langle \rho \rangle)\delta N + 3H\delta\dot{N} \\ &\quad - 2\frac{\nabla^2}{a^2}\delta N + \frac{2}{a}H\partial_k\delta N_k + \frac{1}{a}\partial_k\delta\dot{N}_k - \frac{3}{4m_p^2}\delta p + \frac{3}{4m_p^2}\delta N\langle\dot{\phi}^2\rangle + \frac{3}{4m_p^2}\langle\dot{\phi}\partial_k\phi\rangle\delta N_k \\ &\quad - \frac{3}{8a^2}\gamma_{km}\langle\partial_k\phi\partial_m\phi\rangle + \frac{\gamma}{12m_p^2a^2}\langle(\vec{\nabla}\phi)^2\rangle. \end{aligned} \quad (7.74)$$

Now, we can write an equation of $\delta\ddot{N}$ in terms of only first derivatives

$$\begin{aligned}\delta\ddot{N} = & -3H\delta\dot{N} + \frac{2}{m_p^2}\langle\rho\rangle\delta N + \frac{3}{a}(\partial_k\delta N_k) - \frac{1}{2}H\dot{\gamma} - \frac{1}{2m_p^2}(\langle p\rangle + \langle\rho\rangle)\gamma + \frac{\nabla^2}{4a^2}\gamma \\ & - \frac{1}{4a^2}\partial_k\partial_m\gamma_{km} + 3H^2\delta N - \frac{\nabla^2}{a^2}\delta N - \frac{3}{4m_p^2}\delta p + \frac{3}{4m_p^2}\delta N\langle\dot{\phi}^2\rangle \\ & + \frac{3}{4m_p^2}\langle\dot{\phi}\partial_k\phi\rangle\delta N_k - \frac{3}{8m_p^2}\gamma_{km}\langle\partial_k\phi\partial_m\phi\rangle + \frac{\gamma}{12m_p^2a^2}\langle(\vec{\nabla}\phi)^2\rangle .\end{aligned}\quad (7.75)$$

Equivalently, we can substitute the term $\dot{H}$ in Eq. (7.54) obtaining

$$\begin{aligned}\delta\ddot{N}_i = & -4H\delta\dot{N}_i + \frac{2}{m_p^2}(\langle p\rangle + \langle\rho\rangle)\delta N_i + a^{-2}\dot{a}\partial_i\delta N - a^{-1}\partial_i\delta\dot{N} \\ & - \frac{a^{-2}}{2}\dot{a}[2\partial_k(\gamma_{ki}) - \partial_i\gamma] + \frac{a^{-1}}{2}[2\partial_k(\dot{\gamma}_{ki}) - \partial_i\dot{\gamma}] .\end{aligned}\quad (7.76)$$

Finally, we can also notice that the *eom* for tensor perturbations, given by Eq. (7.54), contains a term proportional to $\ddot{\gamma}_{ij}$, which we need to eliminate in order to obtain *eoms* that contain at most first derivative terms. Let us recall Eq. (7.54) for $\ddot{\gamma}_{ij}$

$$\begin{aligned}\ddot{\gamma}_{ij} = & -3H\dot{\gamma}_{ij} + \frac{\nabla^2}{a^2}\gamma_{ij} + \delta_{ij}\left(\ddot{\gamma} + 3H\dot{\gamma} - \frac{\nabla^2}{a^2}\gamma\right) \\ & + \mathcal{T}_{ij}[\gamma, \dot{\gamma}] + \mathcal{S}_{ij}[\delta N, \delta\dot{N}] + \mathcal{V}_{ij}[\delta N_i, \delta\dot{N}_i] + \mathcal{M}_{ij}[\phi, \dot{\phi}] ,\end{aligned}\quad (7.77)$$

For instance, the *eom* for γ_{11} reads

$$\begin{aligned}\ddot{\gamma}_{11} = & -3H\dot{\gamma}_{11} + \frac{\nabla^2}{a^2}\gamma_{11} \\ & + \left(\ddot{\gamma}_{11} + \ddot{\gamma}_{22} + \ddot{\gamma}_{33} + 3H(\dot{\gamma}_{11} + \dot{\gamma}_{22} + \dot{\gamma}_{33}) - \frac{\nabla^2}{a^2}(\gamma_{11} + \gamma_{22} + \gamma_{33})\right) \\ & + \mathcal{T}_{11}[\gamma, \dot{\gamma}] + \mathcal{S}_{11}[\delta N, \delta\dot{N}] + \mathcal{V}_{11}[\delta N_i, \delta\dot{N}_i] + \mathcal{M}_{11}[\phi, \dot{\phi}] ,\end{aligned}\quad (7.78)$$

This clearly makes the equation depend on both $\ddot{\gamma}_{22}$ and $\ddot{\gamma}_{33}$. We can use these relations to write the *eoms* for γ_{ii} independently of the other components. In doing so, we obtain the *eoms* for the diagonal components of γ_{ij}

$$\ddot{\gamma}_{ij} = -3H\dot{\gamma}_{ij} + \frac{\nabla^2}{a^2}\gamma_{ij} - \frac{1}{2}(\mathcal{X} - 2\mathcal{X}_{ij}) , \quad \text{for } i = j , \quad (7.79)$$

where we have defined $\mathcal{X}_{ij} \equiv \mathcal{T}_{ij}[\gamma, \dot{\gamma}] + \mathcal{S}_{ij}[\delta N, \delta\dot{N}] + \mathcal{V}_{ij}[\delta N_i, \delta\dot{N}_i] + \mathcal{M}_{ij}[\phi, \dot{\phi}]$ and $\mathcal{X} = \delta_{ij}\mathcal{X}_{ij}$. Meanwhile, for the off-diagonal components, the *eoms* read

$$\ddot{\gamma}_{ij} = -3H\dot{\gamma}_{ij} + \frac{\nabla^2}{a^2}\gamma_{ij} + \mathcal{X}_{ij} , \quad \text{for } i \neq j . \quad (7.80)$$

Now we are in a position to write down a lattice formulation for Eqs. (7.75), (7.76), (7.79), and (7.80). First, we will specify where cosmological perturbations reside on the lattice. We will consider them to live at lattice sites $\mathbf{n}$, for which we must define the derivative operators accordingly. In this regard, spatial gradients will be computed using the neutral lattice derivative, defined in Eq. (2.46), and denoted by ∇_0 .

7.2.2 Lattice formulation

Finally, in order to implement the equations on the lattice we need them in program units. We will follow standard procedure of setting units, which is matter fields are measured in units of some relevant field amplitude f_* , and distances and times in units of a relevant energy scale ω_* . We have the following,

$$\tilde{\phi} \equiv \frac{\phi}{f_*}, \quad d\tilde{t} = \omega_* dt, \quad d\tilde{x}_i = \omega_* dx_i. \quad (7.81)$$

where we define $' \equiv \partial/\partial\tilde{t}$ and $\tilde{\nabla}_i \equiv \partial/\partial\tilde{x}_i$. Given this definitions we can define the program version of the following quantities

$$\tilde{V}(\tilde{\phi}) = \frac{V(\phi)}{f_*^2 \omega_*^2}, \quad \tilde{\rho} = \frac{\rho}{f_*^2 \omega_*^2}, \quad \tilde{p} = \frac{p}{f_*^2 \omega_*^2}. \quad (7.82)$$

If we apply the variables transformations to our equations the terms that contain either p/m_p^2 or ρ/m_p^2 , or any quadratic combination of derivatives of matter fields e.g. $\langle \dot{\phi}^2 \rangle / m_p^2$, will transform to the following

$$\frac{(p, \rho)}{m_p^2} = \left(\frac{f_* \omega_*}{m_p} \right)^2 (\tilde{p}, \tilde{\rho}). \quad (7.83)$$

We notice metric perturbations are already dimensionless.

Given the aforementioned lattice operators, we can write down the lattice formulations of the system of Eqs. (7.75), (7.76), (7.79) and (7.80). For δN

$$\begin{cases} \delta N' = \pi_{\delta N} \\ \pi'_{\delta N} = \mathcal{K}_{\delta N}[\tilde{\phi}, \tilde{\phi}', a, \delta g] \end{cases}, \quad (7.84)$$

with

$$\begin{aligned}
\delta\mathcal{K}_{\delta N}[\tilde{\phi}, \tilde{\phi}, a, \delta g] &= -3\tilde{H}\delta N' + 2\left(\frac{f_*}{m_p^2}\right)\langle\tilde{\rho}\rangle\delta N + \frac{3}{a}(\tilde{\nabla}_k^0\delta N_k) - \frac{1}{2}\tilde{H}\gamma' \\
&\quad - \frac{1}{2}\left(\frac{f_*}{m_p^2}\right)(\langle\tilde{p}\rangle + \langle\tilde{\rho}\rangle)\gamma + \frac{\tilde{\nabla}_0^2}{4a^2}\gamma - \frac{1}{4a^2}\tilde{\nabla}_k^0\tilde{\nabla}_m^0\gamma_{km} + 3\tilde{H}^2\delta N \\
&\quad - \frac{\tilde{\nabla}_0^2}{a^2}\delta N - \frac{3}{4}\left(\frac{f_*}{m_p}\right)^2\delta\tilde{p} + \frac{3}{4}\left(\frac{f_*}{m_p}\right)^2\delta N\langle\tilde{\phi}^2\rangle \\
&\quad + \frac{3}{4}\left(\frac{f_*}{m_p}\right)^2\langle\tilde{\phi}'\tilde{\nabla}_k^0\tilde{\phi}\rangle\delta N_k - \frac{3}{8}\left(\frac{f_*}{m_p}\right)^2\gamma_{km}\langle\tilde{\nabla}_k^0\phi\tilde{\nabla}_m^0\phi\rangle \\
&\quad + \frac{\gamma}{12a^2}\left(\frac{f_*}{m_p}\right)^2\left\langle\left(\sum_i\tilde{\nabla}_i^0\phi\right)^2\right\rangle. \tag{7.85}
\end{aligned}$$

For δN_i

$$\begin{cases} \delta N'_i = \pi_{\delta N_i, i} \\ \pi'_{\delta N_i, i} = \mathcal{K}_{\delta N_i, i}[\tilde{\phi}, \tilde{\phi}', a, \delta g] \end{cases}, \tag{7.86}$$

with

$$\begin{aligned}
\mathcal{K}_{\delta N_i, i}[\tilde{\phi}, \tilde{\phi}', a, \delta g] &= -4\tilde{H}\delta N'_i + 2\left(\frac{f_*}{m_p}\right)^2(\langle\tilde{p}\rangle + \langle\tilde{\rho}\rangle)\delta N_i + a^{-2}a'\tilde{\nabla}_i^0\delta N - a^{-1}\tilde{\nabla}_i^0\delta N' \\
&\quad - \frac{a^{-2}}{2}a'[2\tilde{\nabla}_k^0(\gamma_{ki}) - \tilde{\nabla}_i^0\gamma] + \frac{a^{-1}}{2}[2\tilde{\nabla}_k^0(\gamma'_{ki}) - \tilde{\nabla}_i^0\gamma']. \tag{7.87}
\end{aligned}$$

For γ_{ij}

$$\begin{cases} \gamma'_{ij} = \pi_{\gamma, ij} \\ \pi'_{\gamma, ij} = \mathcal{K}_{\gamma, ij}[\tilde{\phi}, \tilde{\phi}', a, \delta g] \end{cases}, \tag{7.88}$$

with

$$\mathcal{K}_{\gamma, ij}[\tilde{\phi}, \tilde{\phi}', a, \delta g] = -3\tilde{H}\gamma'_{ij} + \frac{\tilde{\nabla}_0^2}{a^2}\gamma_{ij} - \frac{1}{2}(\tilde{\mathcal{X}} - 2\tilde{\mathcal{X}}_{ij}), \quad \text{given } i = j, \tag{7.89}$$

$$\mathcal{K}_{\gamma, ij}[\tilde{\phi}, \tilde{\phi}', a, \delta g] = -3\tilde{H}\gamma'_{ij} + \frac{\tilde{\nabla}_0^2}{a^2}\gamma_{ij} + \tilde{\mathcal{X}}_{ij}, \quad \text{given } i \neq j. \tag{7.90}$$

and

$$\tilde{\mathcal{X}}_{ij} \equiv \tilde{\mathcal{T}}_{ij}[\gamma, \gamma'] + \tilde{\mathcal{S}}_{ij}[\delta N, \delta N'] + \tilde{\mathcal{V}}_{ij}[\delta N_i, \delta N'_i] + \tilde{\mathcal{M}}_{ij}[\tilde{\phi}, \tilde{\phi}'] \tag{7.91}$$

$$\tilde{\mathcal{T}}_{ij}[\gamma, \gamma'] \equiv a^{-2}\tilde{\nabla}_i^0\tilde{\nabla}_j^0\gamma - a^{-2}(\tilde{\nabla}_k^0\tilde{\nabla}_i^0\gamma_{kj} + \tilde{\nabla}_k^0\tilde{\nabla}_j^0\gamma_{ki}) + a^{-2}(\tilde{\nabla}_k^0\tilde{\nabla}_m^0\gamma_{km})\delta_{ij}, \tag{7.92}$$

$$\begin{aligned}\tilde{\mathcal{S}}_{ij}[\delta N, \delta N'] &\equiv -\delta_{ij} \left(4\tilde{H}^2 \delta N - \frac{1}{4} \left(\frac{f_*}{m_p} \right)^2 \left(\langle \tilde{p} \rangle + \frac{1}{3} \langle \tilde{\rho} \rangle \right) \delta N + 4\tilde{H} \delta N' + 2 \frac{\tilde{\nabla}_0^2}{a^2} \delta N \right) \\ &\quad + \frac{2}{a^2} \tilde{\nabla}_i^0 \tilde{\nabla}_j^0 \delta N, \end{aligned} \quad (7.93)$$

$$\begin{aligned}\tilde{\mathcal{V}}_{ij}[\delta N_i, \delta N'_i] &\equiv -\delta_{ij} (4a^{-1} \tilde{H} (\tilde{\nabla}_k^0 \delta N_k) + 2a^{-1} (\tilde{\nabla}_k^0 \delta N'_k)) \\ &\quad + a^{-1} (\tilde{\nabla}_i^0 \delta N'_j + \tilde{\nabla}_j^0 \delta N'_i) + 2a^{-1} \tilde{H} (\tilde{\nabla}_i^0 \delta N_j + \tilde{\nabla}_j^0 \delta N_i), \end{aligned} \quad (7.94)$$

$$\begin{aligned}\tilde{\mathcal{M}}_{ij}[\tilde{\phi}, \tilde{\phi}'] &\equiv \left(\frac{f_*}{m_p} \right)^2 \left[a^{-2} \left(\tilde{\nabla}_i^0 \phi \tilde{\nabla}_j^0 \phi - \frac{\delta_{ij}}{3} \left(\sum_i \tilde{\nabla}_i^0 \phi \right)^2 \right) + \delta_{ij} \delta \tilde{p} \right] \\ &\quad - \left(\frac{f_*}{m_p} \right)^2 \left[\delta_{ij} \left(\delta N \langle \tilde{\phi}'^2 \rangle + a^{-1} \langle \tilde{\phi}' (\tilde{\nabla}_k^0 \tilde{\phi}) \rangle \delta N_k - \frac{1}{2} a^{-2} \gamma_{km} \langle \tilde{\nabla}_k^0 \phi \tilde{\nabla}_m^0 \phi \rangle \right) \right] \\ &\quad - \left(\frac{f_*}{m_p} \right)^2 \left[\gamma_{ij} \left\langle \frac{1}{3} a^{-2} \left(\sum_i \tilde{\nabla}_i^0 \phi \right)^2 \right\rangle \right]. \end{aligned} \quad (7.95)$$

Finally for ϕ

$$\begin{cases} \phi' = a^{-3} \tilde{\pi}_\phi \\ \tilde{\pi}'_\phi = a^3 \mathcal{K}[\tilde{\phi}, \tilde{\phi}', a, \delta g] \end{cases}, \quad (7.96)$$

with

$$\begin{aligned}\mathcal{K}[\tilde{\phi}, \tilde{\phi}', a, \delta g] &= (1 + 2\delta N) \frac{\tilde{\nabla}_0^2}{a^2} \tilde{\phi} + (\delta N' - \gamma' + a^{-1} \tilde{\nabla}_i^0 \delta N_i) \tilde{\phi}' + 2a^{-1} \delta N_i \tilde{\nabla}_i^0 \tilde{\phi}' \\ &\quad - a^{-1} \tilde{H} \delta N_i (\tilde{\nabla}_i^0 \tilde{\phi}) + a^{-1} \delta N'_i (\tilde{\nabla}_i^0 \tilde{\phi}) + \frac{\gamma_{ij}}{a^2} \tilde{\nabla}_i^0 \tilde{\nabla}_j^0 \phi \\ &\quad - (1 + 2\delta N) \frac{\partial \tilde{V}}{\partial \tilde{\phi}}. \end{aligned} \quad (7.97)$$

Chapter 8

Conclusions and Future directions

The physics of the early universe offers a unique window to explore physics operating at energy scales far beyond those achievable in terrestrial experiments. Moreover, several extensions of the Standard Model of particle physics could lead to distinctive observable signatures in the form of gravitational wave backgrounds (GWBs). The goal of the early universe community is to make increasingly accurate predictions that will soon be tested by upcoming GW experiments. In this context, we have seen that many of the phenomena leading to observable signatures are inherently non-linear. The emerging field of **Lattice Cosmology** is thus positioned as a powerful tool to assess the complex dynamics involved and to make reliable predictions aimed at improving our understanding of the origins of the universe.

In light of the findings presented in this thesis, several future directions can be pursued to expand our knowledge in these areas. In the following, we will discuss these future paths.

- **Particle couplings spectroscopy with GWs:** The appearance of the stairway signature in the GWB spectrum from a preheating scenario makes us believe that similar effect is expected for several other types of scenarios. Among these, we can consider tachyonic preheating [50–53, 63, 64], axion preheating [65–68], or geometric preheating [46–49]. Additionally, In the case of strong first-order phase transitions, gauge fields coupled to the corresponding Higgs sector of the theory will inevitably be excited during the nucleation and collision of bubbles. Arguably, this could lead to new peaks in the GWB spectrum, probing the Higgs–gauge couplings. We could also consider the possibility of interactions among the daughter species and analyze how the peak structure is affected, and whether it provides a way to measure these couplings as well.

- **GWs from preheating via trilinear interactions:** Preheating scenarios with an inflaton monomial or polynomial potential, and a trilinear coupling between the inflaton and a daughter field, lead to very efficient GW production. However, the resulting GW signal lies outside the sensitivity of current and planned direct and indirect probes of GWs. One direction to explore in future work is to consider the same type of coupling in lower-scale inflationary scenarios, in which the peak frequency of the GWB could be shifted to smaller values. It is not clear whether the efficiency of GW production will be significantly suppressed in such scenarios.
- **Geometric reheating of the universe:** Preheating through non-minimal couplings to gravity has been explored in various scenarios, such as Ricci Reheating [45]. Nevertheless, the idea of particle production via gravitational interactions could be relevant in many other contexts in the early universe, including dark matter, GWs, dark radiation, or baryon asymmetry; see, for instance, the review [326]. These mechanisms can be highly efficient and may lead to exponential particle production, resulting in non-linear dynamics that quickly invalidate the assumptions of analytical approaches. In this context, lattice formulations of these scenarios will become necessary to fully address the implications of such interactions in the early universe.
- **Axion Inflation:** Axion inflation is a well-motivated and phenomenologically rich scenario in the early universe. Therefore, a proper description of the dynamics under strong backreaction must involve a fully inhomogeneous treatment, including the local fluctuations of $F\tilde{F}$ and the presence of inflaton gradients. This is essential in order to make truly reliable predictions of the associated signatures. For example, the amplitude and frequency today of the gravitational wave background sourced during inflation by the excited gauge field strongly depend on how long inflation is prolonged. Furthermore, the production of scalar metric perturbations and the resulting abundance of primordial black holes are highly sensitive to the inherently local nature of the non-linear dynamics.
- **Cosmological perturbations sourced by non-linear dynamics:** The ability to probe early universe scenarios will greatly depend on the correct assessment of their phenomenological consequences. We have seen there may be very energetic processes, characterized by non-linear dynamics, that are efficient sources of cosmological perturbations. These perturbations can later source observable signatures in our current observations of the Universe. For instance, the appearance of non-Gaussian correlations in the power spectrum of CMB anisotropies or in the matter power spectrum of Large Scale Structure. Furthermore, the possibility of primordial black holes being created due to

a particular shape of the scalar power spectrum is another phenomenological consequence of such dynamics. It is, consequently, essential to have precise control over the exact mechanism through which cosmological perturbations are created.

The early universe offers a wide and unique window into physics that is beyond our current knowledge, and describe scenarios at energies out of reach from our technological abilities. The possibility that this era was governed by very energetic non-linear dynamics introduces the necessity to develop tools able to investigate such scenarios. The most precise tool that is capable of describing the dynamics of systems characterized by non-linear coupled systems of PDEs are lattice techniques. It is, therefore, necessary to start looking at research avenues in our current theoretical cosmological program that include computational simulations of the field theory dynamics in a cosmological context. In this way, precise and reliable predictions can be made that can later be confronted with more precise and wide-ranging observations from gravitational wave backgrounds, or precise observations of the CMB and LSS. This is how we will be able to answer some primordial questions about the origin of our Universe, and perhaps what its ultimate fate will be.

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Part II

Resumen de la Tesis

Prefacio

Nuestras observaciones actuales sobre el universo, a escalas cosmológicas, apoyan la idea de que este es isótropo y homogéneo en un grado muy alto. El paradigma principal para explicar tales propiedades es la *inflación cósmica*, que propone la expansión exponencial del universo hace 13.800 millones de años. La imagen más temprana del universo, a la cual tenemos acceso a través de observaciones electromagnéticas, es el Fondo Cósmico de Microondas (CMB por sus siglas en inglés). A partir de su observación varias cuestiones surgieron sobre las propiedades de dicha imagen. Entre los problemas que surgieron están: el problema del horizonte cosmológico y el problema de planitud del universo. Es precisamente el paradigma de inflación que resuelve estas dudas, y no sólo eso, además provee un mecanismo natural de producción de las perturbaciones necesarias que sirven como semillas para explicar la estructura del universo. A los fenómenos que ocurren durante estos momentos del universo se los conoce ampliamente como el **Universo Temprano**. A pesar del éxito del paradigma inflacionario para explicar varias cualidades del cosmos, este permanece como una hipótesis y no un hecho demostrado del universo. El reto principal yace en encontrar una teoría inspirada en la física de partículas, es decir conectar la cosmología con el Modelo Estándar de física de partículas. Muchos modelos han sido propuestos; sin embargo, nuestras observaciones no son lo suficientemente precisas para distinguir a un único modelo entre todos ellos. Además, varios modelos de física de partículas que pretendan explicar inflación deben hacerlo a escalas muy altas, cercanas a la escala de Planck, para el cual no conocemos una teoría definitiva.

Adicionalmente, el éxito de la teoría de síntesis de núcleos del *Big Bang* (BBN por sus siglas en inglés) para explicar la abundancia de los elementos químicos livianos en el universo requiere de ciertas condiciones iniciales muy específicas. Se requiere el equilibrio térmico entre ciertos grados de libertad del modelo estándar para poder predecir las abundancias observadas. Esto sugiere inmediatamente en como podemos conectar estos dos momentos del universo, es decir, como la energía almacenada en el grado de libertad que sostiene la época inflacionaria se transfiere a los grados de libertad relativistas y que luego alcancen el equilibrio térmico. Esta transición del universo se la conoce como recalentamiento (*reheating* en inglés) y está ampliamente no constreñida principalmente porque no la podemos observar de forma directa. Esto se debe a que el universo durante su etapa temprana es básicamente opaco a la luz, y esto no cambia hasta que los primeros átomos se forman, es precisamente este momento en los cuales los fotones son libres y forman el fondo cósmico de microondas. Mecanismos para recalentar el universo propuestos se basan en el decaimiento perturbativo del campo inflacionario en los capos del modelo estándar, de todos modos pronto se entendió que los fenómenos

no-perturbativos pueden ser relevantes en las etapas tempranas, esto se conoce como el precalentamiento del universo (*preheating* en inglés).

El universo temprano no está limitado a inflación y recalentamiento; esta puede ser también la época para la cual muchos fenómenos pueden ocurrir. En particular, transiciones de fase y la creación de defectos topológicos pueden haber marcado el universo temprano y dejado señales; siendo estos fenómenos una ventana hacia física más allá de nuestro actual entendimiento. En este contexto podríamos preguntarnos: ¿Cómo podemos observar estos fenómenos? Una posible respuesta es través de las Ondas Gravitacionales. La observación de una señal de ondas gravitacionales, en 2015, hecha por el observatorio LIGO marcó el inicio de la astronomía de ondas gravitacionales y abrió una oportunidad única para observar el universo temprano. Esto es porque se conoce que los fenómenos que hemos descrito anteriormente producen fondos cósmicos de ondas gravitacionales. Además, las recientes observaciones del *Pulsar Timing Array* (PTA) sugiere la existencia de un fondo de ondas gravitacionales a bajas frecuencias. Futuros observatorios intentarán sondear otras regiones del espectro de ondas gravitacionales, sobretodo en altas frecuencias. Es así necesario y esencial describir estos fenómenos y producir predicciones confiables sobre estos fondos cósmicos.

La física del universo temprano es inherentemente no-lineal, y el enfoque analítico usualmente simplifica el escenario para poder hacer los cálculos manejables. A pesar de que las predicciones y el conocimiento que tenemos sobre esta etapa viene de este enfoque, el mismo falla en la mayoría de escenarios realistas. La razón principal proviene de la imposibilidad de resolver analíticamente los sistemas de ecuaciones acopladas en derivadas parciales que describen a estas teorías. Una manera de resolver estas ecuaciones es hacerlo en redes tridimensionales. El conjunto de técnicas y herramientas para estudiar el universo temprano, o de manera general la dinámica de teorías de campos interactuantes en tiempo real, se refiere como **Lattice Cosmology** (nombre que mantenemos en español a falta de una traducción idónea). El objetivo de la presente disertación es presentar ejemplos donde las simulaciones en redes tridimensionales son esenciales para describir la dinámica no-lineal de los escenarios de inflación y recalentamiento, además de demostrar como las ondas gravitacionales pueden ayudarnos a observar y constreñir estos fenómenos. Para este fin, un código de acceso público llamado *CosmoLattice* ha sido desarrollado para simular la dinámica de campos interactuantes en fondos que se expanden. Durante el curso de mi investigación pre-doctoral, dos módulos de el código fueron publicadas las cuales fui co-desarrollador de. La versión más actual del código, el cual puede encontrarse en el repositorio virtual [GitHub](#). Información adicional sobre la estructura del código y el manual de usuario puede encontrarse en la página web de [CosmoLattice](#).

El Universo temprano

El universo a escalas cosmológicas es homogéneo e isótropo a un grado muy alto, por lo tanto podemos evocar el *Principio Cosmológico* como fundamento para describir sus propiedades. Según la relatividad general, la métrica en su forma más general que se ajusta al principio cosmológico esta dada por

$$ds^2 = g_{\mu\nu}dx^\mu dx^\nu = -dt^2 + a^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2(d\theta^2 + \sin^2\theta d\phi^2) \right], \quad (1)$$

donde r, θ, ϕ son las coordenadas polares, $a(t)$ es el factor de escala y k es la curvatura espacial. Nuestro universo es prácticamente plano, con curvatura cero, por lo tanto tomamos $k = 0$. La métrica describe la geometría de nuestro universo en coordenadas comóviles, las que siguen la expansión del universo. Mientras tanto las coordenadas físicas, que son aquellas que observamos, se obtienen mediante $r_{\text{fis}} \equiv a(t)r$. El factor de escala no es una entidad física por lo cual para medir la expansión del universo se introduce el objeto matemático conocido como el parámetro de Hubble, definido como

$$H \equiv \frac{1}{a} \frac{da}{dt}. \quad (2)$$

Según el tipo de contenido de materia en el universo este evolucionan de distinta manera. Esto se obtiene al escribir las ecuaciones de Einstein para un fluido perfecto, cuya descripción esta dada por

$$T_{\mu\nu} = pg_{\mu\nu} + (p + \rho)u_\mu u_\nu. \quad (3)$$

donde p es la presión, ρ la densidad de energía homogénea del fluido y u la velocidad del fluido medida por un observado comóvil. El resultado de estas ecuaciones son las llamadas ecuaciones de Friedmann, junto con la ecuación de continuidad

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p), \quad \text{1er ecuación de Friedmann.} \quad (4)$$

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho, \quad \text{2da ecuación de Friedmann.} \quad (5)$$

$$\dot{\rho} + 3H(\rho + p) = 0, \quad \text{Ecuación de continuidad.} \quad (6)$$

Los tipos de materia para los cuales soluciones de estas ecuaciones existen se basan en su ecuación de estado barotrópica, dada por la expresión $w = p/\rho$. La ecuación de estado

w para cada tipo de materia es

$$w = \begin{cases} 0 & \text{Radiación } (E \ll m) \\ \frac{1}{3} & \text{Materia } (E \gg m) \\ -1 & \text{Constante cosmológica} \\ w(t) & \text{Fluido genérico} \end{cases}, \quad (7)$$

Los tipos de soluciones para el factor de escala y la densidad de energía respectivamente son

$$a(t) \propto \begin{cases} t^{1/2} \\ t^{2/3} \\ e^{Ht} \\ e^{\int dt H} \end{cases}, \quad \rho(a) \propto \begin{cases} a^{-4} & \text{Radiación} \\ a^{-3} & \text{Materia} \\ \text{const} & \text{Constante cosmológica} \\ e^{-3 \int da(1+w(a))/a} & \text{Fluido genérico} \end{cases}, \quad (8)$$

Inflación

La fase de expansión exponencial del universo, llamada inflación cósmica, es el paradigma principal para explicar las propiedades del universo a escalas cosmológicas. Para que esta fase ocurra durante el tiempo suficiente se deben cumplir dos condiciones, las cuales se basan en los parámetros ϵ y η definidos como

$$\epsilon \equiv -\frac{\dot{H}}{H^2} = -\frac{d \ln H}{dN}, \quad \eta \equiv \frac{d \ln \epsilon}{dN} = \frac{\dot{\epsilon}}{\epsilon H}. \quad (9)$$

Estas condiciones que deben cumplirse simultáneamente son $\epsilon < 1$ y $|\eta| < 1$. Para obtener una solución satisfactoria del problema del horizonte cósmico se necesita mínimamente que el universo haya crecido un factor e^{50-60} . Una posible realización de esta fase inflacionaria, proveniente de la física de partículas, es la de un campo escalar (ϕ) que rueda hacia el mínimo de su potencial de manera lenta, efecto conocido en inglés como *slow-roll*. En dicho caso, inflación ocurre siempre cuando su energía potencial sea mucho mayor que su energía cinética $V(\phi) \gg \dot{\phi}^2$. Asimismo, inflación provee de un mecanismo natural para producir las perturbaciones necesarias en el fluido primordial que sirven de semilla para la formación de estructura. El mecanismo se puede entender como la conversión de fluctuaciones cuánticas de los campos de materia y radiación presentes en esta época, y como inflación las convierte en perturbaciones clásicas. Durante inflación las fluctuaciones cuánticas de los campos existen a escalas mucho más pequeñas que el tamaño del horizonte causal. Debido a la expansión exponencial estas fluctuaciones pequeñas son alargadas a la misma magnitud que el tamaño del horizonte, y eventualmente son más grandes y quedan congeladas adquiriendo

la amplitud imprimida en ellas al salir del horizonte. Una vez inflación termina y el universo se expande estas fluctuaciones congeladas anteriormente re-entran en el horizonte. Aquellas fluctuaciones cuya longitud de onda son del tamaño del universo observable son las mismas que entraron al horizonte más tarde y son fósiles de esta época tan lejana. Las observaciones de la fluctuaciones de temperatura en el mapa del fondo cósmico de microondas se ajustan de manera muy precisa a esta descripción, y son las cuales nos brindan los parámetros que permiten constreñir o falsear los distintos modelos inflacionarios. El espectro de las fluctuaciones de temperatura en el *CMB* sigue aproximadamente la parametrización

$$\Delta_s^2 = A_s(k_*) \left(\frac{k}{k_*} \right)^{n_s-1} \quad (10)$$

donde A_s es la amplitud de fluctuaciones escalares, n_s es la inclinación espectral y k_* la escala pivote. En este caso la inclinación espectral mide la dependencia de la amplitud de las fluctuaciones en función de su momento comóvil, o longitud de onda. Para el caso de inflación esta dependencia es prácticamente nula ($n_s \approx 1$) ya que la amplitud de fluctuaciones es proporcional al parámetro de Hubble durante inflación, el cuál se mantiene prácticamente constante. Por otro lado, no solo se producen fluctuaciones escalares sino también fluctuaciones tensoriales, u ondas gravitacionales primordiales, el ratio entre ambas amplitudes se conoce como el *tensor-to-scalar ratio*, definido como

$$r \equiv \frac{\Delta_t^2(k)}{\Delta_s^2(k)} . \quad (11)$$

Por un lado, las observaciones del CMB nos han permitido observar el espectro de fluctuaciones escalares. Por otro lado, nuestras capacidades tecnológicas actuales aún no nos han permitido medir ondas gravitacionales primordiales, para lo cual tenemos un límite superior. Los valores medidos más actualizados provienen del *Atacama Cosmology Telescope* (ACT) [7] que usan $k_* = 0.05 \text{Mpc}^{-1}$ y miden:

$$n_s = 0.974 \pm 0.003 \text{ (68\% CL)} , \quad (12)$$

mientras que Planck 2018 combinado con datos de BICEP2/Keck Array BK15 [25] suponen un límite superior:

$$r < 0.032 \text{ (95\% CL)} . \quad (13)$$

Finalmente, la amplitud de fluctuaciones escalares medida es:

$$A_s = 2.099_{-0.292}^{+0.296} \cdot 10^{-9} \text{ (68\% CL)} . \quad (14)$$

Recalentamiento de Universo

Una vez inflación termina la energía almacenada en el campo escalar inflacionario (al cual llamaremos inflatón) debe transferirse a los grados de libertad relativistas, y estos deben eventualmente llegar a equilibrio térmico. La era térmica del universo de alcanzarse mínimamente a temperaturas mayores a MeV, condición necesaria para la síntesis de los núcleos livianos. Inicialmente se consideró que el decaimiento perturbativo del inflatón a grados de libertad escalares o fermiónicos mediante acoples del tipo

$$\frac{1}{2}g^2\phi\chi^2, \quad \text{y} \quad h\phi\bar{\psi}\psi. \quad (15)$$

Se puede calcular la tasa de decaimiento del inflatón a los campos acoplados, usando técnicas perturbativas de teoría cuántica de campos, y obtener el momento al cual la energía del inflatón se transfiere a estos grados de libertad. En ese momento se dice que el universo está recalentado y la ecuación de estado del mismo será en la cuál la radiación domina, $w = 1/3$. Sin embargo, este método simplifica un escenario en el cual puede transcurrir la transferencia de energía. El campo inflacionario es una conjunto homogéneo de estados que oscilan coherentemente, son un condensado de Bose-Einstein. Por lo tanto hay efectos colectivos que se ven aumentados. Consideremos el Lagrangiano de interacción de dos grados de libertad: un campo inflacionario ϕ y un campo acoplado a este, el cual llamaremos campo hija, χ . La ecuación del campo χ en este contexto sería

$$\ddot{\chi} - \frac{1}{a^2}\nabla^2\chi + 3H\dot{\chi} + g^2\phi^2\chi = 0. \quad (16)$$

Ahora suponemos que el campo ϕ es un campo homogéneo, por lo tanto no depende de las coordenadas espaciales. Si escribimos la ecuación de cada modo de Fourier de campo χ , a los cuales denotaremos como χ_k , sería

$$\ddot{\chi}_k + 3H\dot{\chi}_k + \left(\frac{\mathbf{k}^2}{a^2} + g^2\phi^2\right)\chi_k = 0. \quad (17)$$

El campo ϕ después de finalizar inflación oscilará alrededor del mínimo de su potencial. Si el potencial es del tipo $V(\phi) \sim m^2\phi^2$ entonces las oscilaciones serán armónicas. Analizando la Eq. (17), esta sugiere que la frecuencia de oscilación efectiva de cada modo χ_k es $\omega_k^2 = k^2/a^2 + g^2\phi^2$, por lo tanto es una frecuencia que cambia con el tiempo. Si la tasa de cambio de esta sucede más rápido que su amplitud entramos en un régimen no adiabático. Las consecuencias de este régimen implican soluciones inestables para χ_k , es decir que crecen exponencialmente siempre y cuando la condición no adiabática se cumpla. El crecimiento exponencial del campo χ mediante este proceso se conoce como

resonancia paramétrica y es un efecto que tiene mucha relevancia en la transferencia de energía en el universo temprano, por lo tanto es un efecto que no puede ser obviado. Este efecto es no perturbativo ya que no depende analíticamente del acople g^2 , sin embargo en el límite en que la amplitud del campo χ no tiene ningún efecto sobre ϕ se está en el régimen lineal. Si la excitación exponencial del campo χ ocurre de manera eficiente y prolongada, este tendrá un efecto sobre la dinámica del campo inflacionario. La reacción que tiene sobre su entorno incitará sobretodo la fragmentación del inflatón, es decir la aproximación de un campo homogéneo ya no será válida. En este punto es necesario acudir a simulaciones en redes tridimensionales.

Ondas gravitacionales

Los sucesos del universo temprano se caracterizan por ser muy energéticos y violentos. Como se ha visto anteriormente, inflación es ya una fuente de ondas gravitacionales que conforman un fondo cósmico de modos tensoriales con un espectro casi invariante ante el cambio de escala. Por otro lado, el precalentamiento del universo se basa en la excitación exponencial de campos escalares. Este crecimiento desmedido libera grandes cantidades de energía y anisotropías en el sector de materia. Son precisamente estas anisotropías la fuente de ondas gravitacionales. Nuestra capacidad de detectar ondas gravitacionales es aún incipiente, y aún no se ha confirmado la observación de un fondo cósmico; sin embargo, existen varios planes futuros para extender esta capacidad a muchos rangos de frecuencia y mejorar la sensibilidad de los detectores.

Las ondas gravitacionales son identificadas con las perturbaciones tensoriales de la métrica espacial, denotadas como h_{ij} , con la propiedad de ser transversas y libre de traza

$$ds^2 = -dt^2 + a^2(t)(\delta_{ij} + h_{ij})dx^i dx^j, \quad \text{con } \partial_i h_{ij} = 0 \text{ y } h_{ii} = 0. \quad (18)$$

La dinámica de la ondas gravitacionales está descrita por la siguiente ecuación de movimiento

$$\ddot{h}_{ij} + 3H\dot{h}_{ij} - \frac{\nabla^2}{a^2}h_{ij} = \frac{2}{m_{\text{p}}^2 a^2}\Pi_{ij}^{\text{TT}}, \quad (19)$$

donde Π_{ij}^{TT} es la parte transversa y libre de traza (TT) del tensor anisótropo Π_{ij} . El cual se obtienen de extraer la parte espacial del tensor de energía momento del sector de materia. La fuente de ondas gravitacionales puede ser escrita como

$$\Pi_{ij}^{\text{TT}}(\mathbf{k}, t) = \Lambda_{ijkl}(\hat{\mathbf{k}})\Pi_{kl}(\mathbf{k}, t), \quad (20)$$

donde $\Lambda_{ij,kl}$ es un proyector definido como

$$\Lambda_{ij,lm}(\hat{\mathbf{k}}) \equiv P_{il}(\hat{\mathbf{k}})P_{jm}(\hat{\mathbf{k}}) - \frac{1}{2}P_{ij}(\hat{\mathbf{k}})P_{lm}(\hat{\mathbf{k}}), \quad (21)$$

$$P_{ij}(\mathbf{k}) = \delta_{ij} - \hat{k}_i \hat{k}_j, \quad \hat{k}_i = k_i/k. \quad (22)$$

Objetivos

Los objetivos de esta disertación son:

- Estudiar la dinámica e implicaciones fenomenológicas de los escenarios del universo temprano.
- Entender las implicaciones de distintos modelos inflacionarios en los observables que permitirán distinguirlos entre ellos.
- Comprender el régimen de precalentamiento del universo, y evaluar las implicaciones de los fenómenos no perturbativos en la eficiencia de la transferencia de energía.
- Usar la información de los fondos de ondas gravitacionales para entender, estudiar y constreñir los escenarios del universo temprano.
- Desarrollar técnicas y métodos de simulaciones en redes tridimensionales.

Metodología: Simulaciones en redes tridimensionales

Para el propósito de simular la dinámica de campos interactuantes en un universo en expansión acudiremos al uso de redes tridimensionales. Esto consiste básicamente en describir las ecuaciones de movimiento en términos de campos que dependen de coordenadas continuas $f(t, \mathbf{x})$ en funciones que dependen en coordenadas discretas $f(n_0 \delta t, \delta x \mathbf{n})$, donde δt es el paso temporal, δx es la distancia mínima de la red tridimensional, n_0 es el número de iteraciones y $\mathbf{n} = (n_1, n_2, n_3)$ es la ubicación de un punto de la red tridimensional. Esta red es evidentemente finita por lo tanto caracterizada por el número de puntos por dimensión $\mathcal{N}$. Por lo tanto el vector posición en la red se escribe

$$\mathbf{n} = (n_1, n_2, n_3) \quad \text{con} \quad n_i = 0, 1, \dots, \mathcal{N} - 1. \quad (23)$$

La longitud de la caja está dada por $L = \delta x \mathcal{N}$. Forzamos que las funciones definidas en la red tienen condiciones de frontera periódicas $f(\mathbf{n} + \hat{i}\mathcal{N}) = f(\mathbf{n})$. De la misma manera, podemos definir la red recíproca

$$\tilde{\mathbf{n}} = (\tilde{n}_1, \tilde{n}_2, \tilde{n}_3) \quad \text{with} \quad \tilde{n}_i = -\frac{\mathcal{N}}{2} + 1, -\frac{\mathcal{N}}{2}, \dots, -1, 0, 1, \dots, \frac{\mathcal{N}}{2}, \quad (24)$$

que es donde vive la transformada de Fourier discreta de un campo, la cual está definida por

$$f(\mathbf{n}) = \frac{1}{\mathcal{N}^3} \sum_{\tilde{\mathbf{n}}} e^{-i\frac{2\pi}{\mathcal{N}}\mathbf{n}\tilde{\mathbf{n}}} f(\tilde{\mathbf{n}}) \quad \Leftrightarrow \quad f(\tilde{\mathbf{n}}) = \sum_{\mathbf{n}} e^{+i\frac{2\pi}{\mathcal{N}}\mathbf{n}\tilde{\mathbf{n}}} f(\mathbf{n}) . \quad (25)$$

La red recíproca respeta las mismas condiciones de frontera periódicas y además está discretizada, lo cual implica la existencia de un momento mínimo, momento infrarrojo k_{IR} , definido como

$$k_{\text{IR}} = \frac{2\pi}{L} = \frac{2\pi}{\mathcal{N}\delta x} , \quad (26)$$

donde el momento en cada punto de la red recíproca está definido como $\mathbf{k} = \tilde{\mathbf{n}}k_{\text{IR}}$. Existe también un momento máximo, momento ultravioleta, definido como

$$k_{\text{UV}} = \frac{\mathcal{N}}{2}k_{\text{IR}} = \frac{\pi}{\delta x} . \quad (27)$$

Para poder estudiar la dinámica de los campos interactuantes se necesita escribir las ecuaciones de movimiento, la cuál necesita una noción de derivadas. En este caso, se usan el método de diferencias finitas en donde una derivada espacial o temporal se define como la diferencia entre la amplitud del campo entre un punto y su vecino, o un momento y el siguiente, normalizado por el espaciado o paso temporal. Al hacer esta identificación de manera sistemática se puede escribir un sistema de iteración. La evolución temporal de las ecuaciones se hace mediante la aplicación de algoritmos de evolución que nos permite saber el valor de un campo a un momento siguiente sabiendo el valor del mismo ahora. Al conjunto de técnicas de discretización y evolución las nombramos como *Lattice Cosmology*.

Formulación de una teoría de campos interactuantes en un fondo en expansión

En esta sección presentamos la formulación de una teoría de campos donde nos restringimos a considerar únicamente campos bosónicos y un sector gauge abeliano U(1). Consideramos la teoría de campos más general compuesta por campos escalares $\{\phi\}$, campos escalares complejos cargados bajo la simetría gauge U(1) $\{\varphi\}$ y un campo vectorial gauge U(1) A_μ . La acción se escribe como

$$\mathcal{S} = - \int dx^4 \sqrt{-g} \left\{ \frac{1}{2} \partial_\mu \phi \partial^\mu \phi + (D_\mu \varphi)^* (D^\mu \varphi) + \frac{1}{4} F_{\mu\nu} F^{\mu\nu} + V(\phi, |\varphi|) \right\} , \quad (28)$$

donde ϕ y φ son campos singletes y cargados respectivamente. El potencial $V(\phi, |\varphi|)$ describe las interacciones de estos

$$\phi \in \mathbb{R} \quad , \quad \varphi = \frac{1}{\sqrt{2}}(\varphi_1 + i\varphi_2) . \quad (29)$$

Introducimos la definición del derivado covariante y del tensor de campo del vector gauge U(1)

$$D_\mu \equiv \partial_\mu - ig_A Q_A A_\mu , \quad (30)$$

$$F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu , \quad (31)$$

donde g_A es el acoplamiento al sector U(1) y Q_A la carga de los campos cargados. Los campos eléctricos y magnéticos se definen como

$$E_i \equiv F_{0i} , \quad \text{y} \quad B_i \equiv \frac{1}{2} \epsilon_{ijk} F^{jk} . \quad (32)$$

El sistema satisface las siguientes transformaciones de simetría gauge bajo las cuales la acción (28) es invariante

$$\phi(x) \rightarrow \phi(x) , \quad (33)$$

$$\varphi(x) \rightarrow e^{-ig_A Q_A \alpha(x)} \varphi(x) , \quad (34)$$

$$A_\mu(x) \rightarrow A_\mu(x) - \partial_\mu \alpha(x) , \quad (35)$$

lo cual garantiza que $F_{\mu\nu}$ es automáticamente invariante bajo transformaciones de gauge.

Queremos seguir la dinámica de los campos en el contexto de un fondo en expansión. Consideramos una métrica FLRW conforme plana:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu = -a(\eta)^2 d\eta^2 + a(\eta)^2 \delta_{ij} dx^i dx^j \quad (36)$$

donde $a(\eta)$ es el factor de escala medido en tiempo α , el cual se caracteriza que para $\alpha = 0$ se recupera el tiempo coordinado/cósmico (t) y para $\alpha = 1$ el tiempo conforme ($d\tau \equiv a^{-1}dt$). Adoptamos la convención $\dot{f} \equiv df/dt$ y $f' \equiv df/d\eta$, y obtenemos el siguiente sistema de ecuaciones en derivadas parciales acopladas para la teoría completa

$$\phi'' - a^{-2(1-\alpha)} \vec{\nabla}^2 \phi + (3-\alpha) \frac{a'}{a} \phi' = -a^{2\alpha} \frac{\partial V}{\partial \phi} , \quad (37)$$

$$\varphi'' - a^{-2(1-\alpha)} \vec{D}_A^2 \varphi + (3-\alpha) \frac{a'}{a} \varphi' = -\frac{a^{2\alpha}}{2} \frac{\partial V}{\partial |\varphi|} \frac{\varphi}{|\varphi|} , \quad (38)$$

$$F'_{0i} - a^{-2(1-\alpha)} \partial_j F_{ji} + (1-\alpha) \frac{a'}{a} F_{0i} = a^{2\alpha} J_i^A = 2a^2 g_A Q_A^{(\varphi)} \text{Im}[\varphi^* \vec{D}_A \varphi] , \quad (39)$$

$$\partial_i F_{0i} = a^{2\alpha} J_0^A = 2a^2 g_A Q_A^{(\varphi)} \text{Im}[\varphi^* D_0^A \varphi] , \quad (40)$$

donde la última ecuación no es una ecuación de movimiento sino una ecuación de restricción: la condición de Gauss, que se asemeja a la ley de Gauss de la electrodinámica. La evolución de la métrica en expansión está dada por las ecuaciones de Friedmann introducidas en el capítulo anterior; aquí escribimos una forma generalizada con fuente en todos los campos de la teoría. Recordamos que el tensor energía-momento para un fluido perfecto en un fondo en expansión es

$$\bar{T}_{\mu\nu} \equiv (\bar{\rho} + \bar{p})u_\mu u_\nu + \bar{p}g_{\mu\nu} , \quad \Rightarrow \quad \begin{cases} \bar{\rho} = a^{-2\alpha} \bar{T}_{00} \\ \bar{p} = \frac{1}{3a^2} \sum_j \bar{T}_{jj} \end{cases} , \quad (41)$$

donde $\bar{p}$ y $\bar{\rho}$ representan sus componentes homogéneas. En este marco, las ecuaciones de Friedmann se escriben como

$$\mathcal{H}^2 \equiv \left(\frac{a'}{a} \right)^2 = a^{2\alpha} \frac{\bar{\rho}}{3m_p^2} , \quad \frac{a''}{a} = \frac{a^{2\alpha}}{6m_p^2} [(2\alpha - 1)\bar{\rho} - 3\bar{p}] . \quad (42)$$

Si el fluido tiene una ecuación de estado barotrópica $w = \bar{p}/\bar{\rho}$ y w es constante, la solución a las ecuaciones de Friedmann es

$$a(\eta) = a_i \left(1 + \frac{1}{p} \mathcal{H}_i (\eta - \eta_i) \right)^p , \quad \text{con} \quad p = \frac{2}{3(1+w) - 2\alpha} , \quad (43)$$

donde $a_i \equiv a(\eta_i)$ y $\mathcal{H}_i \equiv \mathcal{H}(\eta_i)$ para un tiempo inicial η_i . Si la expansión es auto consistente, es decir alimentada por la parte homogénea de los campos de materia, debemos identificar las partes homogéneas de la energía y presión derivadas del tensor energía-momento total de la teoría, dado por

$$\begin{aligned} T_{\mu\nu} &= -\frac{2}{\sqrt{-g}} \frac{\delta(\sqrt{-g}\mathcal{L})}{\delta g^{\mu\nu}} = g_{\mu\nu}\mathcal{L} - 2\frac{\delta\mathcal{L}}{\delta g^{\mu\nu}} \\ &= -g_{\mu\nu} \left(g^{\alpha\beta} \left[\frac{1}{2} \partial_\alpha \phi_a \partial_\beta \phi_a + (D_\alpha^A \varphi_b)^* (D_\beta^A \varphi_b) \right] + \frac{1}{4} g^{\alpha\delta} g^{\beta\lambda} F_{\alpha\beta} F_{\delta\lambda} + V(\phi, \varphi) \right) \\ &\quad + [(\partial_\mu \phi_a)(\partial_\nu \phi_a) + 2\text{Re}\{(D_\mu^A \varphi_b)^* (D_\nu^A \varphi_b)\}] + g^{\alpha\beta} F_{\mu\alpha} F_{\nu\beta} , \end{aligned} \quad (44)$$

de donde identificamos las densidades de energía y presión como

$$\rho = E_{K,\phi} + E_{G,\phi} + E_{K,\varphi} + E_{G,\varphi} + E_{K,U(1)} + E_{G,U(1)} + V , \quad (45)$$

$$p = E_{K,\phi} + E_{K,\varphi} + \frac{1}{3}(E_{G,\phi} + E_{G,\varphi}) + \frac{1}{3}(E_{K,U(1)} + E_{G,U(1)}) - V \quad (46)$$

con las siguientes definiciones para las distintas componentes energéticas

$$E_{K,\phi} \equiv \frac{1}{2a^{2\alpha}} \phi'^2, \quad E_{G,\phi} \equiv \frac{1}{2a^{2\alpha}} (\vec{\nabla}\phi)^2, \quad (47)$$

$$E_{K,\varphi} \equiv \frac{1}{2a^{2\alpha}} |D_0^A \varphi|^2, \quad E_{G,\varphi} \equiv \frac{1}{2a^{2\alpha}} |\vec{D}_A \varphi|^2, \quad (48)$$

$$E_{K,U(1)} \equiv \frac{1}{2a^{2+2\alpha}} |\vec{A}'|^2, \quad E_{G,U(1)} \equiv \frac{1}{2a^4} |\vec{B}|^2. \quad (49)$$

$$(50)$$

Las ecuaciones de Friedmann quedan entonces como

$$\left(\frac{a'}{a}\right)^2 = \frac{a^{2\alpha}}{3m_p^2} \langle E_{K,\phi} + E_{G,\phi} + E_{K,\varphi} + E_{G,\varphi} + E_{K,U(1)} + E_{G,U(1)} + V \rangle, \quad (51)$$

$$\begin{aligned} \frac{a''}{a} = & \frac{a^{2\alpha}}{3m_p^2} \langle (\alpha - 2)(E_{K,\phi} + E_{K,\varphi}) + \alpha(E_{G,\phi} + E_{G,\varphi}) \\ & + (\alpha - 1)(E_{K,U(1)} + E_{G,U(1)}) + (\alpha + 1)V \rangle, \end{aligned} \quad (52)$$

donde $\langle \dots \rangle$ denota el promedio espacial. Esto implica que la evolución del fondo está gobernada por la parte homogénea de los campos materia, garantizando que el volumen de promedio abarque todas las longitudes de onda relevantes.

Resultados

En este apartado, se resumirán los resultados más relevantes presentados en la disertación:

Espectroscopía de acoples de partículas con espectros de ondas gravitacionales

En este trabajo, demostramos la capacidad de inferir los acoples de partículas observando el espectro de ondas gravitacionales que generan. Para tal propósito, como prueba de concepto simulamos en redes tridimensionales un escenario de precalentamiento entre el inflatón y un campo hija, cuyo acople es descrito por $\frac{1}{2}g^2\phi^2\chi^2$. Medimos el espectro de ondas gravitacionales e identificamos la amplitud y posición del pico de excitación. Esto lo hacemos para distintos valores de acople g . Con esto obtenemos una parametrización que nos permite inferir el acople g en función de la ubicación del pico de excitación f_p . Luego procedemos a simular casos de precalentamiento donde existen múltiples campos hija acoplados al inflatón, $\frac{1}{2}g^2\phi^2\chi_j^2$, donde $j = 1, 2, 3, \dots$. El espectro resultante tiene una estructura similar a una escalera, donde la ubicación de cada pico de excitación de la escalera se corresponde con la ubicación del mismo en el caso aislado. Demostramos además que los distintos observatorios de ondas gravitacionales programados tienen la capacidad de distinguir esta forma espectral y medir la ubicación de los picos de

excitación. Es así que con la información de ubicación del pico de excitación y la parametrización encontrada previamente se puede inferir los acoples entre las partículas.

Producción de ondas gravitacionales durante precalentamiento con interacciones trilineales

En este trabajo, investigamos la producción de ondas gravitacionales durante el precalentamiento con potenciales inflacionarios del tipo monomial y polinomial, considerando un acoplamiento trilineal $\phi\chi^2$ entre el inflatón ϕ y un campo escalar hija χ , $\frac{1}{2}\sigma\phi\chi^2$. Encontramos que dicha interacción trilineal conduce a una producción exponencial de partículas χ . Esto se debe principalmente a la naturaleza del acoplamiento, el cual al ser trilineal induce una masa efectiva en el campo χ del tipo $m_\chi^2 \approx \sigma\phi$. Esto significa que cada vez que el campo escalar oscile en la rama negativa del potencial la masa efectiva al cuadrado del campo hija es negativa, induciendo una inestabilidad taquiónica y crecimiento exponencial. Como hemos observado anteriormente, estos procesos son eficientes al generar un fondo estocástico de ondas gravitacionales a lo largo del proceso. Estudiamos la dinámica lineal y no lineal del precalentamiento mediante simulaciones en redes tridimensionales, siguiendo la producción de GWs en todas las etapas relevantes. Estos fondos cósmicos están ubicados a frecuencias altas, lo que los hace indetectables por los observatorios de ondas gravitacionales actuales o planeados. Dado que la cantidad de ondas gravitacionales producida es notable en cualquier caso, discutimos las perspectivas de sondear indirectamente el fondo cósmico utilizando restricciones sobre el número efectivo de especies relativistas en el universo, N_{eff} .

Recalentamiento geométrico del universo

En este trabajo, investigamos la transferencia de energía posterior a inflación desde el inflatón (ϕ) hacia un campo escalar (χ) acoplado no mínimamente a gravedad mediante $\xi R|\chi|^2$, considerando modelos con un potencial inflacionario $V_{\text{inf}} \approx |\phi|^p$ alrededor de $\phi = 0$. Ampliamos nuestro conocimiento sobre los escenarios de precalentamiento con acoplamientos directos entre partículas, al considerar este acoplamiento indirecto. El escenario, como otros en precalentamiento, se basan en una inestabilidad taquiónica inducida por el acoplamiento; es este caso la masa efectiva del campo χ está dada por $m_\chi^2 \approx \xi R$, entonces cada vez que el escalar de Ricci se vuelve negativa el campo χ crece exponencialmente. Extendemos este escenario a su régimen no lineal mediante simulaciones en redes tridimensionales. Considerando potenciales tipo α -attractors como referencia, estudiamos la viabilidad de un recalentamiento adecuado para $p = 2, 4, 6$, determinando si se puede alcanzar una dominación de radiación (RD) debido a la dominancia energética de χ sobre ϕ . Observamos que el caso que supone una transferencia eficaz de energía sucede cuando $p = 6$, esto se debe sobre todo a una

característica de este tipo de oscilaciones y es que para $p > 4$ el escalar de Ricci inducido por el inflatón homogéneo es en promedio negativo. Esta característica es fundamental para que la inestabilidad taquiónica sea activada durante la mayor cantidad de tiempo posible. De la misma manera los casos con $p = 2$ y $p = 4$ no son eficientes ya que las oscilaciones inducidas en el escalar de Ricci no son lo suficientemente negativas. También estudiamos la capacidad de este mecanismo ante la posibilidad de bajar la escala de energía de inflación. Encontramos que al bajar la escala por debajo de un umbral las oscilaciones homogéneas del inflatón son frustradas por la fragmentación del mismo y por lo tanto no puede sostener la transferencia de energía.

Simulaciones en redes tridimensionales de inflación con axiones

En este capítulo presentamos un estudio de la dinámica no lineal completamente inhomogénea de la inflación con axiones. Argumentamos la necesidad de utilizar simulaciones en redes tridimensionales para ir más allá de los métodos analíticos y semi-analíticos desarrollados bajo suposiciones restrictivas. Demostramos que dichos métodos fallan al describir con precisión la dinámica verdaderamente local en los casos donde la retroalimentación es fuerte y no despreciable. Utilizamos técnicas en redes tridimensionales que preservan explícitamente la invariancia *gauge* y la simetría de desplazamiento, característica fundamental de las teorías con axiones, y que validamos frente a otros métodos computacionales aplicados a la dinámica lineal y al régimen homogéneo de retro-alimentación. Investigamos la convergencia de las simulaciones con retro-alimentación local, determinando los requisitos para lograr una descripción precisa de la dinámica, y proporcionando parametrizaciones útiles del retraso en el final de la inflación. Identificamos características clave que emergen de un tratamiento local apropiado de la retro-alimentación fuerte. Observamos que la dominancia de la energía magnética sobre la eléctrica es una característica de esta descripción, sobretudo el campo magnético domina por sobre el eléctrico a las escalas del pico de excitación. Además, descubrimos que dinámica no lineal del inflatón induce una excitación del modo longitudinal. Por último, se puede ver que el (des)balance quirral que en el régimen lineal es totalmente positivo y máximo, se relaja sobretudo a escalas que permanecen fuera del horizonte durante los último momentos de inflación. Estos resultados demuestran la necesidad de acudir a simulaciones en redes tridimensionales para poder predecir de manera adecuada las consecuencias fenomenológicas de estos escenarios.

Conclusiones

La física del universo temprano ofrece una ventana única para explorar física más allá de cualquier escala de energía alcanzable en experimentos terrestres. Además, varias extensiones del Modelo Estándar de la física de partículas podrían dar lugar

a marcas observables distintivas en forma de fondos de ondas gravitacionales. El objetivo de la comunidad de cosmología teórica, y sobre todo del universo temprano, es realizar predicciones cada vez más precisas que pronto serán contrastadas con observaciones mejoradas y más extendidas de señales de ondas gravitacionales. En este contexto, hemos visto que muchos de los fenómenos que conducen a señales observables son intrínsecamente no lineales. El campo emergente de la Cosmología en redes tridimensionales, o *Lattice Cosmology*, se perfila así como una herramienta poderosa para abordar la compleja dinámica involucrada y realizar predicciones fiables orientadas a mejorar nuestra comprensión sobre el origen del universo.

A la luz de los resultados presentados en esta tesis, se pueden seguir varias líneas futuras de investigación para ampliar nuestro conocimiento en estas áreas. A continuación, discutimos estas posibles direcciones.

- **Espectroscopía de acoplamientos mediante ondas gravitacionales:**

Se espera que la aparición de señales características, en forma de escaleras del espectro del fondo de ondas gravitacionales, vistas en el escenario de precalentamiento estén también presente en otros tipos de escenarios. Entre ellos, podemos considerar el pre-recalentamiento taquiónico [50–53, 63, 64], el pre-recalentamiento axiónico [65–68], o el pre-recalentamiento geométrico [46–49]. En el caso de transiciones de fase de primer orden, los campos gauge acoplados al sector de Higgs correspondiente de la teoría serán inevitablemente excitados durante la nucleación y colisión de burbujas. Esto podría conducir a nuevos picos en el espectro de ondas gravitacionales, permitiendo explorar los acoplamientos Higgs–gauge. También se podría considerar la posibilidad de acoplamientos entre las especies hijas y analizar cómo se ve afectada la estructura de picos, y si esta ofrece una vía para medir dichos acoplamientos.

- **GWBs provenientes del pre-recalentamiento vía interacciones trilineales:** Los escenarios de pre-recalentamiento con un potencial monomial o polinómico para el inflatón, junto con un acoplamiento trilineal entre el inflatón y un campo hijo, conducen a una producción muy eficiente de ondas gravitacionales. Sin embargo, la señal resultante se sitúa fuera de la sensibilidad de las sondas actuales y planificadas, tanto directas como indirectas, del fondo de ondas gravitacionales. Una posible línea futura consiste en considerar el mismo tipo de acoplamiento en escenarios inflacionarios de menor escala energética, donde la frecuencia pico del ondas gravitacionales podría desplazarse hacia valores más bajos. Es importante destacar que no hay razones particulares para esperar que la eficiencia de producción de ondas gravitacionales se vea significativamente reducida en tales escenarios.

- **Recalentamiento geométrico del universo:** El pre-recalentamiento mediante acoplamientos no mínimos a la gravedad ha sido explorado en varios contextos, como el Recalentamiento de Ricci. No obstante, la idea de producción de partículas vía interacciones gravitatorias podría ser relevante en muchos otros contextos del universo temprano, incluyendo materia oscura, ondas gravitacionales, radiación oscura o la asimetría bariónica. Estos mecanismos pueden ser altamente eficientes y pueden llevar a una producción exponencial de partículas, resultando en dinámicas no lineales que invalidan rápidamente los supuestos de los tratamientos analíticos. En este contexto, las formulaciones en redes tridimensionales de estos escenarios serán necesarias para abordar plenamente las implicaciones de tales interacciones en el universo primitivo.
- **Inflación axiónica:** La inflación axiónica es un escenario bien motivado y con una fenomenología rica en el universo temprano. Por lo tanto, una descripción adecuada de su dinámica bajo retroalimentación fuerte debe involucrar un tratamiento completamente inhomogéneo, incluyendo las fluctuaciones locales de $F\tilde{F}$ y la presencia de gradientes del inflatón. Esto es esencial para realizar predicciones realmente fiables de las señales asociadas. Por ejemplo, la amplitud y frecuencia actual del fondo de ondas gravitacionales generado durante la inflación por el campo gauge excitado dependen fuertemente de cuánto se prolongue la inflación. Además, la producción de perturbaciones escalares del espacio-tiempo y la abundancia resultante de agujeros negros primordiales son altamente sensibles a la naturaleza inherentemente local de la dinámica no lineal.
- **Perturbaciones cosmológicas originadas por dinámicas no lineales:** La capacidad de explorar escenarios del universo primitivo dependerá en gran medida de la evaluación correcta de sus consecuencias fenomenológicas. Hemos visto que pueden existir procesos muy energéticos, caracterizados por dinámicas no lineales, que son fuentes eficientes de perturbaciones cosmológicas. Estas perturbaciones pueden, a su vez, generar señales observables en nuestras observaciones actuales del Universo. Por ejemplo, la aparición de correlaciones no gaussianas en el espectro de potencia de las anisotropías del CMB o en el espectro de potencia de materia de la estructura a gran escala. Además, la posibilidad de que se formen agujeros negros primordiales debido a una forma particular del espectro de potencia escalar es otra consecuencia fenomenológica de dichas dinámicas. Es, por tanto, esencial tener un control preciso sobre el mecanismo exacto mediante el cual se crean las perturbaciones cosmológicas.

El universo primitivo ofrece una ventana amplia y única hacia una física que va más allá de nuestro conocimiento actual, y describe escenarios a energías fuera del alcance de

nuestras capacidades tecnológicas. La posibilidad de que esta era estuviera gobernada por dinámicas no lineales muy energéticas introduce la necesidad de desarrollar herramientas capaces de investigar tales escenarios. La herramienta más precisa capaz de describir la dinámica de sistemas caracterizados por sistemas acoplados no lineales de ecuaciones en derivadas parciales son las técnicas de redes (lattice). Por lo tanto, es necesario comenzar a explorar líneas de investigación dentro de nuestro programa cosmológico teórico actual que incluyan simulaciones computacionales de la dinámica de teorías de campos en un contexto cosmológico. De esta manera, pueden realizarse predicciones precisas y confiables que luego podrán confrontarse con observaciones más precisas y de mayor alcance, ya sea del fondo de ondas gravitacionales, o de observaciones detalladas del CMB y la estructura a gran escala (LSS). Así es como podremos responder algunas preguntas primordiales sobre el origen de nuestro Universo y, quizás, sobre cuál será su destino final.

